Configuration Summary

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DHW ANALYSIS - CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-10-15 22:17:30
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-- Data Preparation --

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lat_size........................ 64
  target_lon_size........................ 128
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics_64x128.pdf
  output_dir............................. data/processed_64x128/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

-- Model Training --

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 80
  num_layers............................. 5
  num_modes_lat.......................... 16
  num_modes_lon.......................... 32
  mlp_ratio.............................. 2.5
  activation............................. gelu
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norm_type.............................. instance_norm
dropout_rate........................... 0.05
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 8
rollout_epoch_start.................... 10
rollout_epoch_end...................... 60
rollout_loss_decay..................... 0.92
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0002
min_learning_rate...................... 1e-06
batch_size............................. 16
num_epochs............................. 120
weight_decay........................... 1e-05
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 10
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 20
early_stopping_metric.................. val_loss
input_data_path........................ data/processed_64x128/sst_final.nc
checkpoint_dir......................... checkpoints_64x128/
best_model_path........................ checkpoints_64x128/best_model.pt
training_log_path...................... results_64x128/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

-- Model Inference --

model_path............................. checkpoints_64x128/best_model.pt
input_data_path........................ data/processed_64x128/sst_final.nc
run_validation......................... True
run_test............................... True
forecast_lead_steps.................... 90
chain_stride_steps..................... 90
output_dir............................. results_64x128/
val_predictions_file................... predictions_val.nc
test_predictions_file.................. predictions_test.nc

-- Concat Chains --
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```
  predictions_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  predictions_file_template..... predictions_{split}.nc
  std_params_path..... data/processed_64x128/std_params.nc
  trend_params_path..... data/processed_64x128/trend_params.nc
  output_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  actual_file_template..... sst_actual_{split}_continuous.nc
  predicted_file_template..... sst_predicted_{split}_chains.nc
  splits..... ['test', 'val']
  lon_correction_threshold..... 0.6

-- Calc DHW --

  mmm_path..... /home/272/xl5758/ct5km_climatology_v3.1.nc
  results_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  predicted_file_template..... sst_predicted_{split}_chains.nc
  actual_file_template..... sst_actual_{split}_continuous.nc
  dhw_output_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_64x128
  output_file_template..... dhw_products_{split}.nc
  run_test..... True
  run_validation..... True
  window_days..... 84
  hotspot_threshold..... 1

-- DHW Plots --

  dhw_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_64x128
  output_pdf..... dhw_analysis_combined.pdf
  output_csv..... dhw_metrics_combined.csv
  spatial_lead_steps..... [1, 7, 14, 30, 60, 90]
  n_sample_chains..... 6
  plot_init_dates..... ['2016-05-20', '2016-10-15', '2017-02-10', '2017-07-09', '20
  dhw_vmax..... 20
  dhw_diff_vmax..... 8
  tropic_lat..... 30
  dhw_cmap..... ['#CCe6F0', '#441C94', '#6F47BF', '#815CCC', '#A383E6', '#ED
  error_cmap..... ['#053061', '#2166AC', '#92C5DE', '#F7F7F7', '#F4A582', '#D6
  rmse_cmap..... YlOrRd
  bias_cmap..... RdBu_r
  bleaching_thresholds..... [(0, 4, 'No Stress', '#27AE60'), (4, 8, 'Warning', '#F39C12'
  gbr_lat_bounds..... (-25.0, -10.0)
  gbr_lon_bounds..... (142.0, 157.0)
  reef_boxes..... [{'name': 'Great Barrier Reef', 'lat': (-25.0, -10.0), 'lon'
```

DHW Analysis - Summary

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DEGREE HEATING WEEKS (DHW) - COMBINED ANALYSIS
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METHODOLOGY  (NOAA Coral Reef Watch v3.1)
  SSTA      = SST - DC              (DC = daily climatology)
  HotSpot   = max(SST - MMM, 0)     (MMM = Maximum Monthly Mean)
  DHW_i     = (1/7) x sum(HotSpot_j) for j=i-83..i, HotSpot_j >= 1.0C

SLIDING WINDOW CONSTRUCTION  (always 84 days ending on valid date)
  Lead L:  (84-L) actual + L predicted SST days
  Lead 1:  83 actual +  1 predicted
  Lead 30: 54 actual + 30 predicted
  Lead 84:  0 actual + 84 predicted
  Lead 90: previous 84 predicted days (days 7-90)

VERIFICATION
  Actual DHW  : 84 observed days ending on the forecast valid date
  Persistence : observed DHW on the day before lead 1, held constant
  Sample      : chains defined at every lead step (same chains for
                model and persistence at all leads)

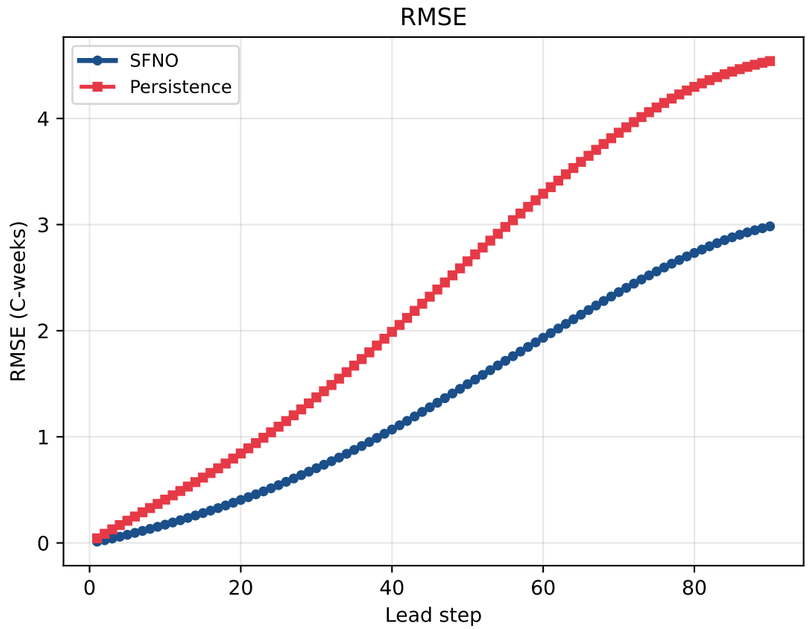
BLEACHING ALERT THRESHOLDS
  DHW  0-4  : No Stress
  DHW  4-8  : Warning
  DHW  8-12 : Alert Level 1
  DHW 12-16 : Alert Level 2
  DHW 16-20 : Alert Level 3
  DHW 20->= : Alert Level 4-5

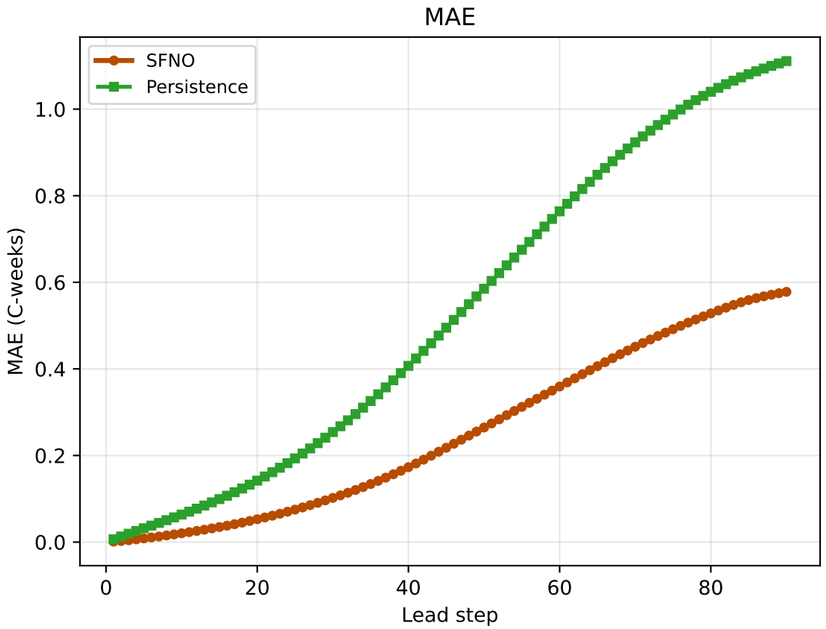
DATASETS
  Test          19 chains x 90 leads x 64 lat x 128 lon
  Val           16 chains x 90 leads x 64 lat x 128 lon
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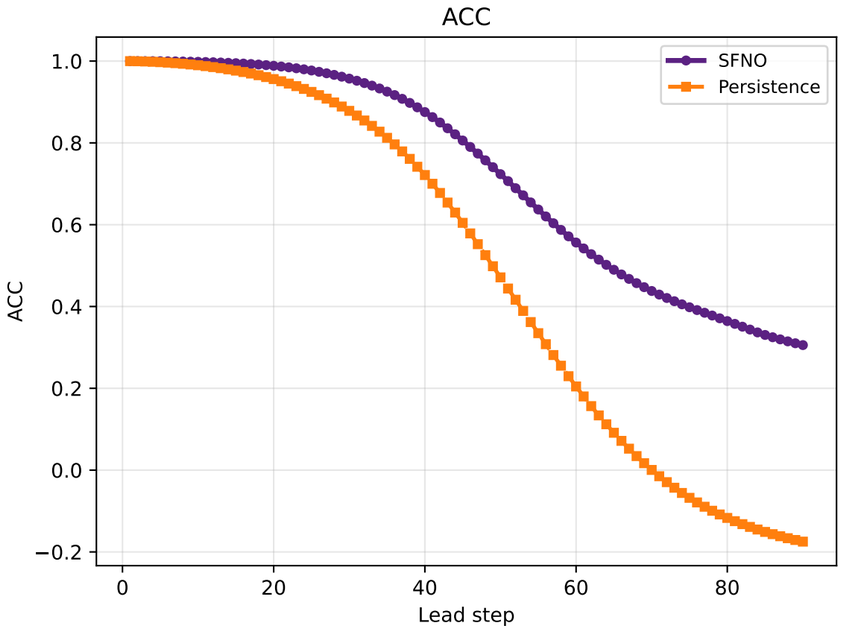
DHW Metrics by Lead Step - Test (18 chains scored at every lead)

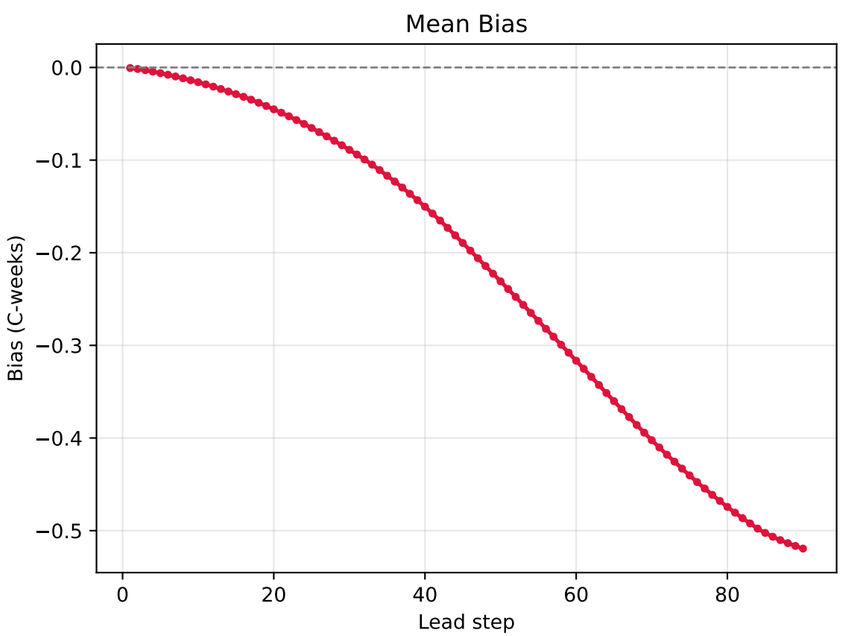
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0123	0.0437	71.9%	0.0011	0.0065	-0.0006	1.0000	0.9999	1.0000
2	0.0258	0.0862	70.1%	0.0025	0.0130	-0.0016	1.0000	0.9996	0.9999
3	0.0415	0.1278	67.5%	0.0043	0.0193	-0.0029	0.9999	0.9990	0.9998
4	0.0586	0.1687	65.2%	0.0064	0.0257	-0.0045	0.9998	0.9983	0.9996
5	0.0766	0.2091	63.4%	0.0085	0.0319	-0.0061	0.9996	0.9973	0.9994
6	0.0947	0.2491	62.0%	0.0107	0.0381	-0.0078	0.9995	0.9962	0.9991
7	0.1132	0.2888	60.8%	0.0130	0.0443	-0.0097	0.9992	0.9949	0.9987
8	0.1323	0.3285	59.7%	0.0153	0.0507	-0.0117	0.9989	0.9934	0.9982
9	0.1520	0.3680	58.7%	0.0178	0.0571	-0.0138	0.9986	0.9917	0.9976
10	0.1721	0.4079	57.8%	0.0203	0.0635	-0.0159	0.9982	0.9898	0.9969
11	0.1927	0.4484	57.0%	0.0230	0.0702	-0.0183	0.9977	0.9877	0.9961
12	0.2139	0.4893	56.3%	0.0257	0.0772	-0.0207	0.9971	0.9853	0.9952
13	0.2359	0.5310	55.6%	0.0286	0.0844	-0.0233	0.9965	0.9827	0.9941
14	0.2585	0.5733	54.9%	0.0316	0.0918	-0.0260	0.9957	0.9798	0.9928
15	0.2811	0.6158	54.3%	0.0347	0.0993	-0.0288	0.9949	0.9767	0.9914
16	0.3044	0.6590	53.8%	0.0379	0.1071	-0.0317	0.9939	0.9733	0.9898
17	0.3284	0.7031	53.3%	0.0413	0.1152	-0.0348	0.9928	0.9695	0.9880
18	0.3531	0.7484	52.8%	0.0449	0.1237	-0.0381	0.9915	0.9654	0.9859
19	0.3787	0.7951	52.4%	0.0488	0.1326	-0.0416	0.9901	0.9609	0.9836
20	0.4049	0.8429	52.0%	0.0528	0.1419	-0.0452	0.9885	0.9560	0.9810
21	0.4312	0.8913	51.6%	0.0568	0.1515	-0.0489	0.9867	0.9507	0.9781
22	0.4580	0.9406	51.3%	0.0610	0.1614	-0.0527	0.9847	0.9450	0.9749
23	0.4859	0.9912	51.0%	0.0655	0.1717	-0.0568	0.9824	0.9387	0.9713
24	0.5148	1.0427	50.6%	0.0702	0.1824	-0.0610	0.9799	0.9319	0.9673
25	0.5447	1.0951	50.3%	0.0750	0.1933	-0.0653	0.9770	0.9246	0.9628
26	0.5758	1.1486	49.9%	0.0801	0.2046	-0.0698	0.9737	0.9167	0.9578
27	0.6073	1.2029	49.5%	0.0853	0.2163	-0.0744	0.9702	0.9081	0.9523
28	0.6390	1.2585	49.2%	0.0907	0.2285	-0.0791	0.9662	0.8989	0.9463
29	0.6716	1.3152	48.9%	0.0963	0.2411	-0.0840	0.9619	0.8889	0.9397
30	0.7042	1.3721	48.7%	0.1020	0.2541	-0.0889	0.9571	0.8783	0.9326
31	0.7369	1.4298	48.5%	0.1079	0.2675	-0.0940	0.9519	0.8669	0.9250
32	0.7701	1.4883	48.3%	0.1140	0.2813	-0.0993	0.9463	0.8547	0.9167
33	0.8043	1.5482	48.0%	0.1203	0.2956	-0.1049	0.9400	0.8416	0.9076
34	0.8398	1.6090	47.8%	0.1271	0.3104	-0.1107	0.9330	0.8274	0.8976
35	0.8763	1.6707	47.5%	0.1341	0.3257	-0.1169	0.9253	0.8123	0.8868
36	0.9133	1.7328	47.3%	0.1414	0.3414	-0.1231	0.9169	0.7963	0.8751
37	0.9510	1.7960	47.1%	0.1489	0.3573	-0.1296	0.9077	0.7792	0.8625
38	0.9894	1.8599	46.8%	0.1566	0.3737	-0.1363	0.8977	0.7609	0.8491
39	1.0287	1.9244	46.5%	0.1647	0.3902	-0.1433	0.8869	0.7417	0.8347
40	1.0684	1.9890	46.3%	0.1730	0.4070	-0.1503	0.8753	0.7214	0.8195
41	1.1086	2.0544	46.0%	0.1816	0.4241	-0.1577	0.8630	0.7001	0.8035
42	1.1498	2.1205	45.8%	0.1904	0.4416	-0.1653	0.8498	0.6776	0.7868
43	1.1920	2.1870	45.5%	0.1994	0.4592	-0.1731	0.8358	0.6541	0.7693
44	1.2349	2.2540	45.2%	0.2086	0.4771	-0.1812	0.8211	0.6297	0.7513
45	1.2778	2.3209	44.9%	0.2179	0.4951	-0.1894	0.8058	0.6045	0.7329
46	1.3209	2.3878	44.7%	0.2272	0.5132	-0.1977	0.7901	0.5787	0.7142
47	1.3644	2.4546	44.4%	0.2365	0.5313	-0.2059	0.7739	0.5524	0.6953
48	1.4083	2.5213	44.1%	0.2459	0.5494	-0.2143	0.7574	0.5255	0.6762
49	1.4520	2.5876	43.9%	0.2553	0.5673	-0.2226	0.7406	0.4985	0.6572
50	1.4960	2.6535	43.6%	0.2646	0.5852	-0.2309	0.7236	0.4712	0.6381
51	1.5394	2.7188	43.4%	0.2740	0.6031	-0.2392	0.7066	0.4439	0.6193
52	1.5836	2.7840	43.1%	0.2835	0.6211	-0.2477	0.6892	0.4166	0.6005
53	1.6279	2.8487	42.9%	0.2931	0.6391	-0.2563	0.6718	0.3892	0.5819
54	1.6720	2.9131	42.6%	0.3027	0.6572	-0.2649	0.6545	0.3620	0.5635
55	1.7162	2.9772	42.4%	0.3123	0.6752	-0.2735	0.6372	0.3348	0.5454
56	1.7599	3.0409	42.1%	0.3217	0.6931	-0.2821	0.6202	0.3079	0.5278
57	1.8033	3.1042	41.9%	0.3311	0.7110	-0.2906	0.6035	0.2813	0.5108
58	1.8467	3.1668	41.7%	0.3405	0.7287	-0.2993	0.5873	0.2552	0.4943
59	1.8899	3.2292	41.5%	0.3499	0.7463	-0.3079	0.5716	0.2296	0.4784
60	1.9332	3.2912	41.3%	0.3593	0.7638	-0.3166	0.5564	0.2045	0.4632
61	1.9767	3.3528	41.0%	0.3687	0.7813	-0.3252	0.5419	0.1801	0.4486
62	2.0205	3.4139	40.8%	0.3782	0.7985	-0.3340	0.5279	0.1565	0.4346
63	2.0641	3.4741	40.6%	0.3876	0.8155	-0.3427	0.5146	0.1338	0.4213
64	2.1075	3.5329	40.3%	0.3969	0.8320	-0.3514	0.5019	0.1121	0.4086
65	2.1511	3.5910	40.1%	0.4063	0.8483	-0.3602	0.4898	0.0913	0.3965
66	2.1945	3.6481	39.8%	0.4155	0.8641	-0.3688	0.4784	0.0715	0.3851
67	2.2374	3.7045	39.6%	0.4247	0.8795	-0.3774	0.4675	0.0524	0.3742
68	2.2802	3.7596	39.4%	0.4337	0.8945	-0.3859	0.4571	0.0343	0.3638
69	2.3222	3.8136	39.1%	0.4425	0.9091	-0.3942	0.4473	0.0169	0.3539
70	2.3634	3.8659	38.9%	0.4512	0.9233	-0.4023	0.4379	0.0005	0.3446
71	2.4038	3.9166	38.6%	0.4597	0.9370	-0.4102	0.4292	-0.0149	0.3358
72	2.4436	3.9656	38.4%	0.4679	0.9503	-0.4180	0.4208	-0.0294	0.3275
73	2.4827	4.0132	38.1%	0.4759	0.9631	-0.4255	0.4129	-0.0431	0.3196
74	2.5210	4.0591	37.9%	0.4839	0.9755	-0.4330	0.4053	-0.0558	0.3121
75	2.5588	4.1036	37.6%	0.4917	0.9875	-0.4403	0.3982	-0.0678	0.3050
76	2.5957	4.1466	37.4%	0.4994	0.9991	-0.4476	0.3913	-0.0790	0.2982
77	2.6314	4.1876	37.2%	0.5068	1.0101	-0.4545	0.3847	-0.0895	0.2916
78	2.6661	4.2264	36.9%	0.5141	1.0206	-0.4613	0.3779	-0.0993	0.2851
79	2.6999	4.2630	36.7%	0.5212	1.0305	-0.4679	0.3711	-0.1084	0.2785
80	2.7328	4.2978	36.4%	0.5282	1.0400	-0.4744	0.3644	-0.1168	0.2720
81	2.7644	4.3305	36.2%	0.5349	1.0491	-0.4805	0.3575	-0.1248	0.2655
82	2.7952	4.3612	35.9%	0.5414	1.0576	-0.4865	0.3507	-0.1321	0.2590
83	2.8249	4.3898	35.6%	0.5477	1.0657	-0.4922	0.3437	-0.1388	0.2524
84	2.8536	4.4167	35.4%	0.5538	1.0733	-0.4978	0.3367	-0.1452	0.2459
85	2.8799	4.4418	35.2%	0.5589	1.0805	-0.5024	0.3304	-0.1511	0.2402
86	2.9040	4.4649	35.0%	0.5635	1.0873	-0.5066	0.3248	-0.1565	0.2350
87	2.9262	4.4865	34.8%	0.5676	1.0937	-0.5102	0.3197	-0.1616	0.2303
88	2.9466	4.5062	34.6%	0.5713	1.0998	-0.5135	0.3148	-0.1664	0.2259
89	2.9657	4.5244	34.5%	0.5747	1.1055	-0.5165	0.3101	-0.1708	0.2218
90	2.9833	4.5408	34.3%	0.5780	1.1109	-0.5193	0.3057	-0.1748	0.2178

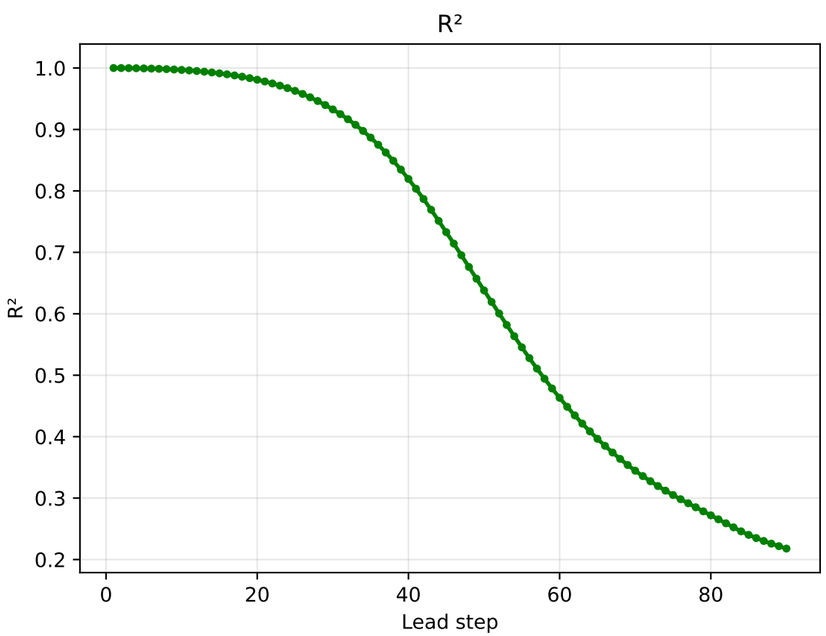
DHW Forecast Skill vs Lead Step - Test

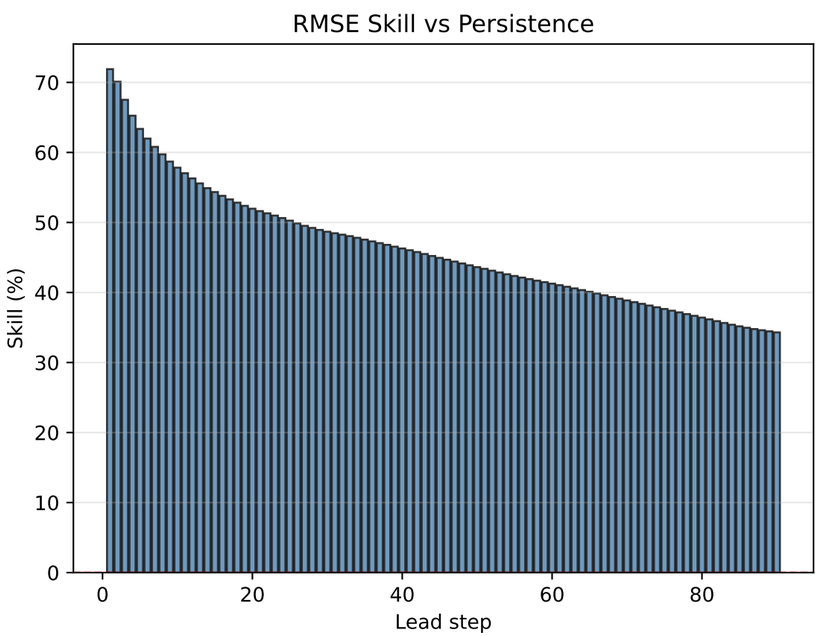


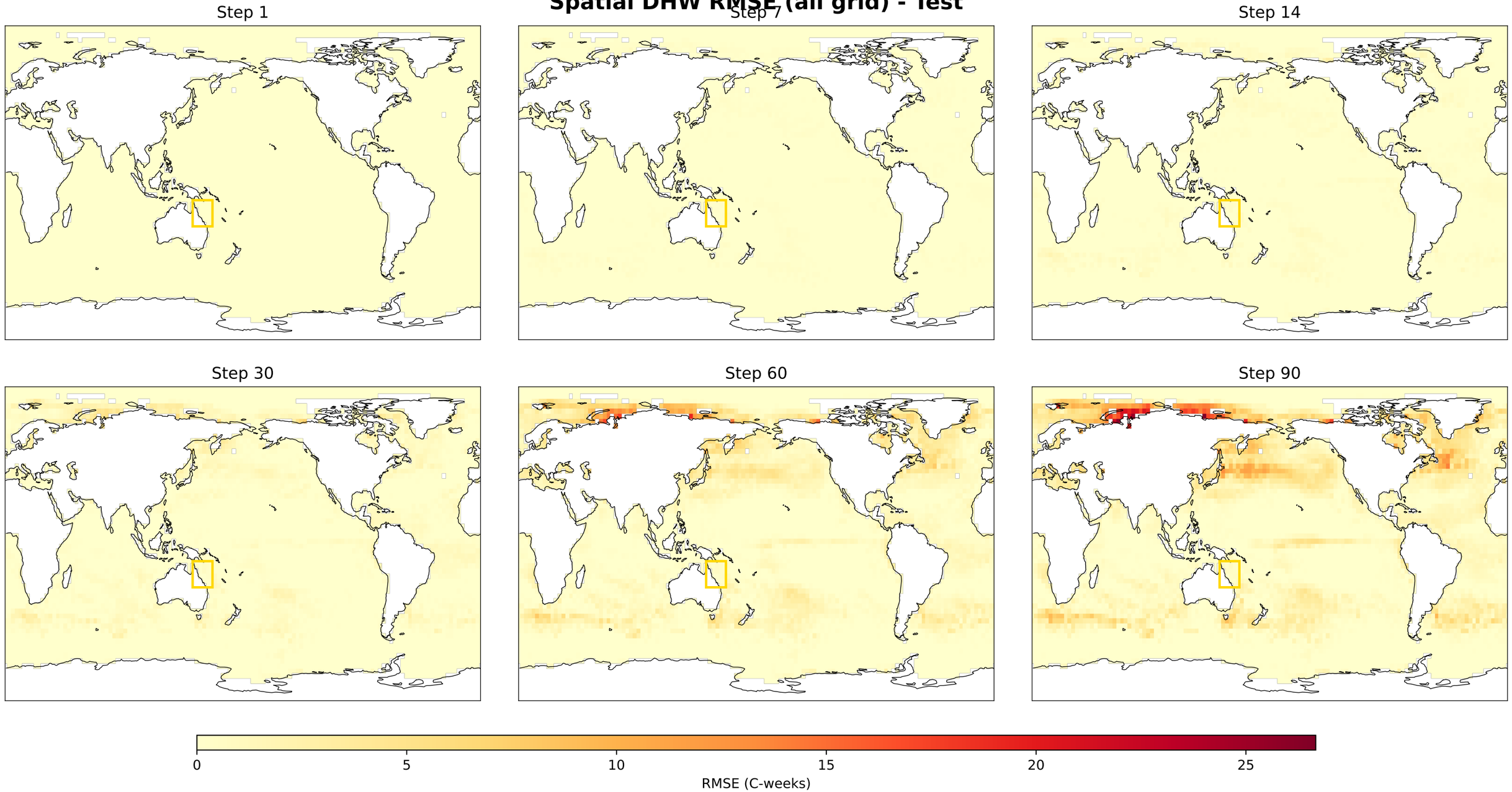


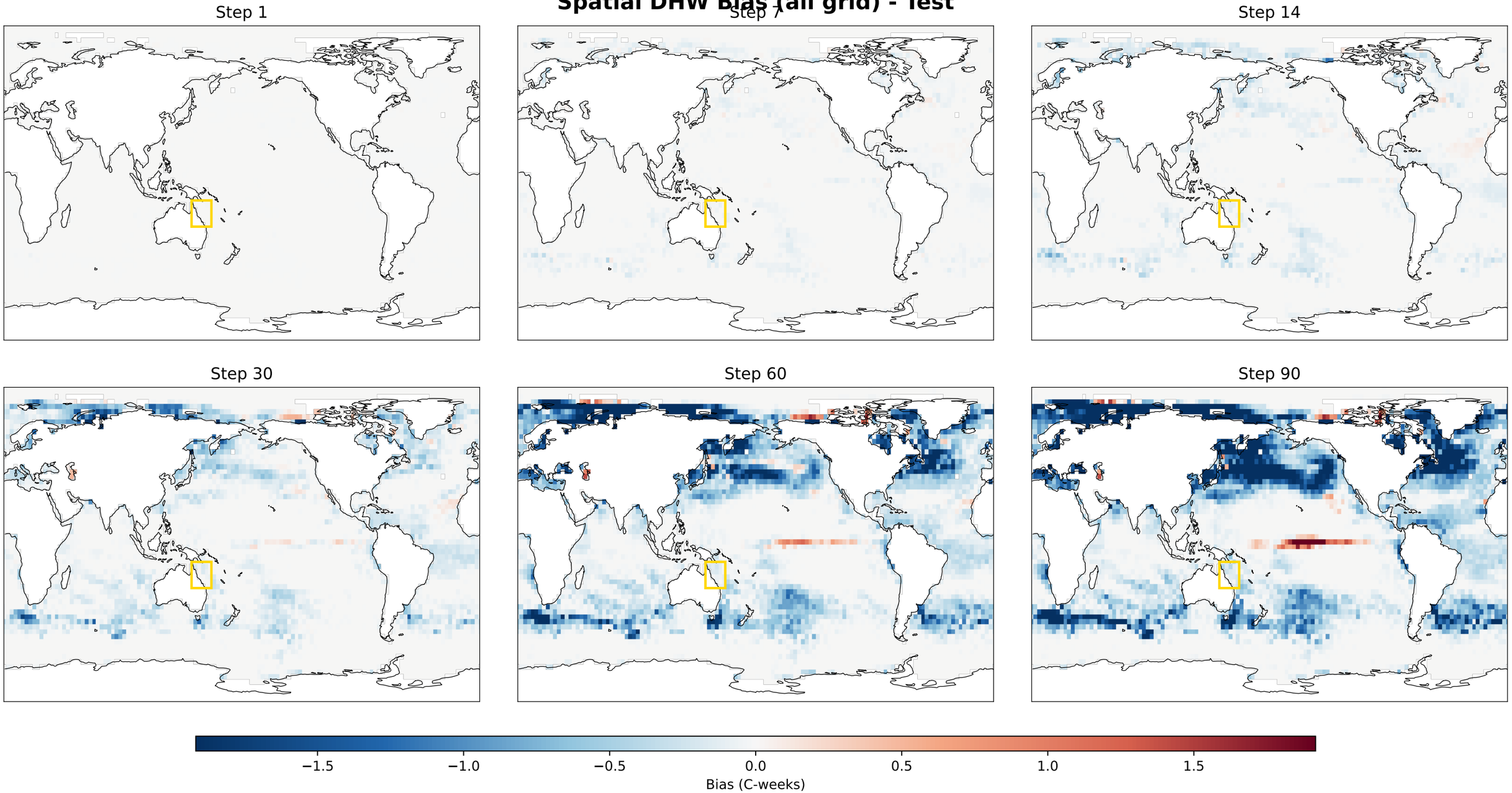


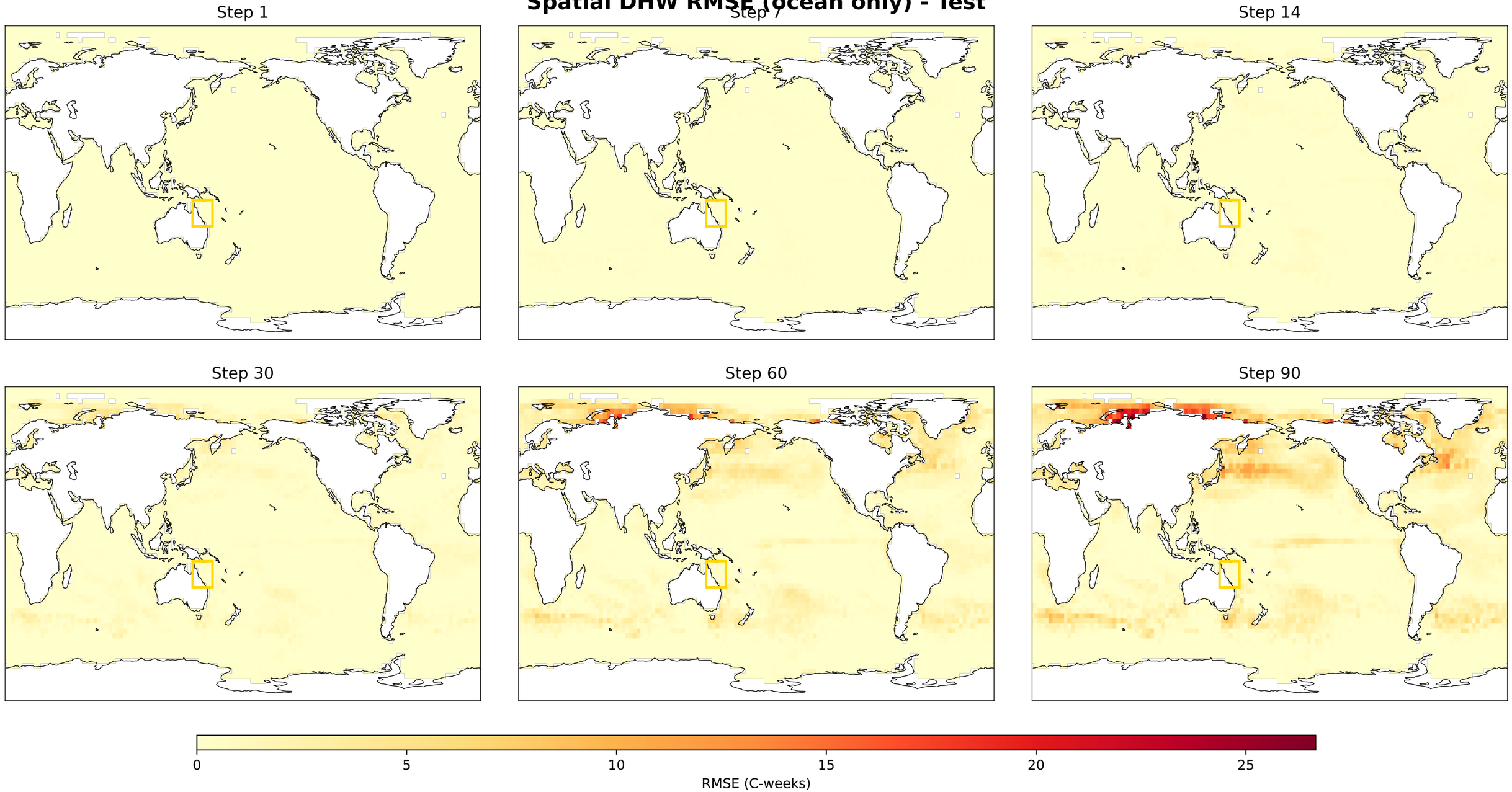


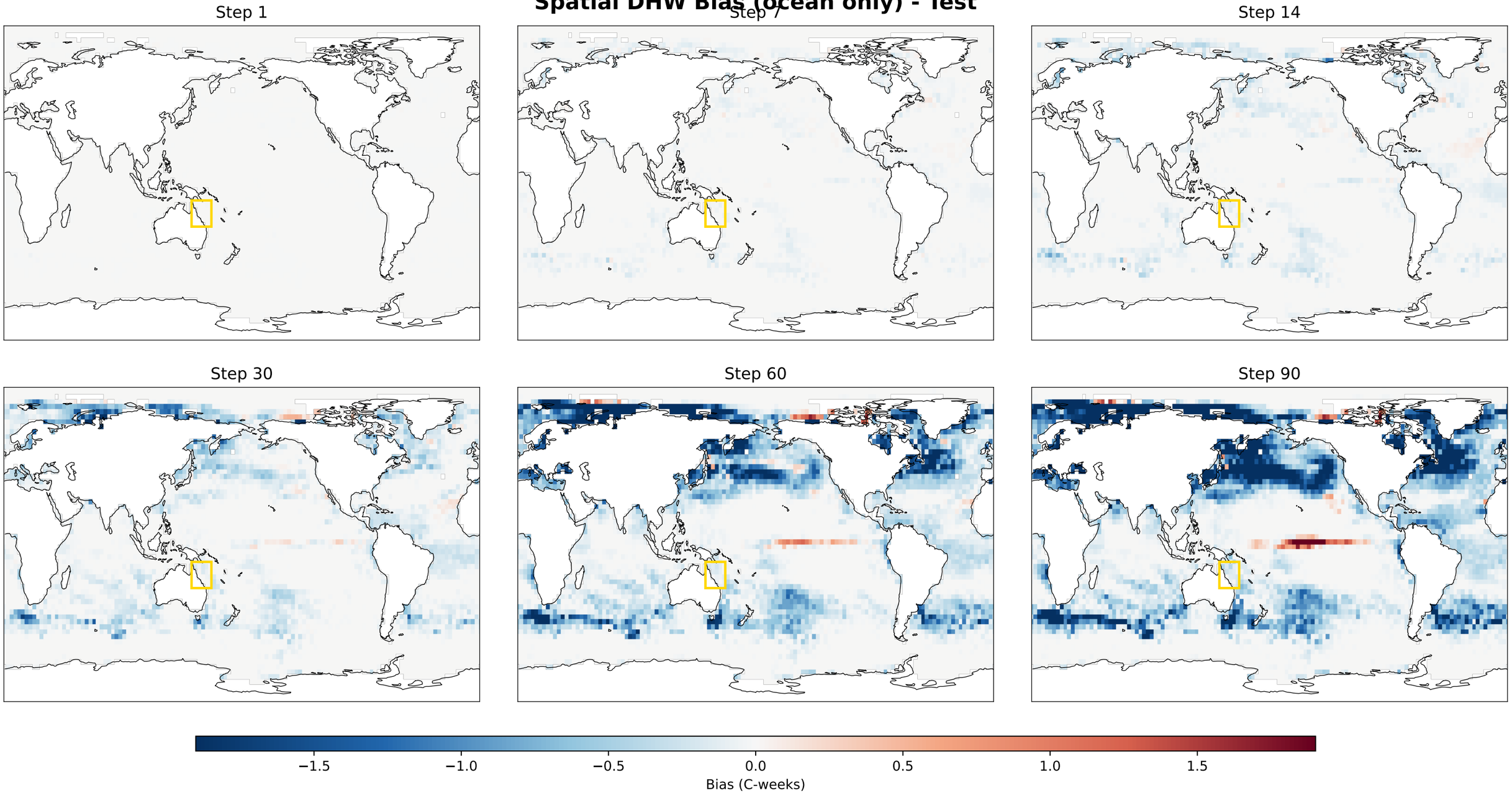





Spatial DHW RMSE (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
5
10
15
20
25
RMSE (C-weeks)

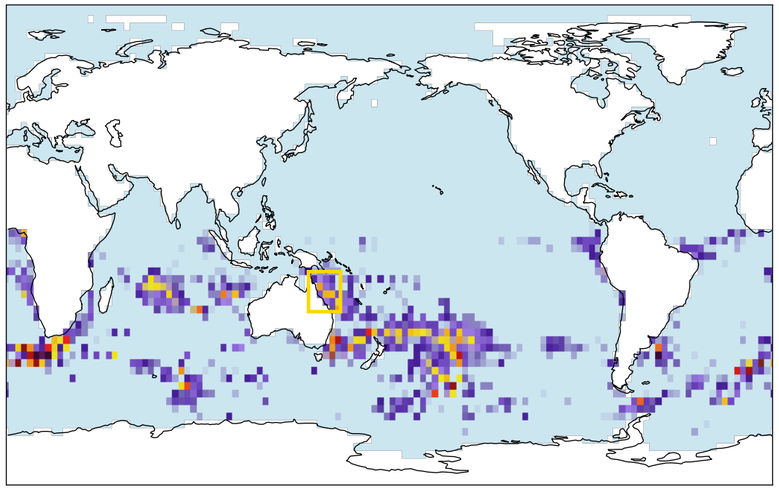

Spatial DHW Bias (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
5
10
15
20
25
RMSE (C-weeks)

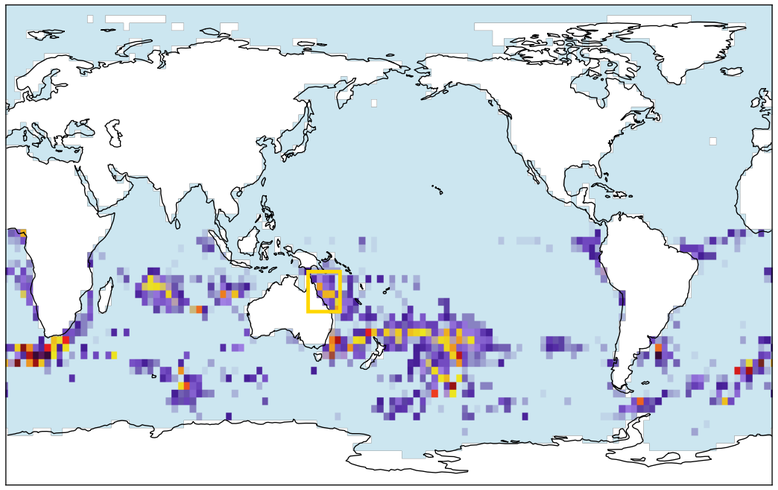

Spatial DHW Bias (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test (C-weeks)

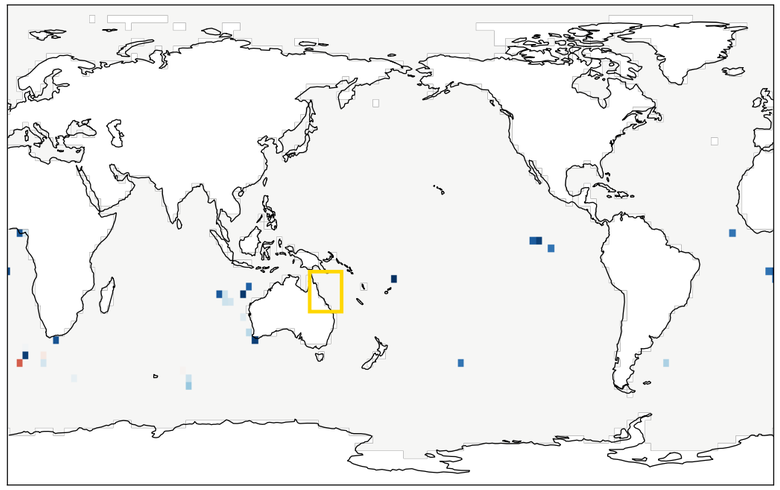
Actual DHW - Step 1



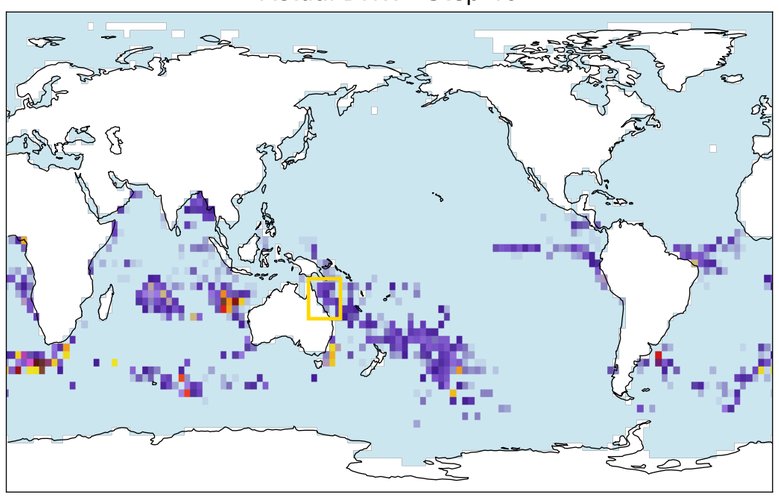
Predicted DHW - Step 1



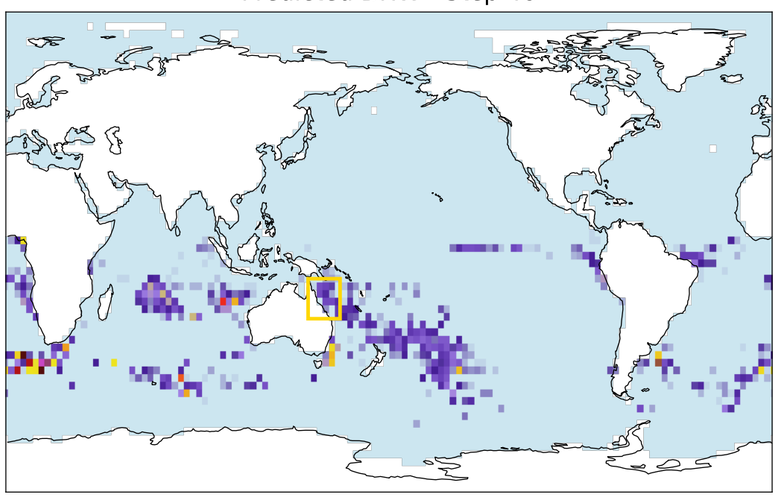
Error (Pred - Actual) - Step 1 [scale ±0.24]



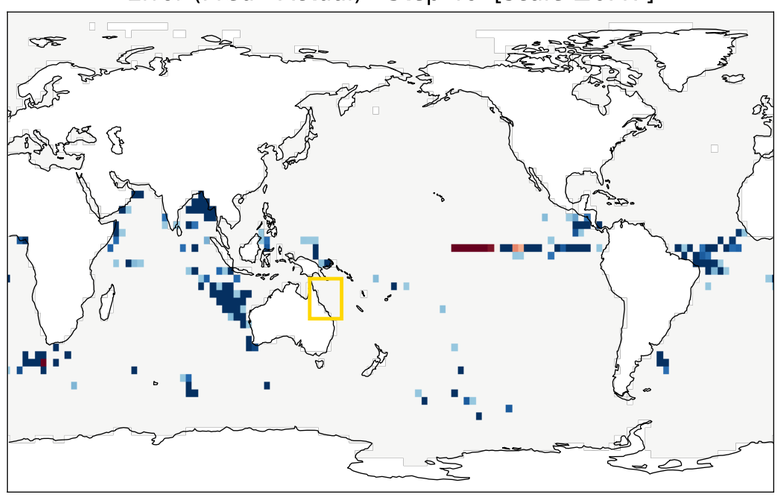
Actual DHW - Step 46



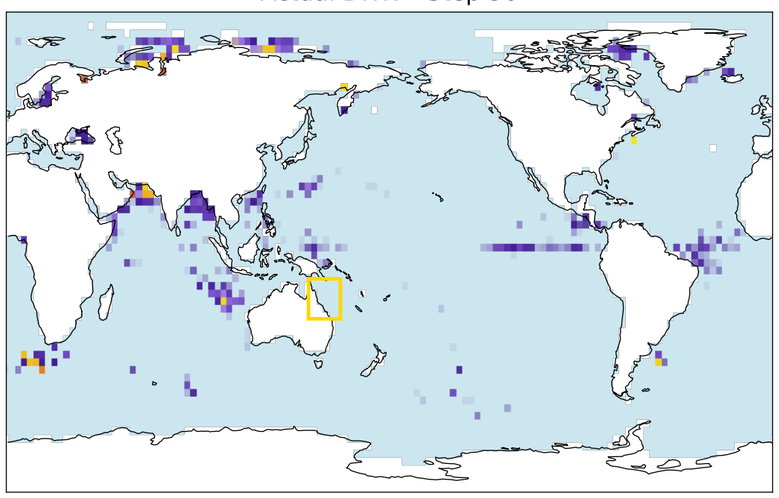
Predicted DHW - Step 46



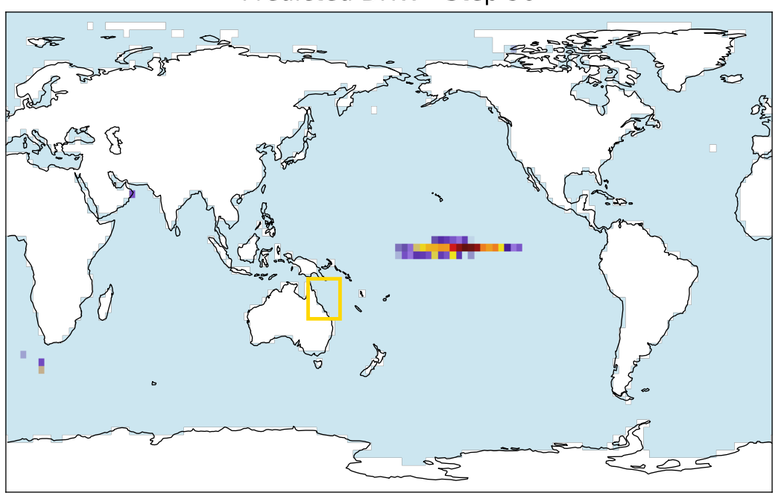
Error (Pred - Actual) - Step 46 [scale ±0.47]



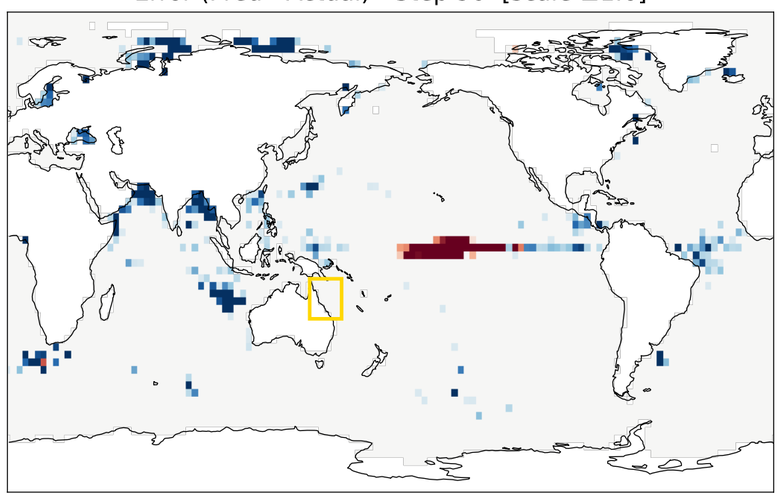
Actual DHW - Step 90

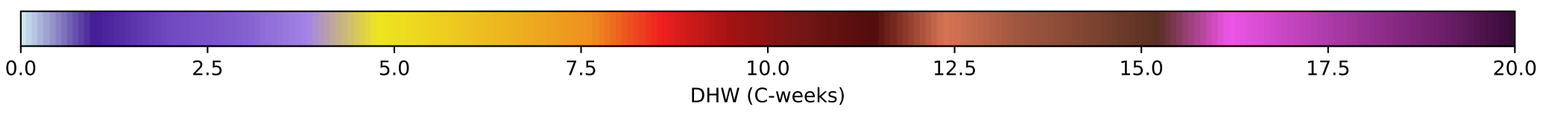


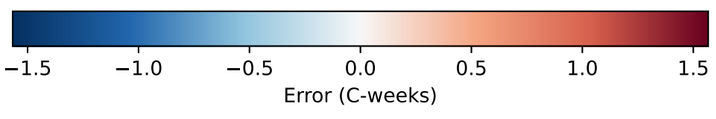
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±1.6]

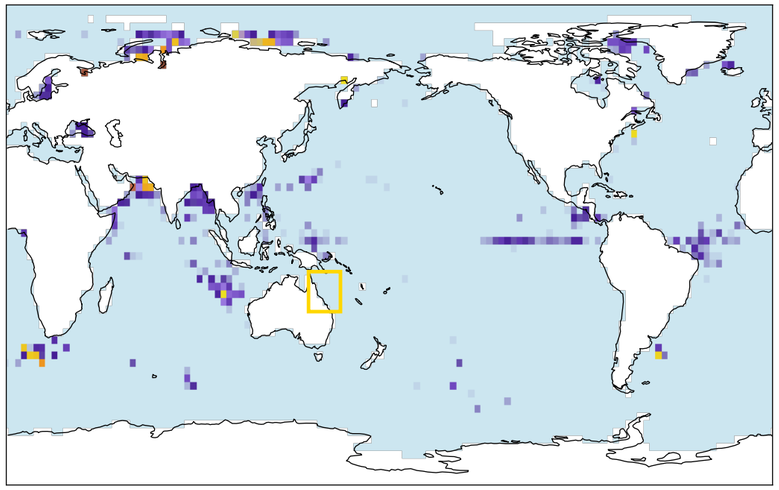




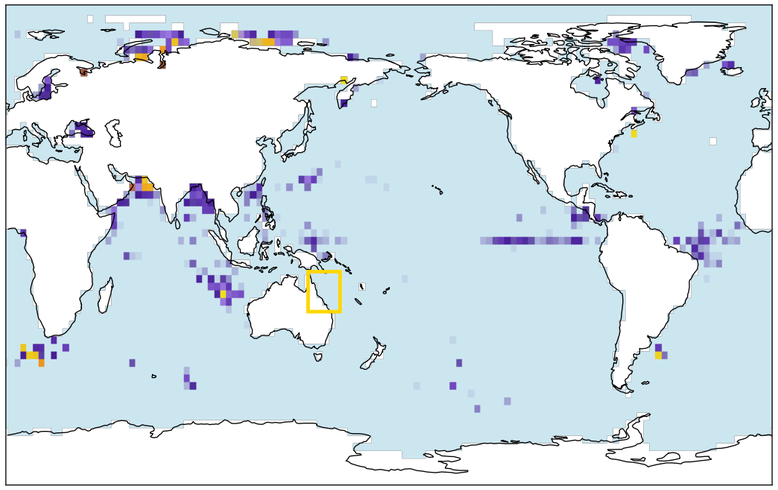


DHW Chain init: 2020-07-09 - Test (C-weeks)

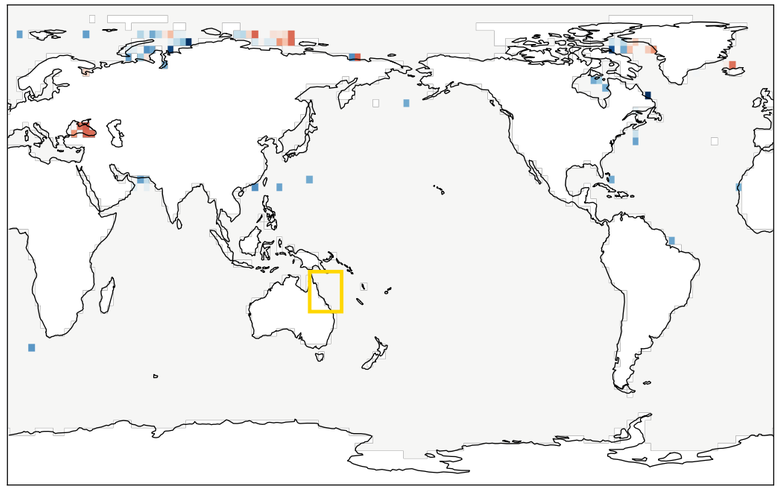
Actual DHW - Step 1



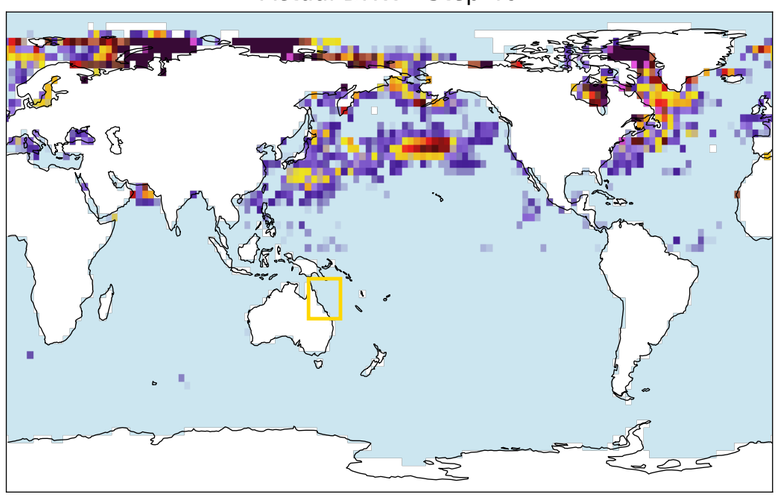
Predicted DHW - Step 1



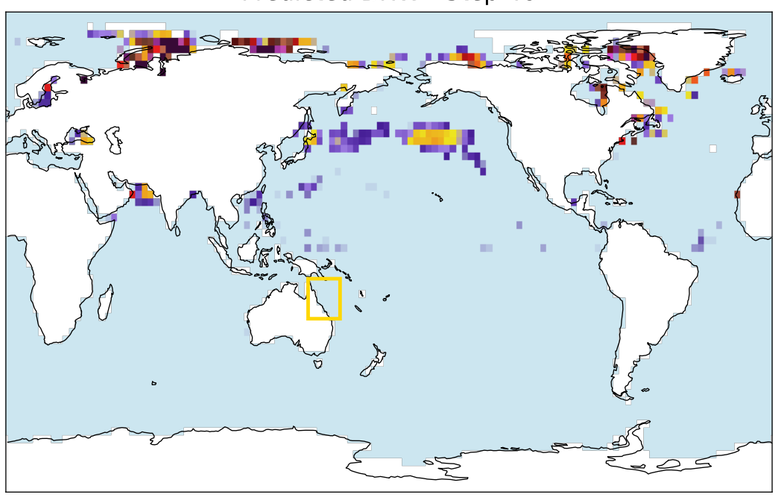
Error (Pred - Actual) - Step 1 [scale ±0.36]



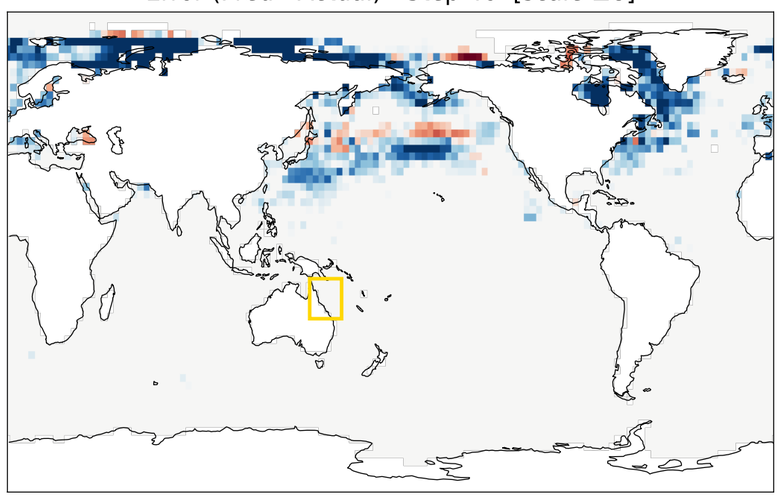
Actual DHW - Step 46



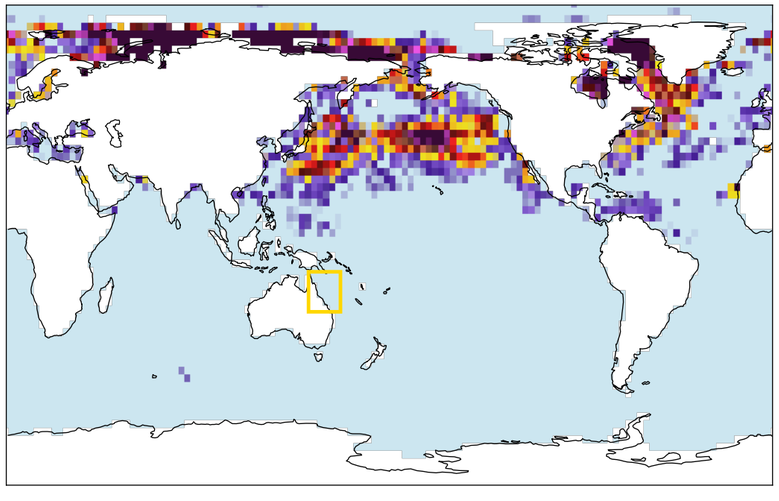
Predicted DHW - Step 46



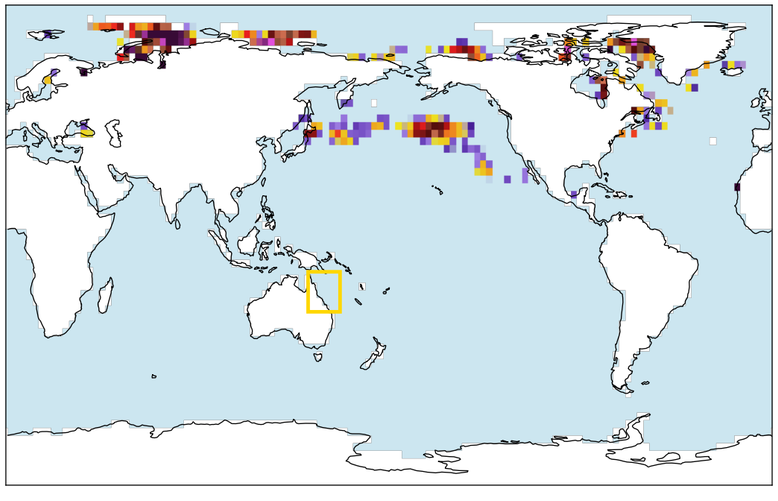
Error (Pred - Actual) - Step 46 [scale ±8]



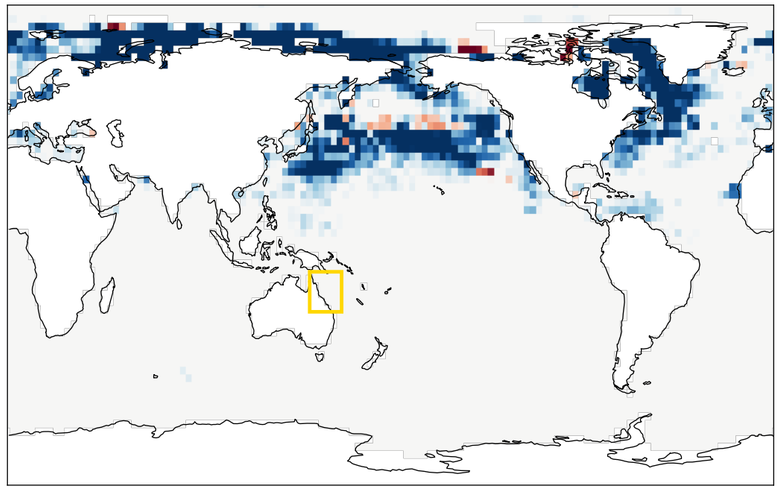
Actual DHW - Step 90

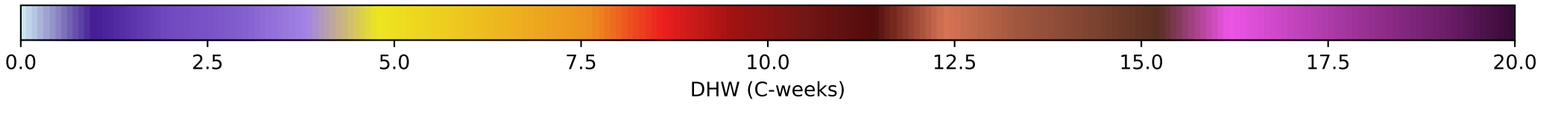


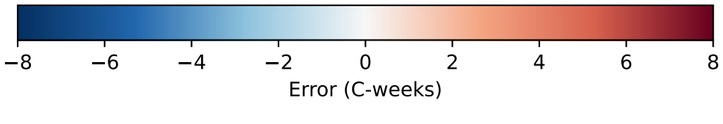
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

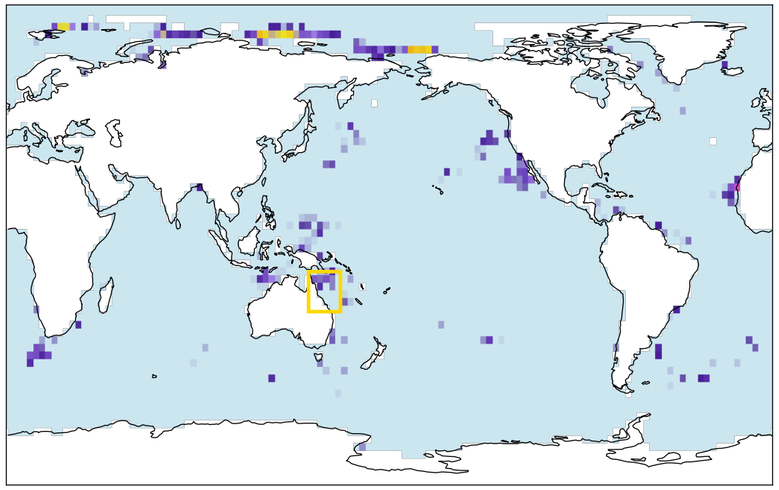




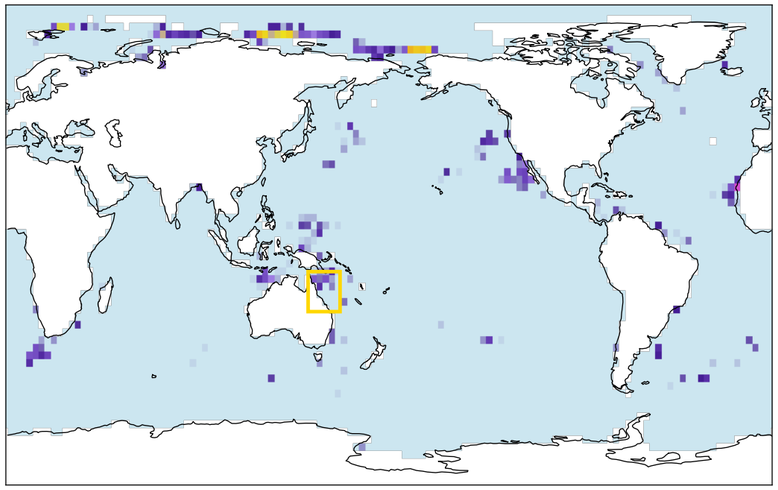


DHW Chain init: 2021-01-06 - Test (C-weeks)

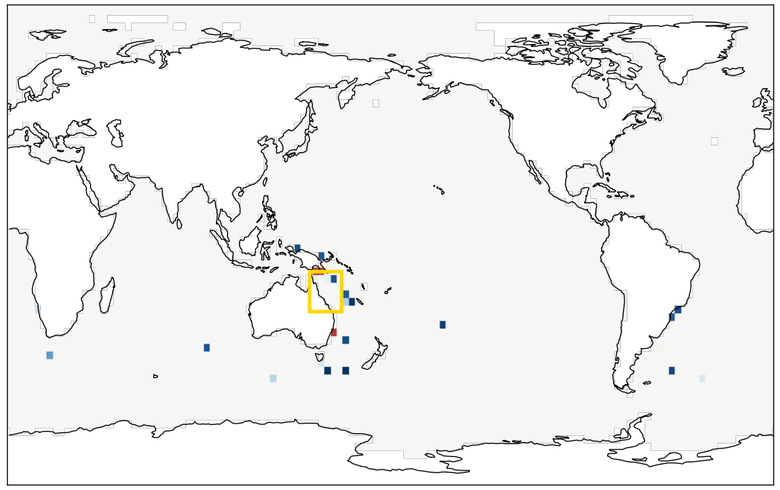
Actual DHW - Step 1



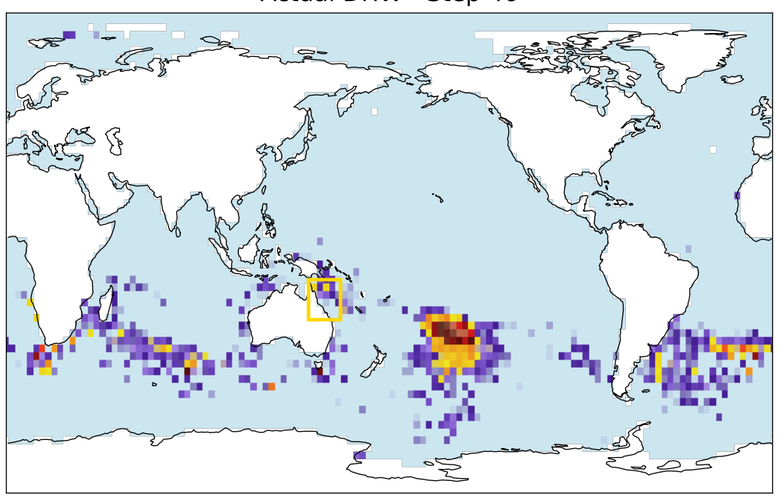
Predicted DHW - Step 1



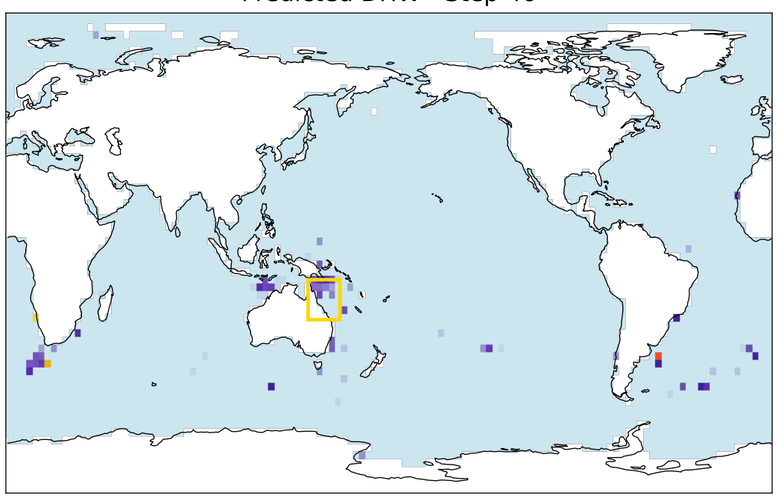
Error (Pred - Actual) - Step 1 [scale ±0.2]



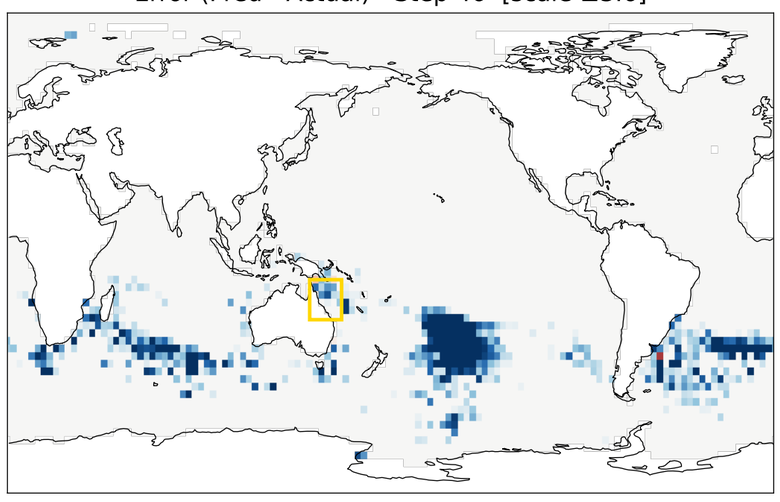
Actual DHW - Step 46



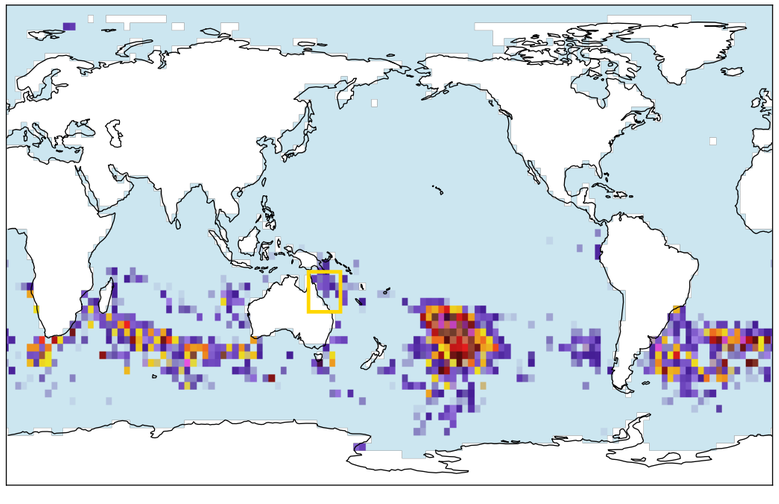
Predicted DHW - Step 46



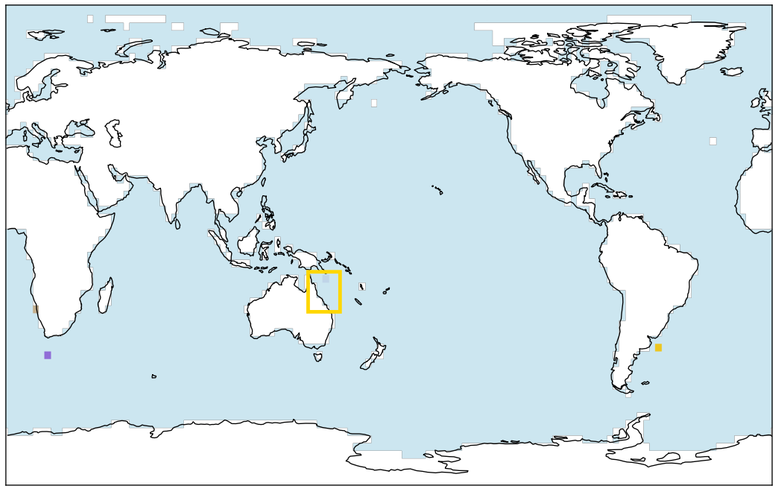
Error (Pred - Actual) - Step 46 [scale ±3.6]



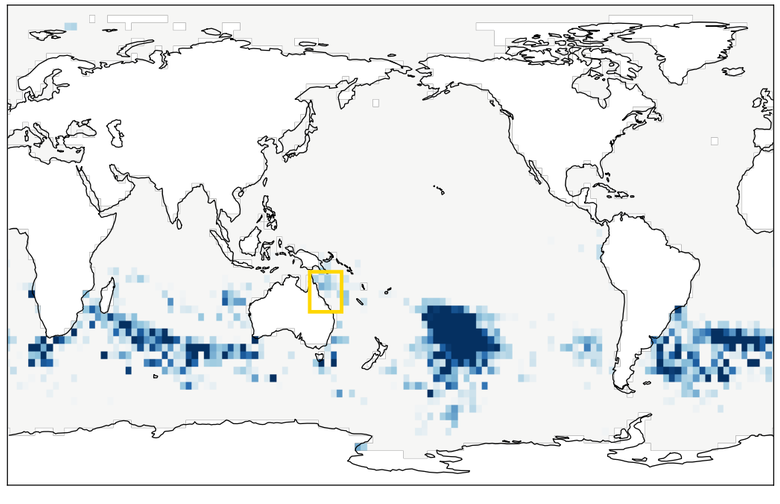
Actual DHW - Step 90

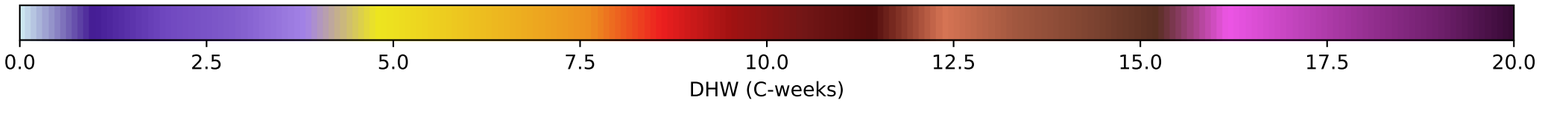


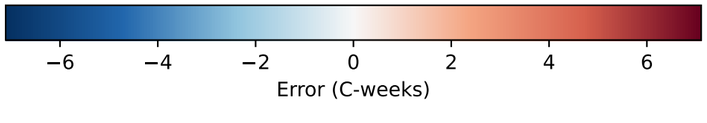
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±7.1]

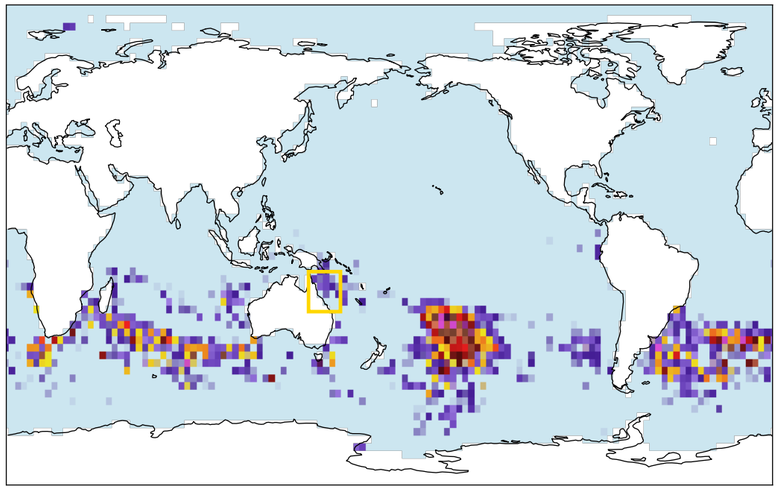




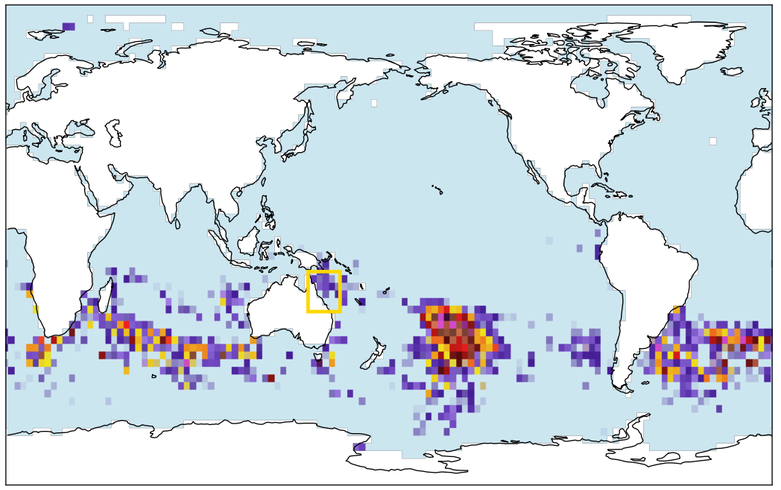


DHW Chain init: 2021-04-06 - Test (C-weeks)

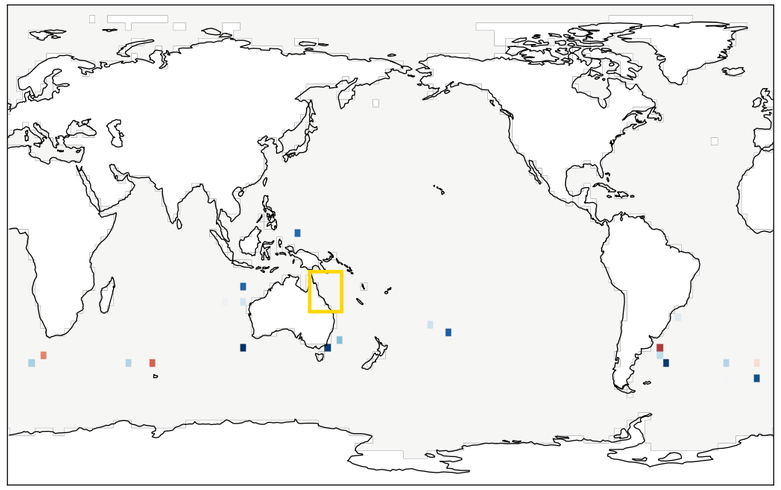
Actual DHW - Step 1



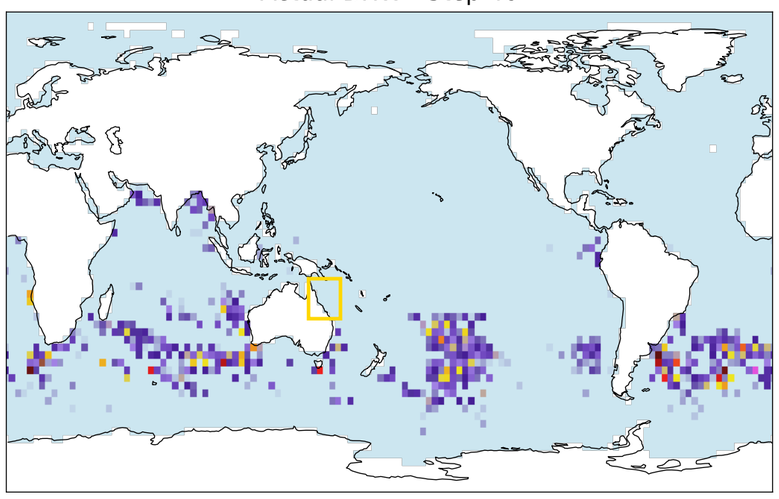
Predicted DHW - Step 1



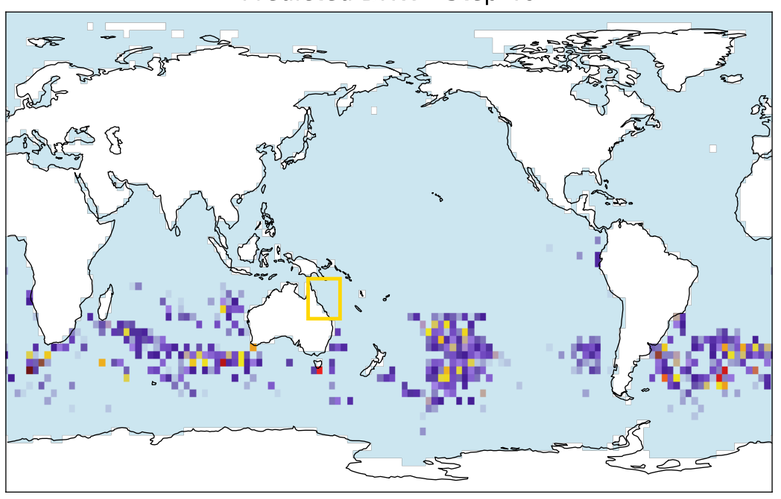
Error (Pred - Actual) - Step 1 [scale ±0.23]



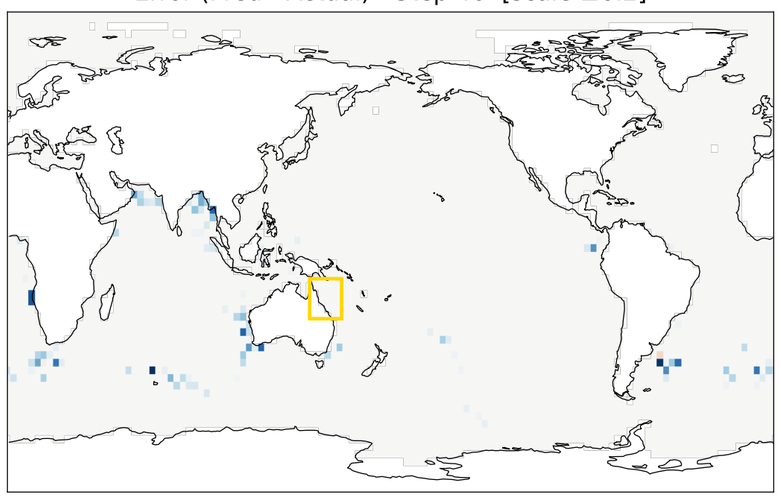
Actual DHW - Step 46



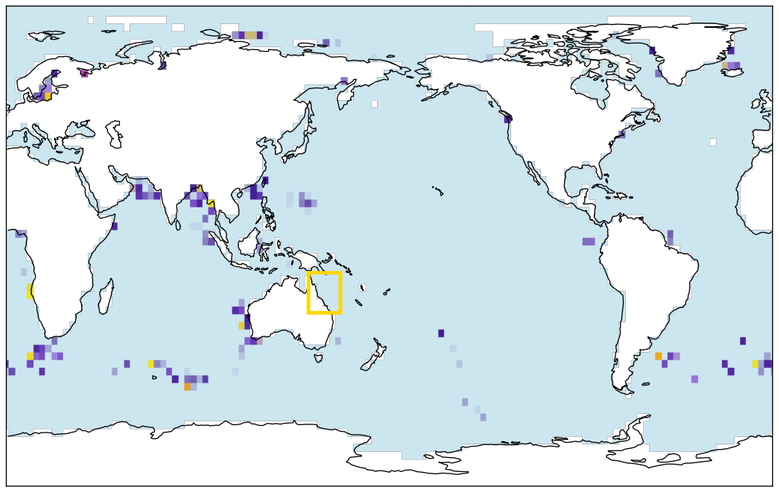
Predicted DHW - Step 46



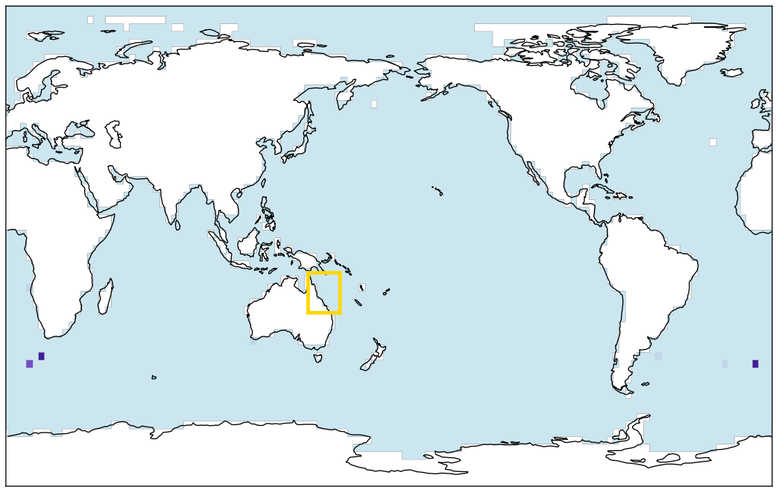
Error (Pred - Actual) - Step 46 [scale ±6.2]



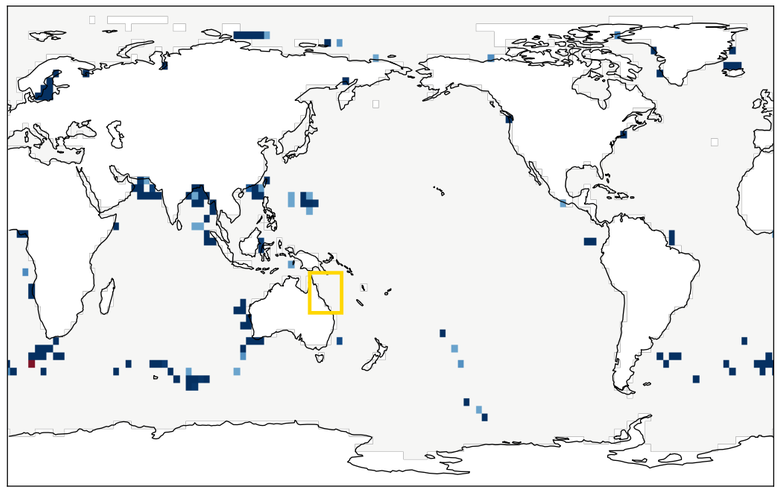
Actual DHW - Step 90

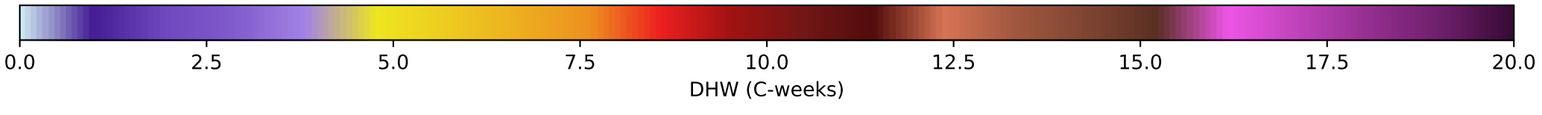


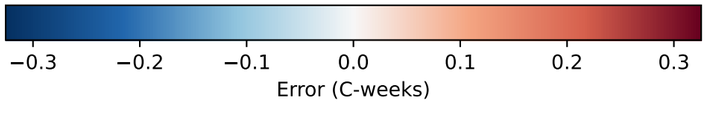
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.33]

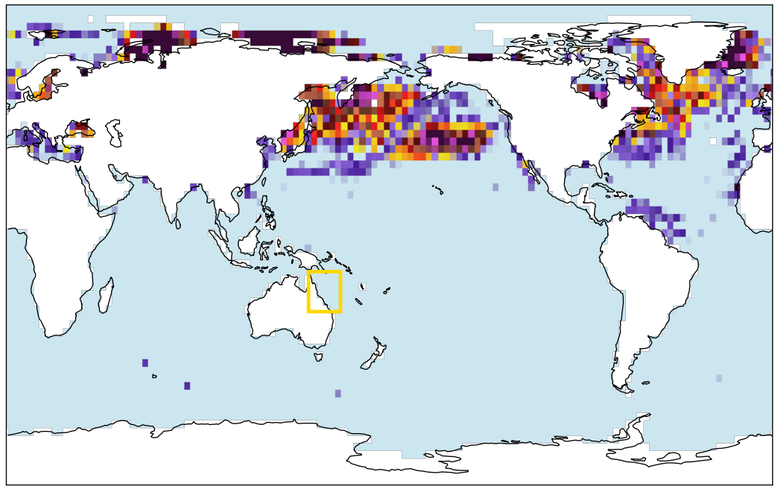




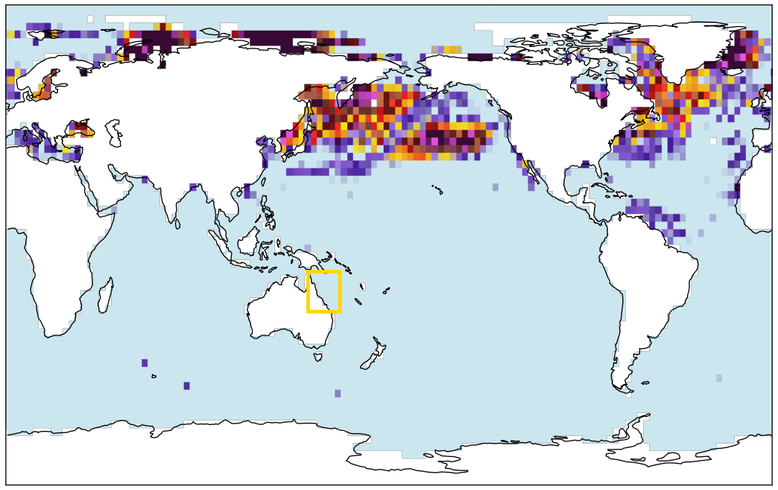


DHW Chain init: 2021-10-03 - Test (C-weeks)

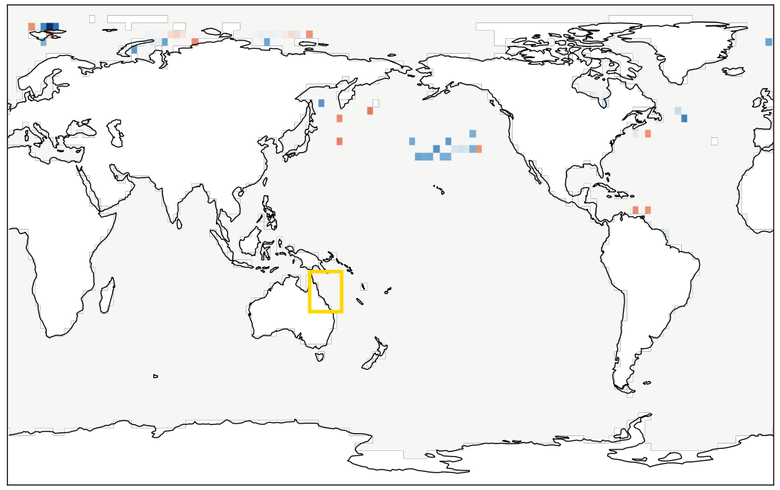
Actual DHW - Step 1



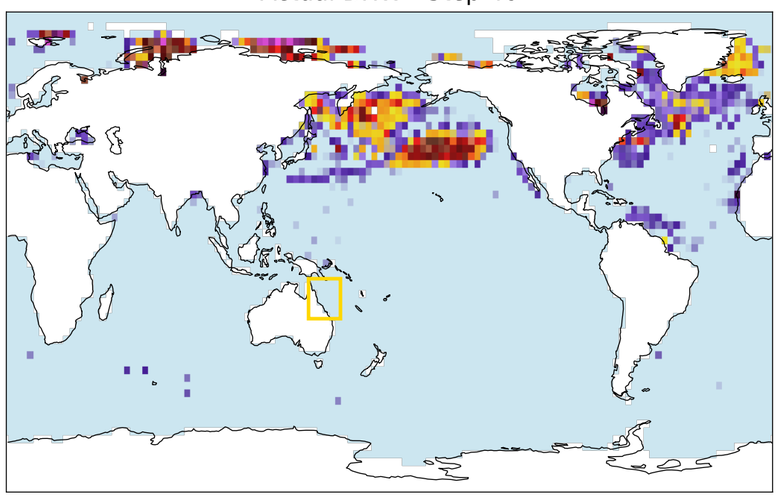
Predicted DHW - Step 1



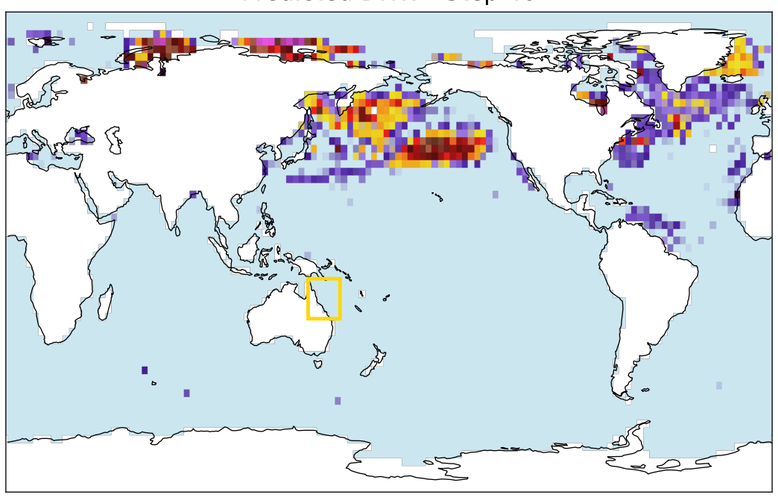
Error (Pred - Actual) - Step 1 [scale ±0.36]



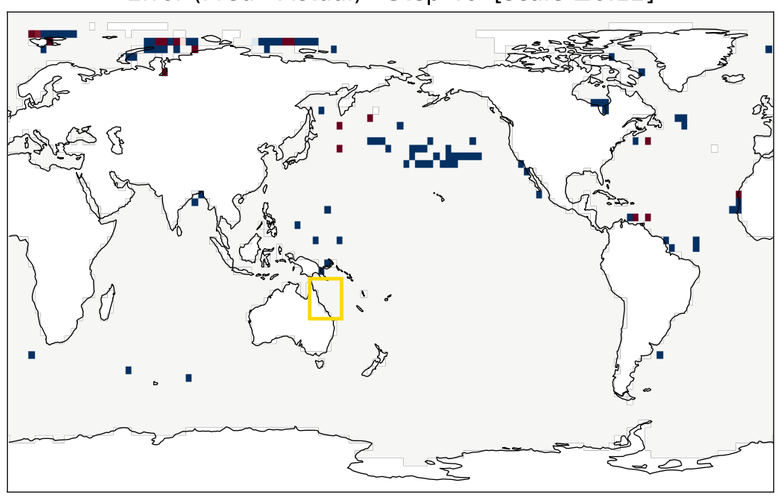
Actual DHW - Step 46



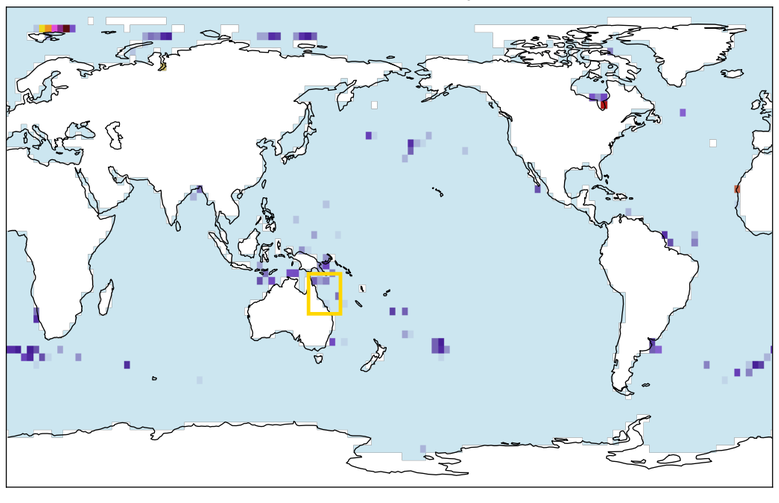
Predicted DHW - Step 46



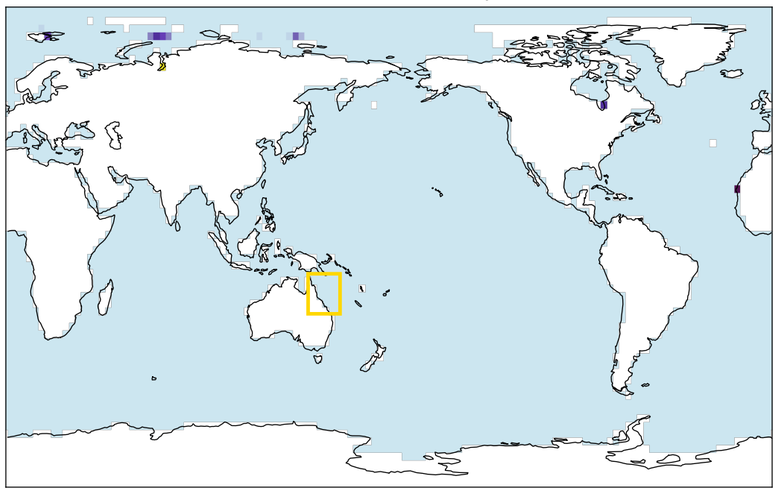
Error (Pred - Actual) - Step 46 [scale ±0.12]



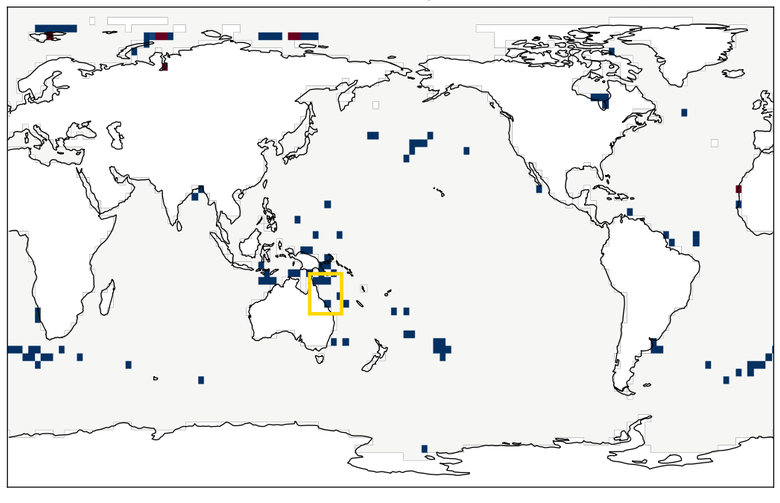
Actual DHW - Step 90

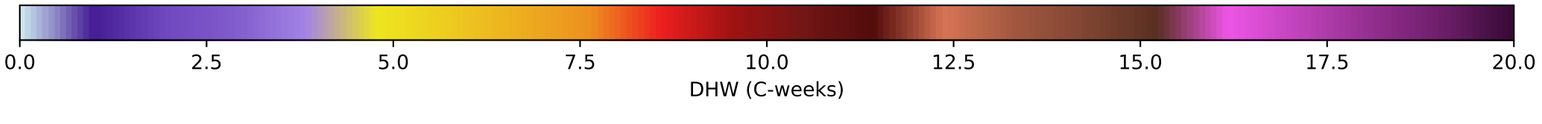


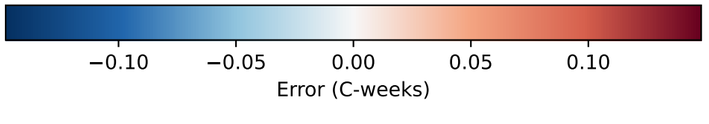
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.15]

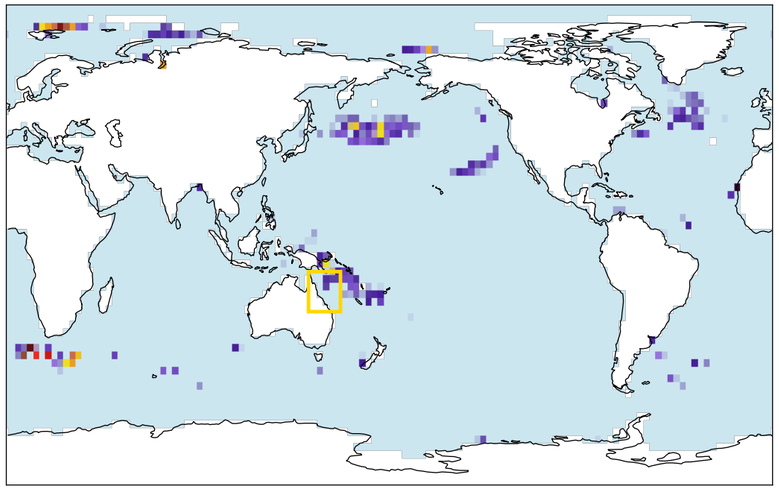




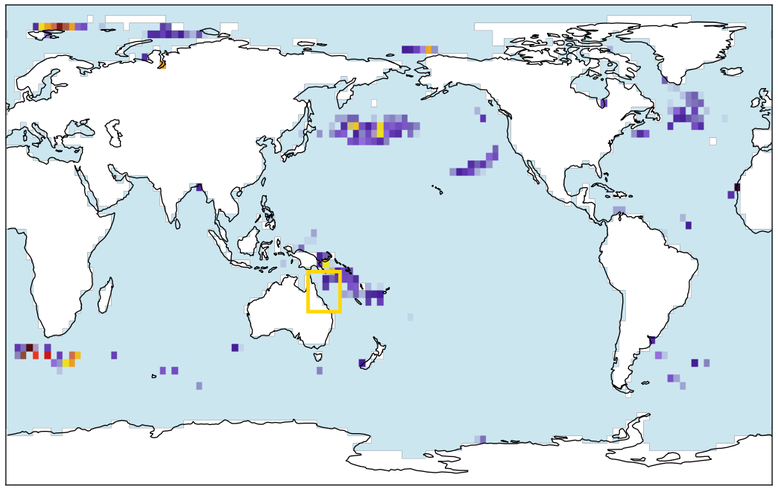


DHW Chain init: 2022-12-27 - Test (C-weeks)

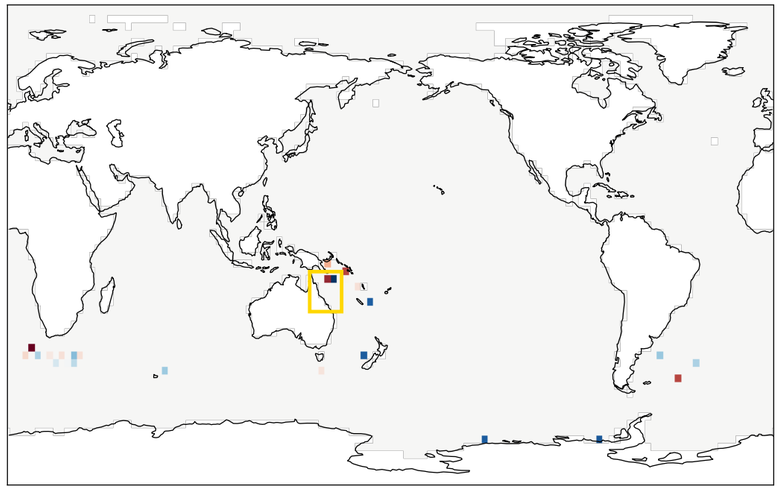
Actual DHW - Step 1



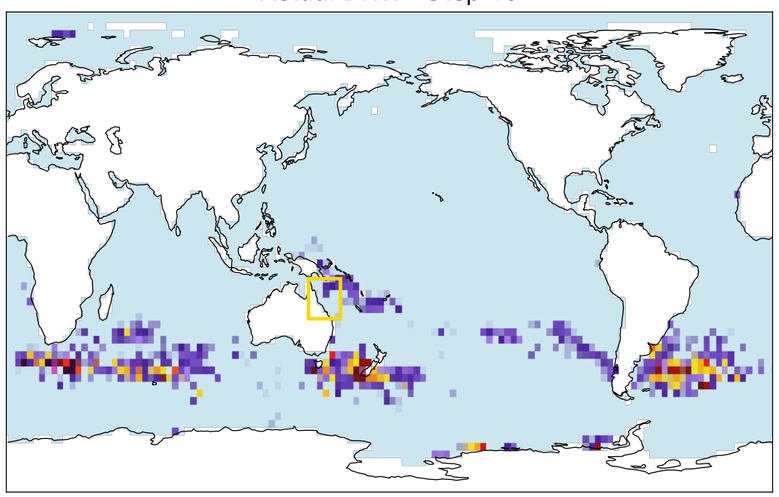
Predicted DHW - Step 1



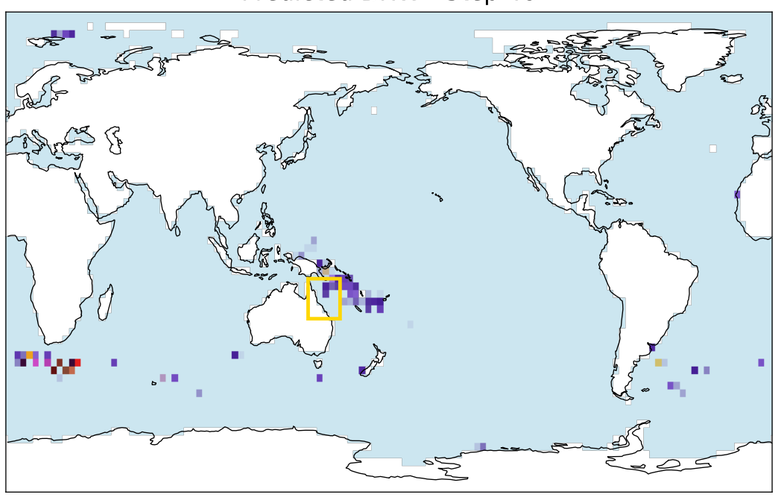
Error (Pred - Actual) - Step 1 [scale ±0.2]



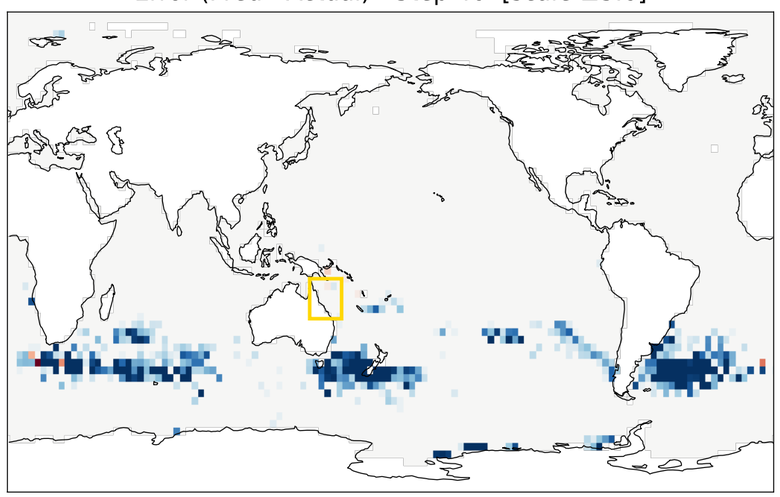
Actual DHW - Step 46



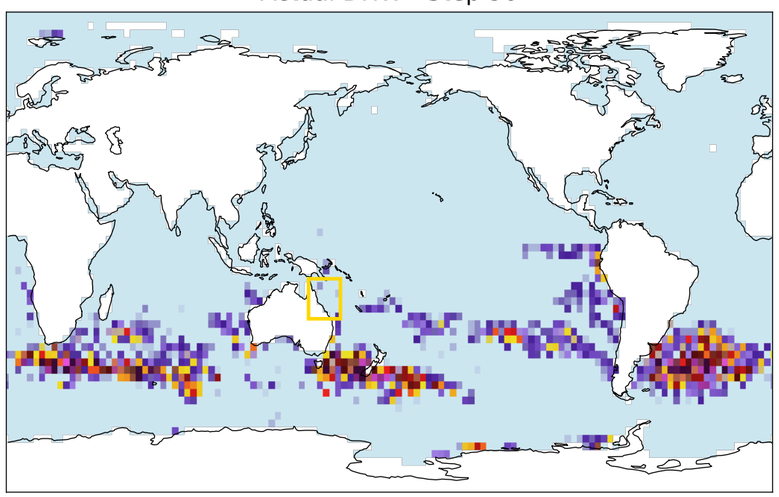
Predicted DHW - Step 46



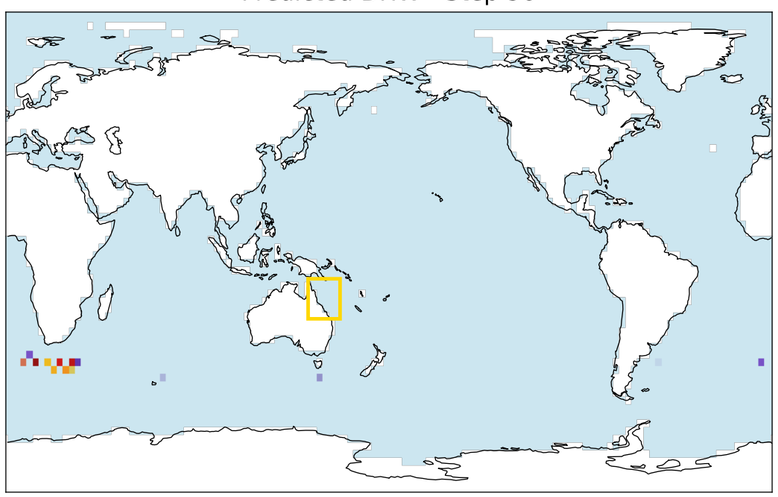
Error (Pred - Actual) - Step 46 [scale ±3.6]



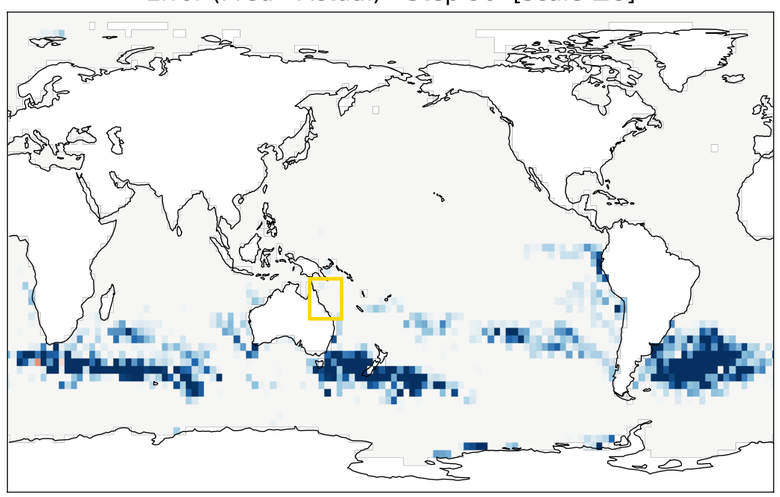
Actual DHW - Step 90

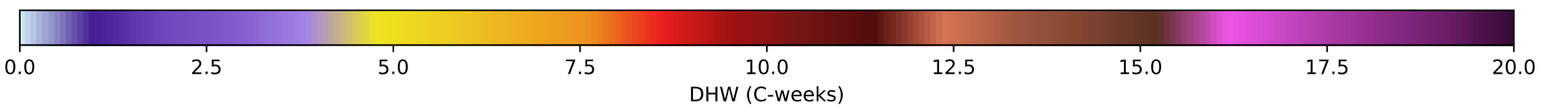


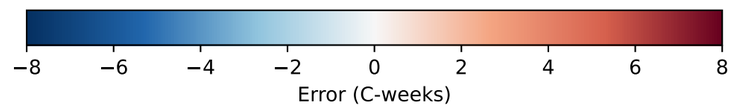
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

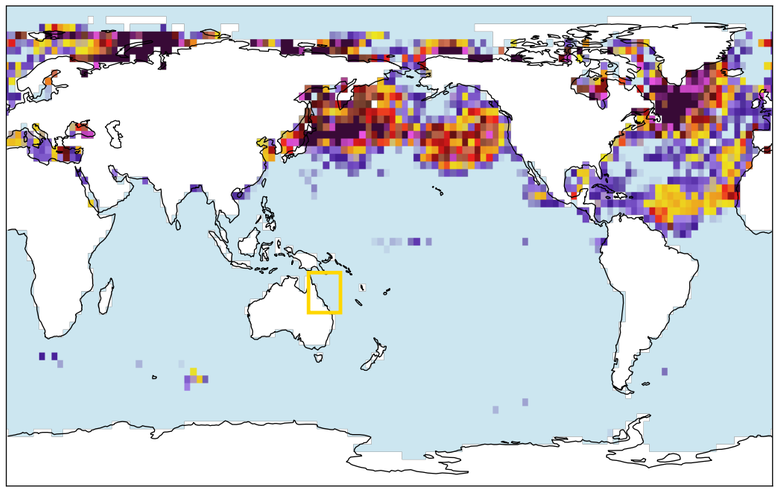




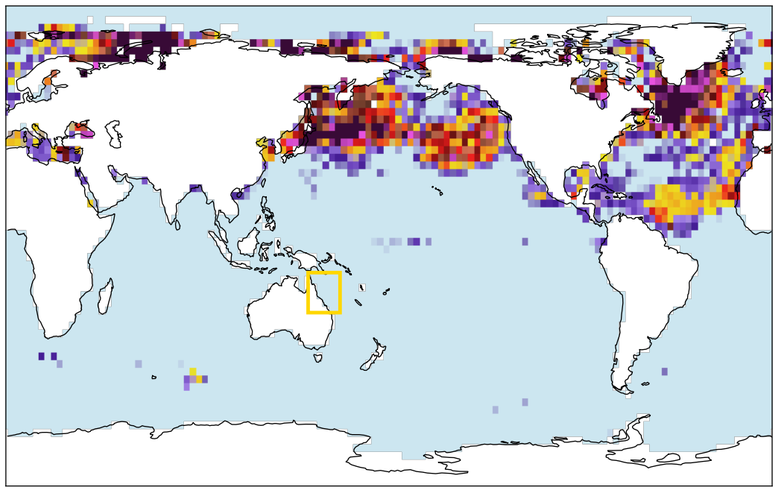


DHW Chain init: 2023-09-23 - Test (C-weeks)

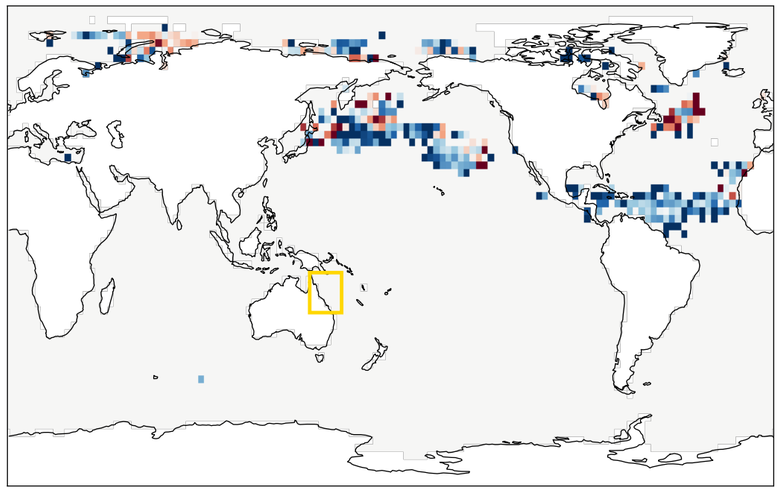
Actual DHW - Step 1



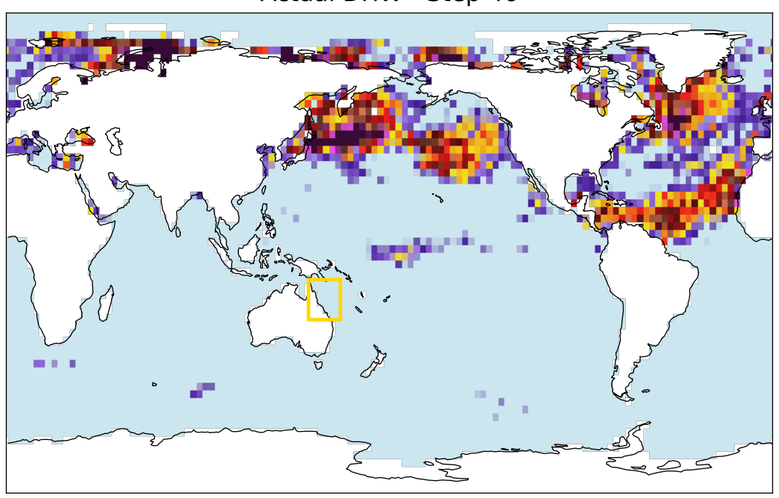
Predicted DHW - Step 1



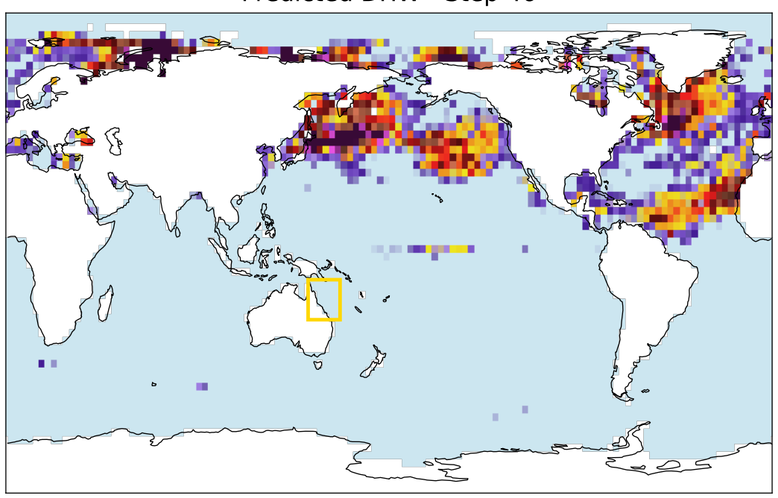
Error (Pred - Actual) - Step 1 [scale ±0.041]



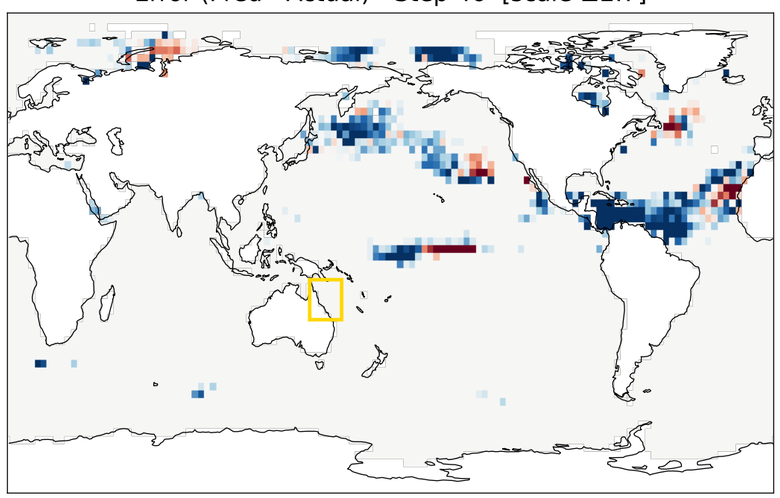
Actual DHW - Step 46



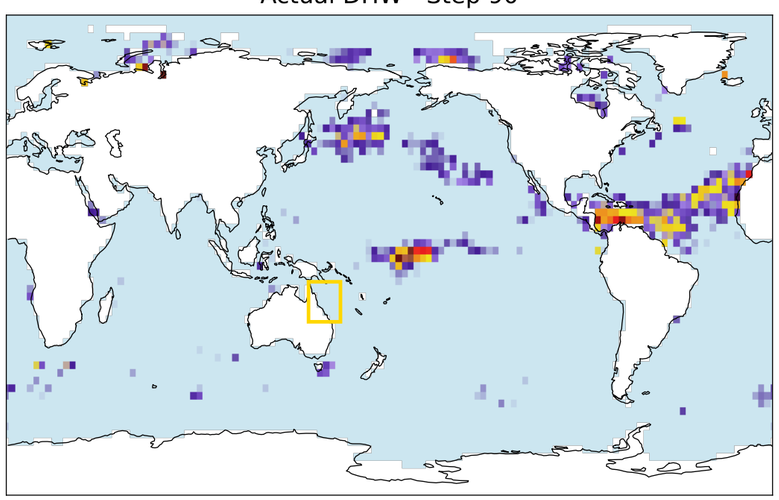
Predicted DHW - Step 46



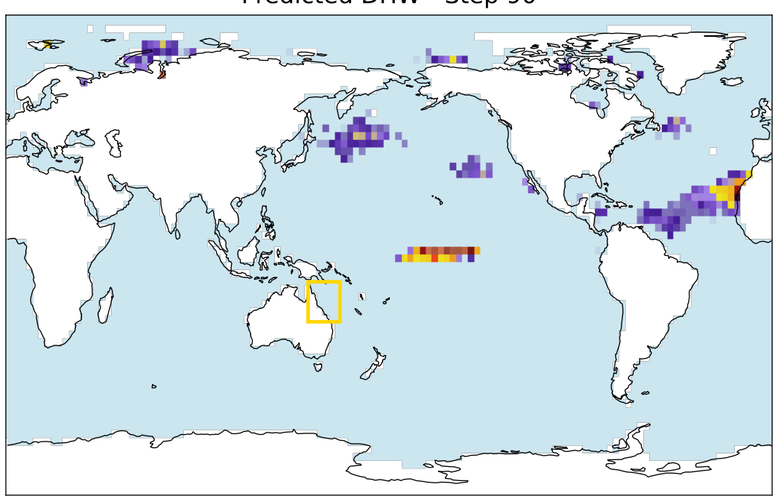
Error (Pred - Actual) - Step 46 [scale ±2.7]



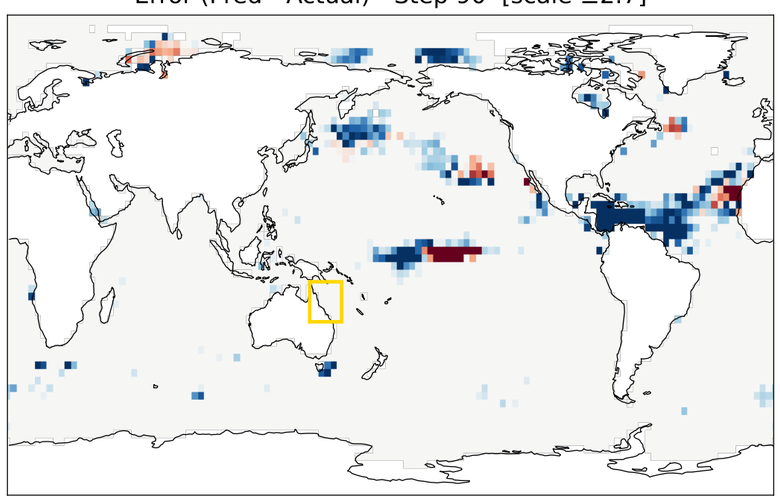
Actual DHW - Step 90

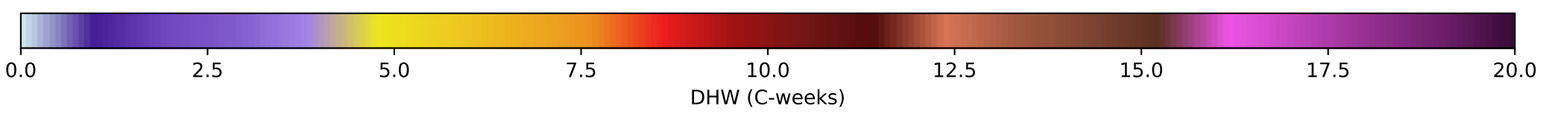


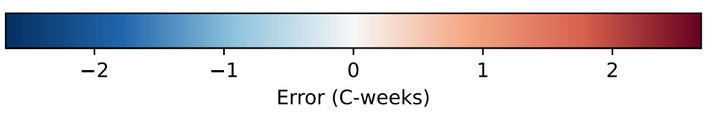
Predicted DHW - Step 90



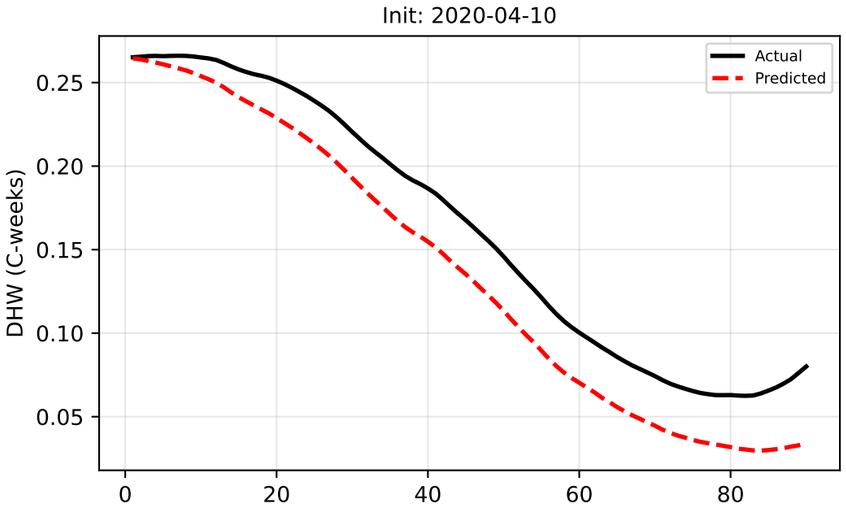
Error (Pred - Actual) - Step 90 [scale ±2.7]

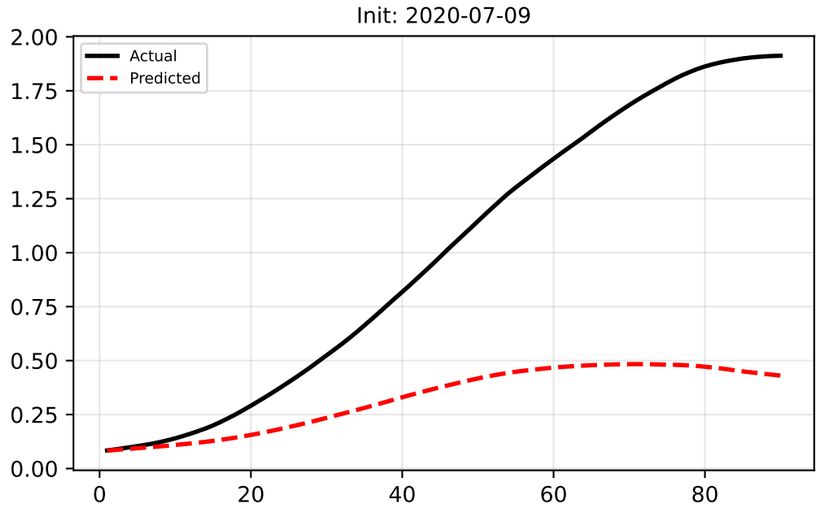


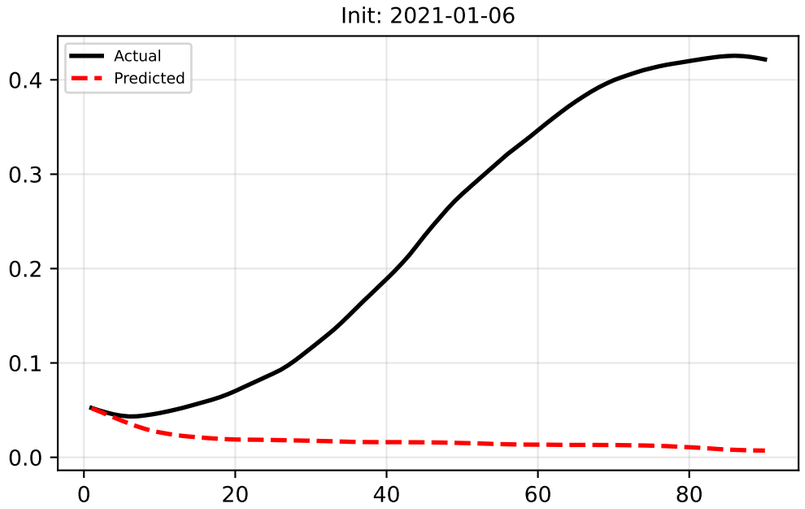


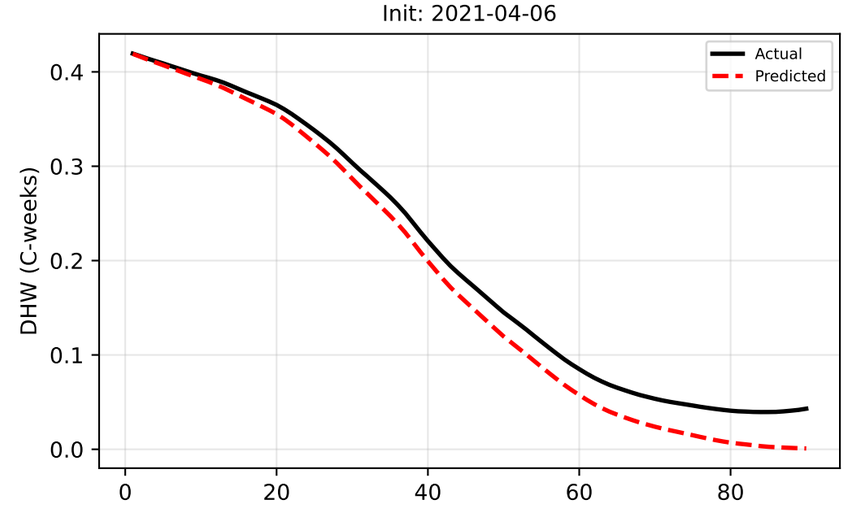


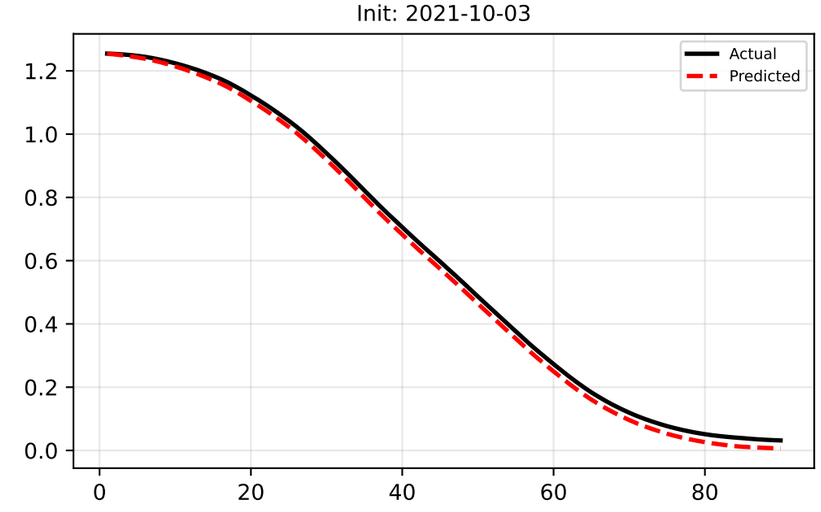
Global-Mean DHW - Test (C-weeks)

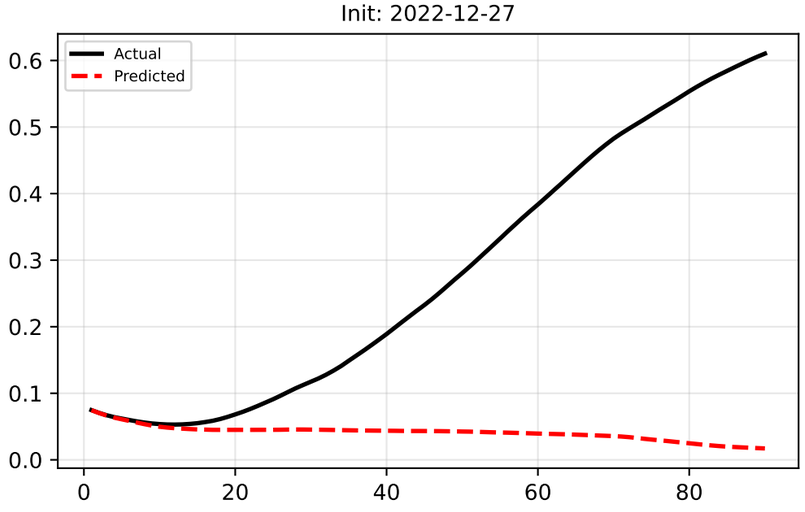


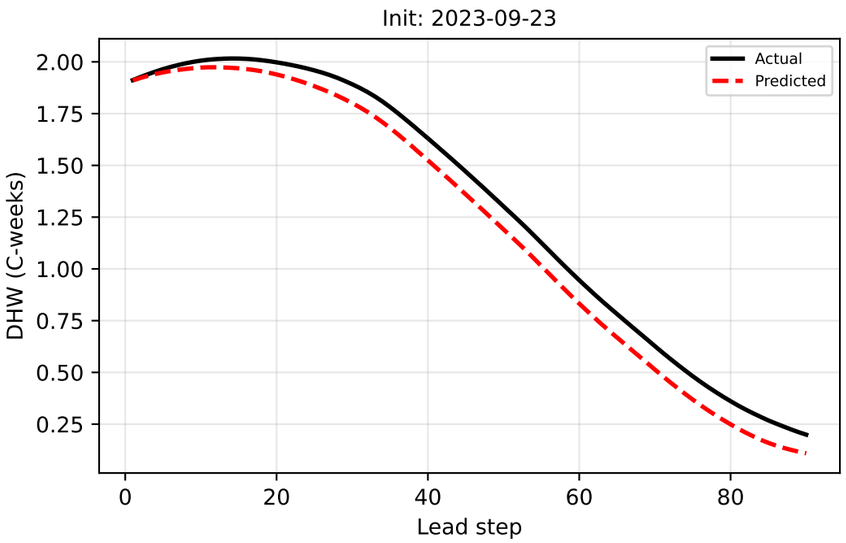




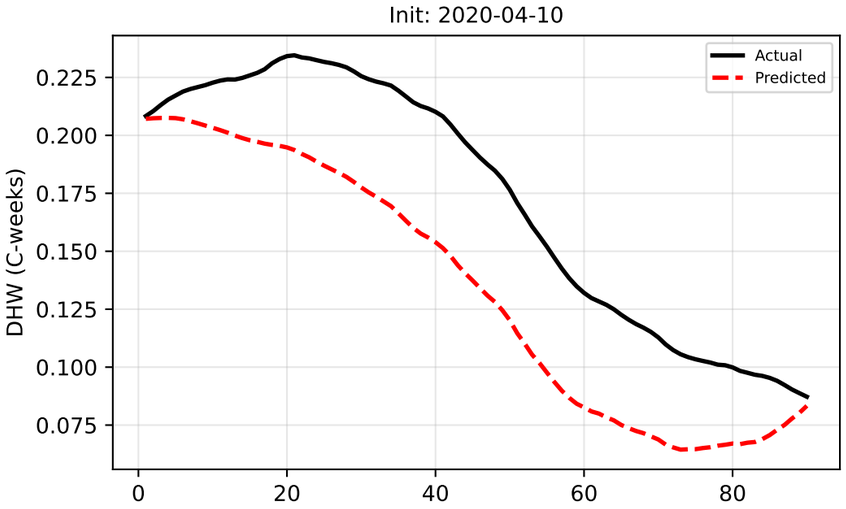


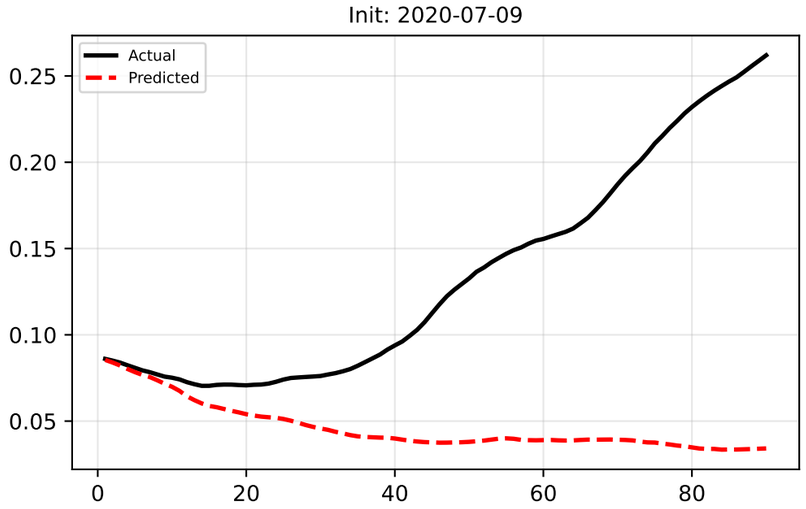


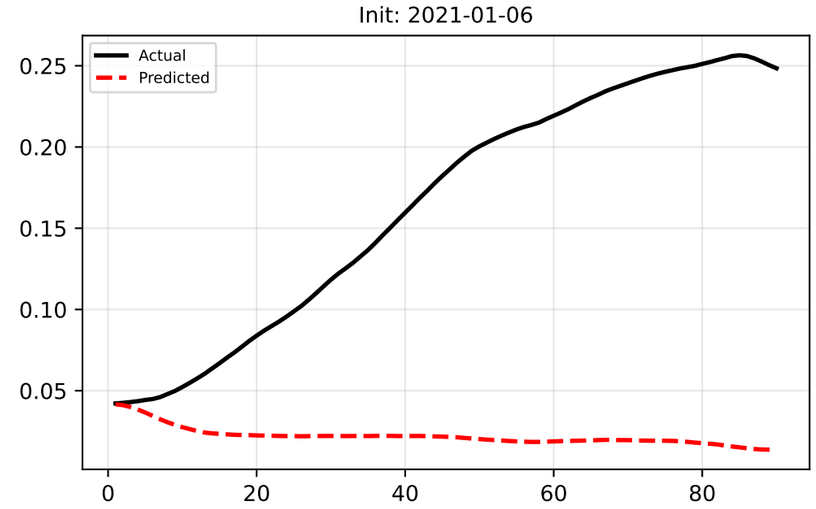


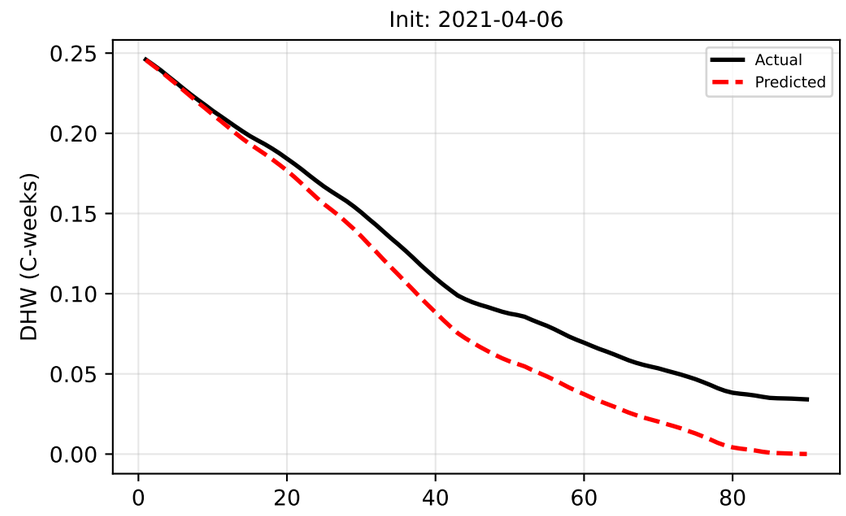


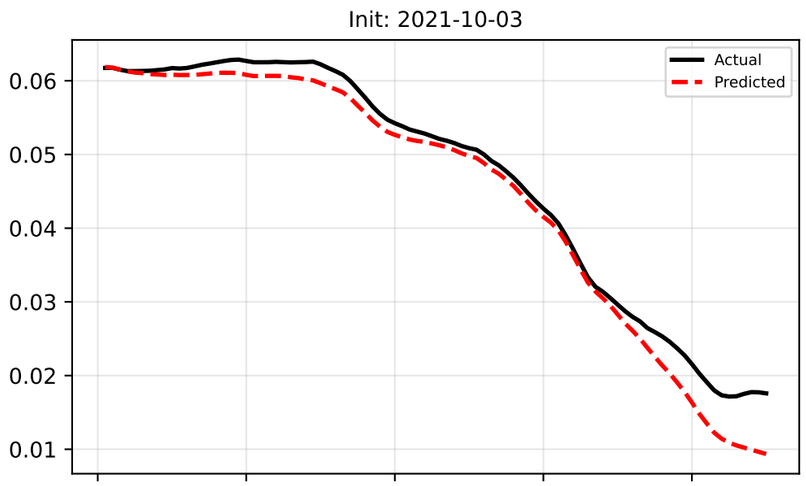
Tropical-Mean DHW (|lat|<=30) - Test (C-weeks)

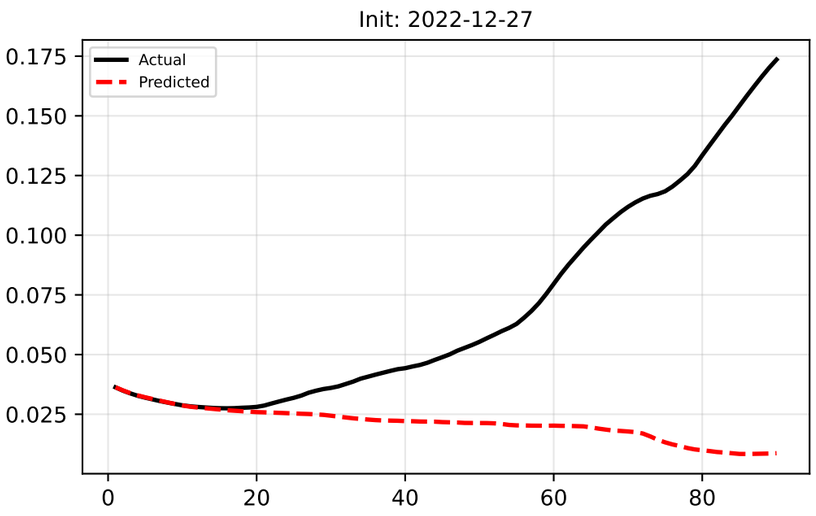


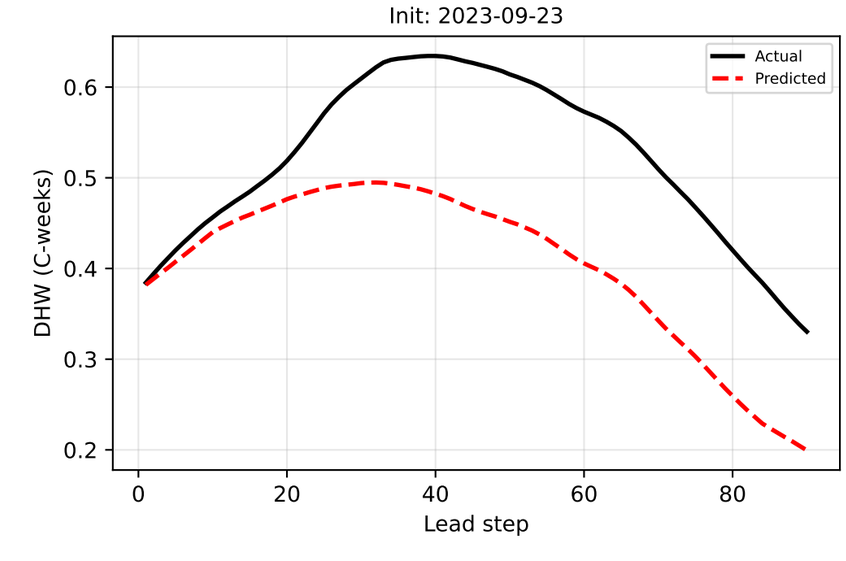


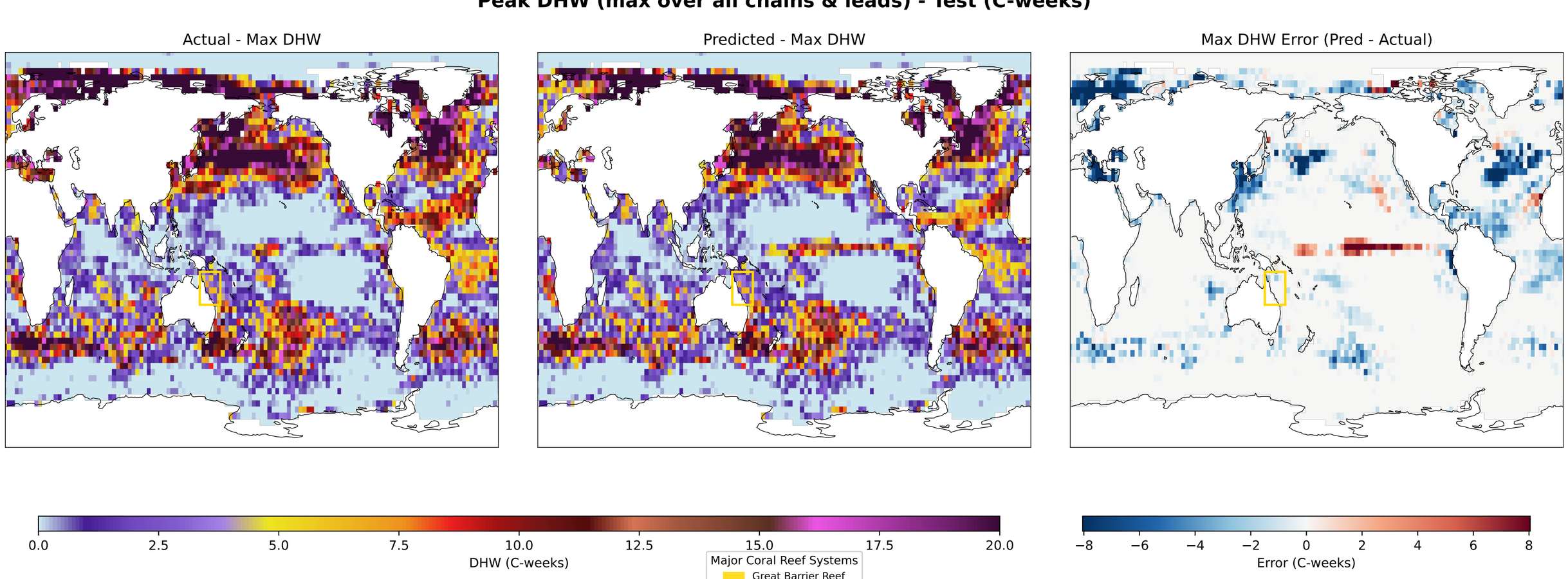


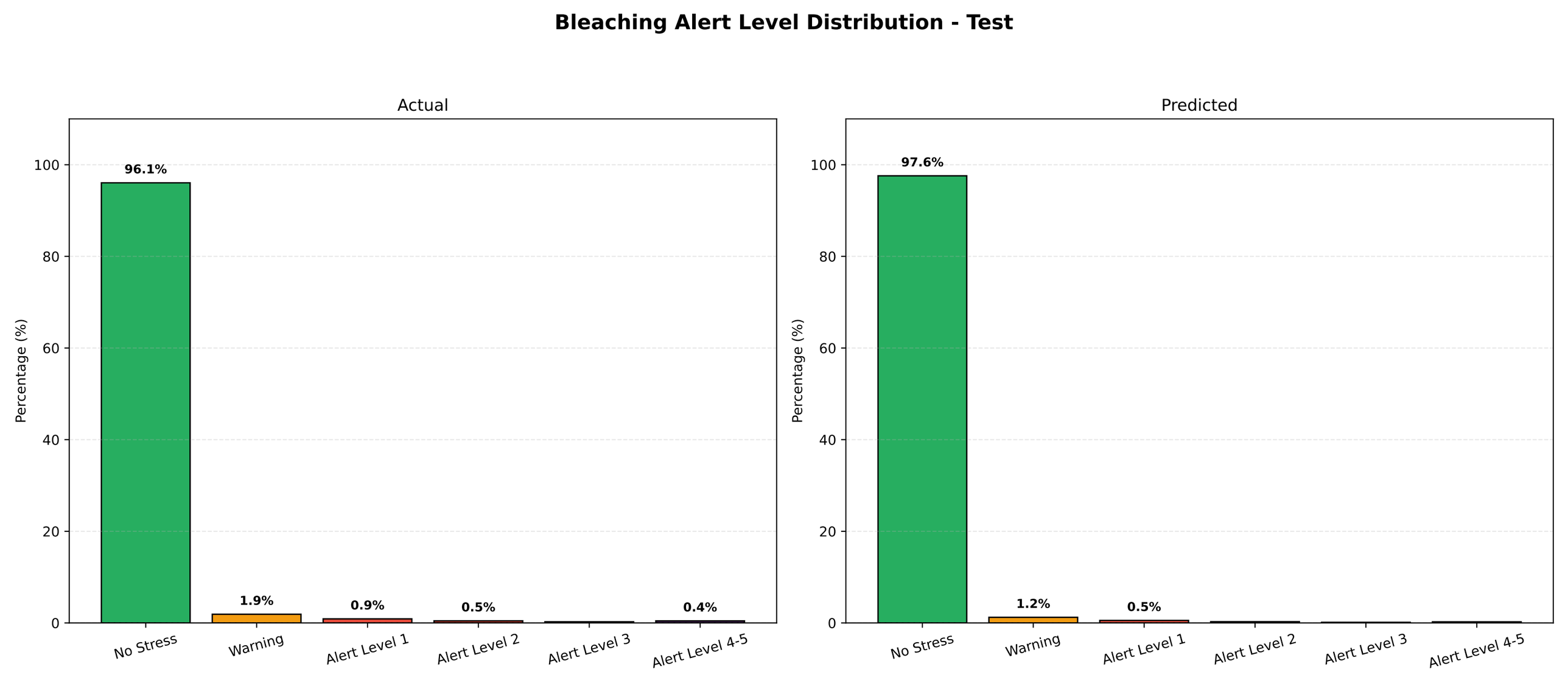






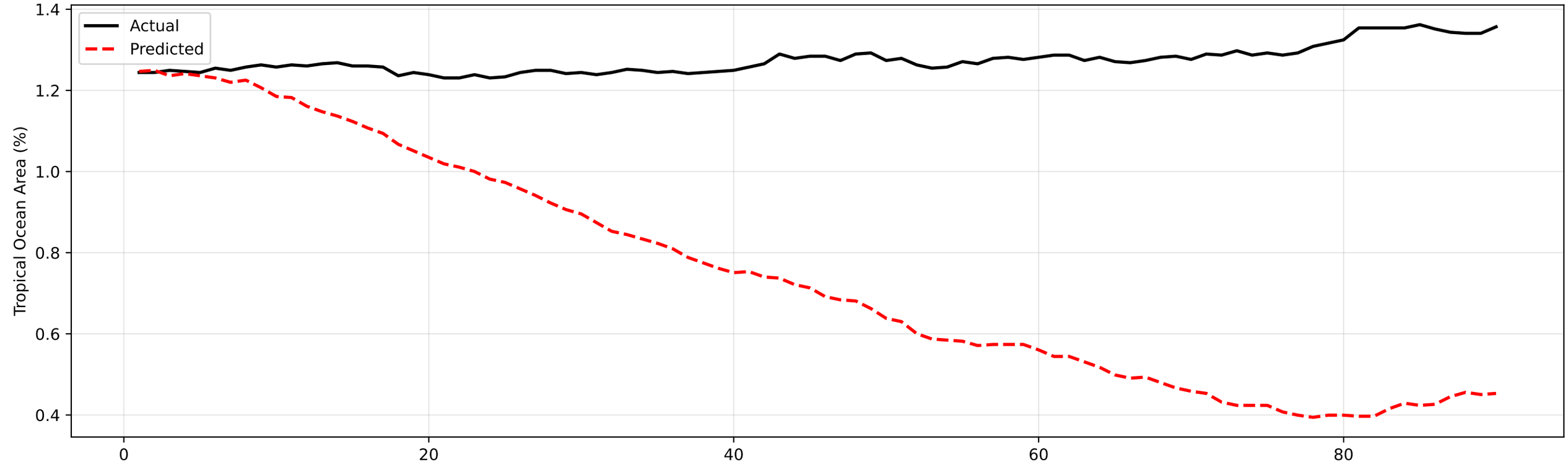



Peak DHW (max over all chains & leads) - Test (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−8
−6
−4
−2
0
2
4
6
8
Error (C-weeks)

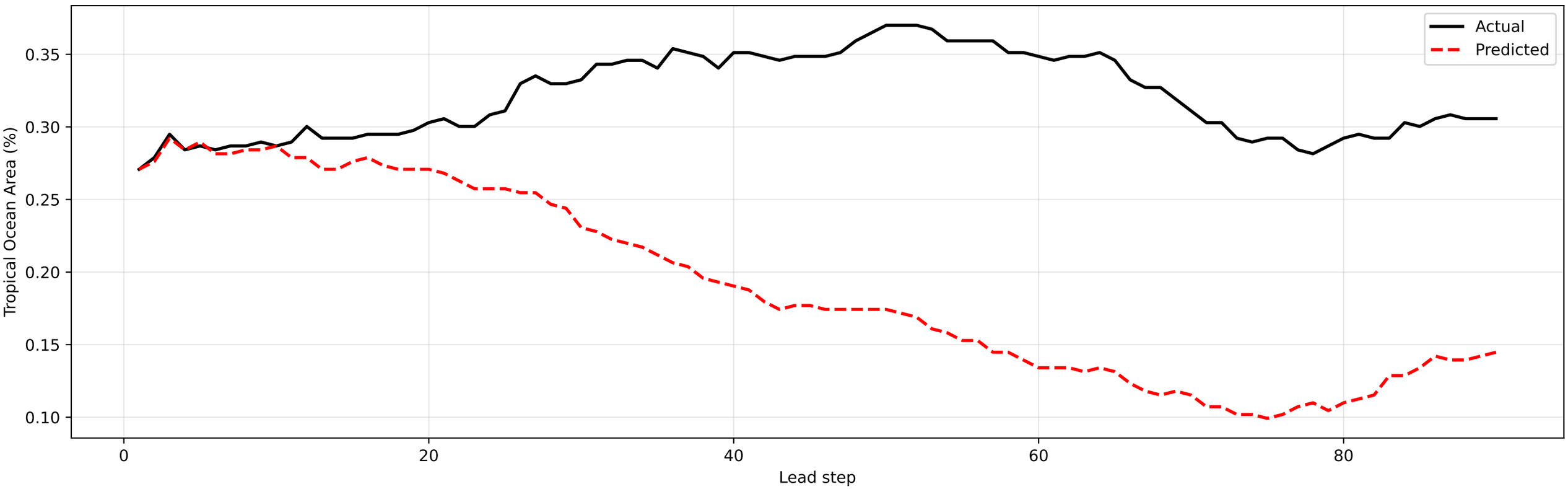

Bleaching Alert Level Distribution - Test
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.1%
1.9%
0.9%
0.5%
0.4%
97.6%
1.2%
0.5%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



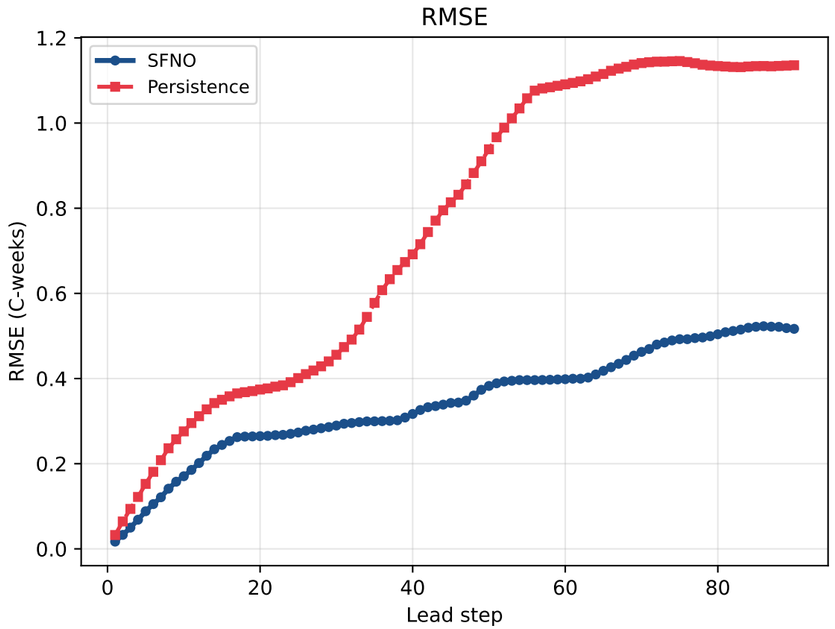
Great Barrier Reef Analysis - Test

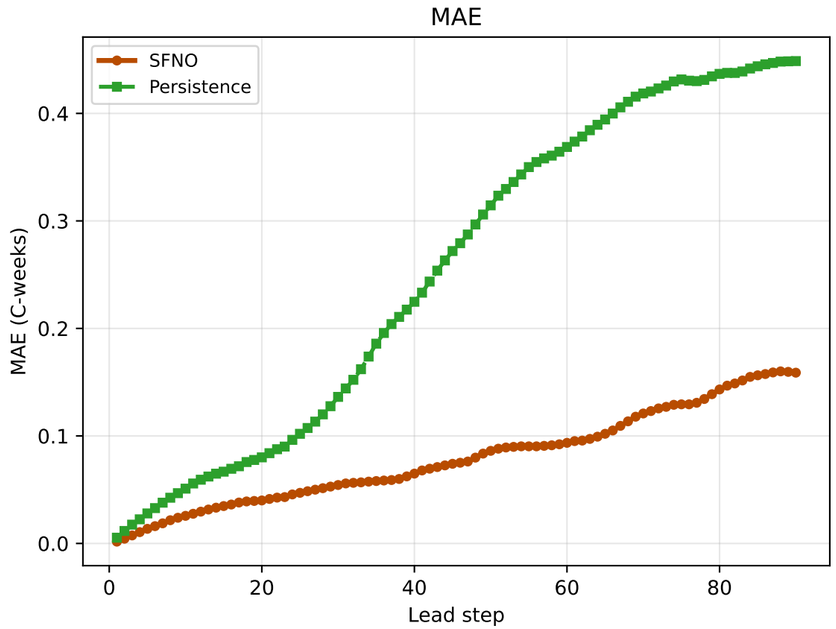
```
                    GREAT BARRIER REEF - TEST
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                      Grid: 5 lat x 5 lon points
```

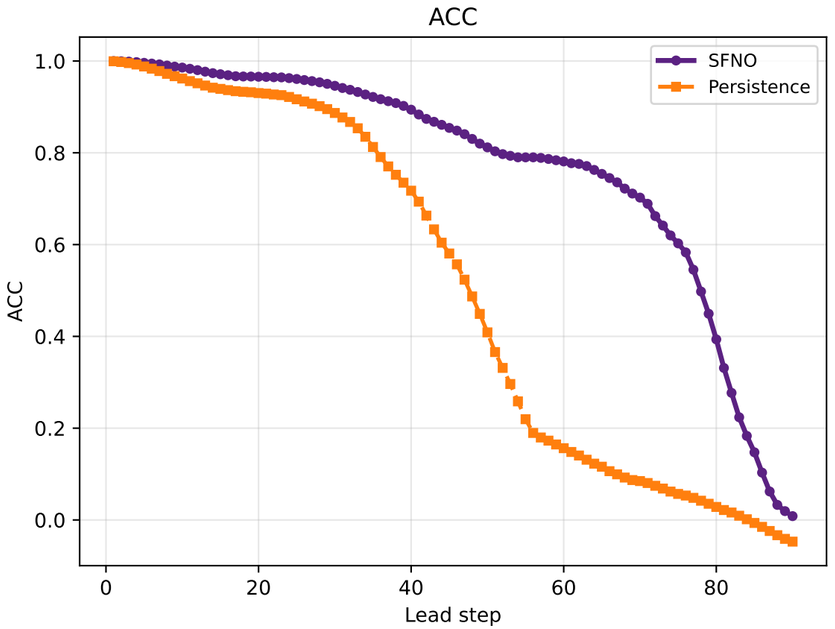
DHW Metrics by Lead Step - Test - GBR (18 chains scored at every lead)

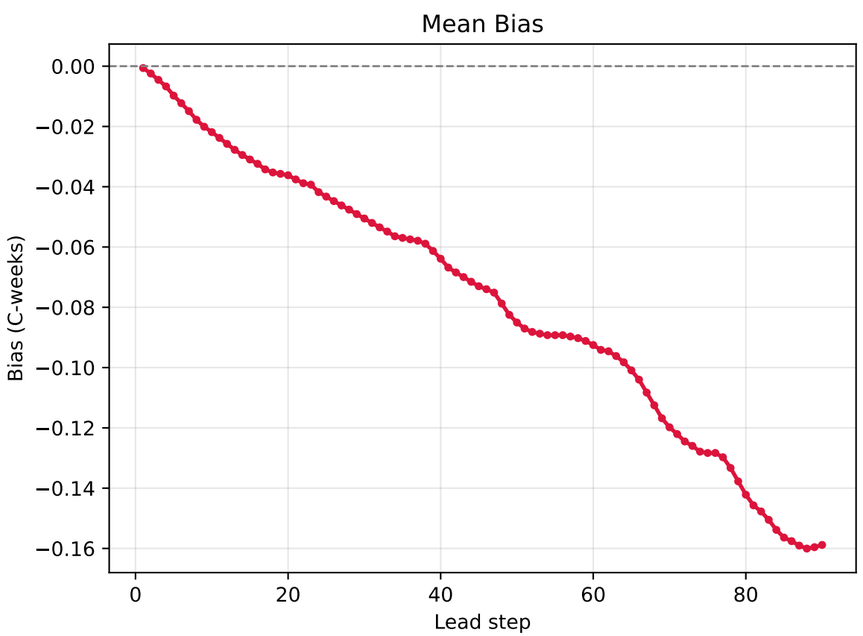
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0170	0.0325	47.7%	0.0017	0.0054	-0.0006	0.9999	0.9995	0.9997
2	0.0328	0.0643	49.0%	0.0044	0.0116	-0.0024	0.9994	0.9979	0.9989
3	0.0499	0.0939	46.9%	0.0074	0.0175	-0.0045	0.9987	0.9955	0.9975
4	0.0685	0.1219	43.8%	0.0105	0.0224	-0.0067	0.9976	0.9924	0.9954
5	0.0884	0.1525	42.1%	0.0136	0.0279	-0.0098	0.9960	0.9881	0.9924
6	0.1052	0.1810	41.9%	0.0161	0.0329	-0.0123	0.9944	0.9834	0.9893
7	0.1212	0.2083	41.8%	0.0187	0.0379	-0.0149	0.9927	0.9781	0.9858
8	0.1413	0.2362	40.2%	0.0216	0.0425	-0.0178	0.9901	0.9721	0.9809
9	0.1577	0.2574	38.7%	0.0239	0.0468	-0.0201	0.9878	0.9669	0.9764
10	0.1706	0.2759	38.2%	0.0257	0.0510	-0.0219	0.9857	0.9619	0.9723
11	0.1854	0.2955	37.2%	0.0276	0.0558	-0.0238	0.9831	0.9562	0.9673
12	0.2016	0.3116	35.3%	0.0296	0.0593	-0.0258	0.9802	0.9515	0.9615
13	0.2186	0.3277	33.3%	0.0316	0.0622	-0.0277	0.9768	0.9466	0.9551
14	0.2339	0.3424	31.7%	0.0333	0.0649	-0.0295	0.9736	0.9419	0.9488
15	0.2441	0.3502	30.3%	0.0348	0.0668	-0.0310	0.9712	0.9393	0.9444
16	0.2534	0.3582	29.3%	0.0362	0.0694	-0.0324	0.9691	0.9366	0.9402
17	0.2624	0.3652	28.1%	0.0381	0.0718	-0.0342	0.9669	0.9343	0.9361
18	0.2637	0.3680	28.3%	0.0391	0.0757	-0.0353	0.9663	0.9329	0.9350
19	0.2641	0.3705	28.7%	0.0395	0.0777	-0.0357	0.9662	0.9319	0.9347
20	0.2648	0.3742	29.2%	0.0400	0.0801	-0.0362	0.9657	0.9302	0.9339
21	0.2654	0.3768	29.6%	0.0414	0.0840	-0.0376	0.9654	0.9289	0.9333
22	0.2671	0.3810	29.9%	0.0426	0.0876	-0.0388	0.9647	0.9271	0.9321
23	0.2677	0.3841	30.3%	0.0431	0.0901	-0.0393	0.9643	0.9256	0.9314
24	0.2702	0.3914	31.0%	0.0456	0.0963	-0.0418	0.9627	0.9217	0.9285
25	0.2732	0.4011	31.9%	0.0471	0.1020	-0.0433	0.9608	0.9167	0.9249
26	0.2775	0.4104	32.4%	0.0486	0.1074	-0.0448	0.9585	0.9118	0.9207
27	0.2802	0.4189	33.1%	0.0500	0.1133	-0.0462	0.9562	0.9070	0.9165
28	0.2832	0.4287	33.9%	0.0514	0.1201	-0.0476	0.9536	0.9017	0.9117
29	0.2860	0.4402	35.0%	0.0529	0.1277	-0.0491	0.9502	0.8952	0.9056
30	0.2896	0.4558	36.5%	0.0543	0.1363	-0.0505	0.9460	0.8868	0.8980
31	0.2938	0.4739	38.0%	0.0558	0.1442	-0.0520	0.9412	0.8770	0.8895
32	0.2955	0.4915	39.9%	0.0564	0.1522	-0.0535	0.9372	0.8672	0.8823
33	0.2977	0.5150	42.2%	0.0568	0.1621	-0.0549	0.9325	0.8535	0.8739
34	0.2993	0.5448	45.1%	0.0575	0.1738	-0.0564	0.9270	0.8349	0.8642
35	0.2995	0.5777	48.1%	0.0580	0.1858	-0.0570	0.9217	0.8127	0.8547
36	0.3000	0.6076	50.6%	0.0585	0.1958	-0.0575	0.9169	0.7908	0.8461
37	0.3007	0.6333	52.5%	0.0590	0.2040	-0.0579	0.9125	0.7703	0.8386
38	0.3020	0.6548	53.9%	0.0600	0.2107	-0.0589	0.9084	0.7521	0.8315
39	0.3084	0.6735	54.2%	0.0624	0.2175	-0.0613	0.9022	0.7349	0.8210
40	0.3168	0.6918	54.2%	0.0650	0.2248	-0.0639	0.8940	0.7175	0.8068
41	0.3261	0.7155	54.4%	0.0679	0.2332	-0.0668	0.8834	0.6935	0.7887
42	0.3326	0.7441	55.3%	0.0695	0.2436	-0.0684	0.8738	0.6632	0.7720
43	0.3354	0.7709	56.5%	0.0710	0.2537	-0.0700	0.8676	0.6331	0.7604
44	0.3387	0.7952	57.4%	0.0726	0.2632	-0.0715	0.8610	0.6042	0.7480
45	0.3426	0.8139	57.9%	0.0741	0.2720	-0.0730	0.8543	0.5806	0.7349
46	0.3436	0.8315	58.7%	0.0751	0.2793	-0.0740	0.8484	0.5570	0.7236
47	0.3481	0.8560	59.3%	0.0762	0.2874	-0.0751	0.8406	0.5236	0.7082
48	0.3601	0.8827	59.2%	0.0798	0.2967	-0.0788	0.8303	0.4870	0.6879
49	0.3736	0.9103	59.0%	0.0836	0.3059	-0.0825	0.8199	0.4490	0.6676
50	0.3826	0.9384	59.2%	0.0861	0.3145	-0.0850	0.8119	0.4087	0.6512
51	0.3888	0.9668	59.8%	0.0881	0.3234	-0.0871	0.8033	0.3658	0.6358
52	0.3931	0.9892	60.3%	0.0892	0.3297	-0.0882	0.7971	0.3313	0.6252
53	0.3948	1.0111	61.0%	0.0898	0.3362	-0.0887	0.7934	0.2963	0.6177
54	0.3964	1.0344	61.7%	0.0903	0.3432	-0.0892	0.7901	0.2584	0.6106
55	0.3964	1.0582	62.5%	0.0903	0.3501	-0.0892	0.7901	0.2195	0.6083
56	0.3964	1.0761	63.2%	0.0903	0.3548	-0.0892	0.7901	0.1896	0.6067
57	0.3966	1.0812	63.3%	0.0908	0.3581	-0.0897	0.7889	0.1797	0.6025
58	0.3971	1.0841	63.4%	0.0913	0.3607	-0.0902	0.7866	0.1728	0.5972
59	0.3979	1.0875	63.4%	0.0922	0.3643	-0.0912	0.7838	0.1645	0.5907
60	0.3985	1.0909	63.5%	0.0936	0.3688	-0.0925	0.7812	0.1564	0.5846
61	0.3996	1.0943	63.5%	0.0952	0.3738	-0.0941	0.7773	0.1481	0.5756
62	0.3997	1.0980	63.6%	0.0957	0.3785	-0.0946	0.7758	0.1404	0.5700
63	0.4023	1.1030	63.5%	0.0972	0.3843	-0.0962	0.7711	0.1313	0.5623
64	0.4096	1.1092	63.1%	0.0993	0.3894	-0.0982	0.7627	0.1227	0.5508
65	0.4181	1.1152	62.5%	0.1020	0.3943	-0.1009	0.7541	0.1160	0.5402
66	0.4266	1.1228	62.0%	0.1051	0.3999	-0.1040	0.7450	0.1063	0.5291
67	0.4347	1.1280	61.5%	0.1094	0.4056	-0.1083	0.7356	0.0994	0.5181
68	0.4437	1.1321	60.8%	0.1136	0.4108	-0.1125	0.7220	0.0924	0.5009
69	0.4539	1.1374	60.1%	0.1179	0.4156	-0.1168	0.7112	0.0870	0.4890
70	0.4626	1.1409	59.5%	0.1209	0.4186	-0.1198	0.7024	0.0847	0.4793
71	0.4693	1.1428	58.9%	0.1231	0.4203	-0.1221	0.6889	0.0805	0.4635
72	0.4798	1.1442	58.1%	0.1256	0.4232	-0.1245	0.6620	0.0744	0.4325
73	0.4846	1.1443	57.6%	0.1271	0.4259	-0.1260	0.6415	0.0684	0.4070
74	0.4895	1.1451	57.3%	0.1290	0.4297	-0.1279	0.6200	0.0619	0.3831
75	0.4925	1.1456	57.0%	0.1294	0.4316	-0.1283	0.6026	0.0567	0.3637
76	0.4925	1.1433	56.9%	0.1294	0.4305	-0.1283	0.5830	0.0532	0.3443
77	0.4950	1.1405	56.6%	0.1308	0.4300	-0.1298	0.5453	0.0482	0.3100
78	0.4964	1.1370	56.3%	0.1344	0.4312	-0.1333	0.4977	0.0419	0.2705
79	0.4994	1.1351	56.0%	0.1388	0.4344	-0.1377	0.4493	0.0353	0.2333
80	0.5039	1.1339	55.6%	0.1432	0.4368	-0.1422	0.3935	0.0284	0.1917
81	0.5089	1.1328	55.1%	0.1468	0.4377	-0.1457	0.3312	0.0216	0.1462
82	0.5117	1.1317	54.8%	0.1488	0.4376	-0.1477	0.2769	0.0162	0.1068
83	0.5150	1.1313	54.5%	0.1516	0.4390	-0.1505	0.2237	0.0092	0.0687
84	0.5193	1.1328	54.2%	0.1549	0.4417	-0.1538	0.1831	0.0015	0.0397
85	0.5215	1.1337	54.0%	0.1564	0.4438	-0.1564	0.1474	-0.0065	0.0135
86	0.5229	1.1340	53.9%	0.1576	0.4457	-0.1576	0.1032	-0.0150	-0.0187
87	0.5218	1.1333	54.0%	0.1591	0.4469	-0.1591	0.0620	-0.0240	-0.0518
88	0.5214	1.1341	54.0%	0.1600	0.4482	-0.1600	0.0328	-0.0332	-0.0761
89	0.5185	1.1349	54.3%	0.1596	0.4484	-0.1596	0.0192	-0.0412	-0.0873
90	0.5167	1.1357	54.5%	0.1588	0.4487	-0.1588	0.0084	-0.0472	-0.0964

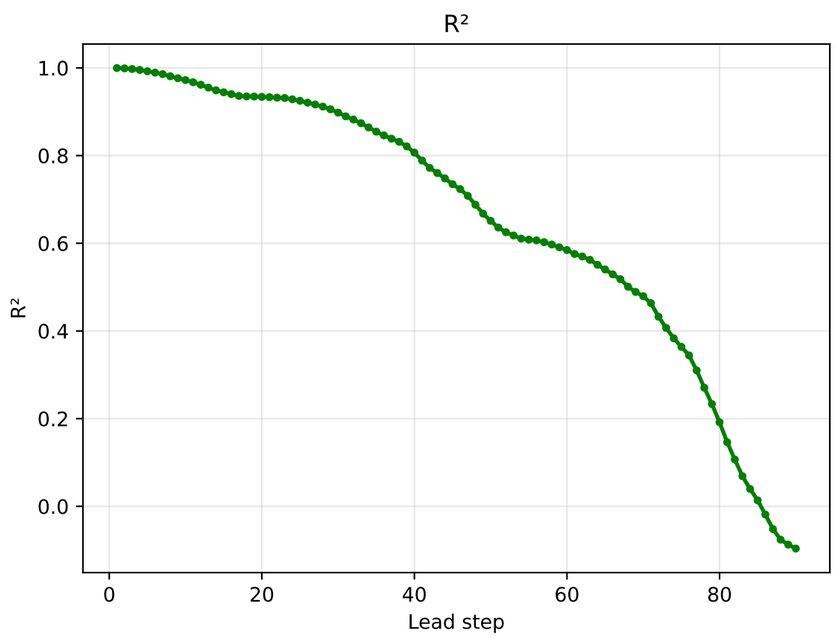
DHW Forecast Skill vs Lead Step - Test - GBR

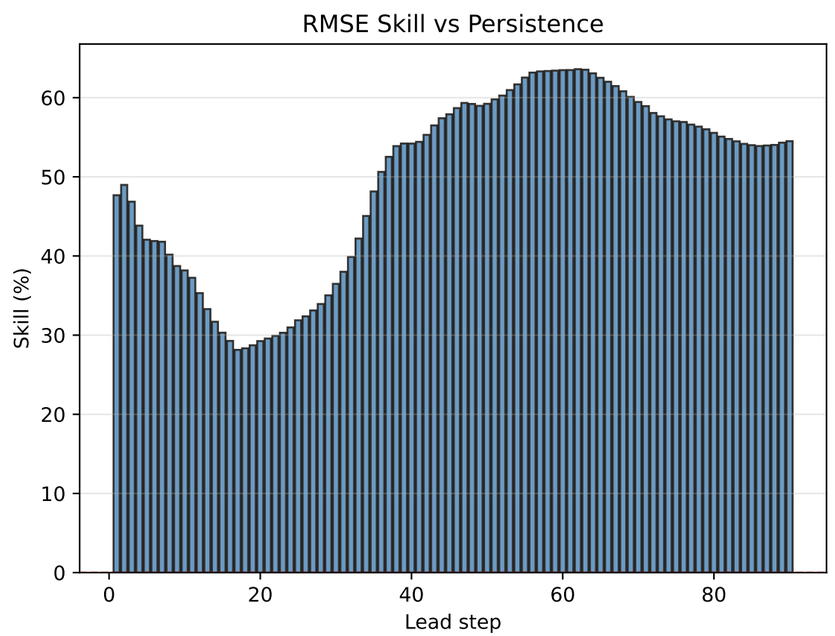


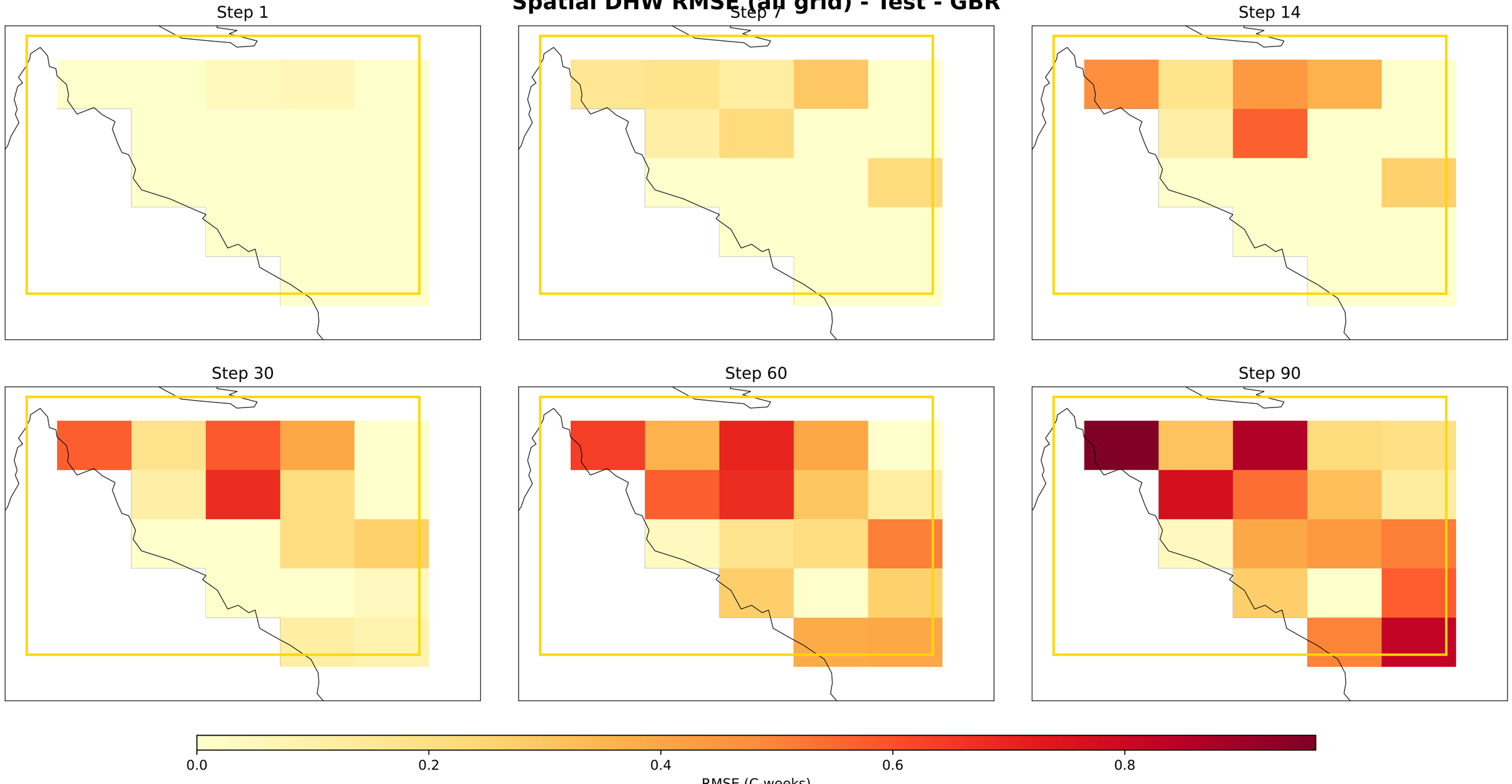


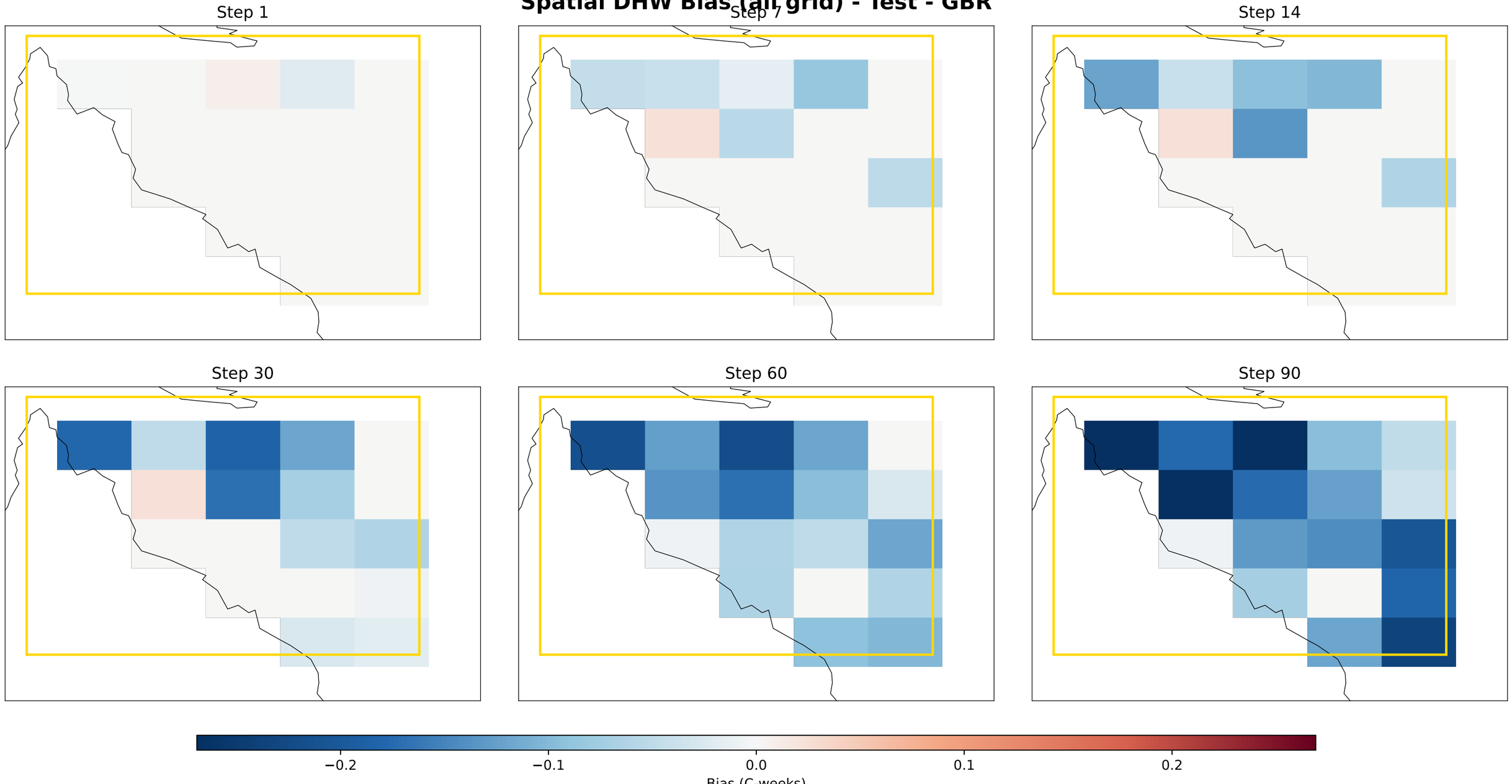


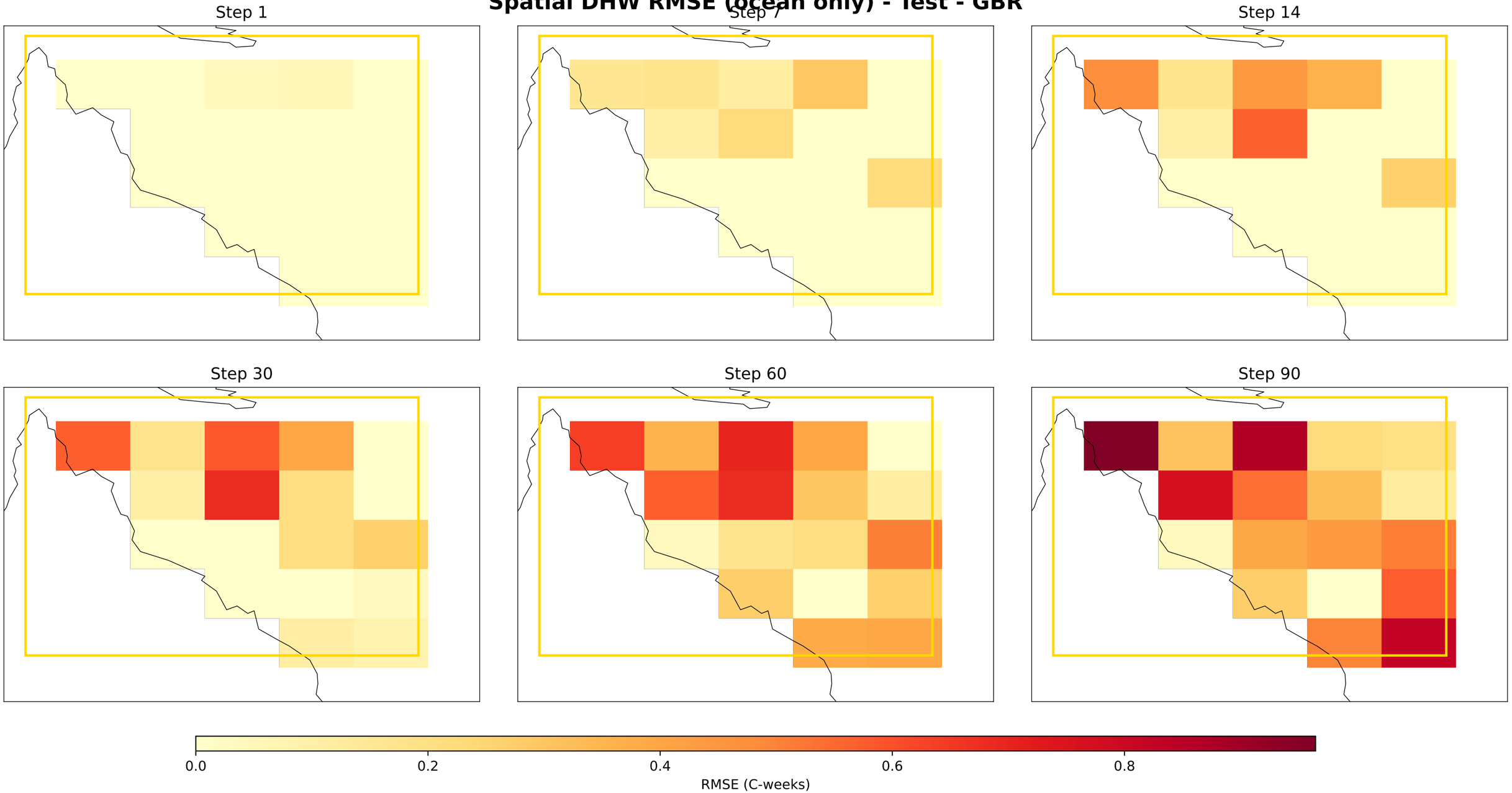


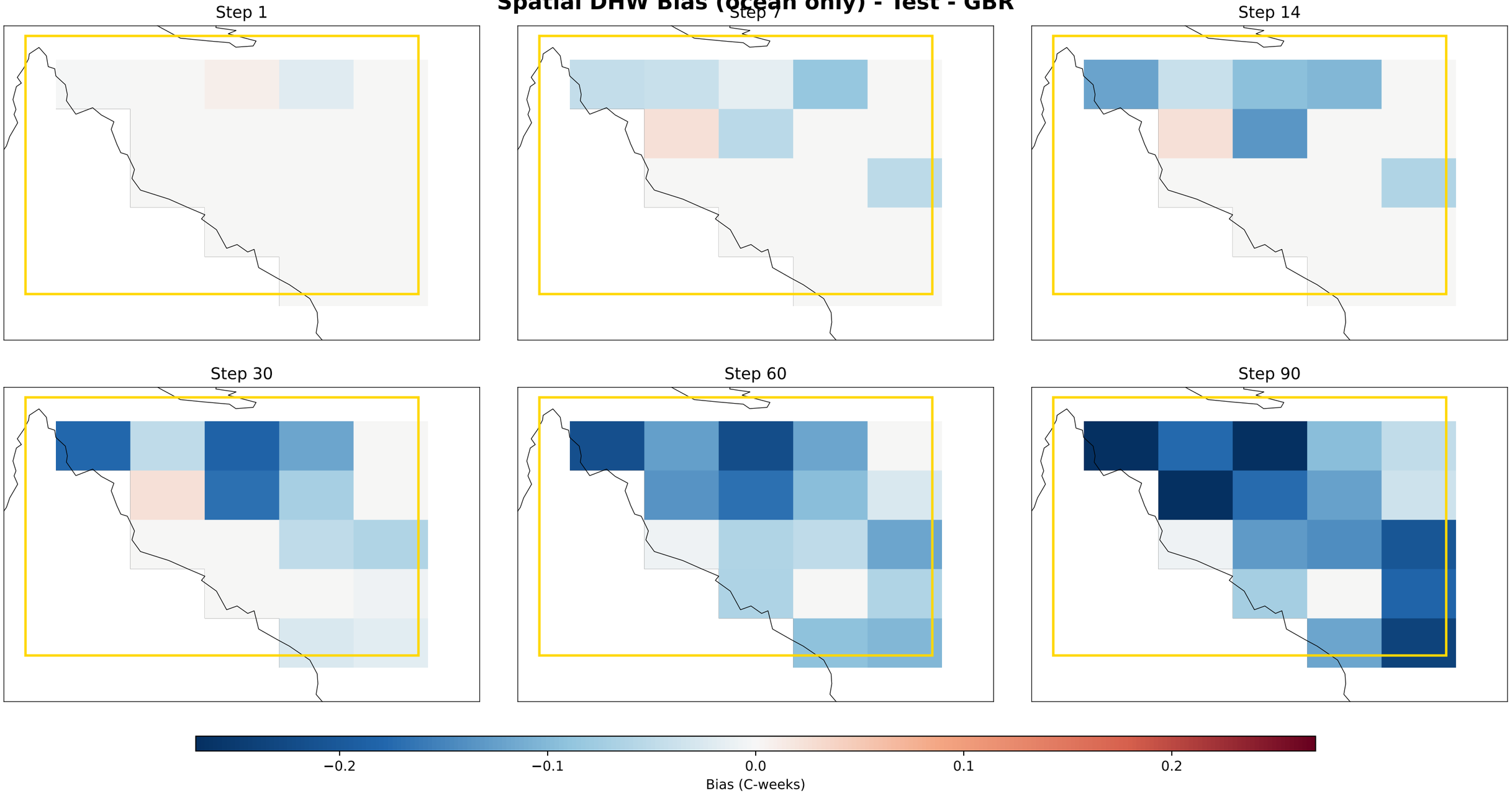





Spatial DHW RMSE (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
RMSE (C-weeks)

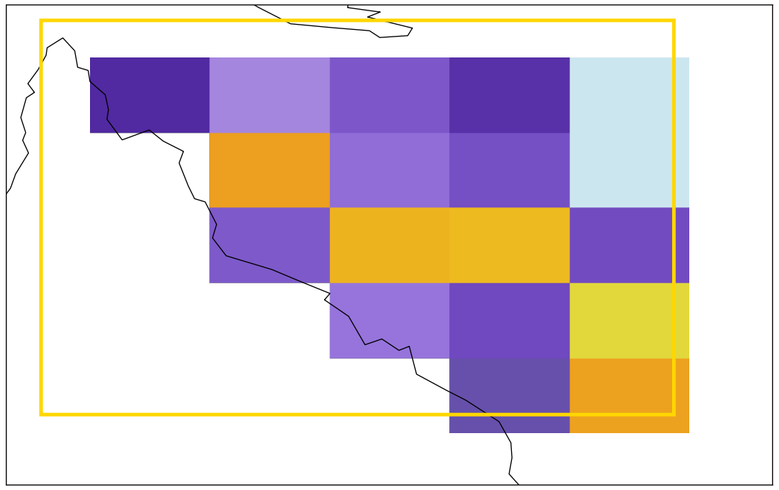

Spatial DHW Bias (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.2
−0.1
0.0
0.1
0.2
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
RMSE (C-weeks)

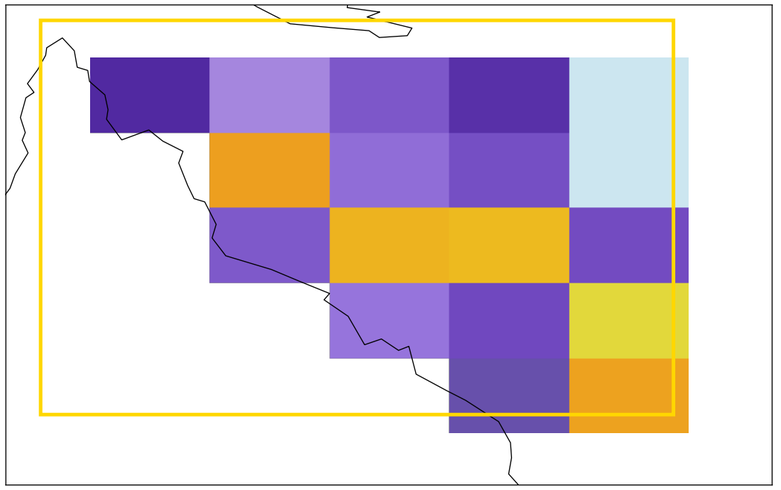

Spatial DHW Bias (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.2
−0.1
0.0
0.1
0.2
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test - GBR (C-weeks)

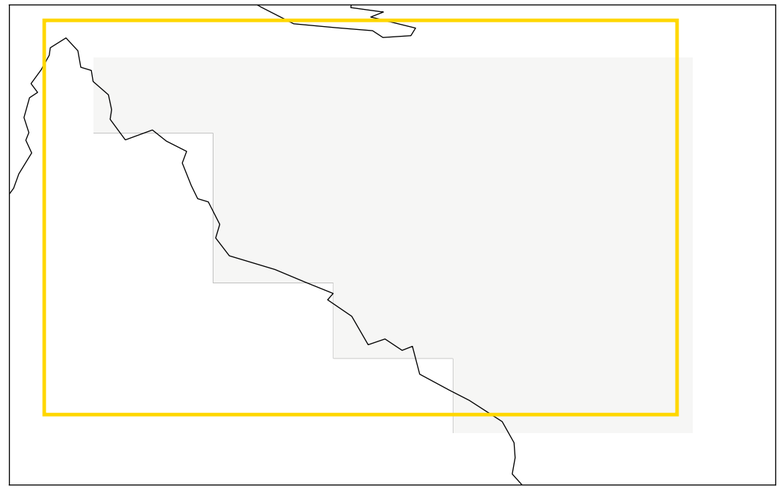
Actual DHW - Step 1



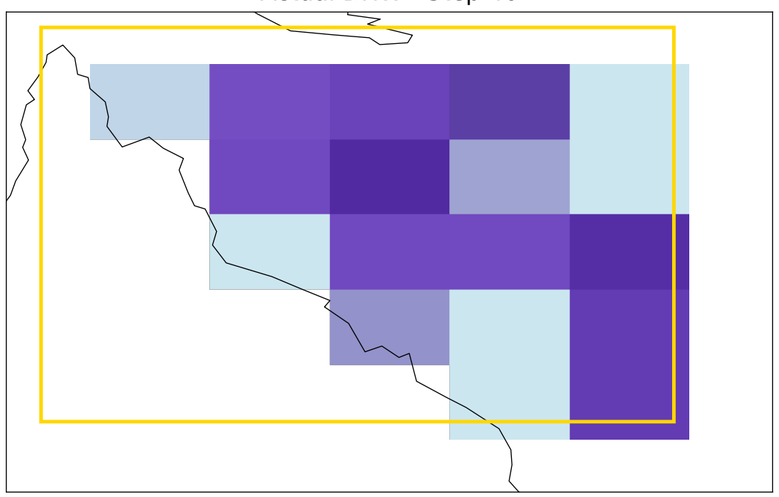
Predicted DHW - Step 1



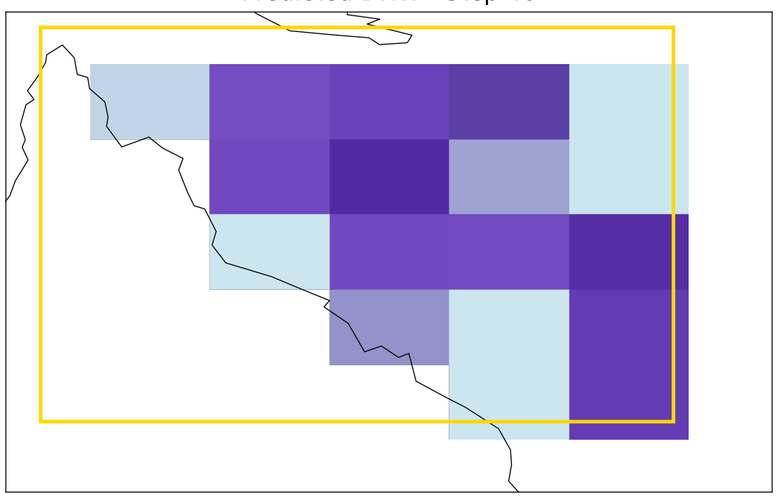
Error (Pred - Actual) - Step 1 [scale ±8]



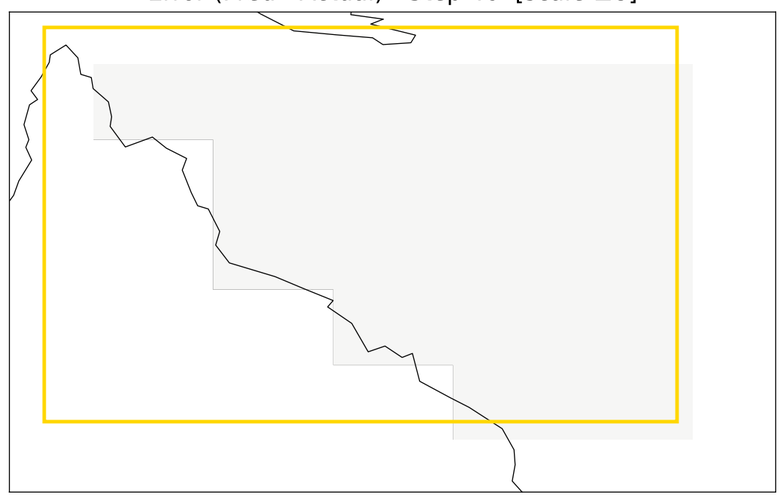
Actual DHW - Step 46



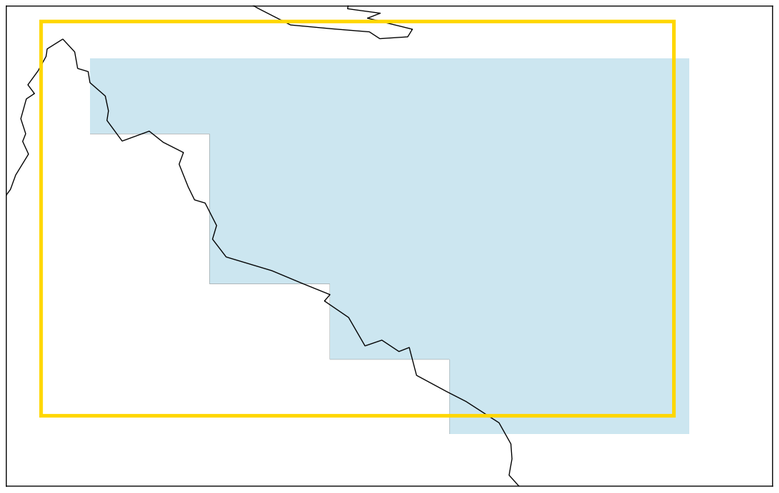
Predicted DHW - Step 46



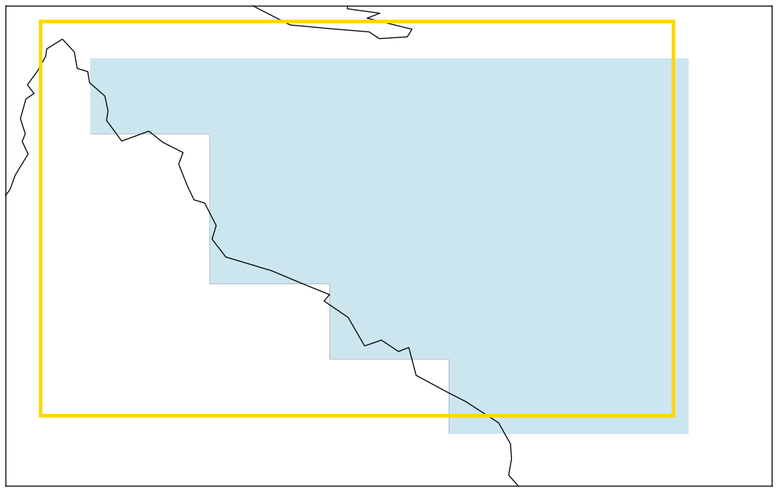
Error (Pred - Actual) - Step 46 [scale ±8]



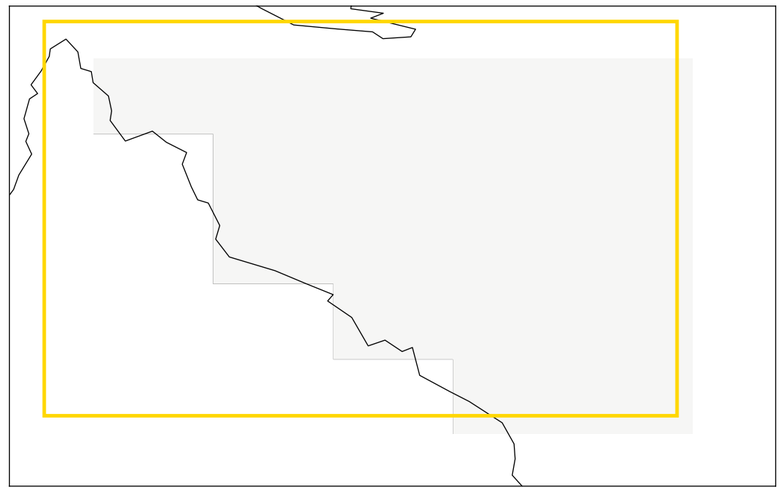
Actual DHW - Step 90

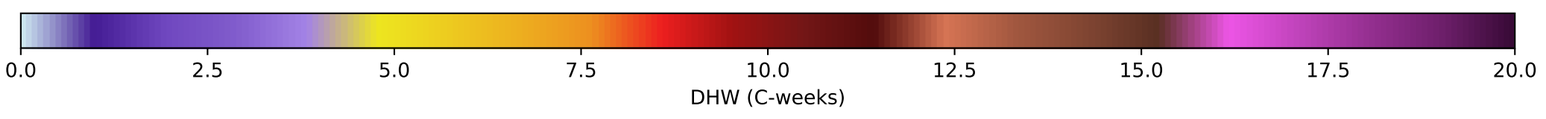


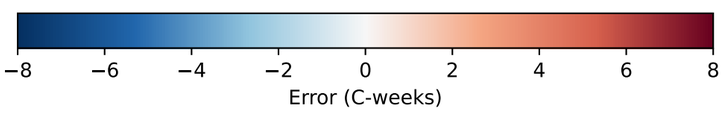
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

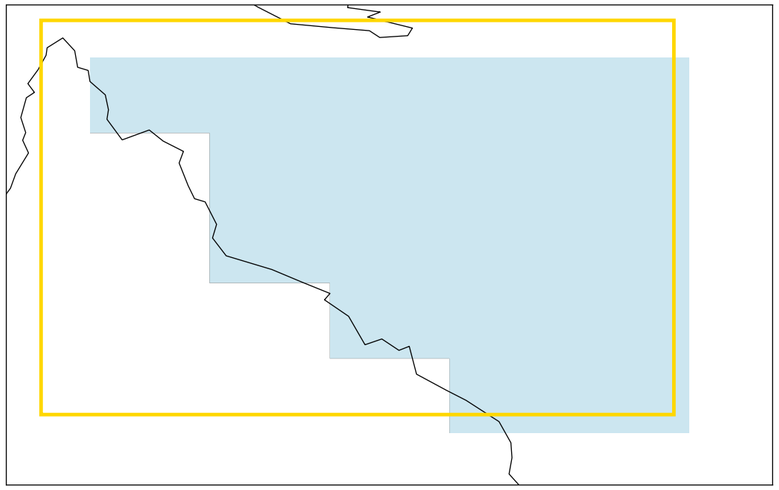




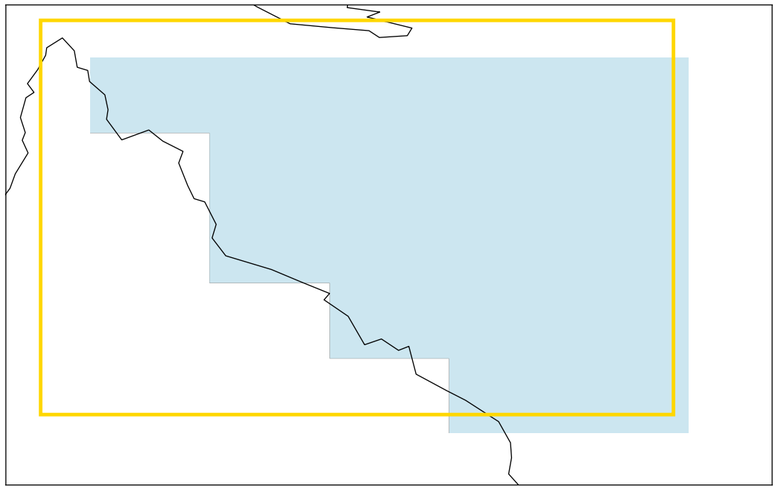


DHW Chain init: 2020-07-09 - Test - GBR (C-weeks)

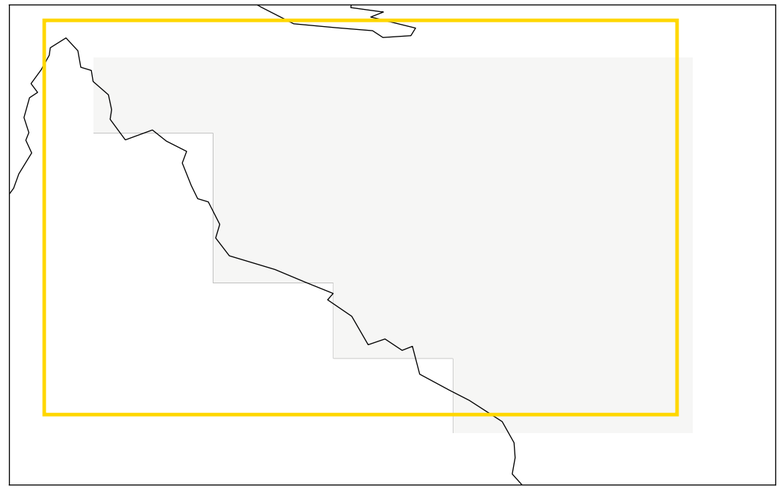
Actual DHW - Step 1



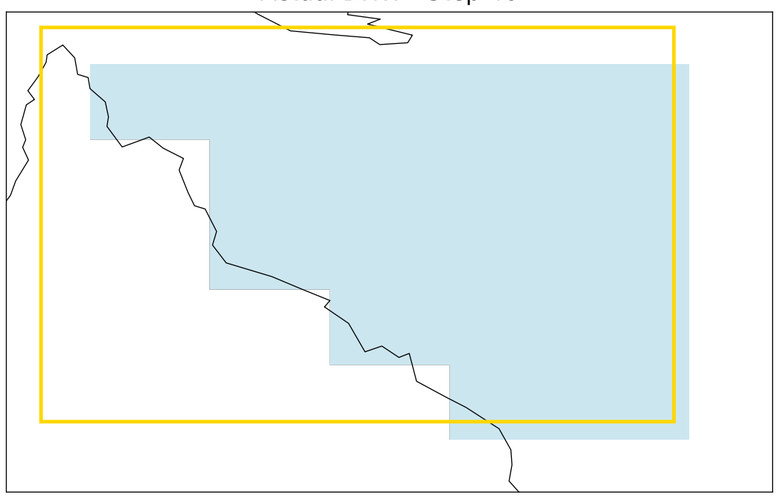
Predicted DHW - Step 1



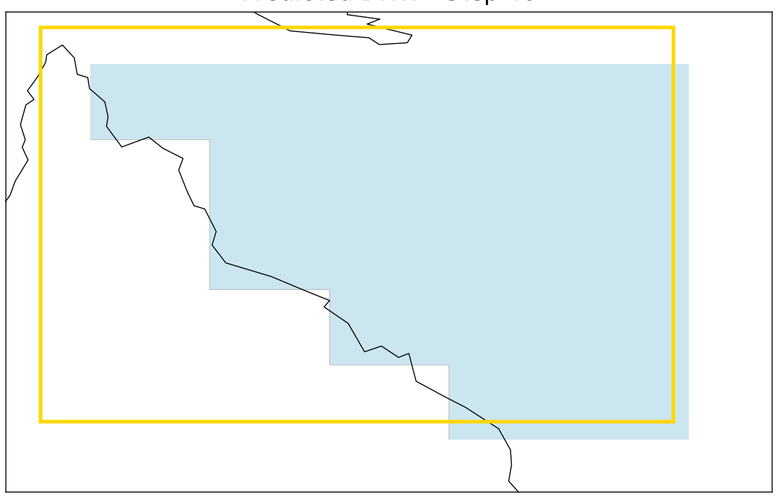
Error (Pred - Actual) - Step 1 [scale ±8]



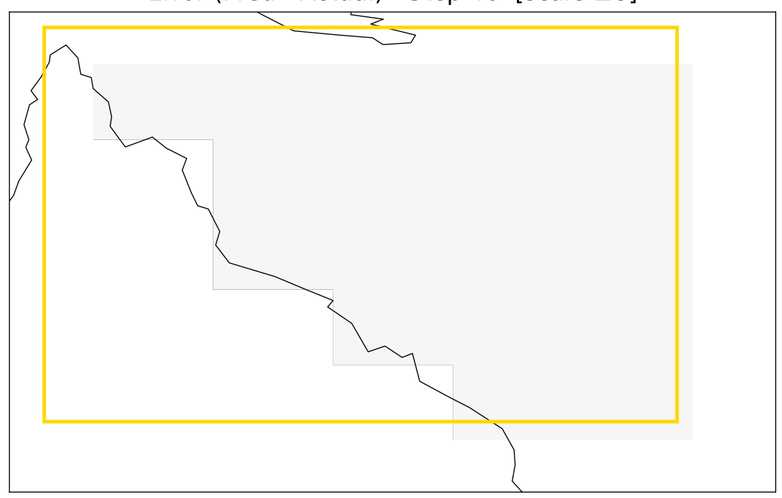
Actual DHW - Step 46



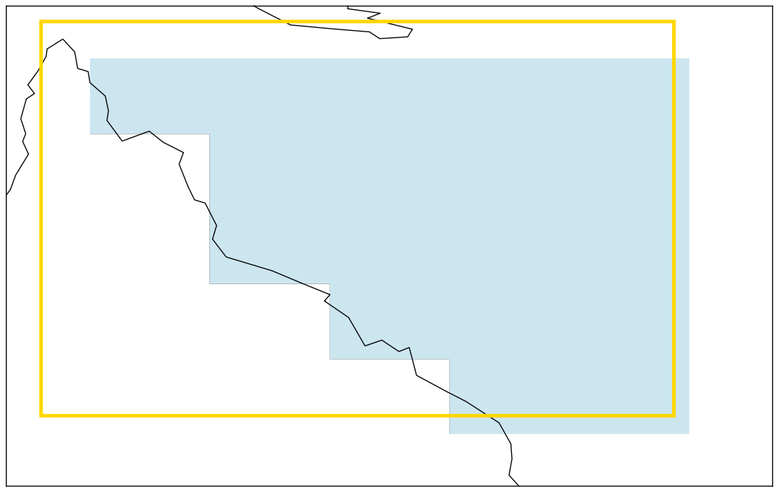
Predicted DHW - Step 46



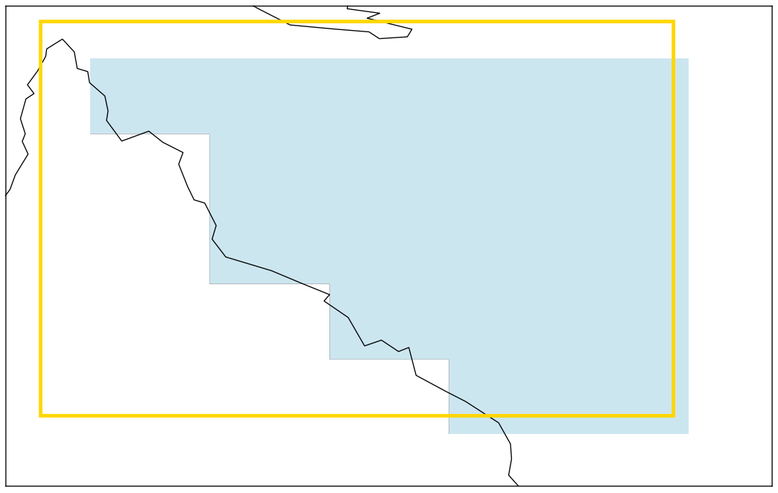
Error (Pred - Actual) - Step 46 [scale ±8]



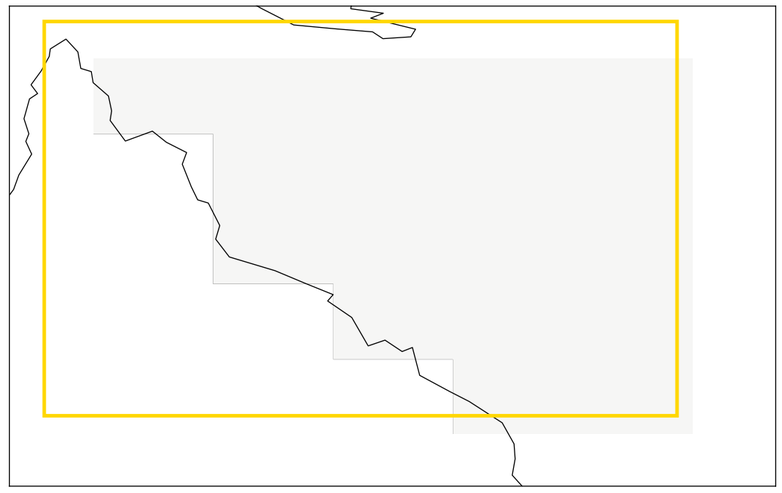
Actual DHW - Step 90

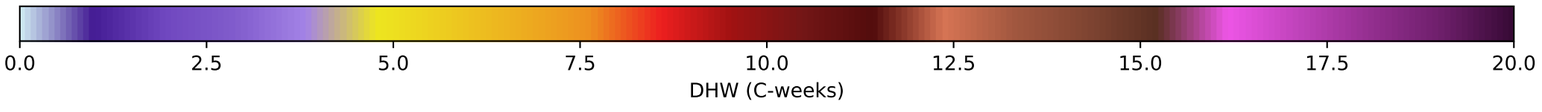


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

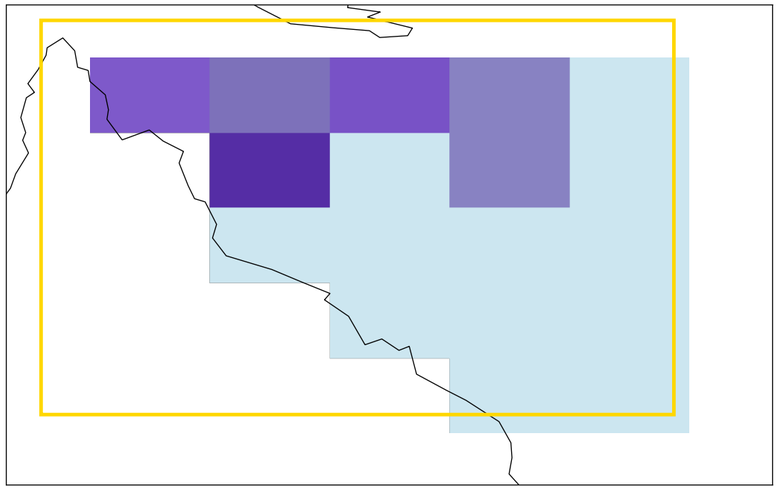




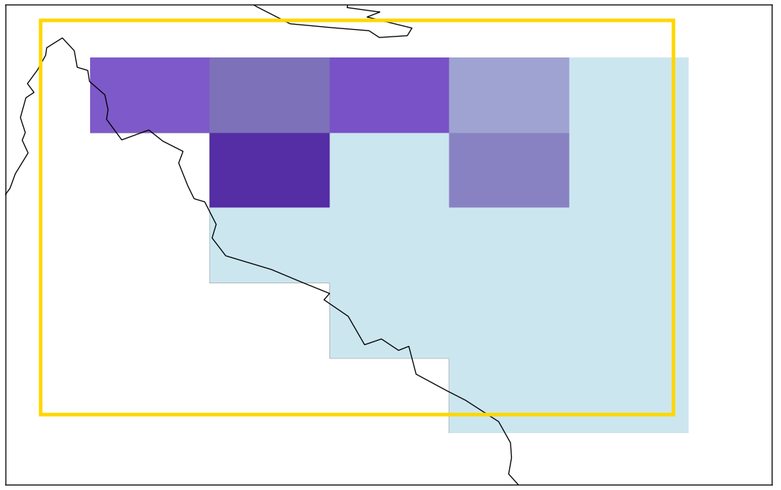


DHW Chain init: 2021-01-06 - Test - GBR (C-weeks)

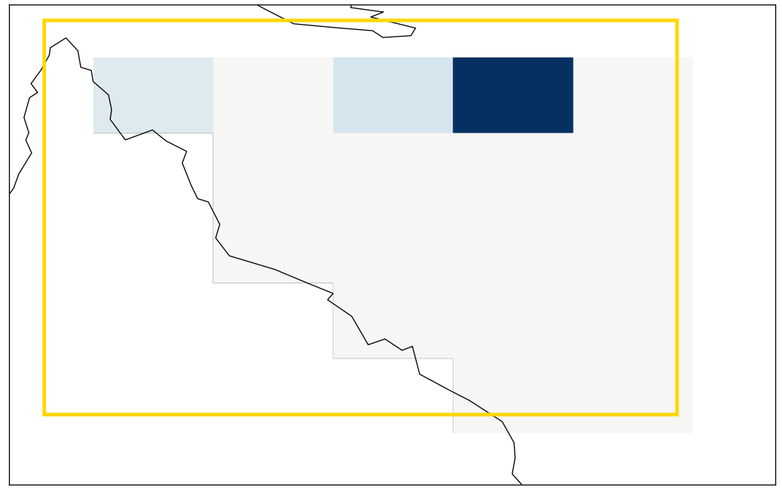
Actual DHW - Step 1



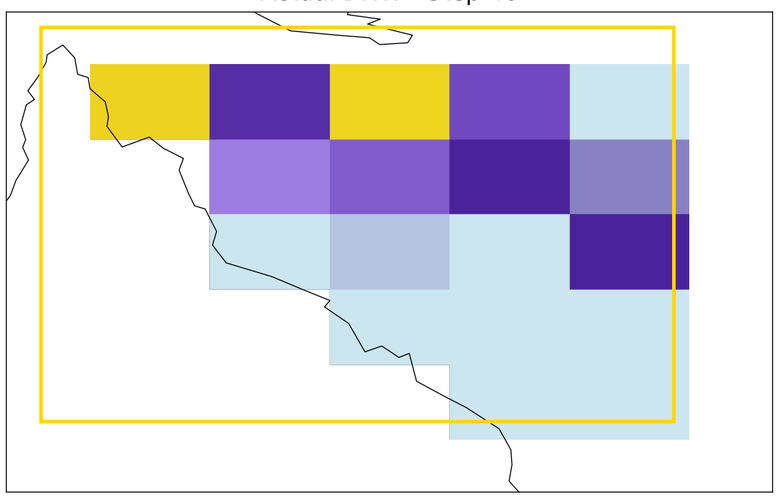
Predicted DHW - Step 1



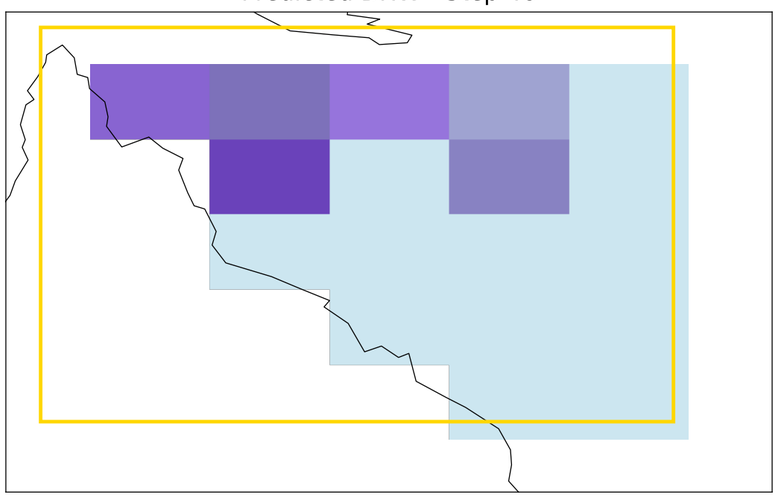
Error (Pred - Actual) - Step 1 [scale ±0.11]



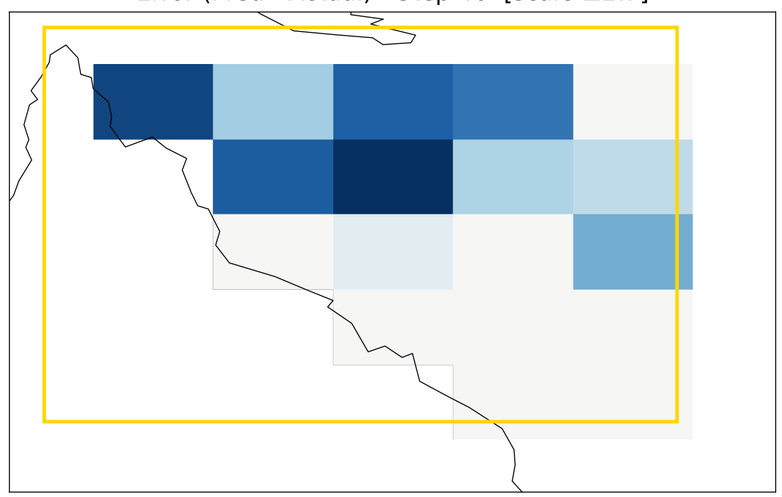
Actual DHW - Step 46



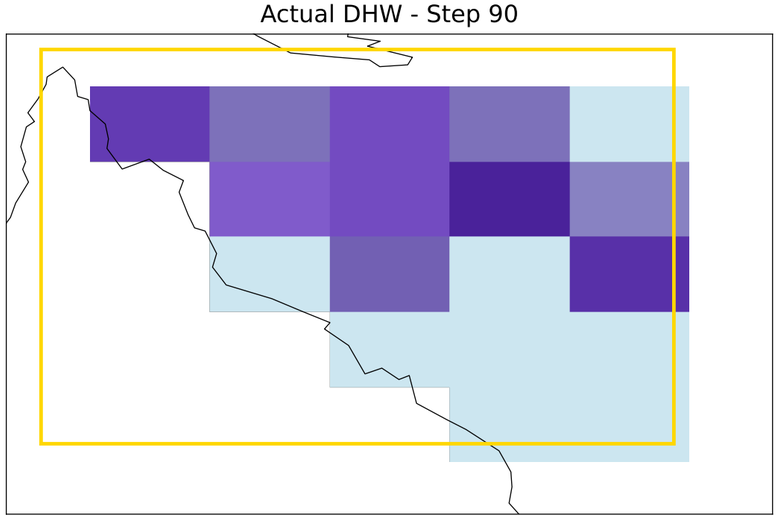
Predicted DHW - Step 46



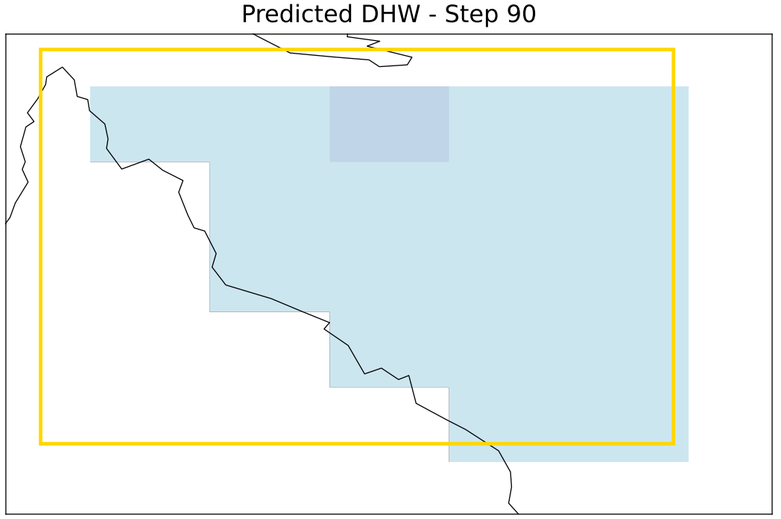
Error (Pred - Actual) - Step 46 [scale ±2.7]



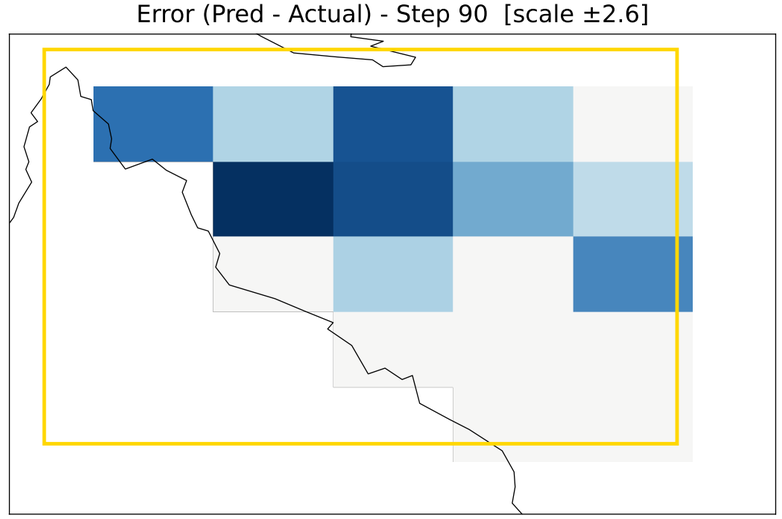
Actual DHW - Step 90

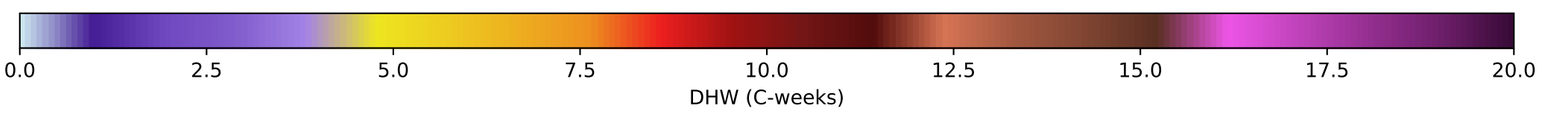


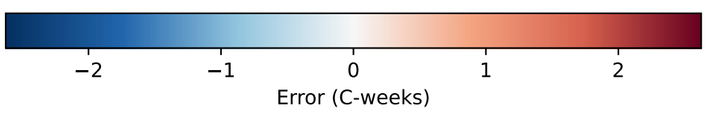
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.6]

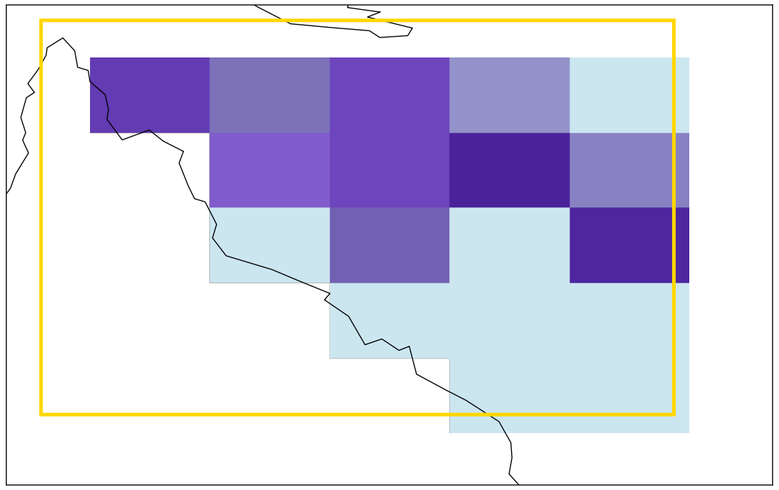




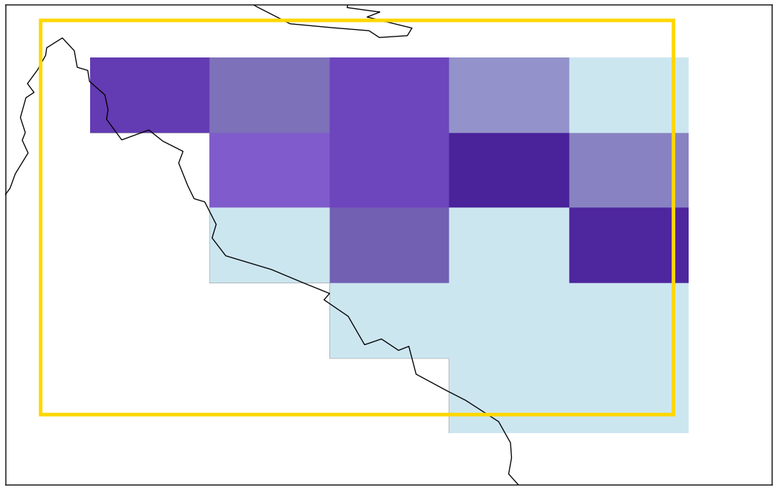


DHW Chain init: 2021-04-06 - Test - GBR (C-weeks)

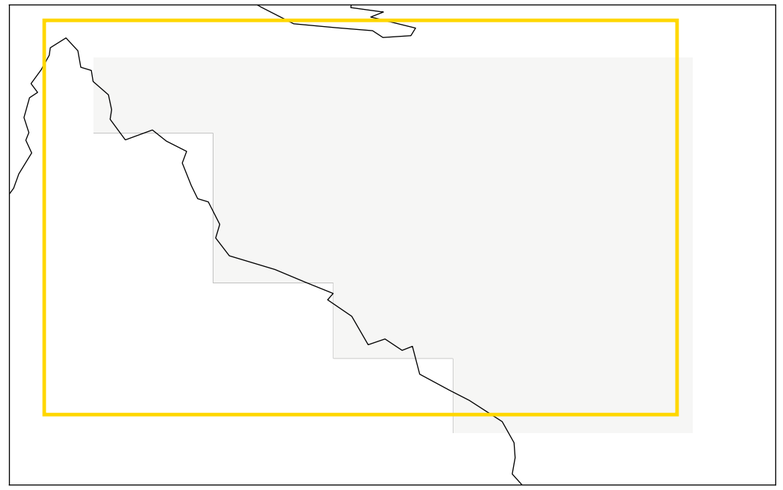
Actual DHW - Step 1



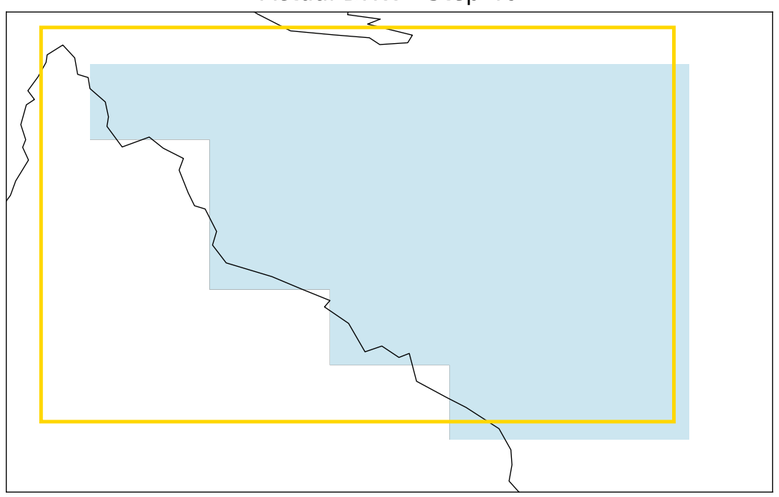
Predicted DHW - Step 1



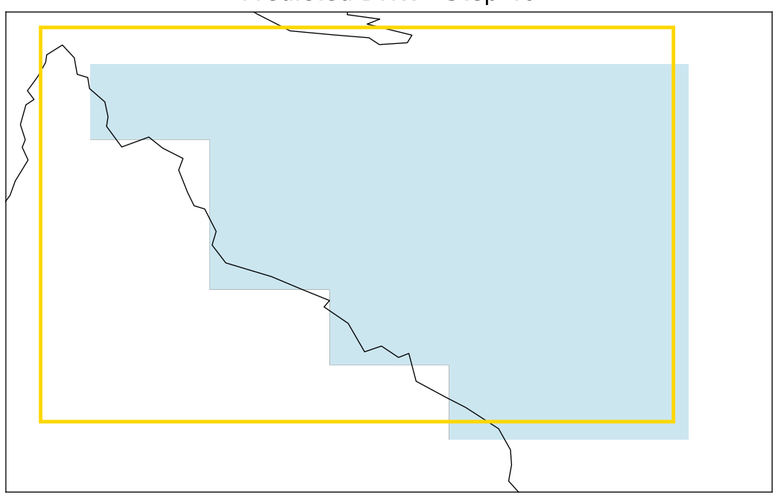
Error (Pred - Actual) - Step 1 [scale ±8]



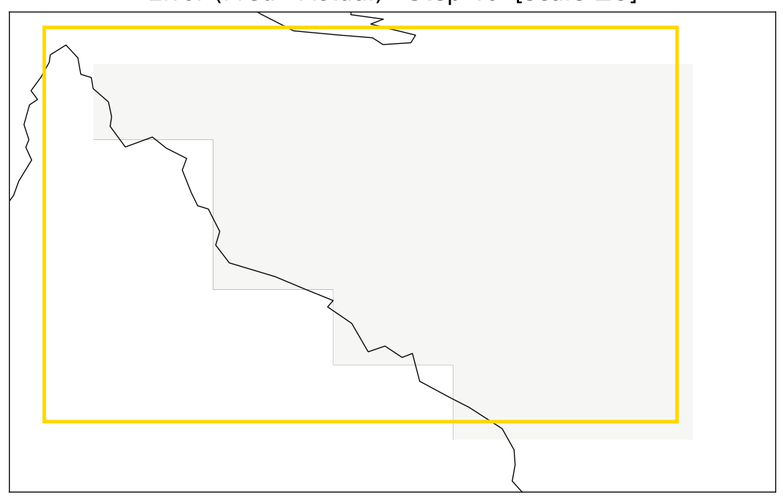
Actual DHW - Step 46



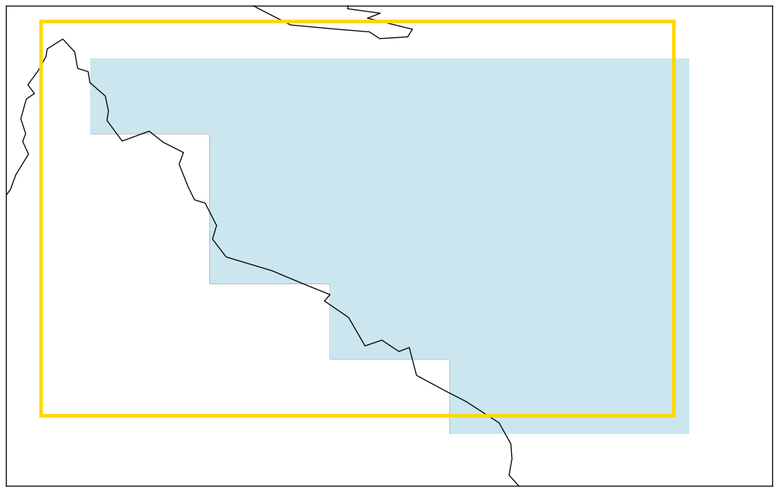
Predicted DHW - Step 46



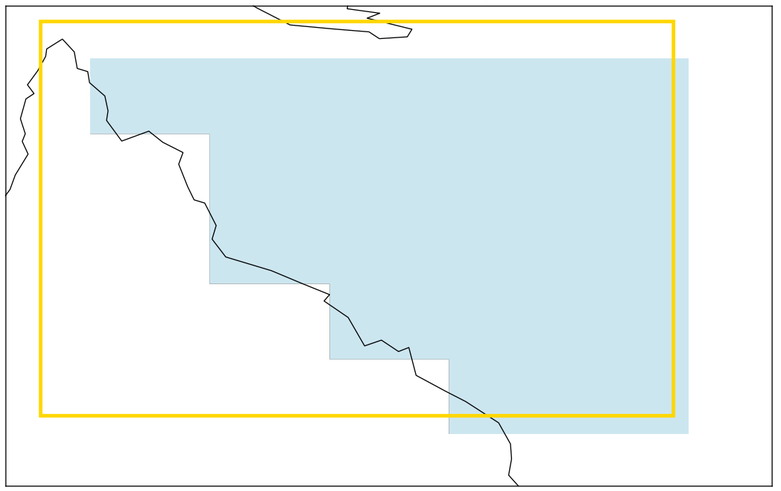
Error (Pred - Actual) - Step 46 [scale ±8]



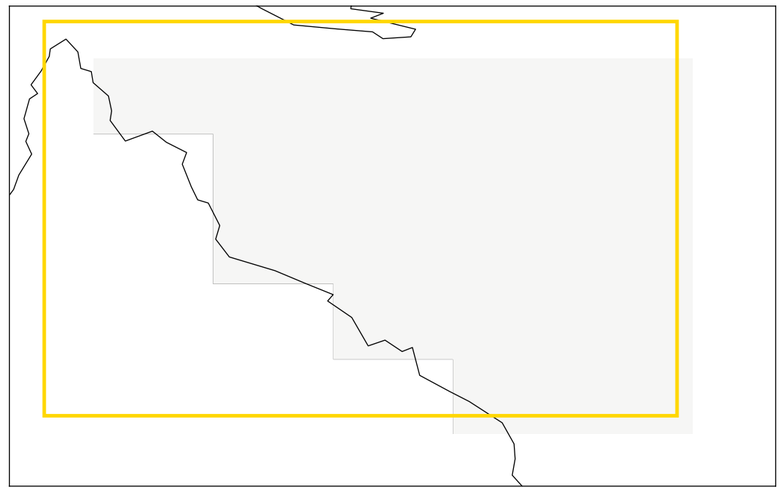
Actual DHW - Step 90

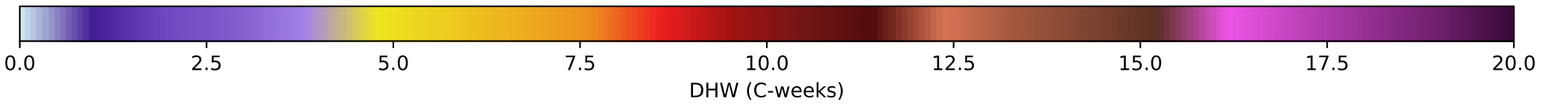


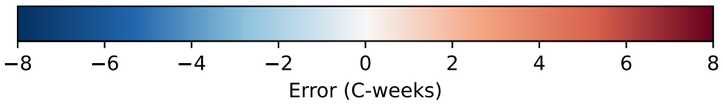
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

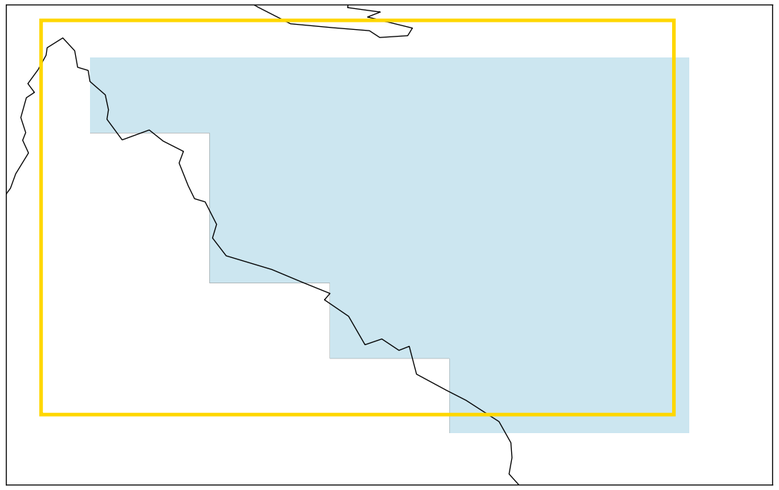




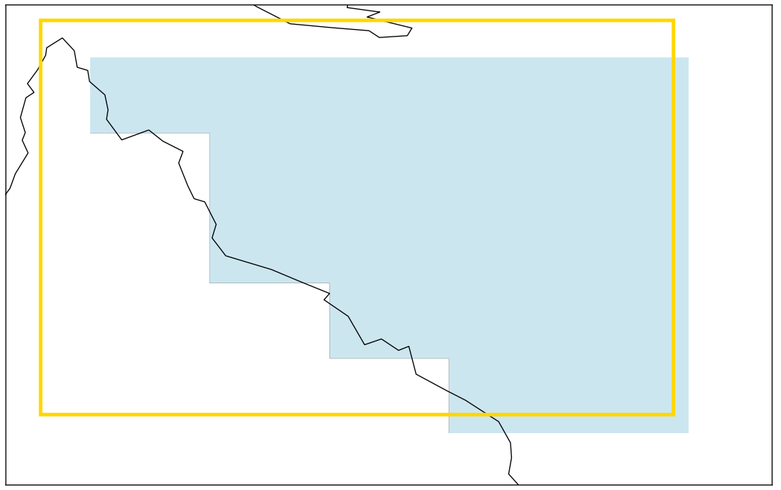


DHW Chain init: 2021-10-03 - Test - GBR (C-weeks)

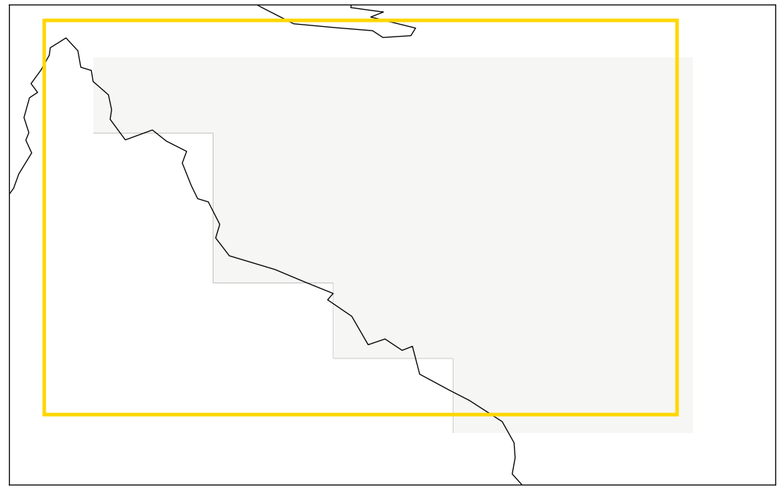
Actual DHW - Step 1



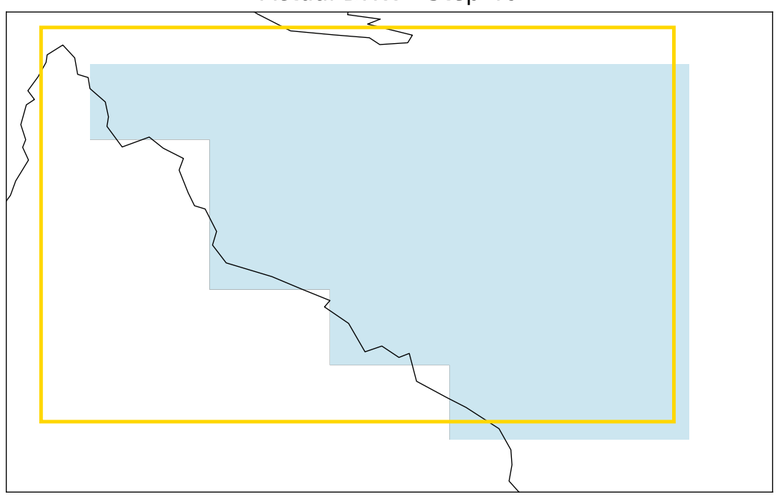
Predicted DHW - Step 1



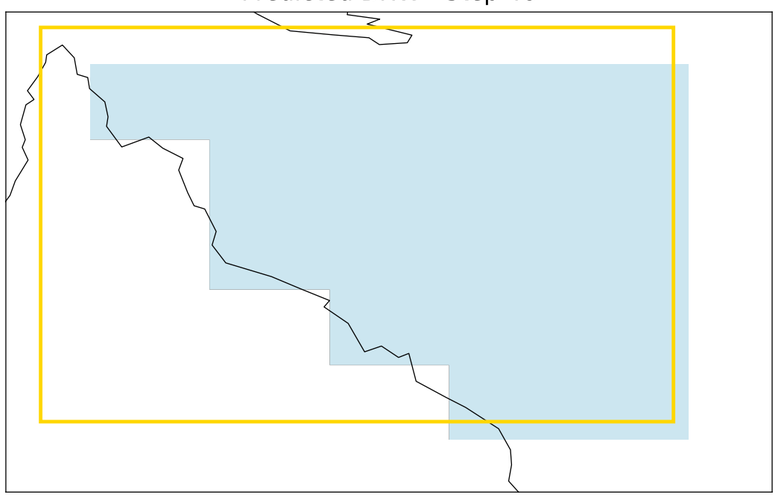
Error (Pred - Actual) - Step 1 [scale ±8]



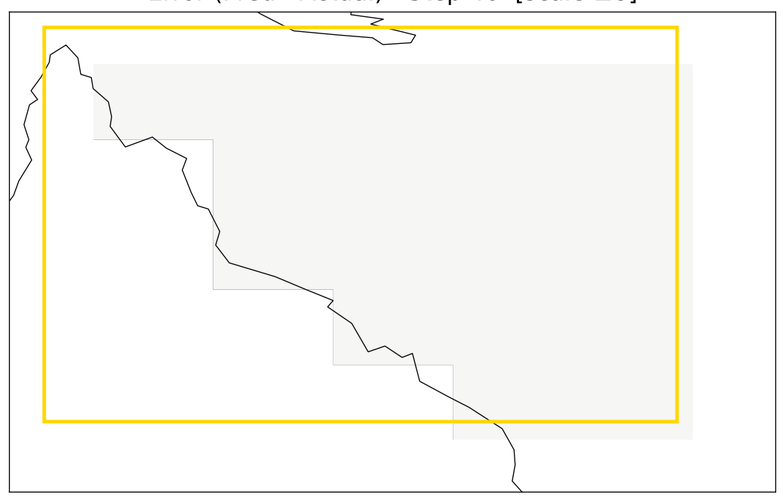
Actual DHW - Step 46



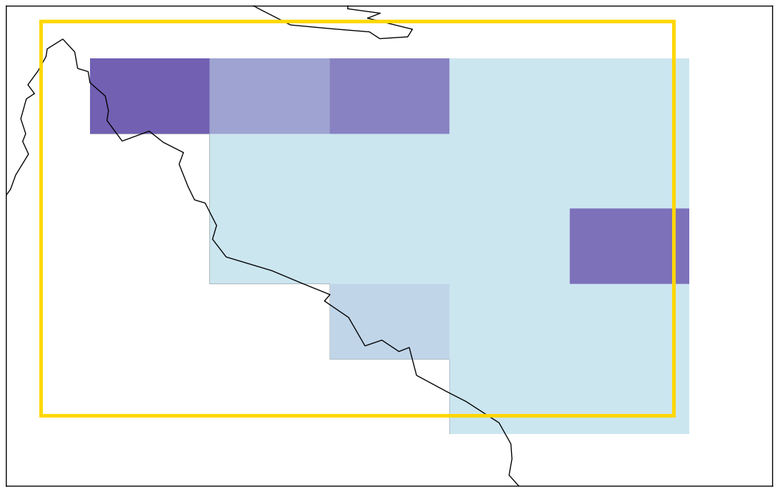
Predicted DHW - Step 46



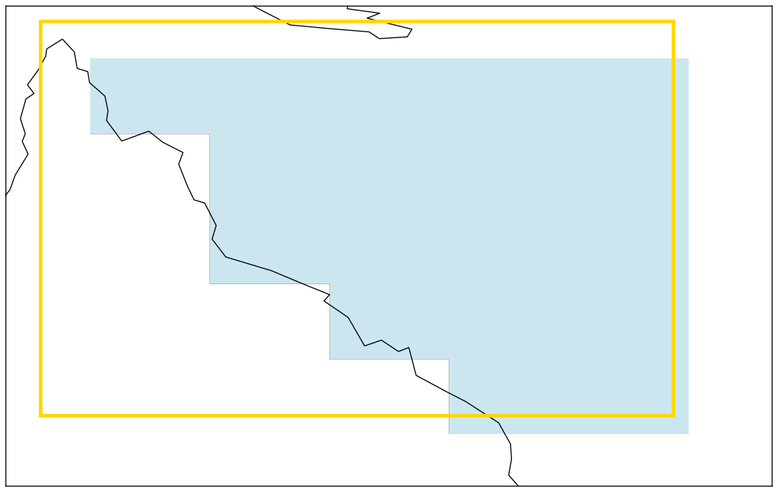
Error (Pred - Actual) - Step 46 [scale ±8]



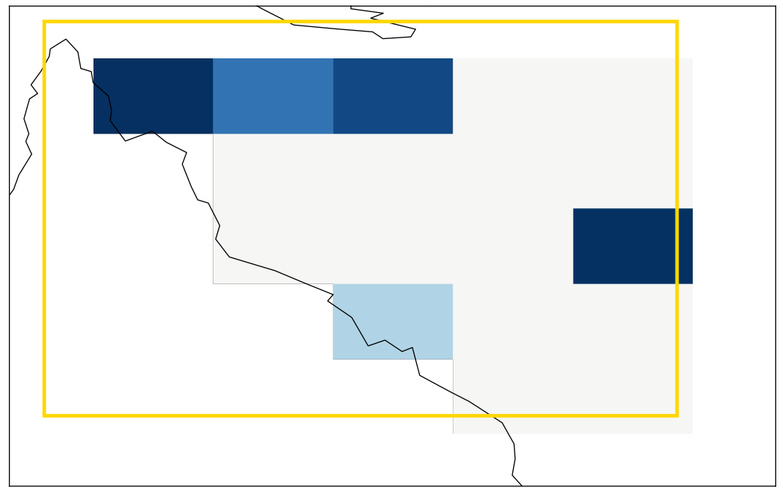
Actual DHW - Step 90

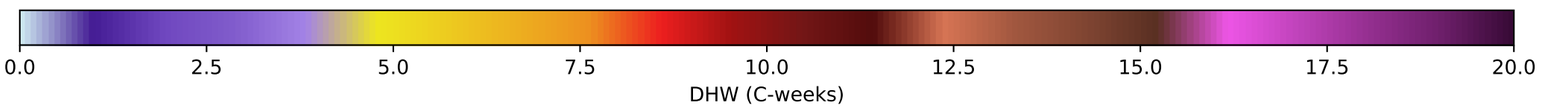


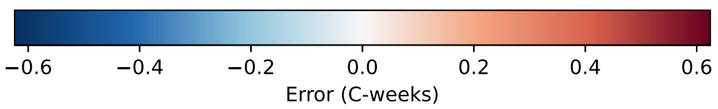
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.62]

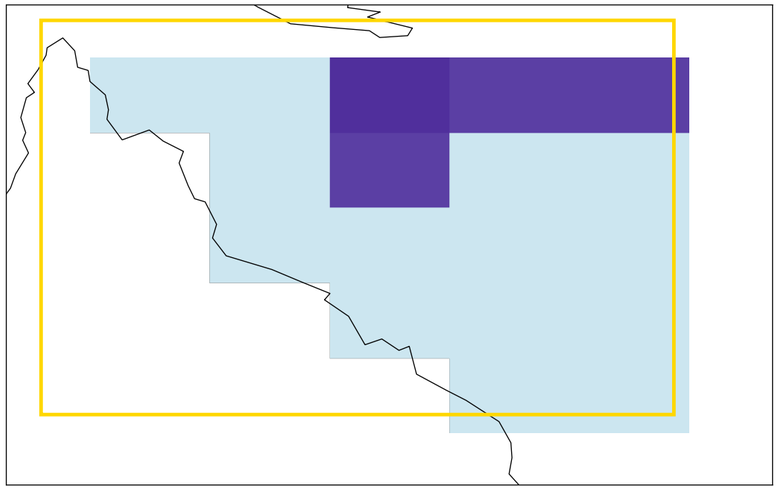




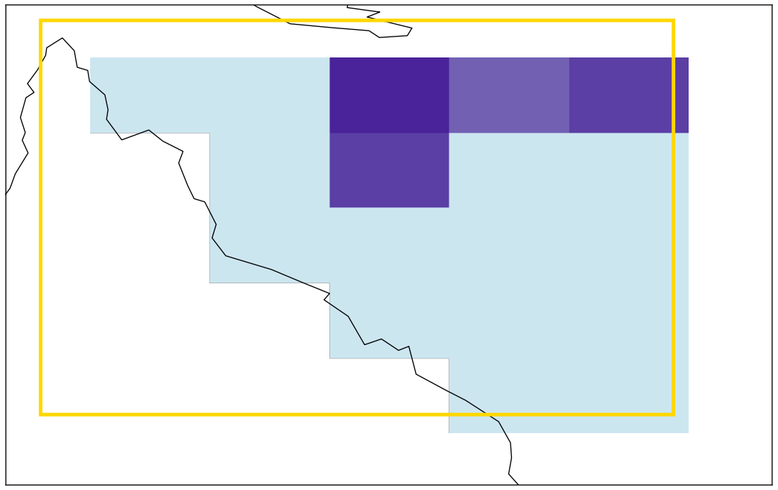


DHW Chain init: 2022-12-27 - Test - GBR (C-weeks)

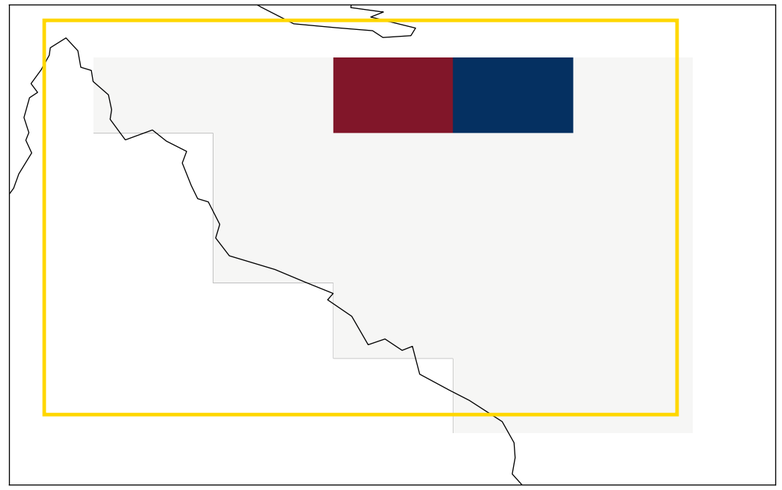
Actual DHW - Step 1



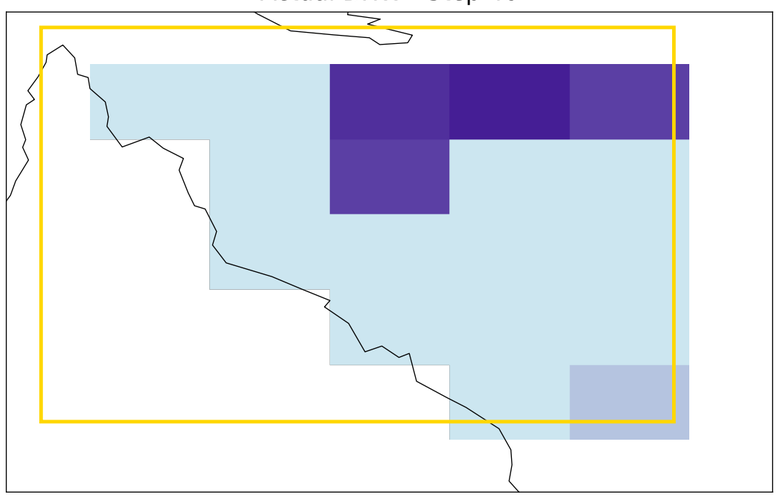
Predicted DHW - Step 1



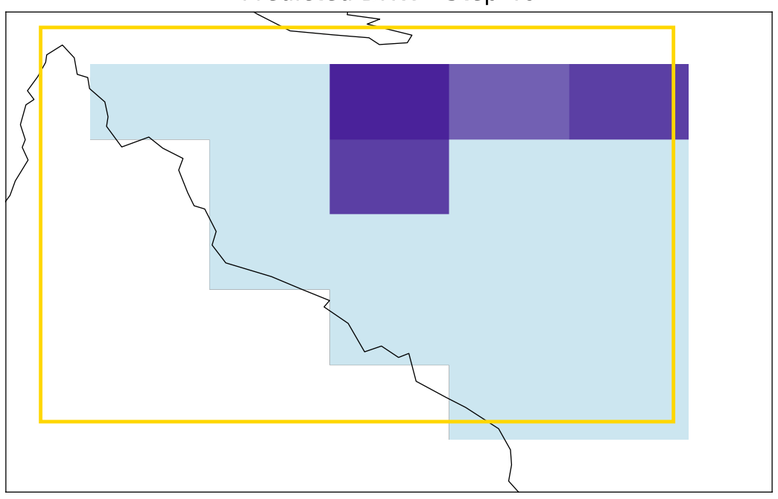
Error (Pred - Actual) - Step 1 [scale ±0.19]



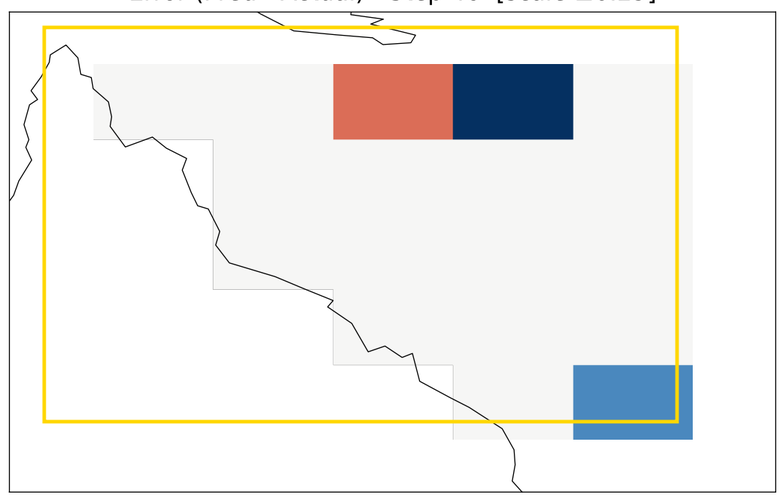
Actual DHW - Step 46



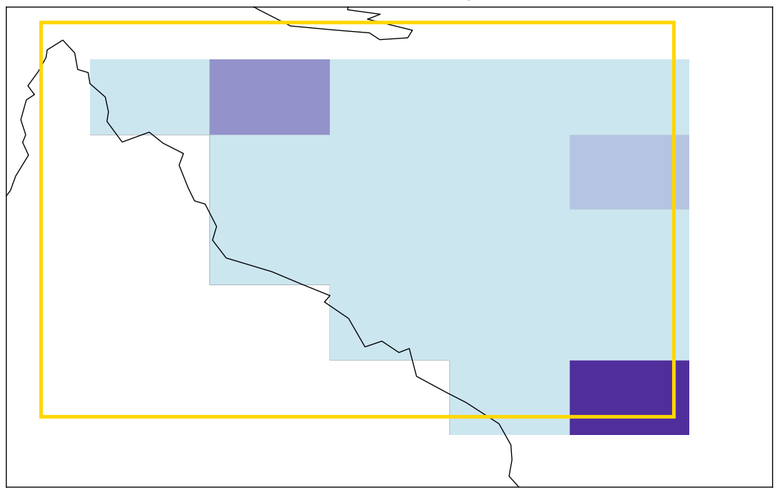
Predicted DHW - Step 46



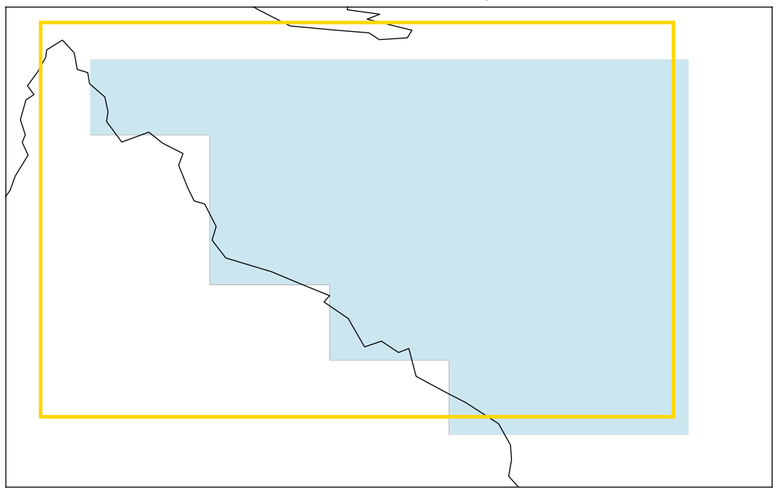
Error (Pred - Actual) - Step 46 [scale ±0.29]



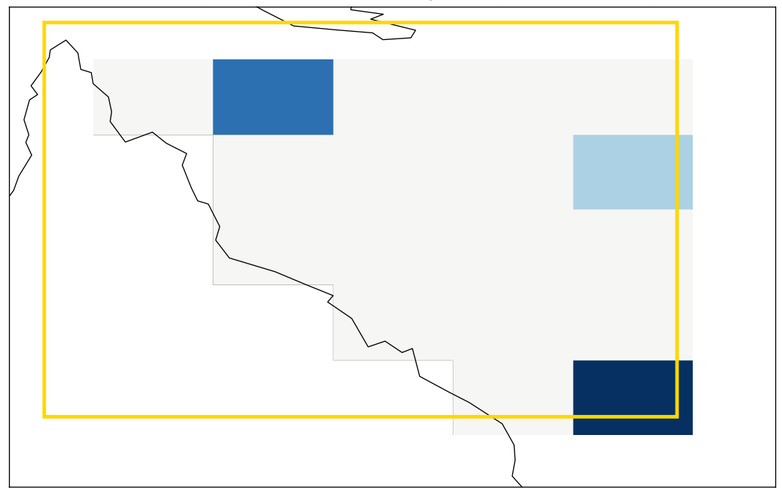
Actual DHW - Step 90

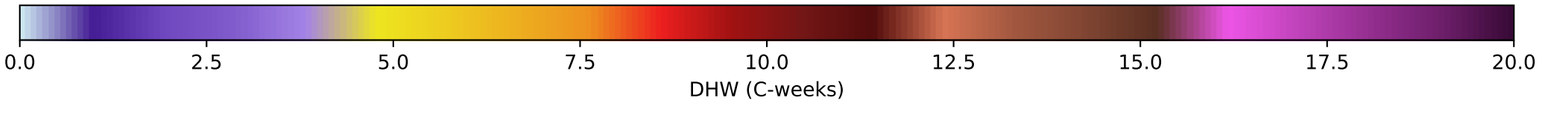


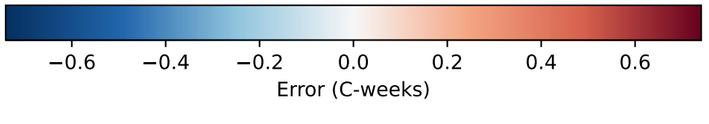
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.74]

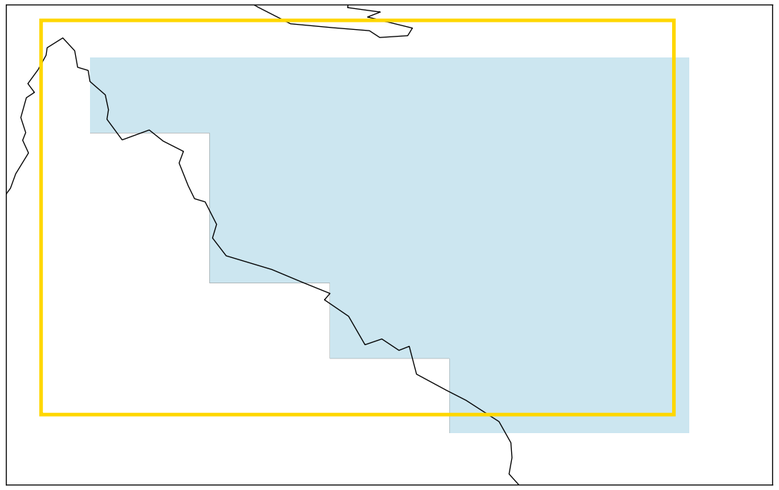




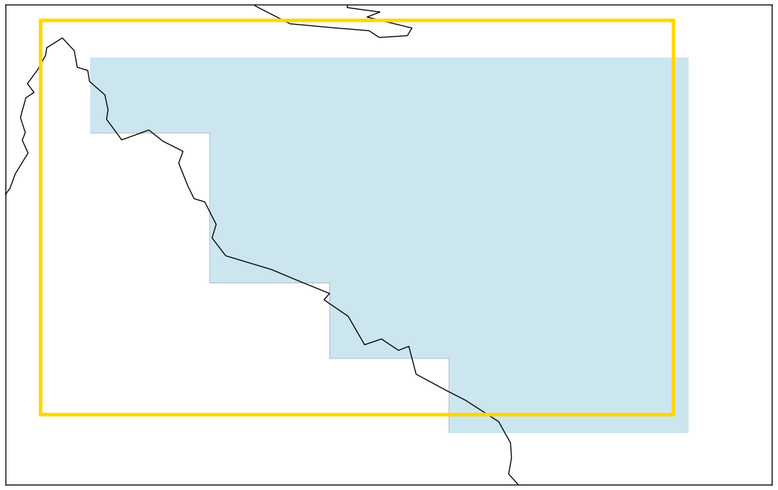


DHW Chain init: 2023-09-23 - Test - GBR (C-weeks)

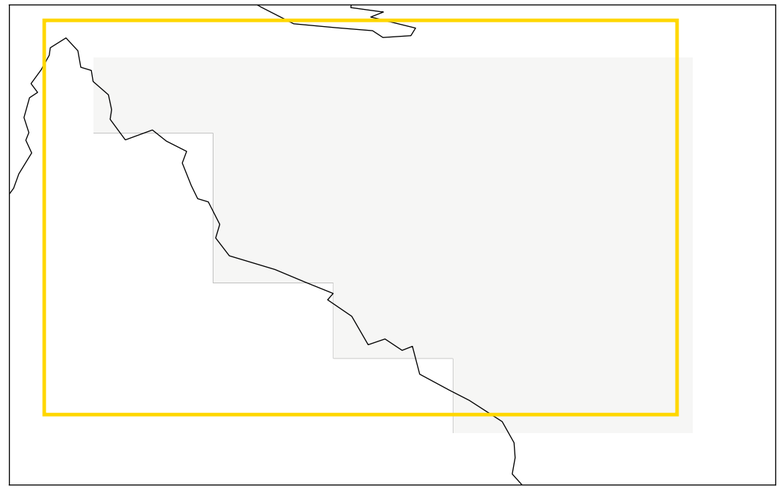
Actual DHW - Step 1



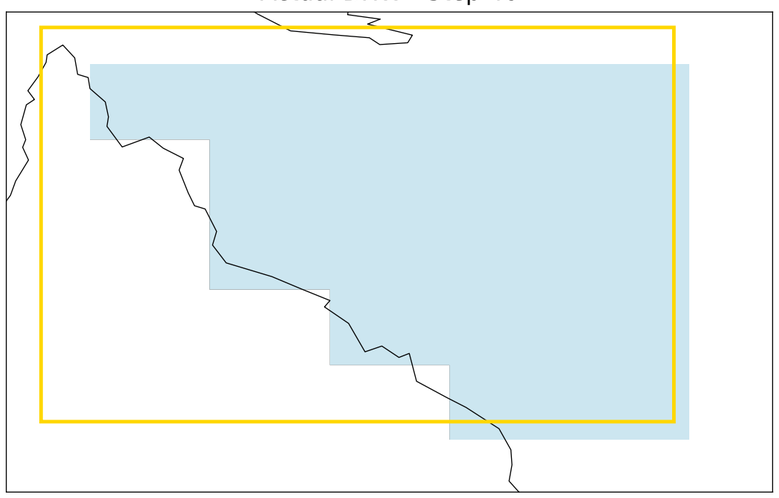
Predicted DHW - Step 1



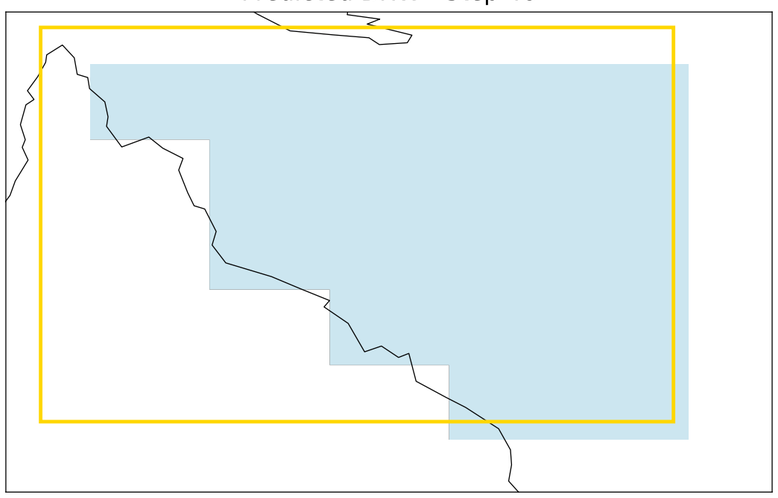
Error (Pred - Actual) - Step 1 [scale ±8]



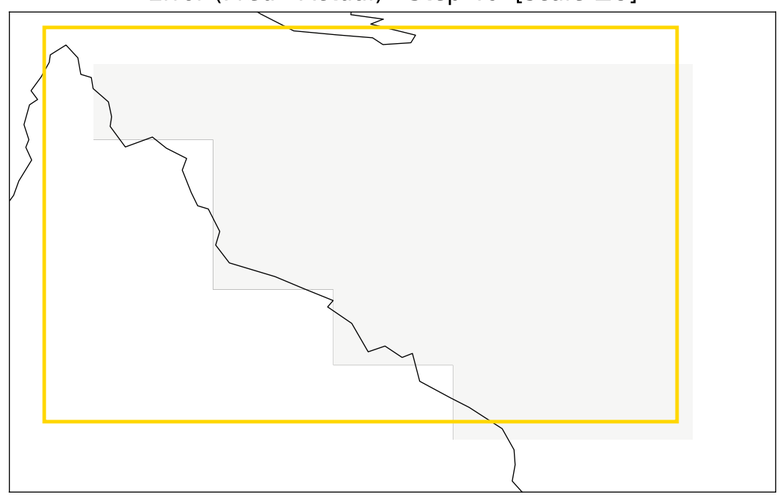
Actual DHW - Step 46



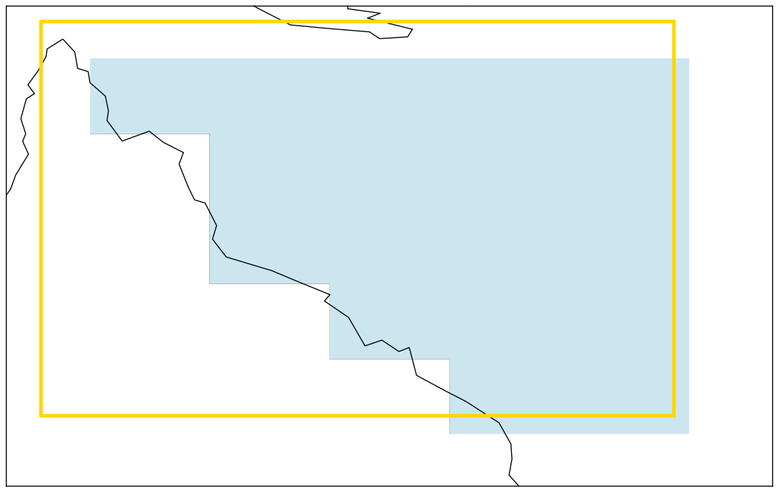
Predicted DHW - Step 46



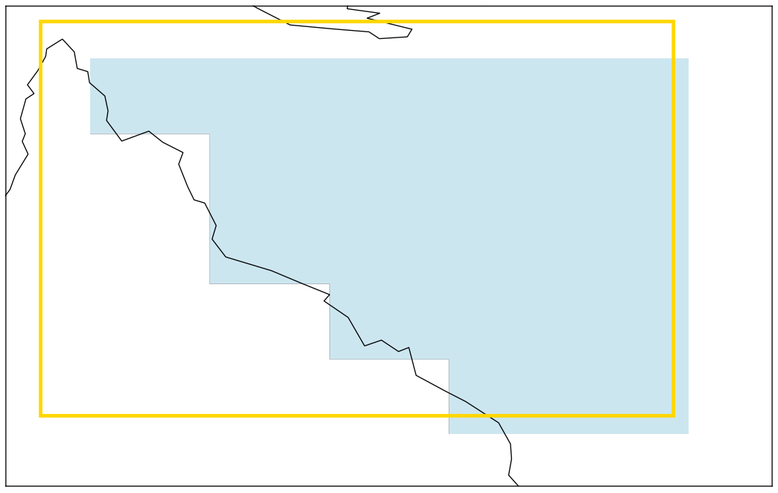
Error (Pred - Actual) - Step 46 [scale ±8]



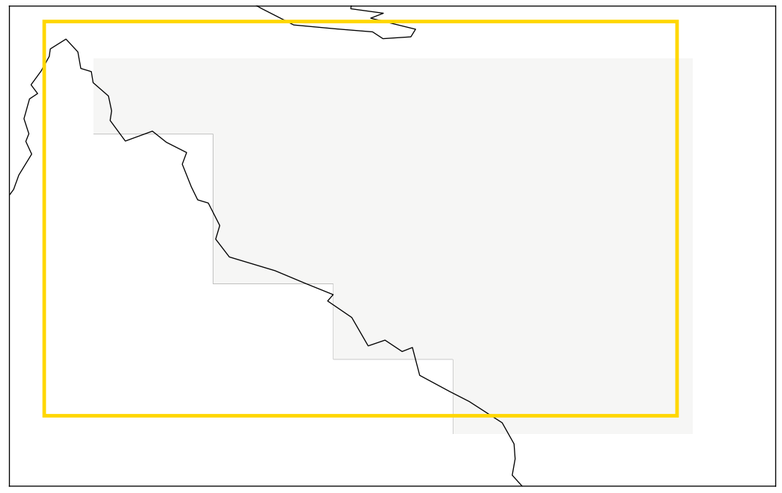
Actual DHW - Step 90

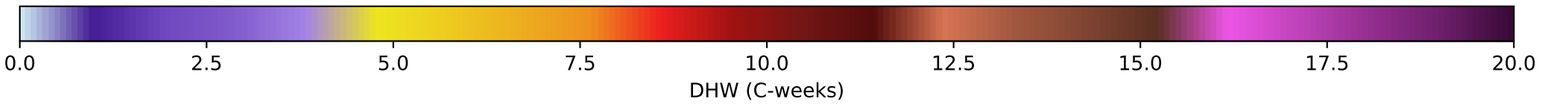


Predicted DHW - Step 90



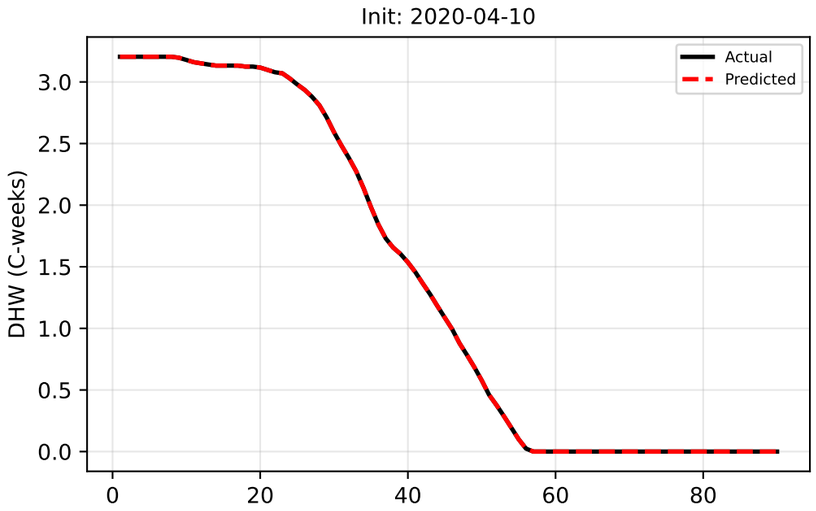
Error (Pred - Actual) - Step 90 [scale ±8]

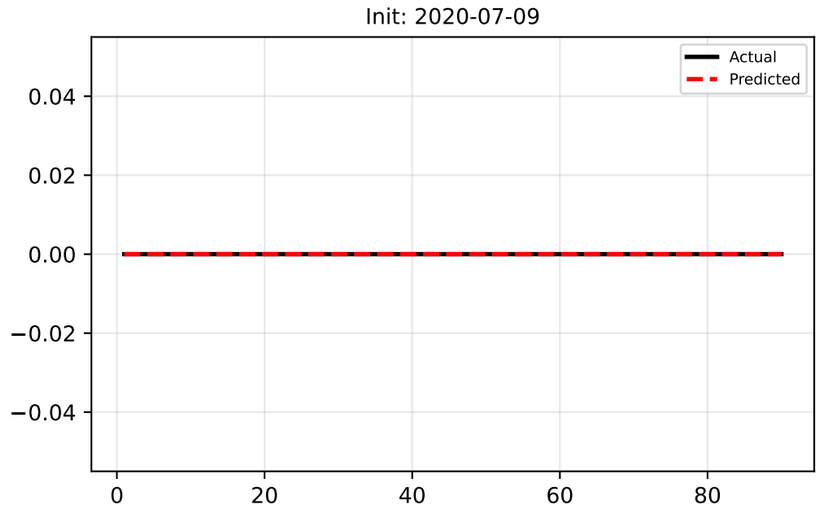


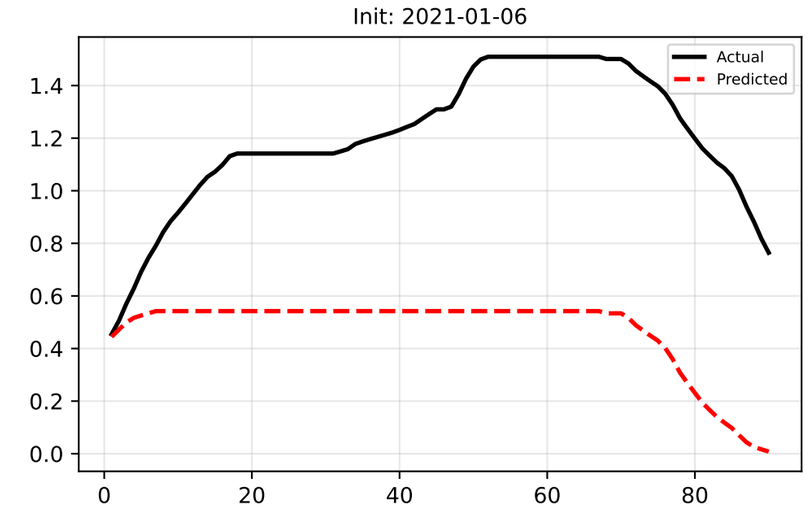


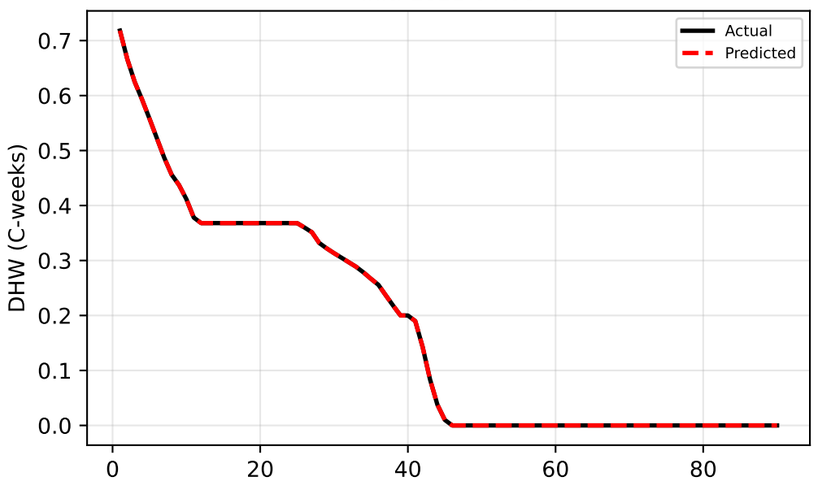


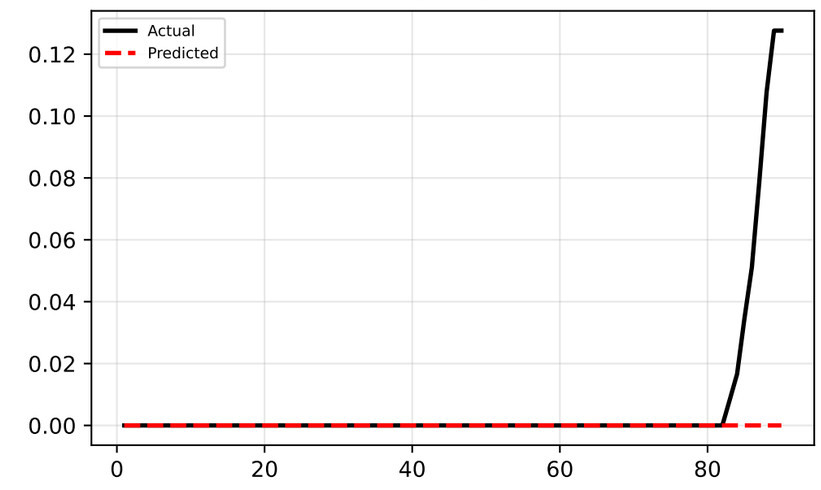
Global-Mean DHW - Test - GBR (C-weeks)

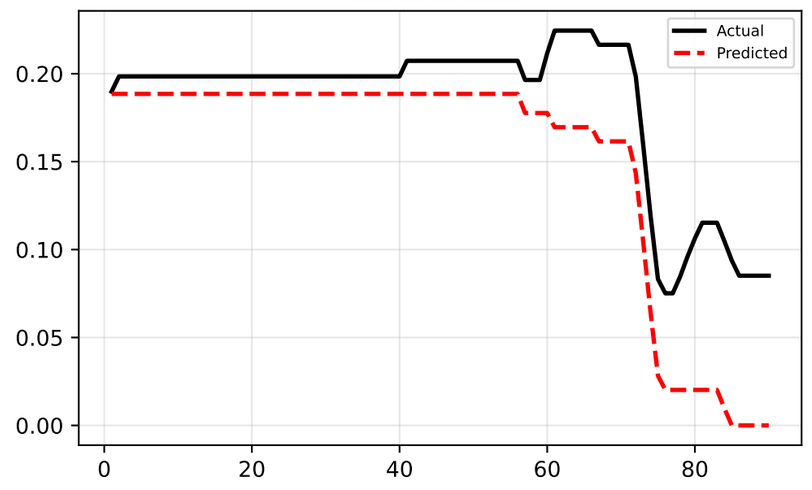


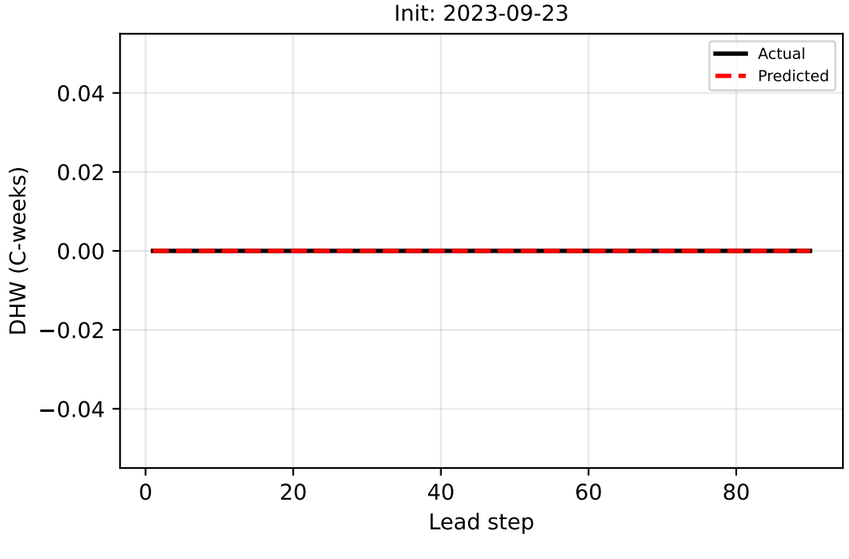


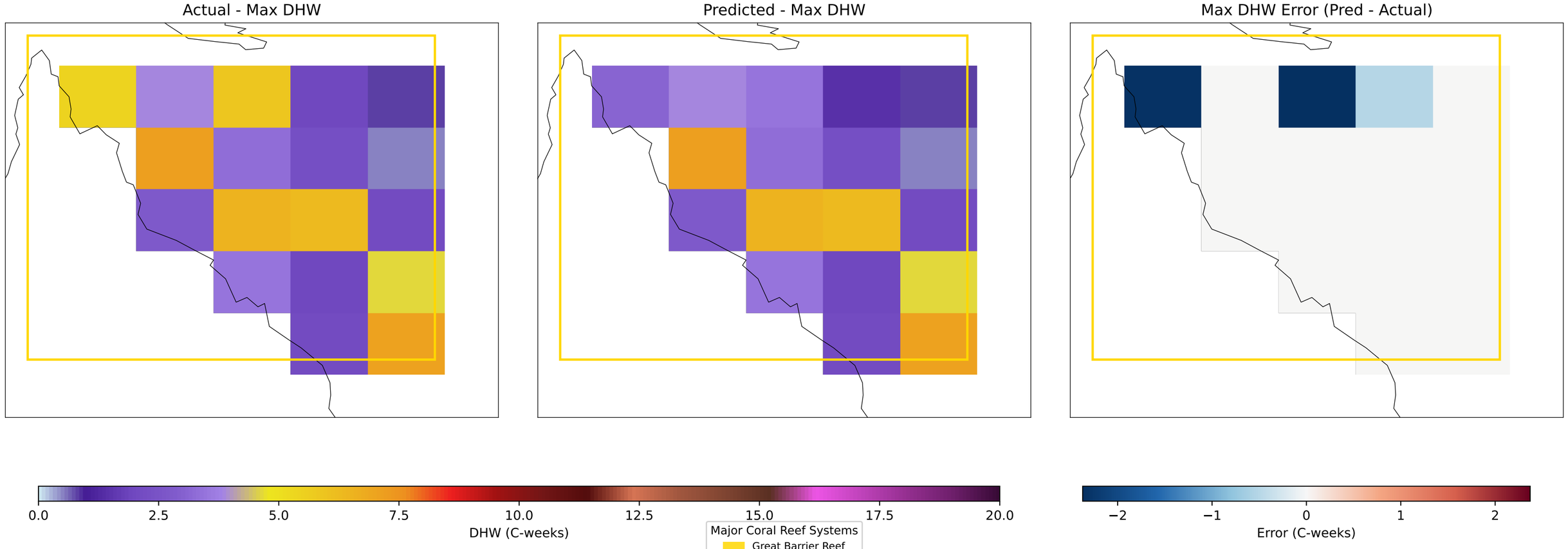


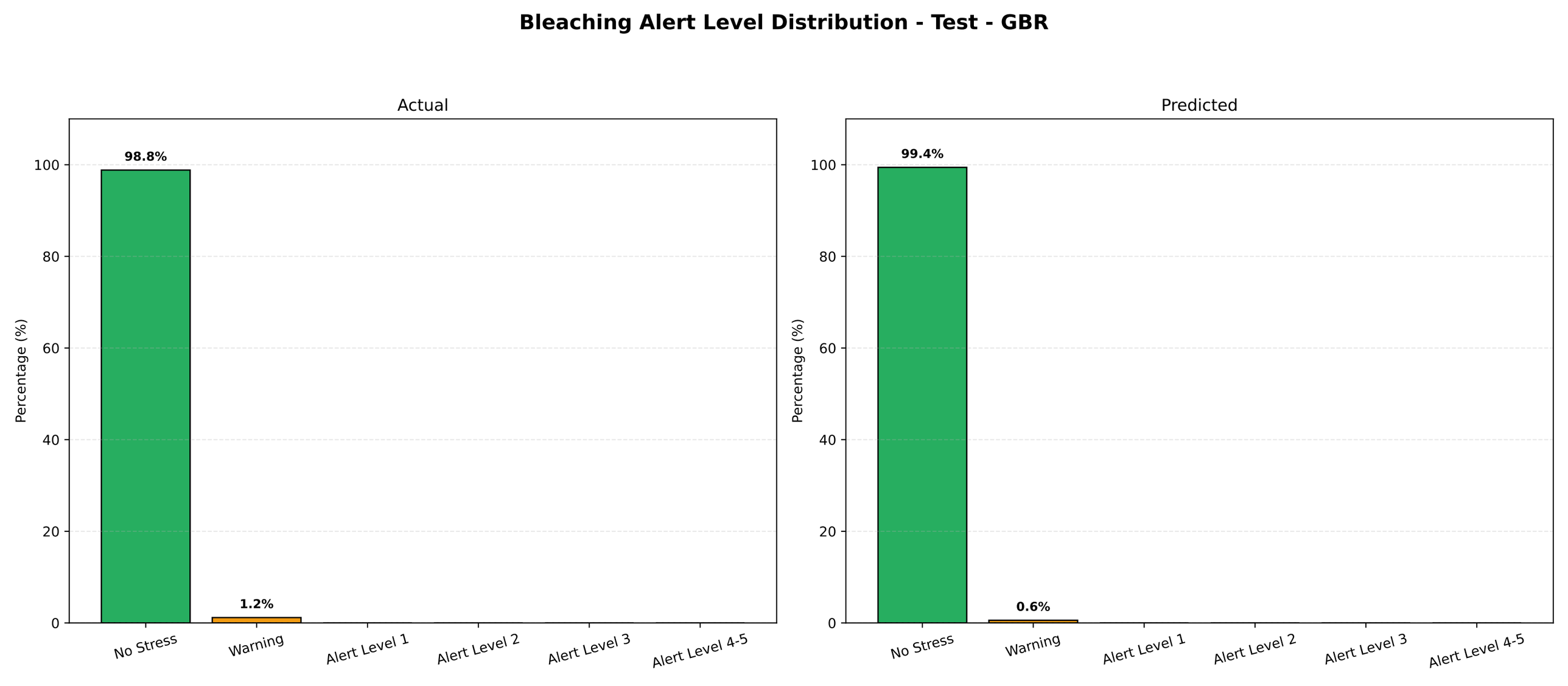




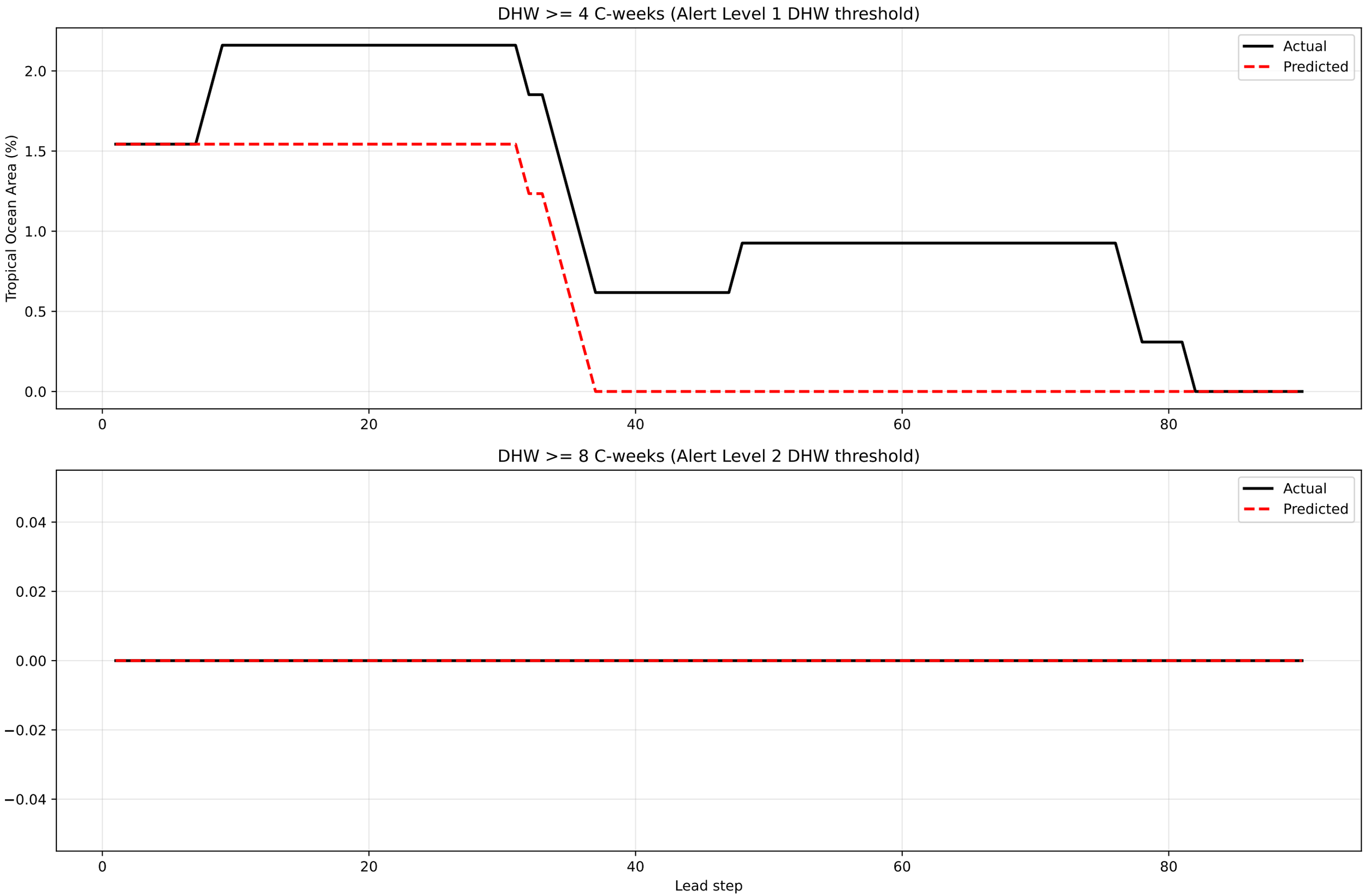





Peak DHW (max over all chains & leads) - Test - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−2
−1
0
1
2
Error (C-weeks)


Bleaching Alert Level Distribution - Test - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
98.8%
1.2%
99.4%
0.6%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

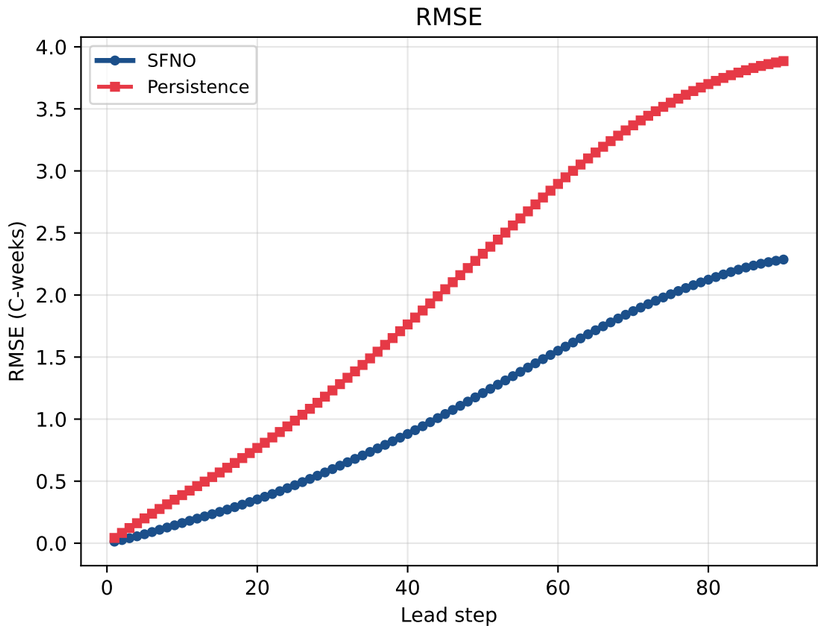
Tropical Ocean Area Under Bleaching Stress - Test - GBR

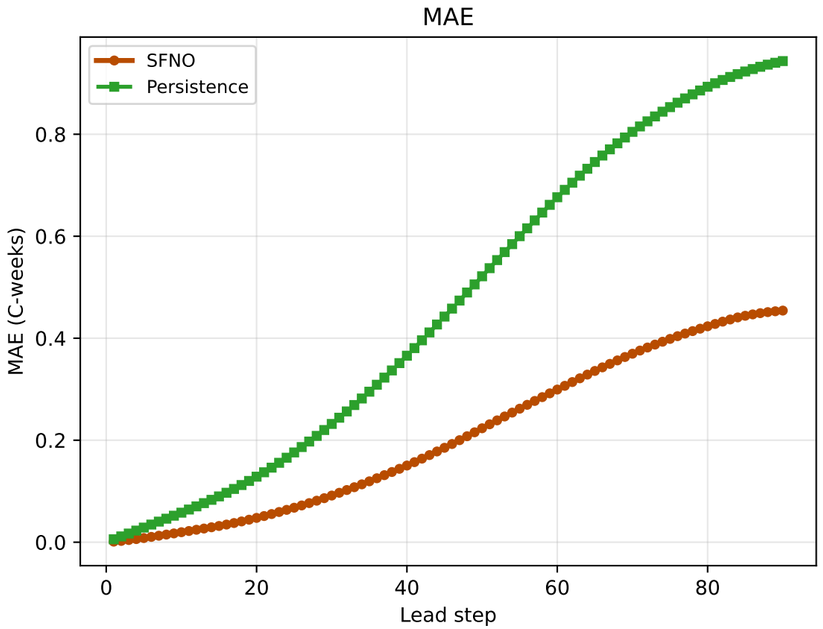


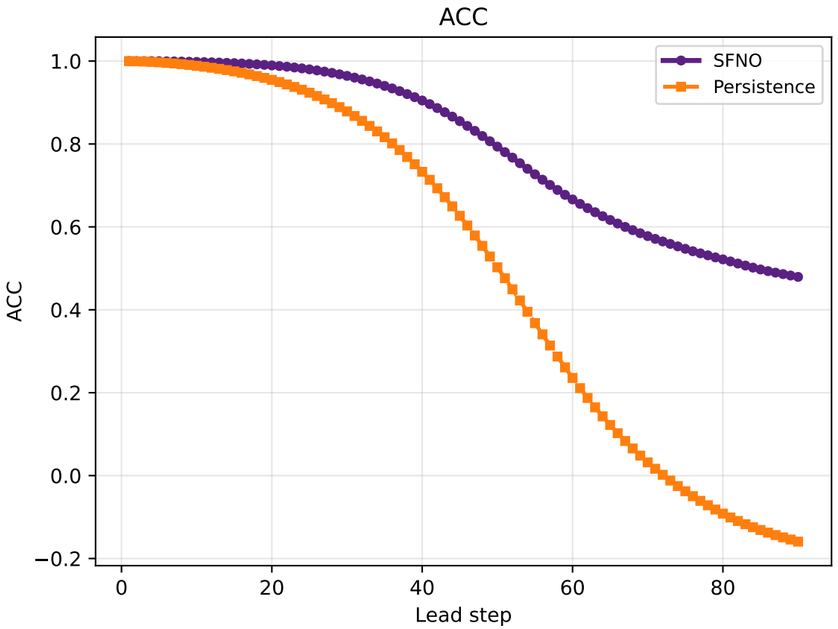
DHW Metrics by Lead Step - Val (15 chains scored at every lead)

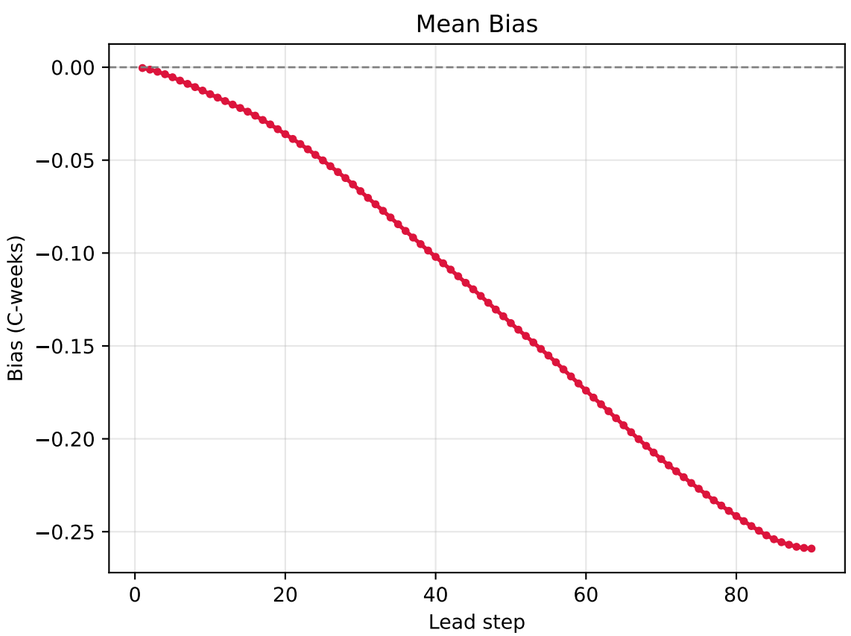
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0128	0.0425	70.0%	0.0011	0.0057	-0.0004	1.0000	0.9999	1.0000
2	0.0257	0.0829	69.0%	0.0025	0.0114	-0.0013	1.0000	0.9995	0.9999
3	0.0406	0.1222	66.8%	0.0043	0.0170	-0.0024	0.9999	0.9989	0.9998
4	0.0558	0.1613	65.4%	0.0061	0.0227	-0.0038	0.9998	0.9980	0.9996
5	0.0723	0.2000	63.9%	0.0082	0.0286	-0.0054	0.9996	0.9970	0.9994
6	0.0900	0.2381	62.2%	0.0104	0.0346	-0.0072	0.9994	0.9957	0.9990
7	0.1083	0.2758	60.8%	0.0127	0.0404	-0.0090	0.9991	0.9943	0.9985
8	0.1264	0.3133	59.7%	0.0149	0.0461	-0.0107	0.9988	0.9926	0.9980
9	0.1443	0.3504	58.8%	0.0172	0.0520	-0.0126	0.9984	0.9907	0.9974
10	0.1626	0.3874	58.0%	0.0196	0.0579	-0.0145	0.9980	0.9886	0.9967
11	0.1805	0.4241	57.4%	0.0220	0.0641	-0.0163	0.9975	0.9863	0.9959
12	0.1984	0.4601	56.9%	0.0244	0.0702	-0.0182	0.9970	0.9839	0.9950
13	0.2162	0.4965	56.5%	0.0268	0.0766	-0.0201	0.9964	0.9812	0.9941
14	0.2342	0.5332	56.1%	0.0292	0.0831	-0.0219	0.9957	0.9783	0.9930
15	0.2523	0.5701	55.7%	0.0318	0.0898	-0.0239	0.9950	0.9752	0.9918
16	0.2711	0.6080	55.4%	0.0346	0.0970	-0.0261	0.9942	0.9717	0.9905
17	0.2909	0.6467	55.0%	0.0376	0.1044	-0.0284	0.9932	0.9680	0.9889
18	0.3108	0.6861	54.7%	0.0407	0.1121	-0.0308	0.9921	0.9639	0.9872
19	0.3318	0.7264	54.3%	0.0441	0.1201	-0.0334	0.9909	0.9594	0.9853
20	0.3531	0.7676	54.0%	0.0476	0.1285	-0.0360	0.9896	0.9546	0.9832
21	0.3744	0.8096	53.7%	0.0513	0.1371	-0.0386	0.9881	0.9494	0.9809
22	0.3965	0.8524	53.5%	0.0552	0.1462	-0.0413	0.9864	0.9437	0.9783
23	0.4194	0.8961	53.2%	0.0592	0.1557	-0.0442	0.9846	0.9376	0.9754
24	0.4434	0.9412	52.9%	0.0634	0.1657	-0.0471	0.9824	0.9309	0.9721
25	0.4676	0.9875	52.7%	0.0676	0.1760	-0.0501	0.9801	0.9237	0.9685
26	0.4924	1.0350	52.4%	0.0721	0.1866	-0.0533	0.9775	0.9158	0.9646
27	0.5180	1.0830	52.2%	0.0767	0.1974	-0.0564	0.9747	0.9074	0.9603
28	0.5439	1.1316	51.9%	0.0814	0.2084	-0.0596	0.9715	0.8984	0.9556
29	0.5703	1.1811	51.7%	0.0863	0.2200	-0.0631	0.9681	0.8888	0.9505
30	0.5973	1.2311	51.5%	0.0915	0.2319	-0.0667	0.9643	0.8785	0.9450
31	0.6249	1.2818	51.2%	0.0969	0.2441	-0.0703	0.9601	0.8675	0.9389
32	0.6524	1.3332	51.1%	0.1023	0.2564	-0.0738	0.9557	0.8558	0.9324
33	0.6795	1.3846	50.9%	0.1078	0.2690	-0.0773	0.9509	0.8434	0.9257
34	0.7070	1.4364	50.8%	0.1134	0.2818	-0.0809	0.9458	0.8304	0.9184
35	0.7354	1.4894	50.6%	0.1194	0.2951	-0.0845	0.9401	0.8164	0.9104
36	0.7639	1.5434	50.5%	0.1254	0.3088	-0.0881	0.9340	0.8014	0.9019
37	0.7925	1.5981	50.4%	0.1315	0.3227	-0.0917	0.9275	0.7854	0.8929
38	0.8215	1.6531	50.3%	0.1376	0.3368	-0.0952	0.9204	0.7687	0.8833
39	0.8506	1.7081	50.2%	0.1439	0.3512	-0.0987	0.9129	0.7511	0.8733
40	0.8802	1.7629	50.1%	0.1503	0.3657	-0.1021	0.9049	0.7329	0.8626
41	0.9110	1.8185	49.9%	0.1570	0.3806	-0.1055	0.8961	0.7136	0.8511
42	0.9427	1.8750	49.7%	0.1639	0.3959	-0.1089	0.8867	0.6932	0.8388
43	0.9754	1.9323	49.5%	0.1709	0.4113	-0.1125	0.8766	0.6717	0.8258
44	1.0082	1.9898	49.3%	0.1779	0.4269	-0.1160	0.8660	0.6495	0.8124
45	1.0410	2.0466	49.1%	0.1851	0.4424	-0.1195	0.8551	0.6267	0.7985
46	1.0739	2.1032	48.9%	0.1925	0.4581	-0.1231	0.8436	0.6033	0.7842
47	1.1073	2.1599	48.7%	0.2001	0.4738	-0.1268	0.8316	0.5792	0.7693
48	1.1411	2.2173	48.5%	0.2078	0.4897	-0.1304	0.8192	0.5542	0.7541
49	1.1752	2.2749	48.3%	0.2155	0.5055	-0.1340	0.8066	0.5286	0.7385
50	1.2096	2.3325	48.1%	0.2234	0.5215	-0.1377	0.7936	0.5026	0.7227
51	1.2438	2.3896	47.9%	0.2312	0.5374	-0.1413	0.7804	0.4762	0.7066
52	1.2777	2.4464	47.8%	0.2389	0.5532	-0.1446	0.7671	0.4495	0.6905
53	1.3118	2.5032	47.6%	0.2465	0.5689	-0.1481	0.7536	0.4226	0.6743
54	1.3463	2.5604	47.4%	0.2541	0.5845	-0.1516	0.7401	0.3952	0.6581
55	1.3809	2.6175	47.2%	0.2618	0.6001	-0.1552	0.7268	0.3679	0.6421
56	1.4154	2.6741	47.1%	0.2694	0.6155	-0.1588	0.7138	0.3407	0.6265
57	1.4497	2.7302	46.9%	0.2770	0.6310	-0.1626	0.7012	0.3138	0.6115
58	1.4835	2.7859	46.7%	0.2845	0.6464	-0.1664	0.6891	0.2872	0.5970
59	1.5172	2.8409	46.6%	0.2919	0.6616	-0.1702	0.6773	0.2611	0.5829
60	1.5508	2.8953	46.4%	0.2993	0.6764	-0.1740	0.6661	0.2357	0.5694
61	1.5842	2.9487	46.3%	0.3067	0.6909	-0.1778	0.6555	0.2110	0.5565
62	1.6174	3.0007	46.1%	0.3141	0.7050	-0.1814	0.6453	0.1873	0.5440
63	1.6505	3.0513	45.9%	0.3214	0.7188	-0.1851	0.6354	0.1647	0.5319
64	1.6835	3.1006	45.7%	0.3287	0.7323	-0.1889	0.6258	0.1429	0.5201
65	1.7160	3.1484	45.5%	0.3358	0.7456	-0.1927	0.6167	0.1222	0.5088
66	1.7480	3.1948	45.3%	0.3429	0.7582	-0.1964	0.6081	0.1024	0.4980
67	1.7796	3.2402	45.1%	0.3497	0.7704	-0.2002	0.5999	0.0835	0.4876
68	1.8107	3.2842	44.9%	0.3565	0.7822	-0.2038	0.5922	0.0655	0.4775
69	1.8409	3.3266	44.7%	0.3632	0.7936	-0.2074	0.5848	0.0483	0.4679
70	1.8704	3.3677	44.5%	0.3696	0.8047	-0.2109	0.5778	0.0319	0.4588
71	1.8989	3.4069	44.3%	0.3758	0.8153	-0.2143	0.5711	0.0165	0.4499
72	1.9266	3.4445	44.1%	0.3818	0.8253	-0.2175	0.5649	0.0020	0.4416
73	1.9538	3.4811	43.9%	0.3875	0.8349	-0.2206	0.5589	-0.0119	0.4337
74	1.9803	3.5164	43.7%	0.3931	0.8443	-0.2238	0.5530	-0.0252	0.4260
75	2.0064	3.5499	43.5%	0.3987	0.8532	-0.2269	0.5470	-0.0377	0.4183
76	2.0320	3.5823	43.3%	0.4040	0.8619	-0.2300	0.5413	-0.0496	0.4108
77	2.0562	3.6139	43.1%	0.4092	0.8703	-0.2331	0.5362	-0.0609	0.4043
78	2.0793	3.6437	42.9%	0.4139	0.8782	-0.2359	0.5314	-0.0716	0.3980
79	2.1018	3.6724	42.8%	0.4187	0.8858	-0.2388	0.5265	-0.0819	0.3917
80	2.1238	3.6998	42.6%	0.4234	0.8931	-0.2416	0.5216	-0.0917	0.3855
81	2.1450	3.7255	42.4%	0.4280	0.9000	-0.2443	0.5167	-0.1009	0.3792
82	2.1657	3.7494	42.2%	0.4325	0.9064	-0.2470	0.5116	-0.1095	0.3728
83	2.1853	3.7716	42.1%	0.4367	0.9123	-0.2495	0.5067	-0.1173	0.3668
84	2.2044	3.7922	41.9%	0.4408	0.9179	-0.2520	0.5018	-0.1245	0.3608
85	2.2218	3.8114	41.7%	0.4440	0.9231	-0.2540	0.4973	-0.1313	0.3555
86	2.2376	3.8295	41.6%	0.4467	0.9278	-0.2556	0.4933	-0.1376	0.3507
87	2.2522	3.8461	41.4%	0.4491	0.9323	-0.2570	0.4897	-0.1434	0.3464
88	2.2650	3.8609	41.3%	0.4512	0.9365	-0.2581	0.4861	-0.1489	0.3422
89	2.2761	3.8738	41.2%	0.4528	0.9402	-0.2587	0.4827	-0.1542	0.3381
90	2.2858	3.8852	41.2%	0.4542	0.9435	-0.2591	0.4794	-0.1593	0.3341

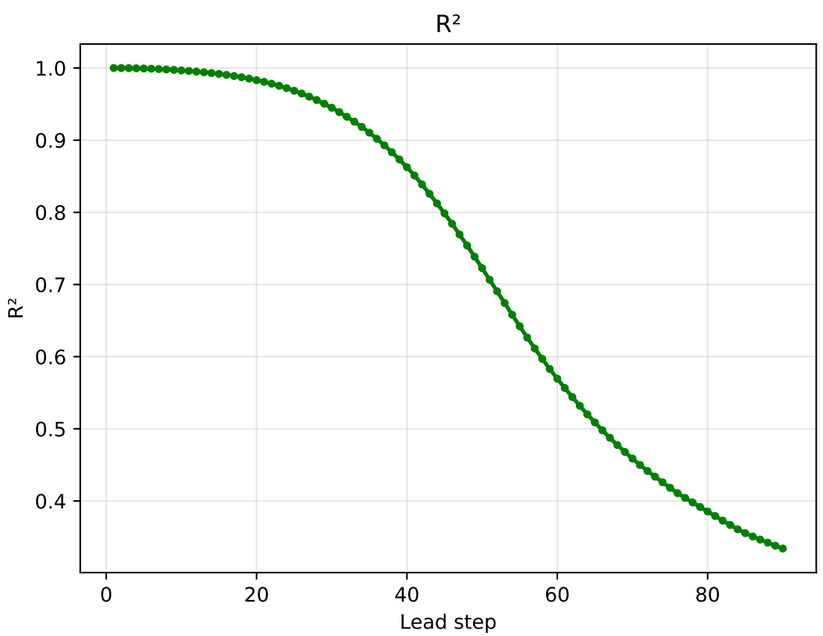
DHW Forecast Skill vs Lead Step - Val

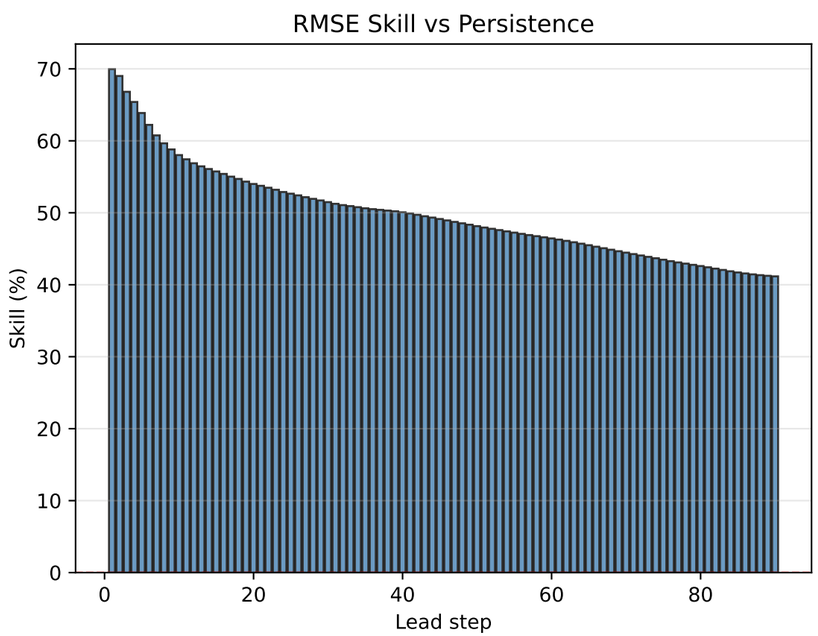


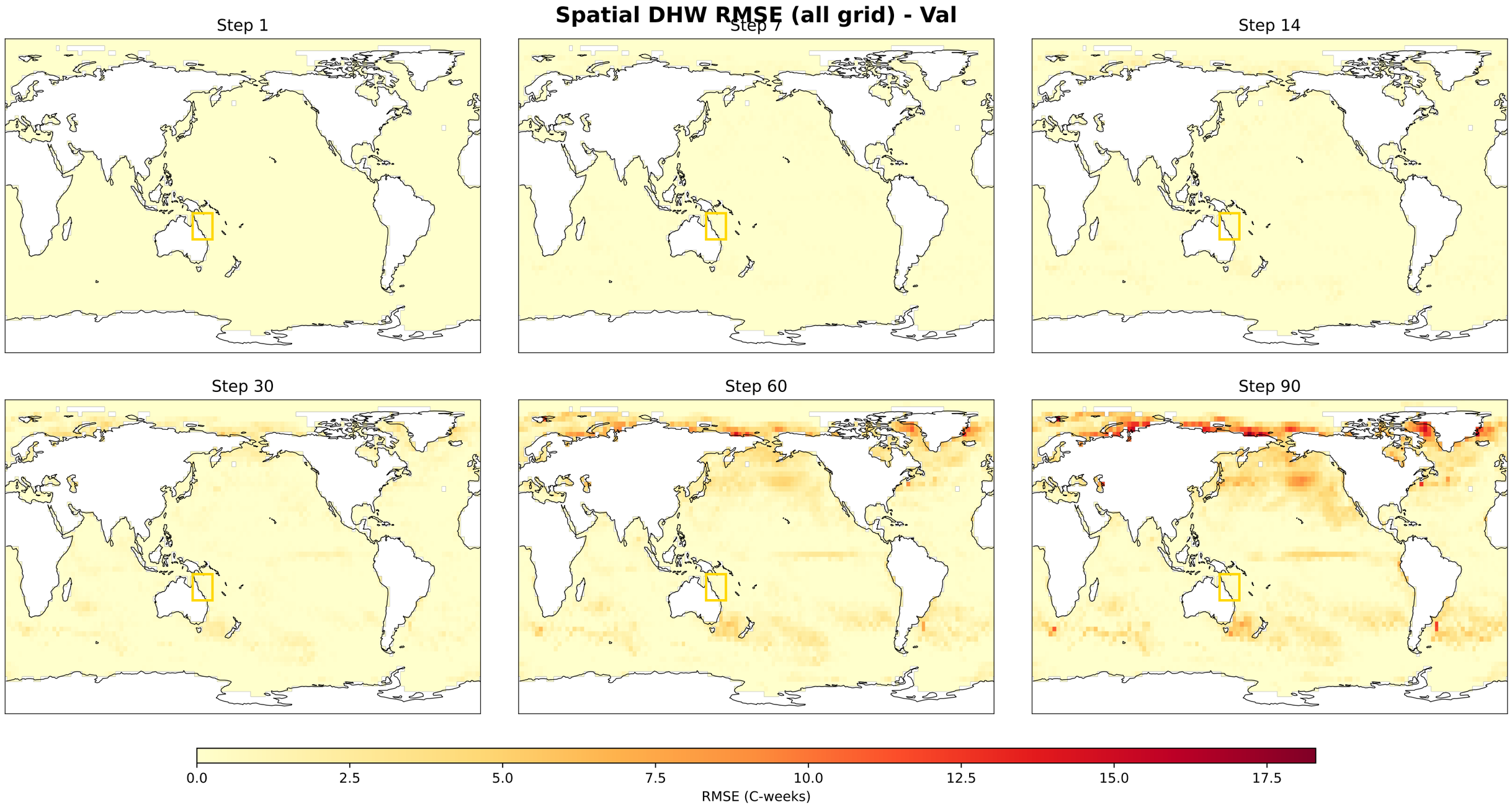


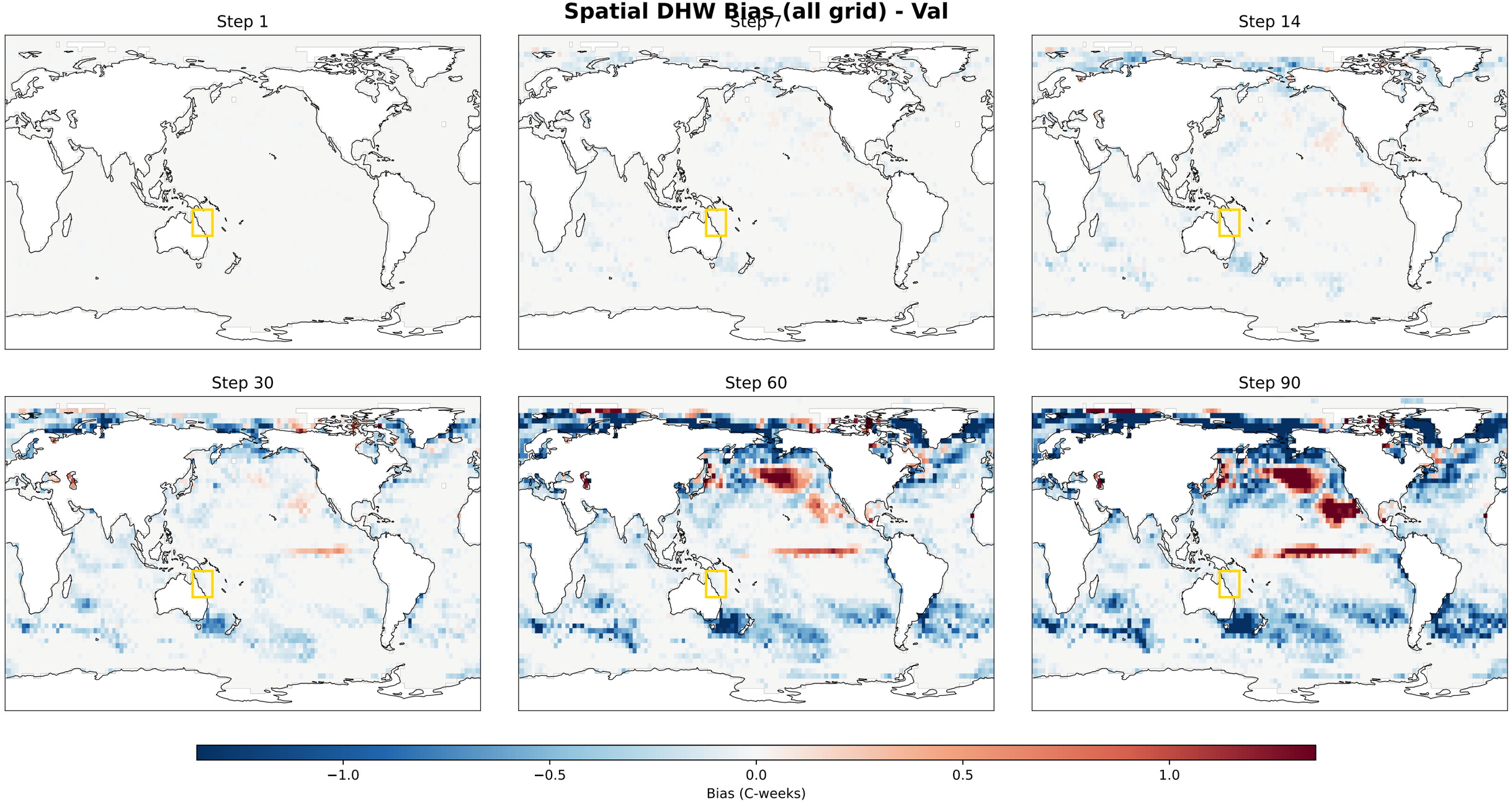


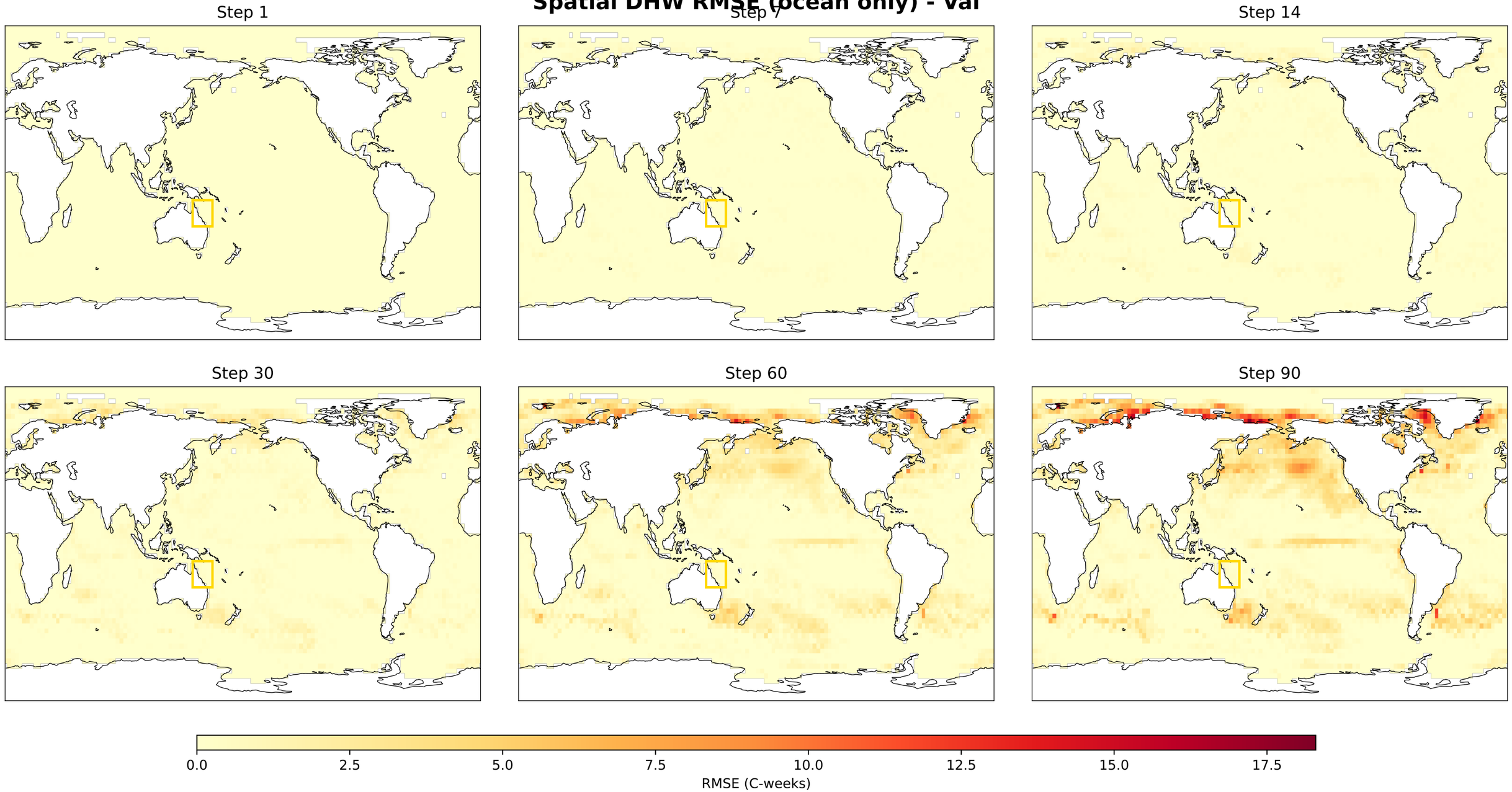


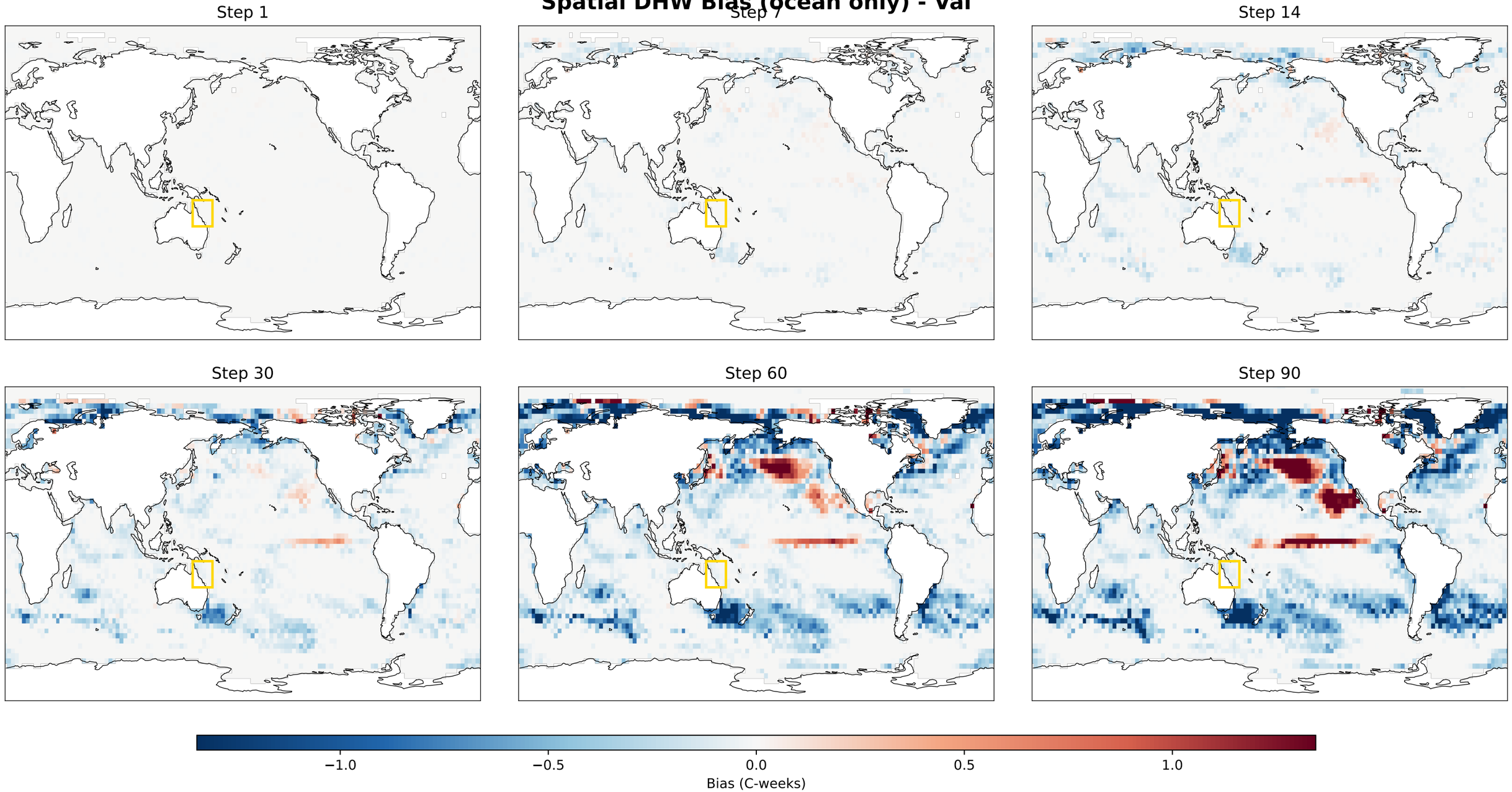





Spatial DHW RMSE (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
RMSE (C-weeks)

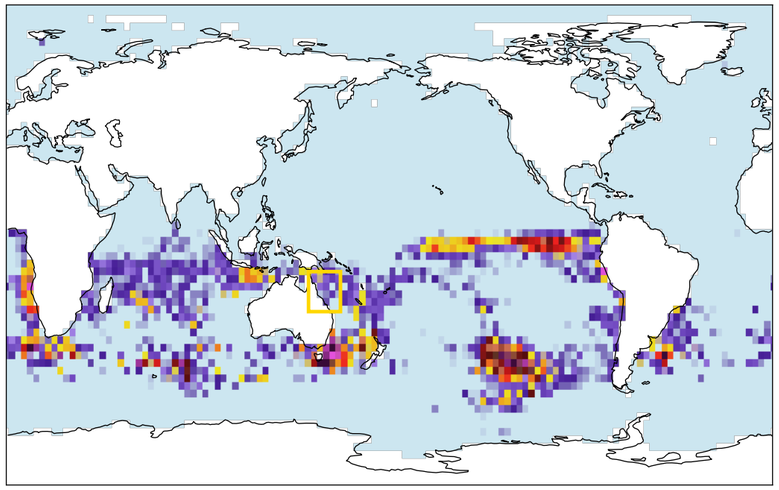

Spatial DHW Bias (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
RMSE (C-weeks)

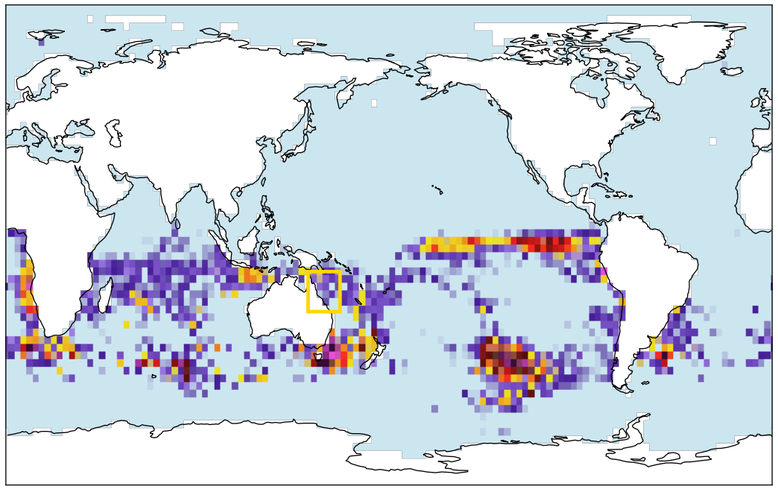

Spatial DHW Bias (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val (C-weeks)

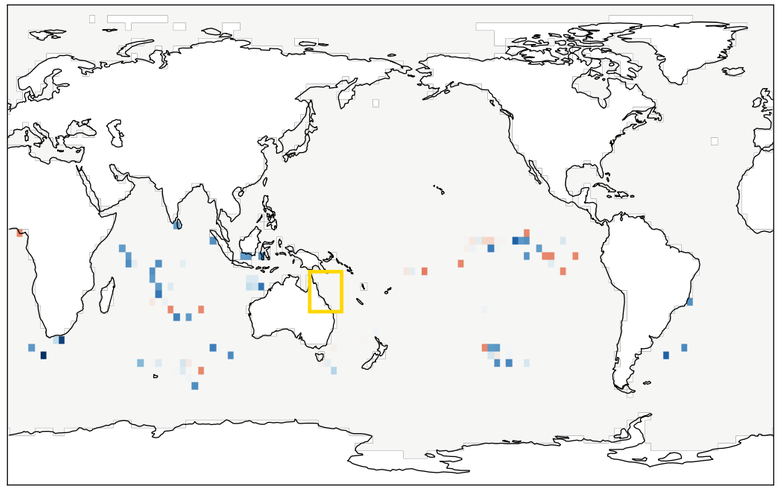
Actual DHW - Step 1



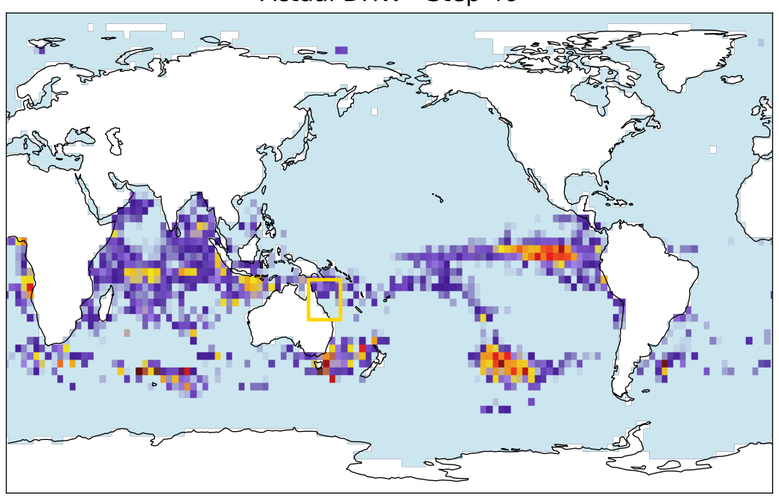
Predicted DHW - Step 1



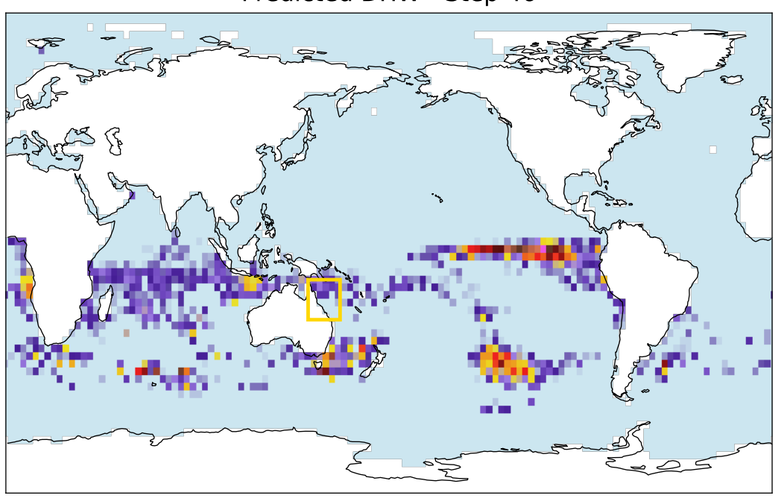
Error (Pred - Actual) - Step 1 [scale ±0.31]



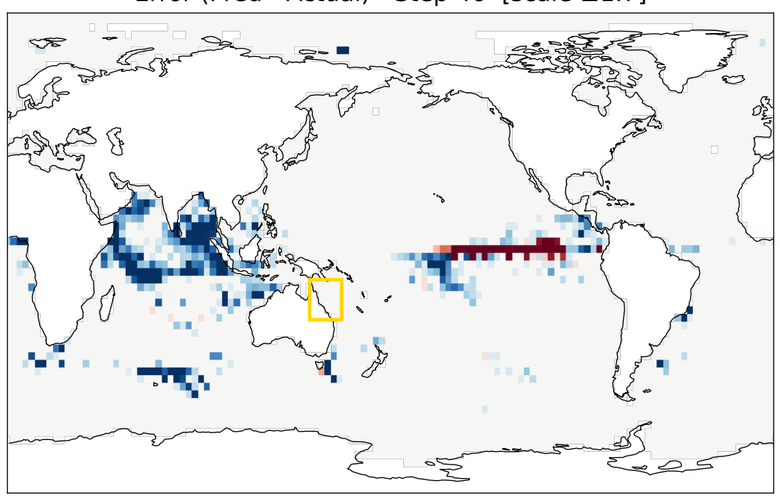
Actual DHW - Step 46



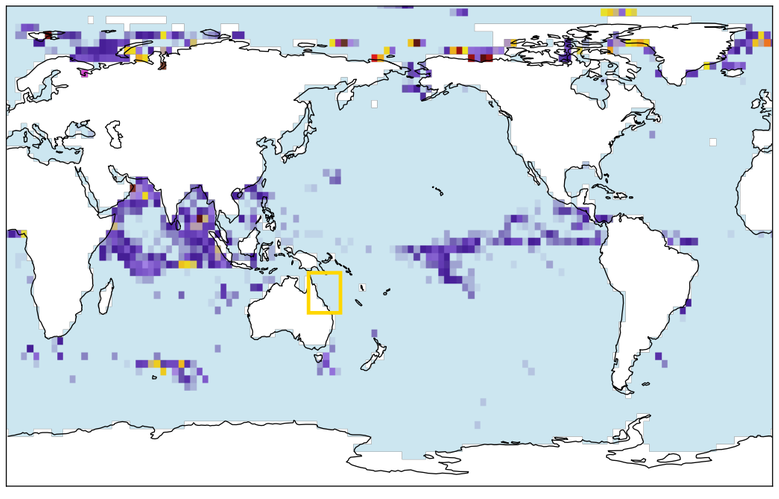
Predicted DHW - Step 46



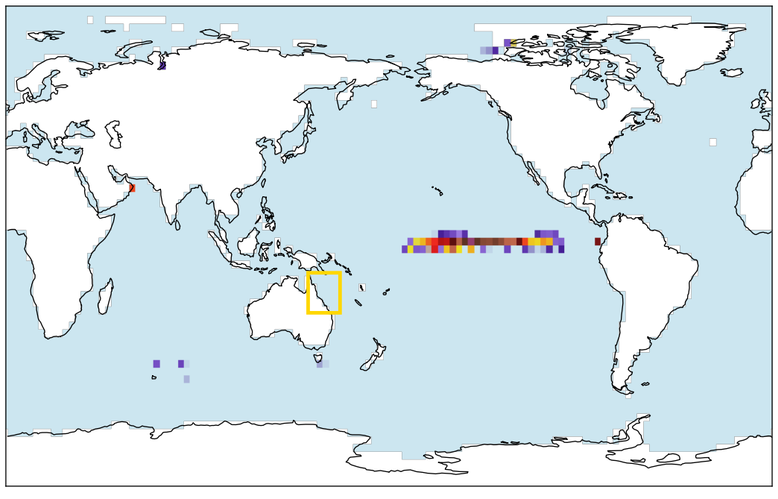
Error (Pred - Actual) - Step 46 [scale ±1.7]



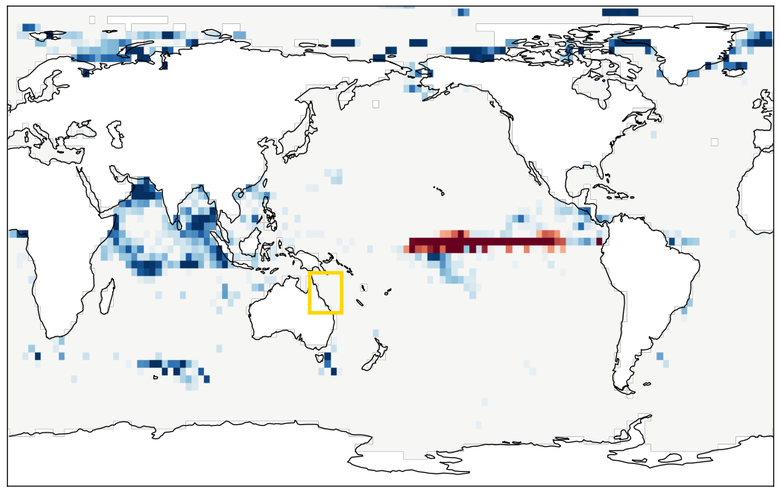
Actual DHW - Step 90

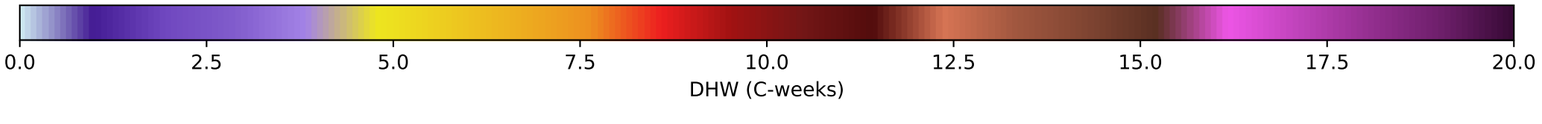


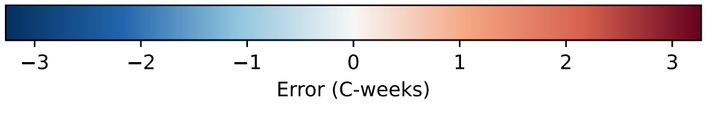
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±3.3]

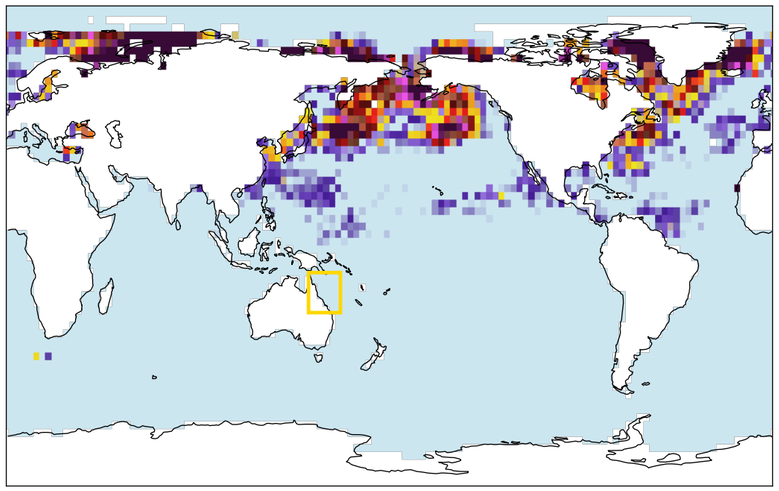




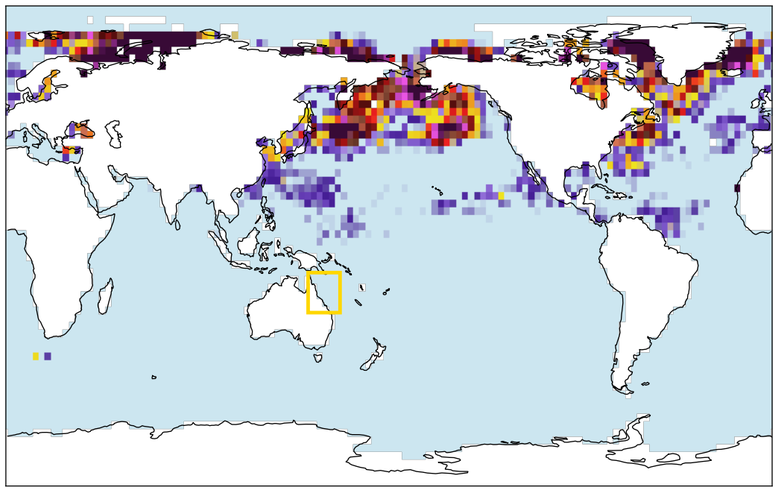


DHW Chain init: 2016-10-07 - Val (C-weeks)

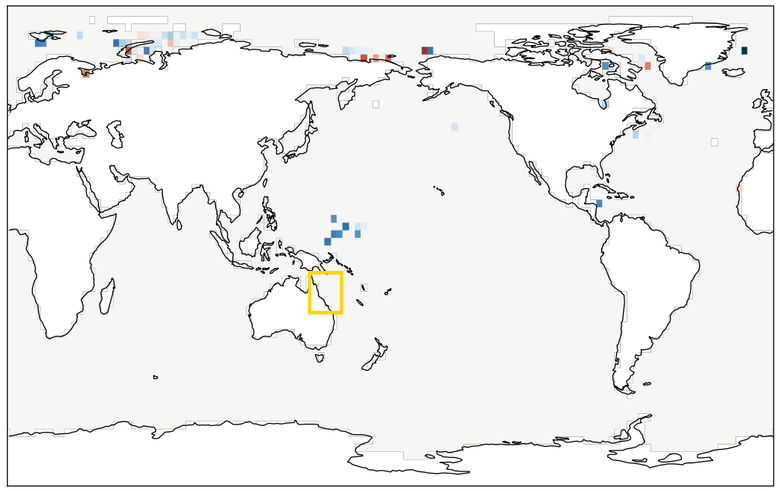
Actual DHW - Step 1



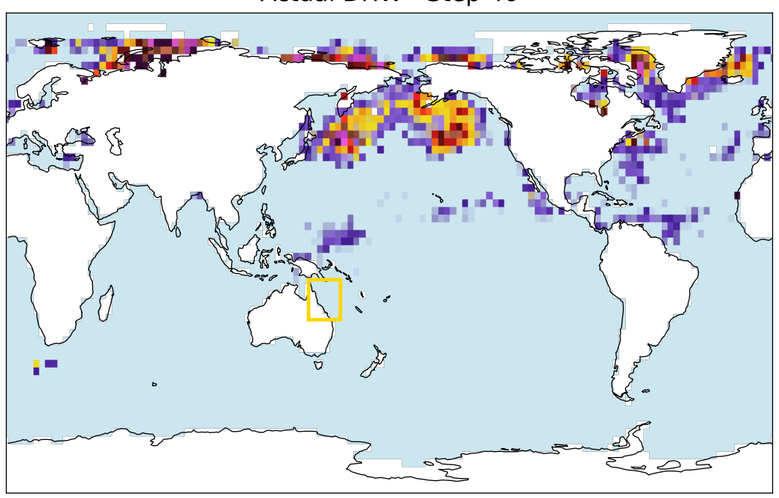
Predicted DHW - Step 1



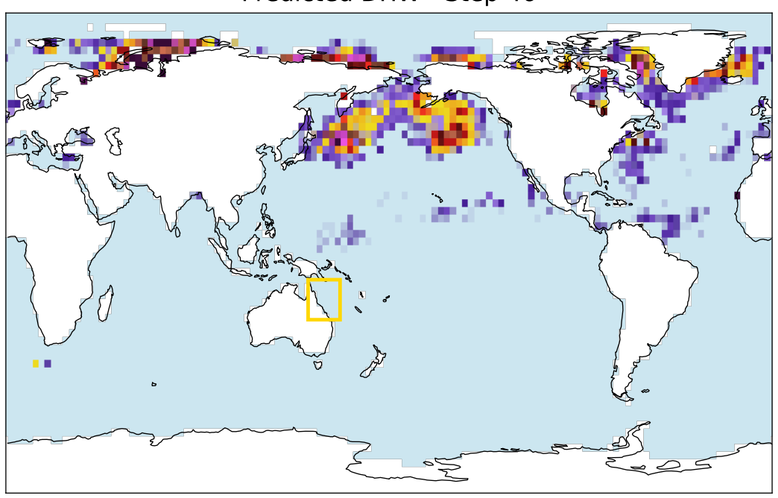
Error (Pred - Actual) - Step 1 [scale ±0.28]



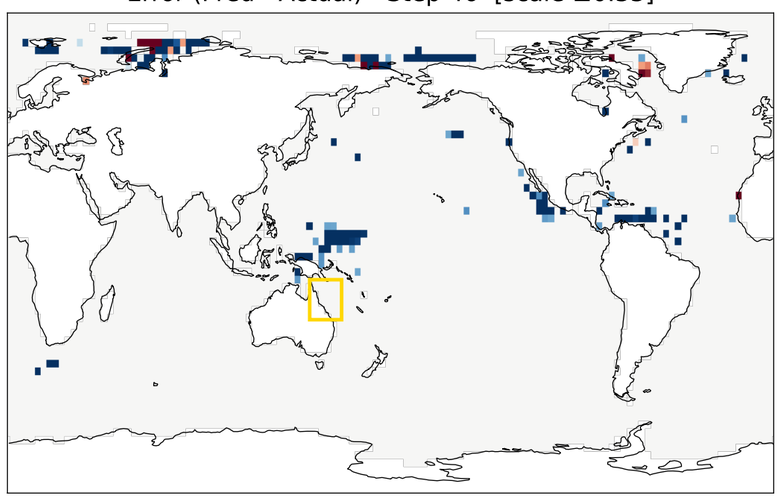
Actual DHW - Step 46



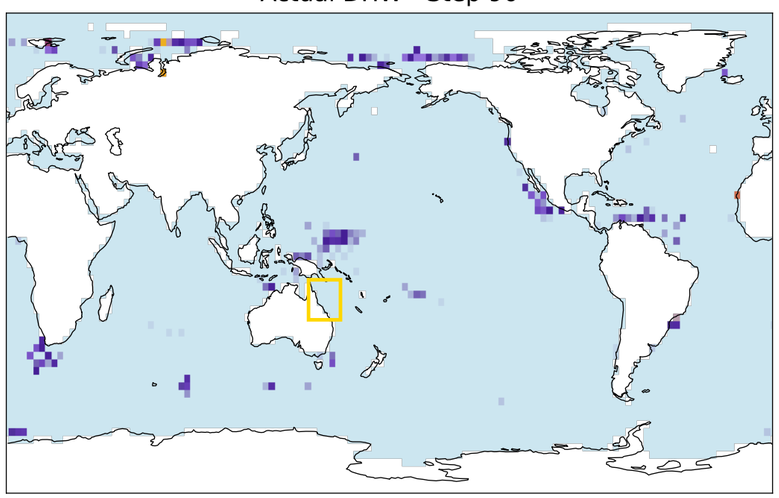
Predicted DHW - Step 46



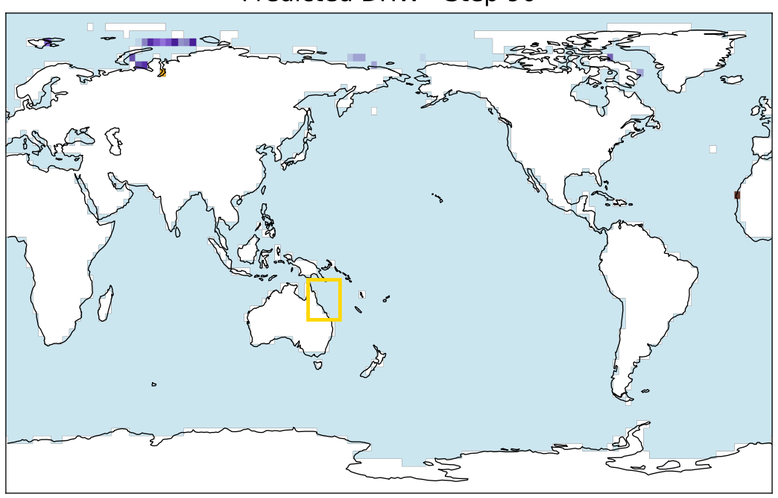
Error (Pred - Actual) - Step 46 [scale ±0.33]



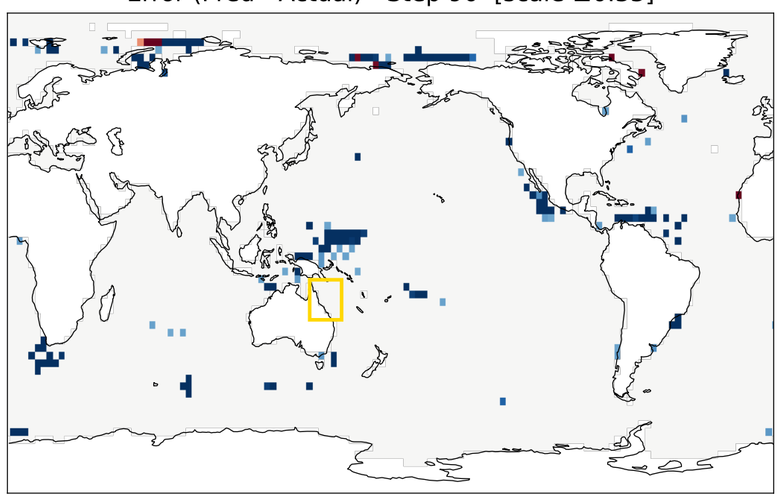
Actual DHW - Step 90

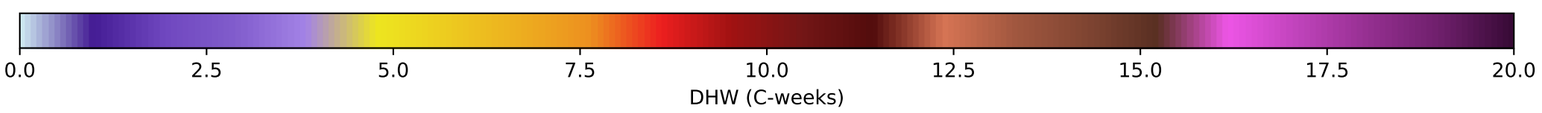


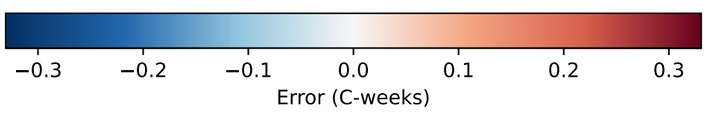
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.33]

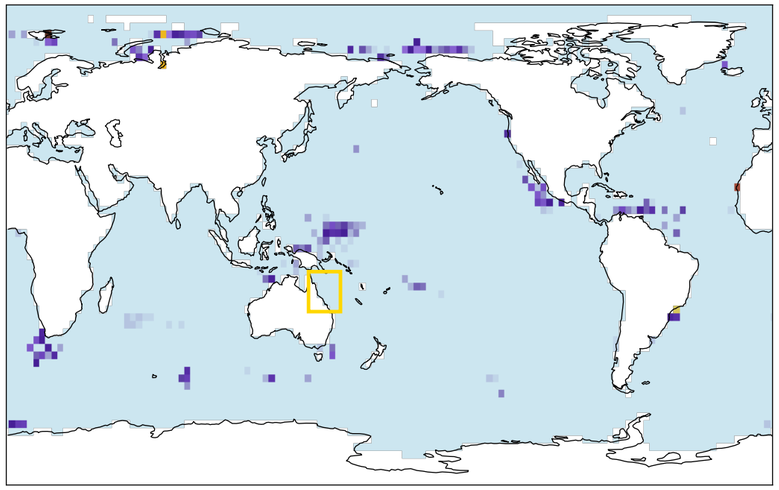




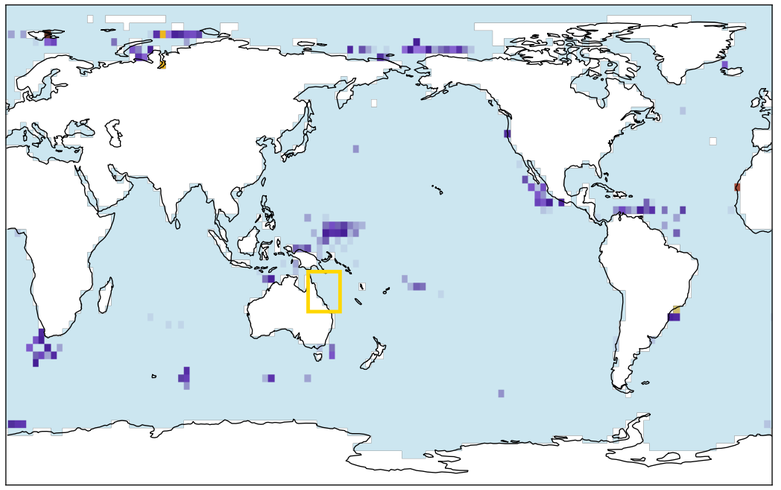


DHW Chain init: 2017-01-05 - Val (C-weeks)

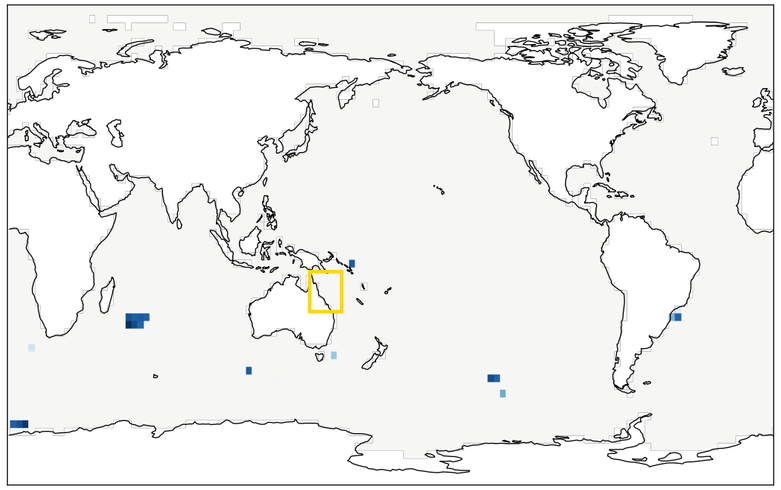
Actual DHW - Step 1



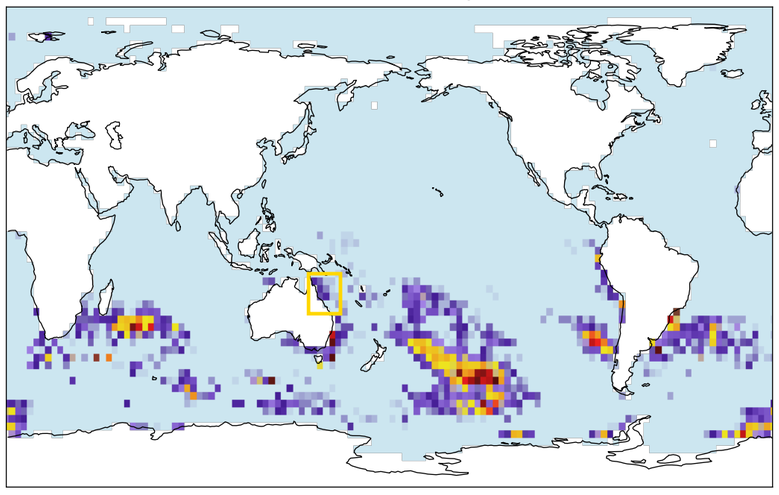
Predicted DHW - Step 1



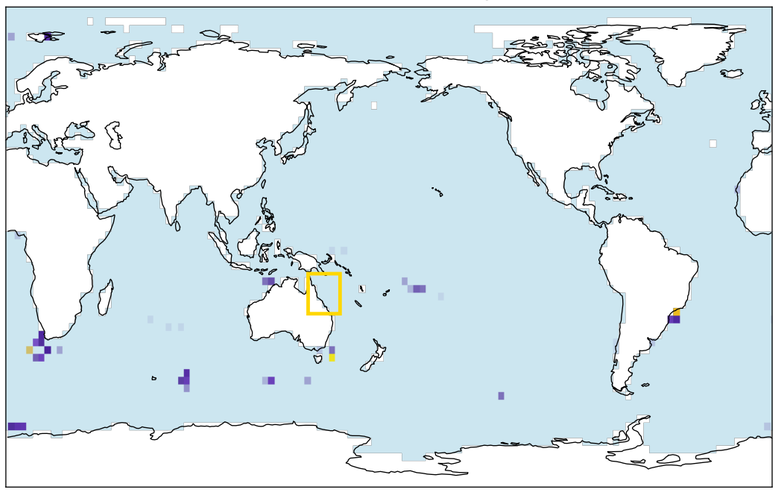
Error (Pred - Actual) - Step 1 [scale ±0.22]



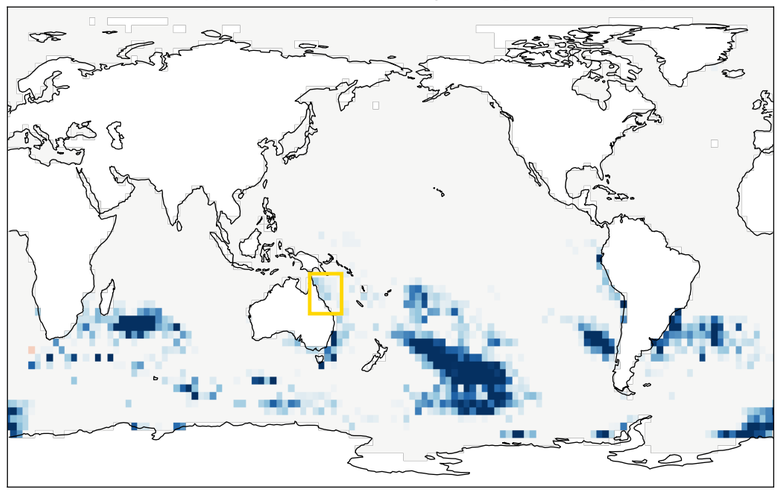
Actual DHW - Step 46



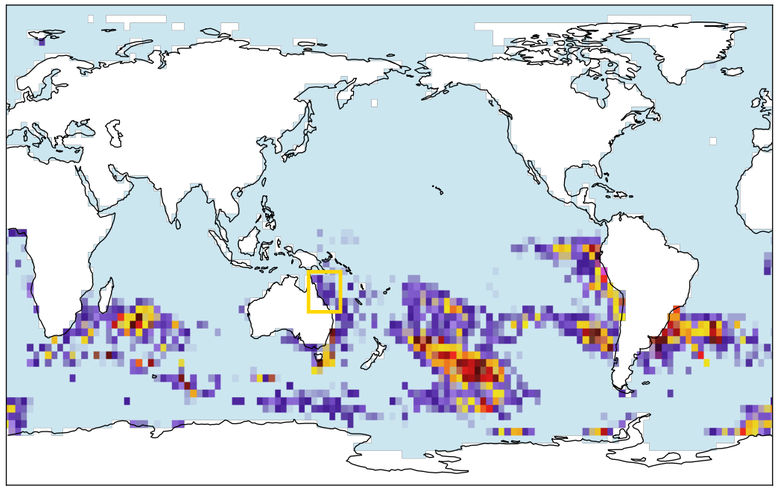
Predicted DHW - Step 46



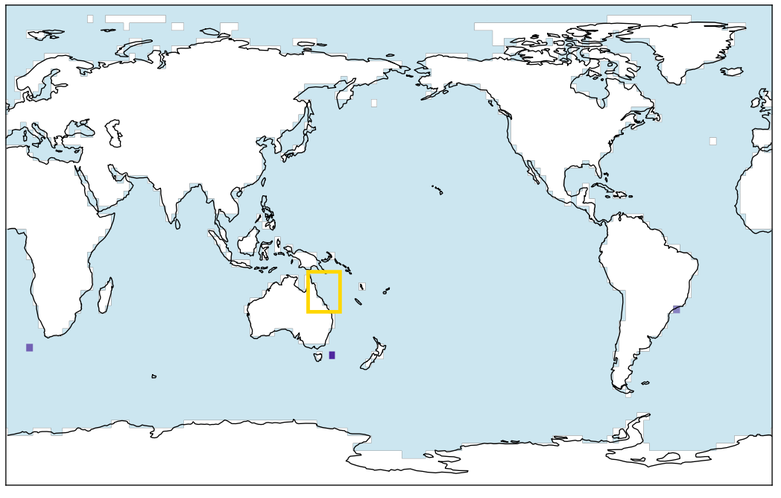
Error (Pred - Actual) - Step 46 [scale ±4.8]



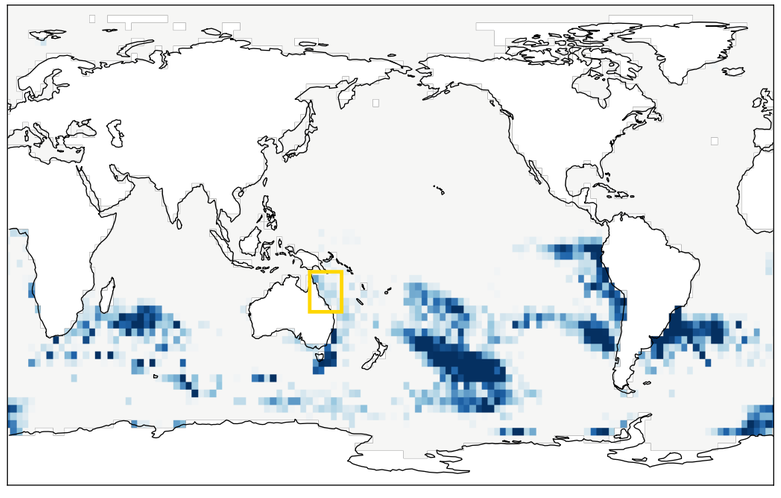
Actual DHW - Step 90

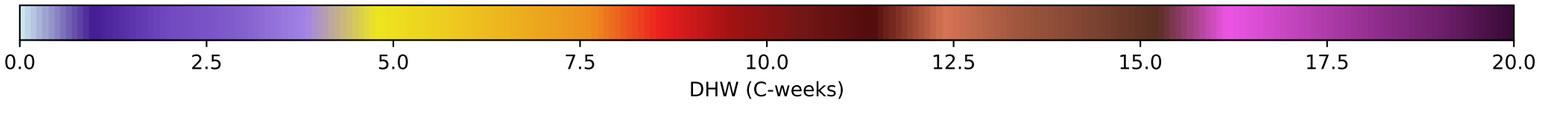


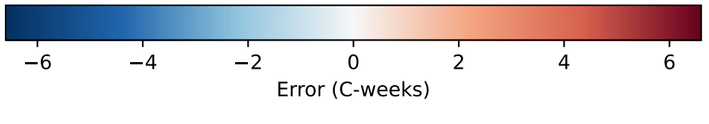
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±6.6]

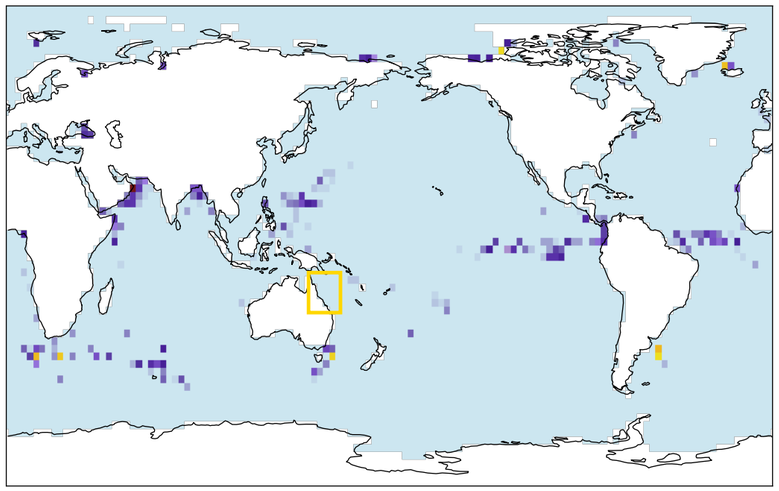




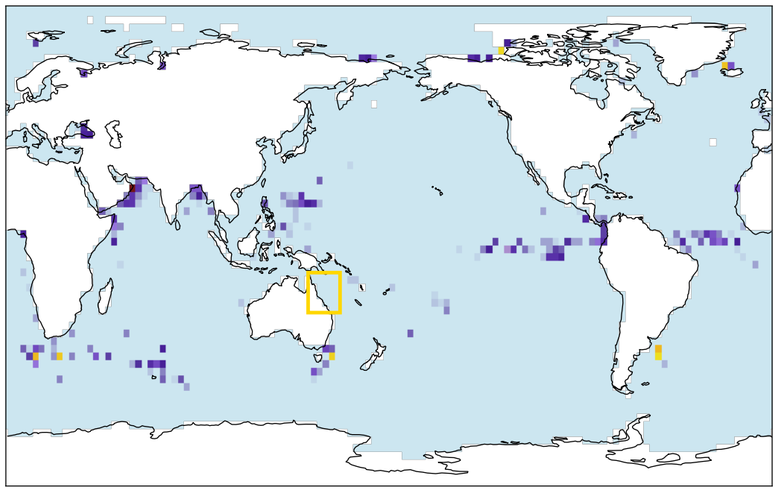


DHW Chain init: 2017-07-04 - Val (C-weeks)

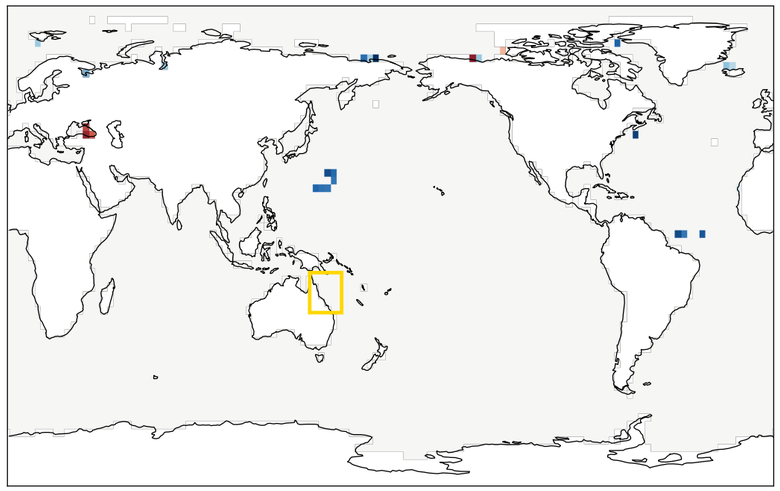
Actual DHW - Step 1



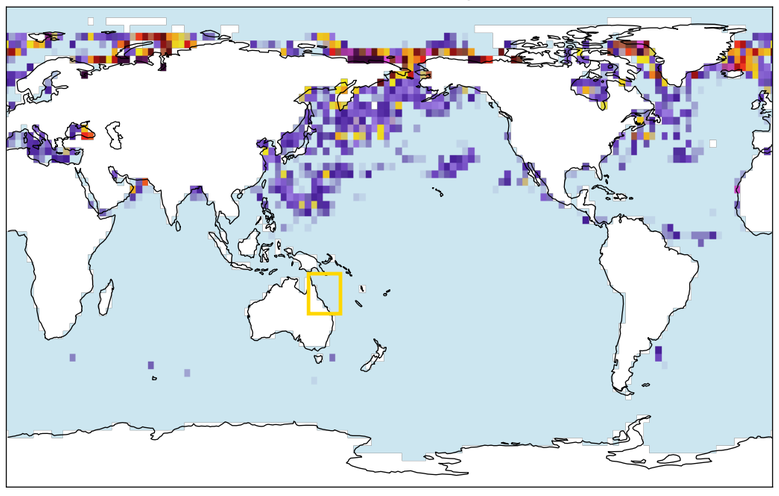
Predicted DHW - Step 1



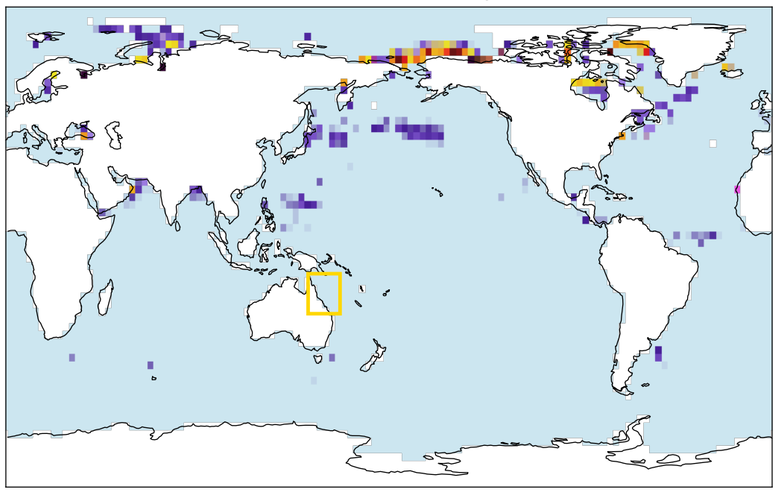
Error (Pred - Actual) - Step 1 [scale ±0.25]



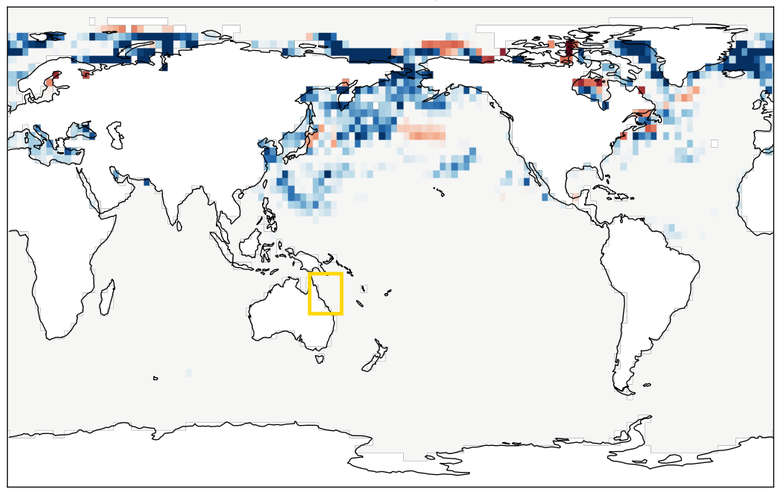
Actual DHW - Step 46



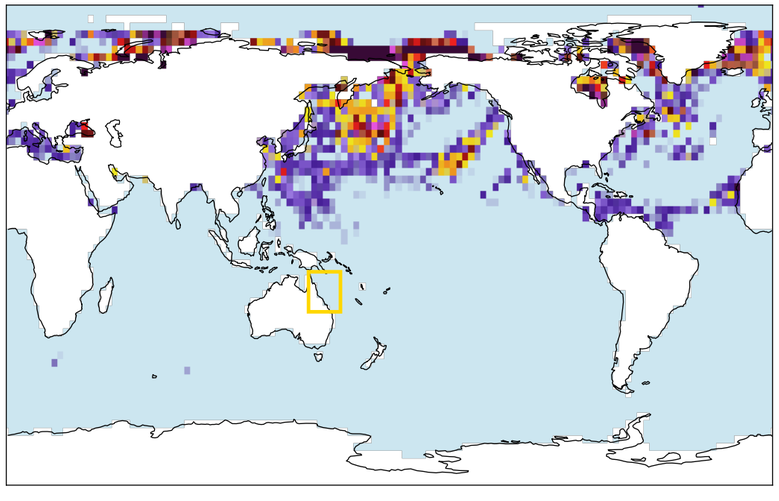
Predicted DHW - Step 46



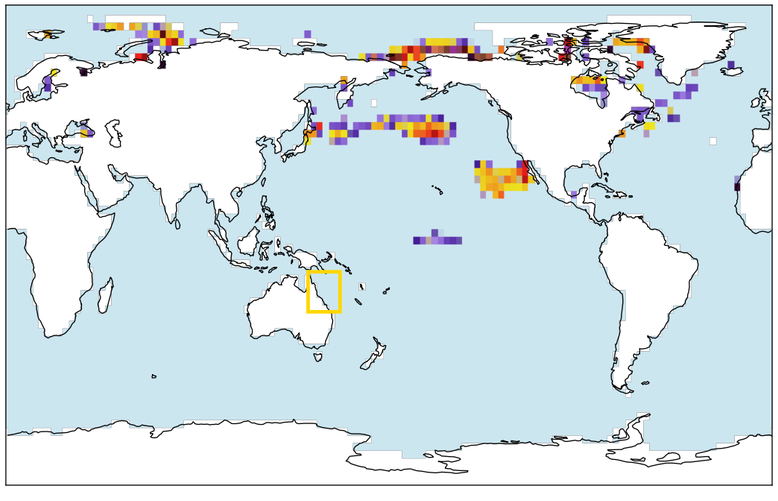
Error (Pred - Actual) - Step 46 [scale ±5.4]



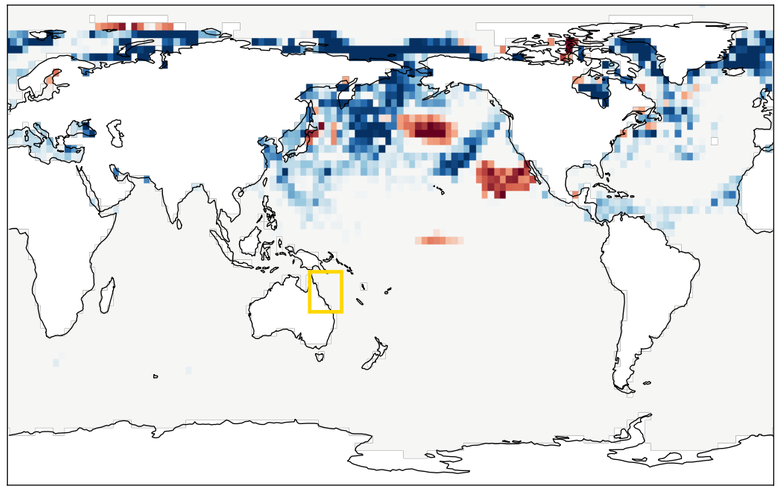
Actual DHW - Step 90

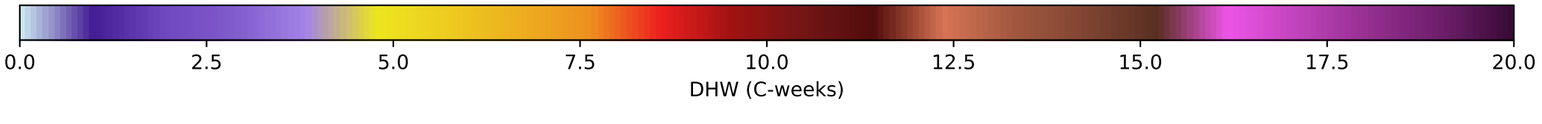


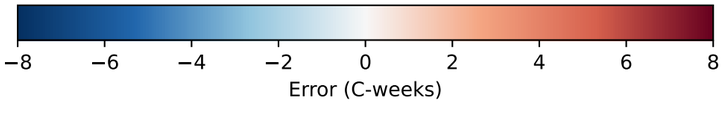
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

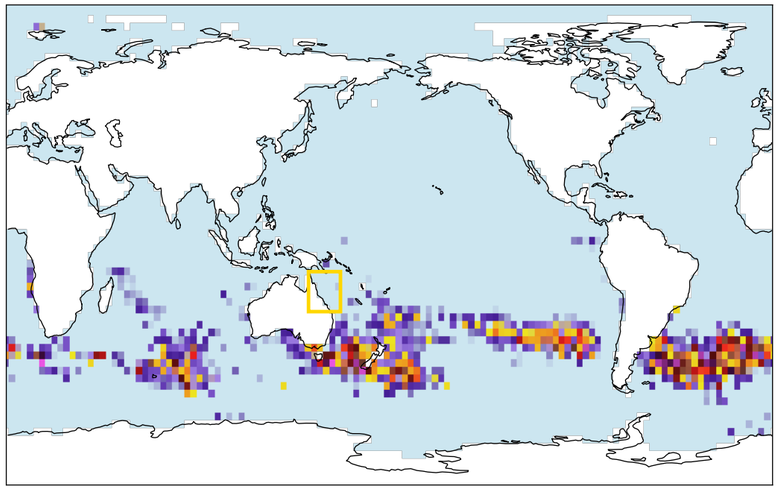




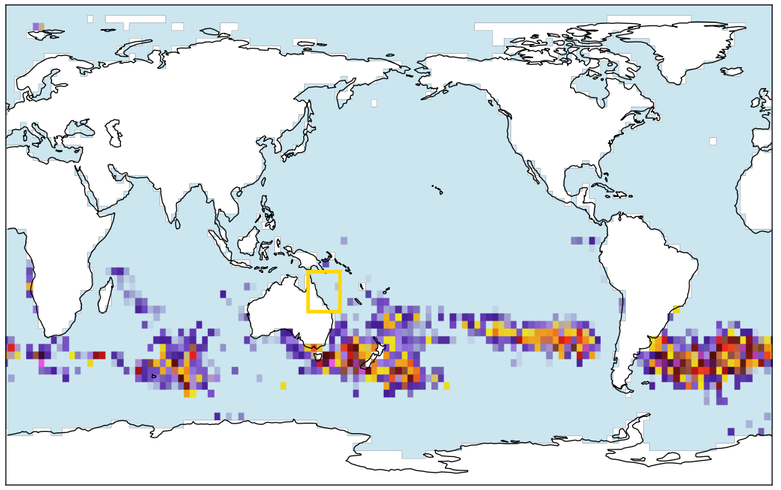


DHW Chain init: 2018-03-31 - Val (C-weeks)

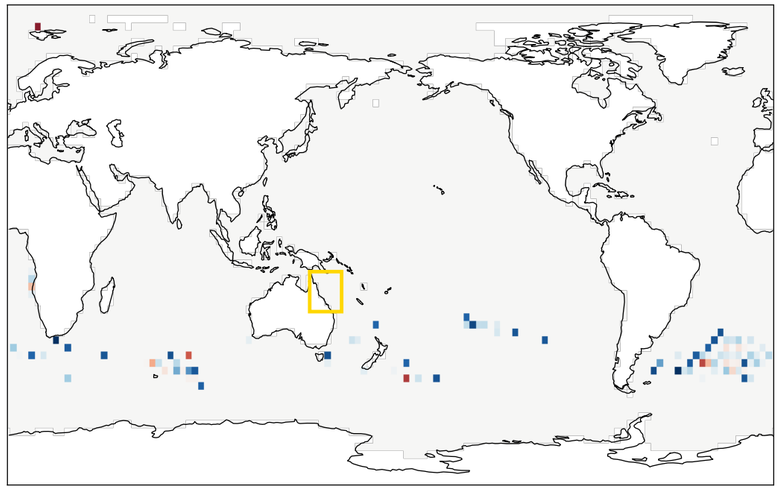
Actual DHW - Step 1



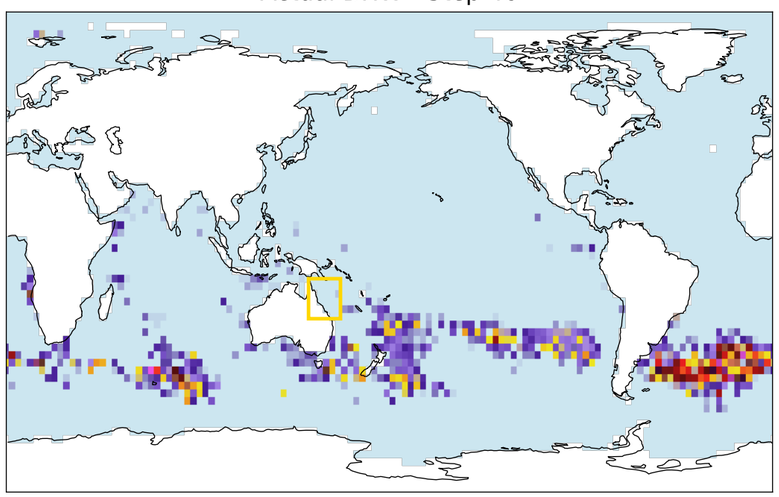
Predicted DHW - Step 1



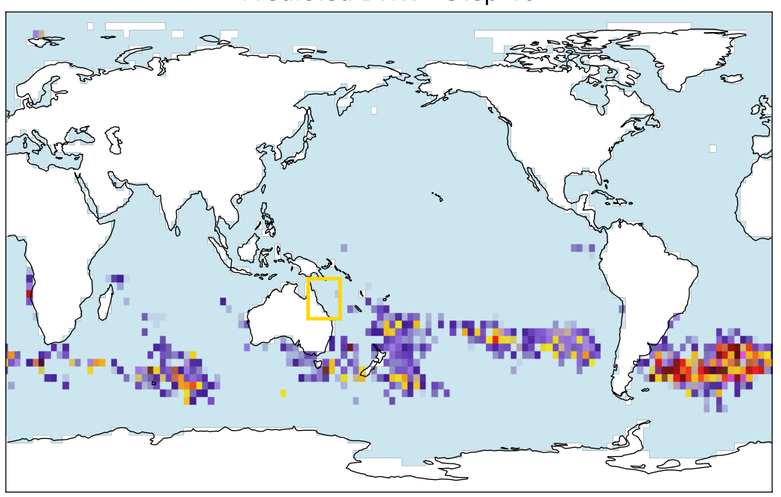
Error (Pred - Actual) - Step 1 [scale ±0.21]



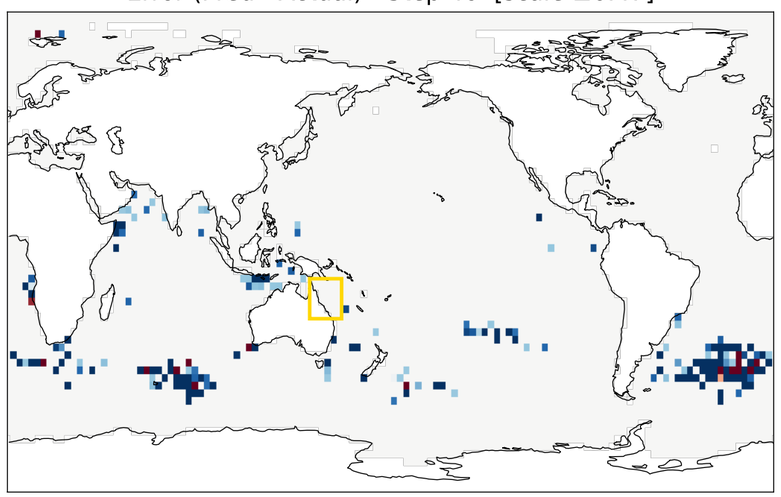
Actual DHW - Step 46



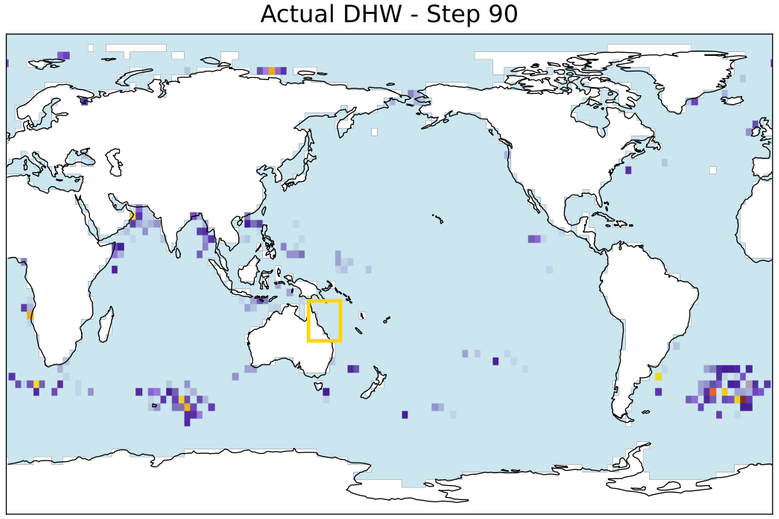
Predicted DHW - Step 46



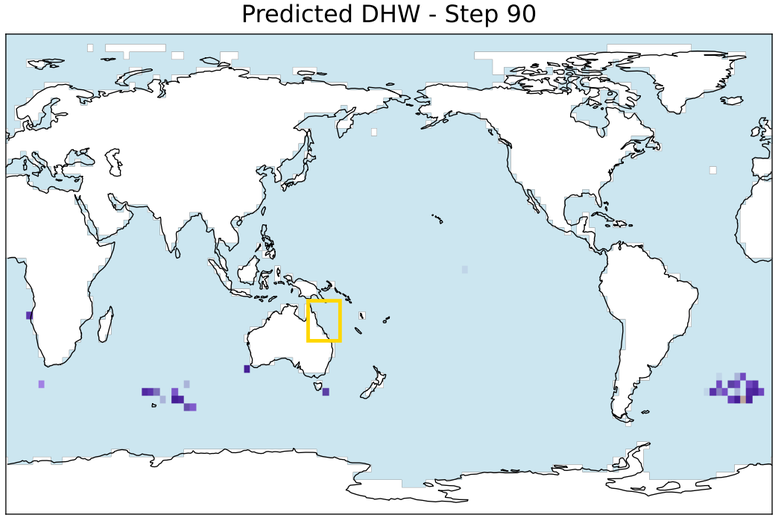
Error (Pred - Actual) - Step 46 [scale ±0.47]



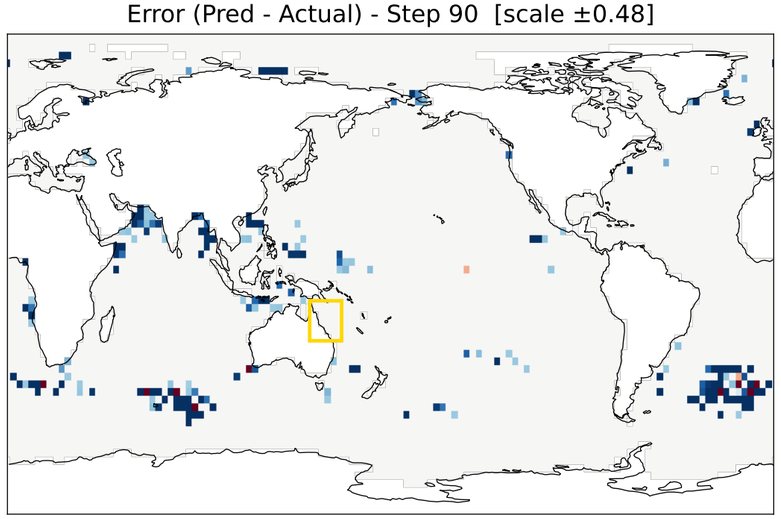
Actual DHW - Step 90

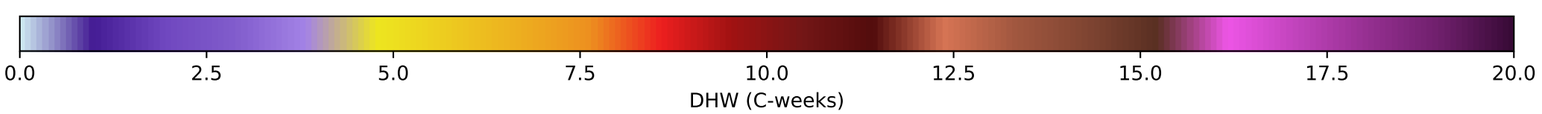


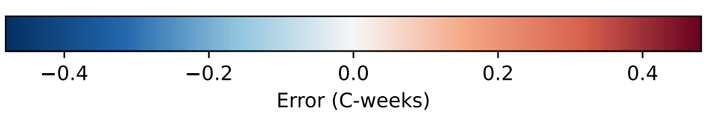
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.48]

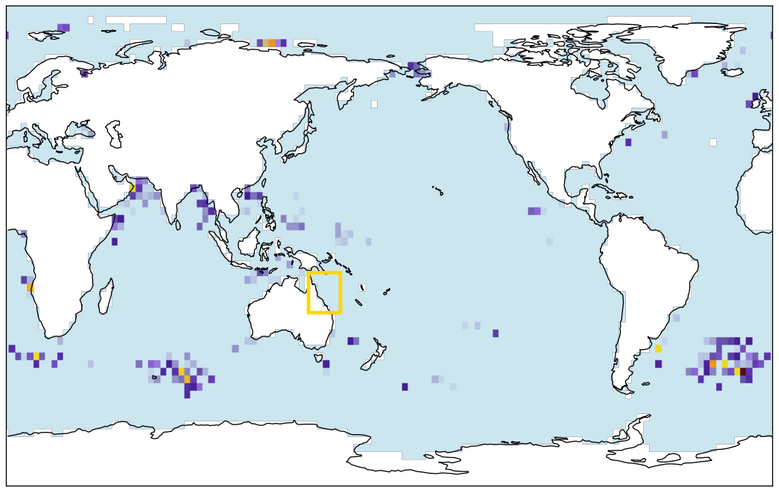




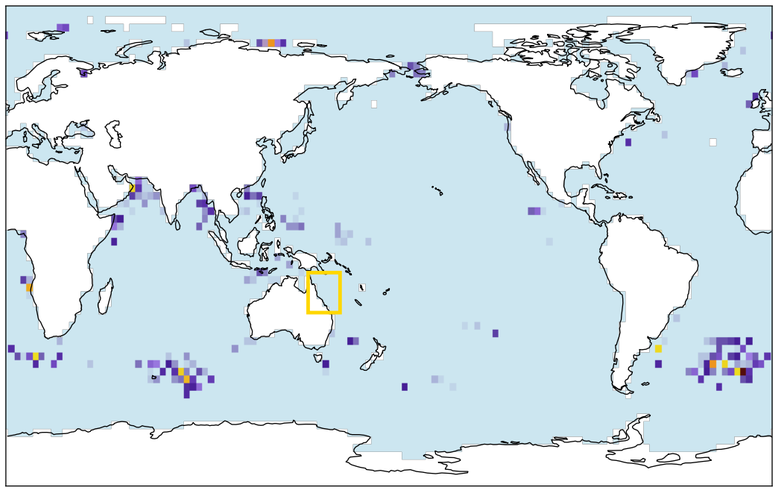


DHW Chain init: 2018-06-29 - Val (C-weeks)

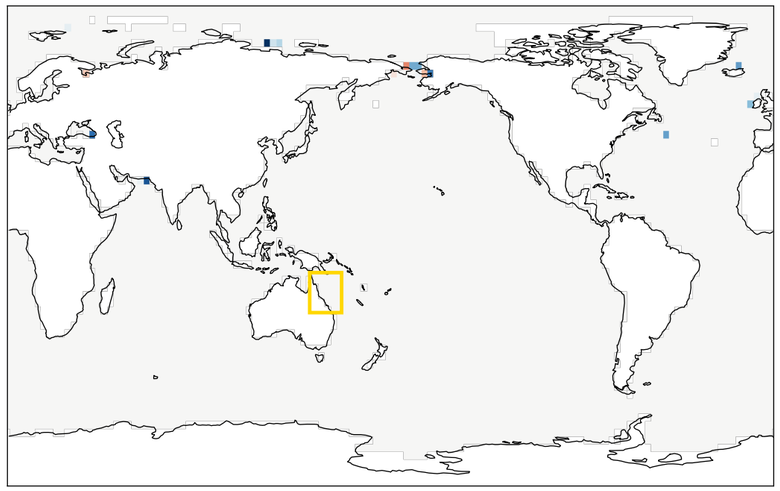
Actual DHW - Step 1



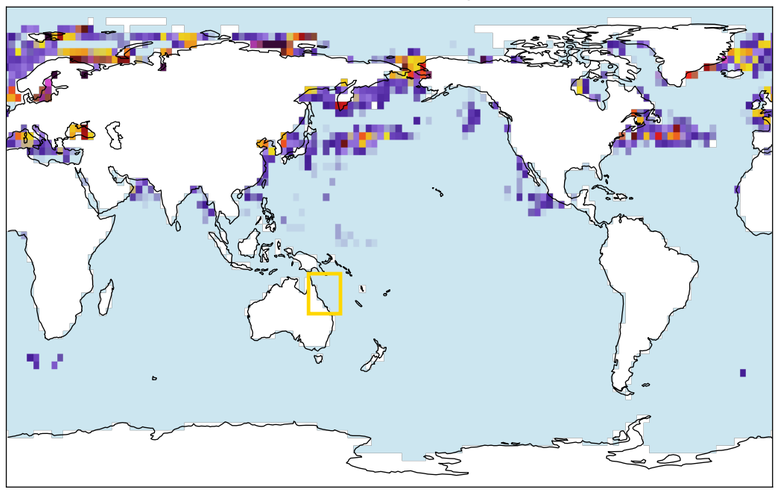
Predicted DHW - Step 1



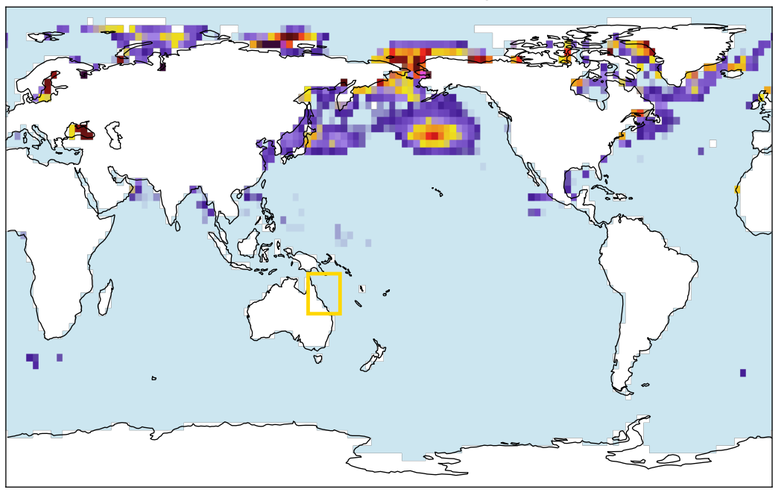
Error (Pred - Actual) - Step 1 [scale ±0.31]



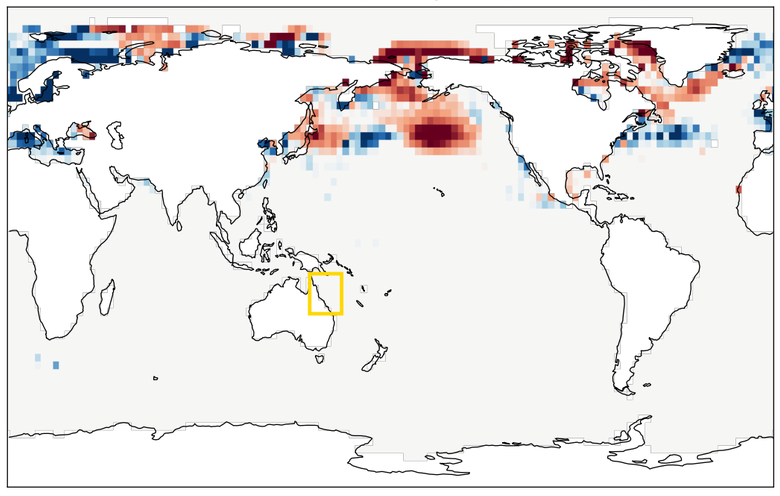
Actual DHW - Step 46



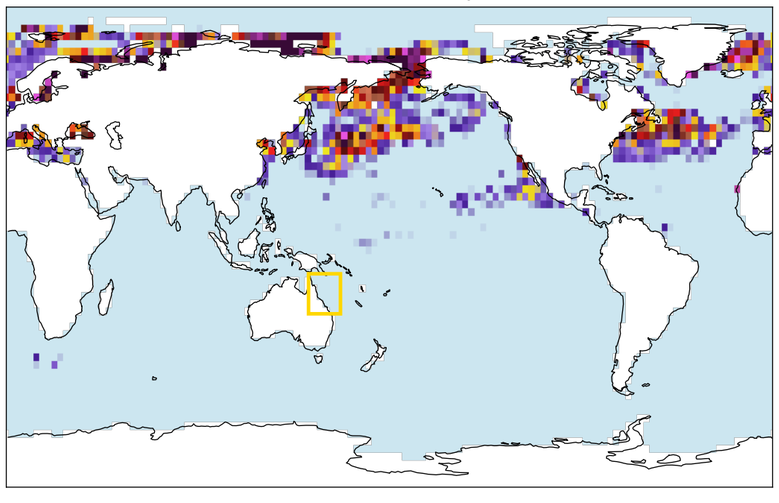
Predicted DHW - Step 46



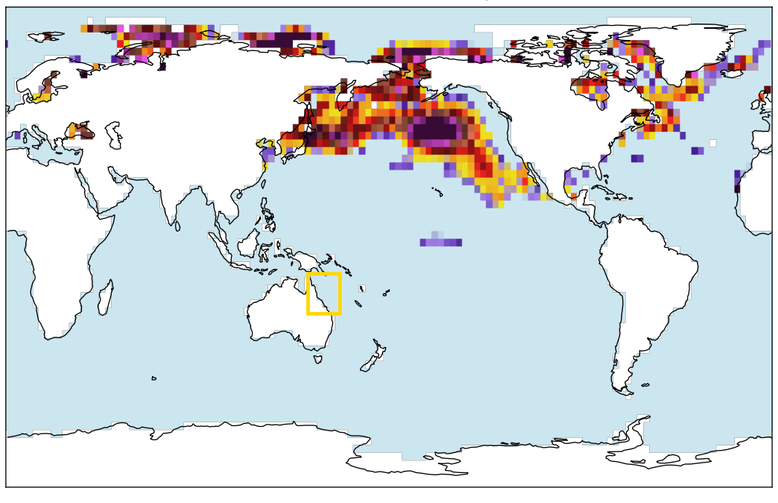
Error (Pred - Actual) - Step 46 [scale ±5.7]



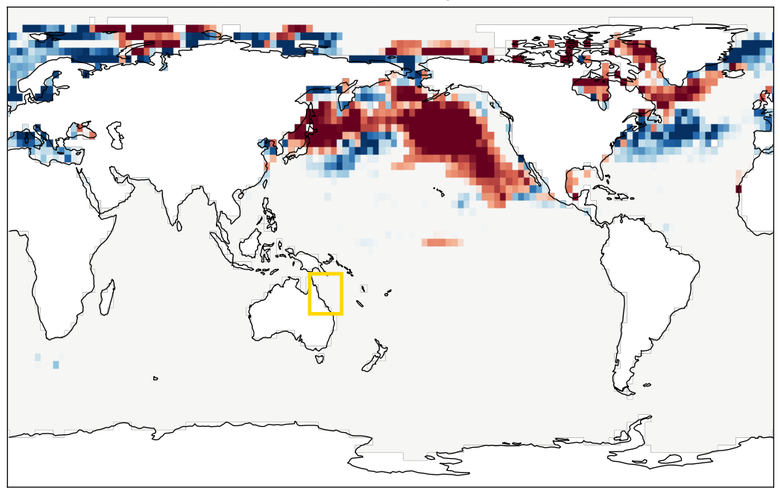
Actual DHW - Step 90

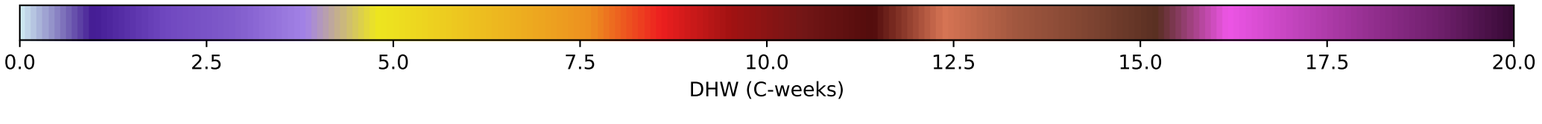


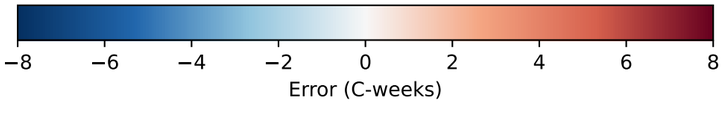
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

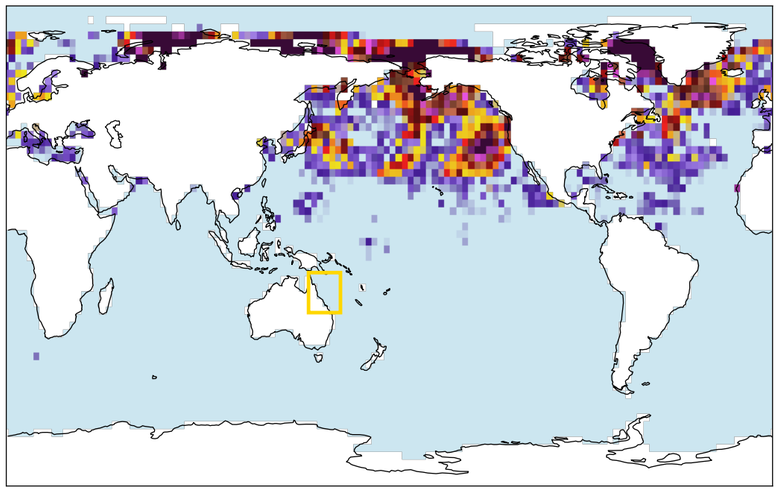




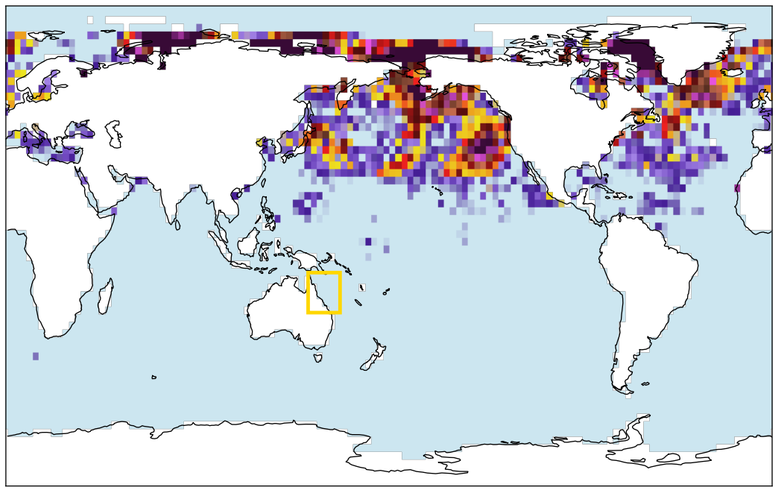


DHW Chain init: 2019-09-22 - Val (C-weeks)

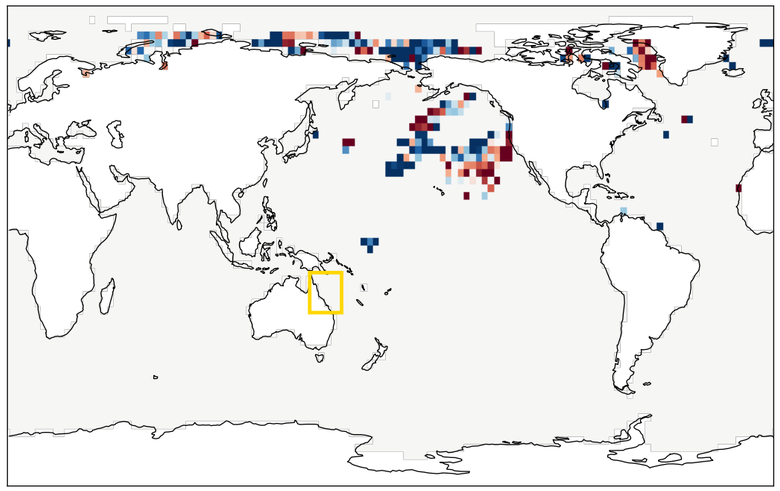
Actual DHW - Step 1



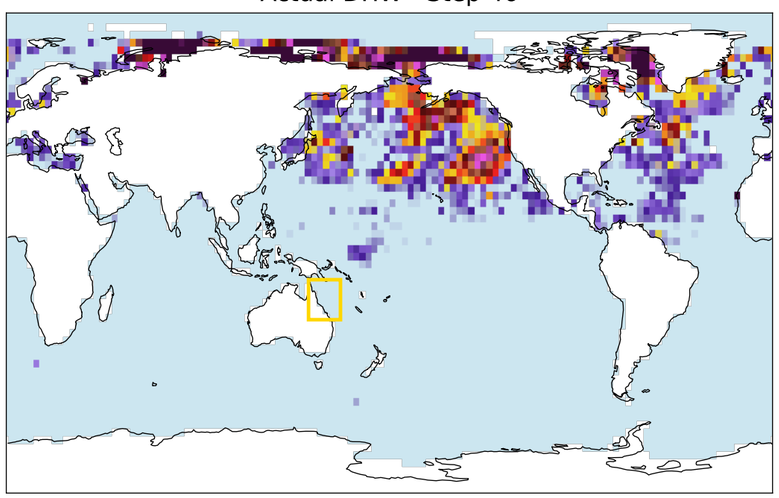
Predicted DHW - Step 1



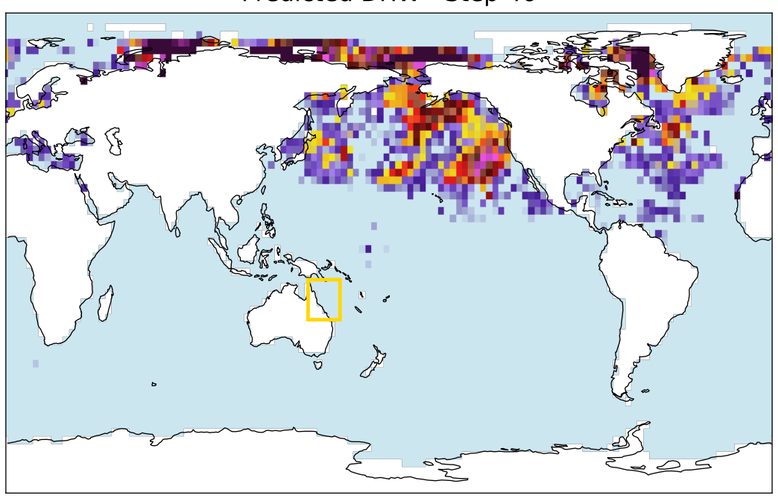
Error (Pred - Actual) - Step 1 [scale ±0.032]



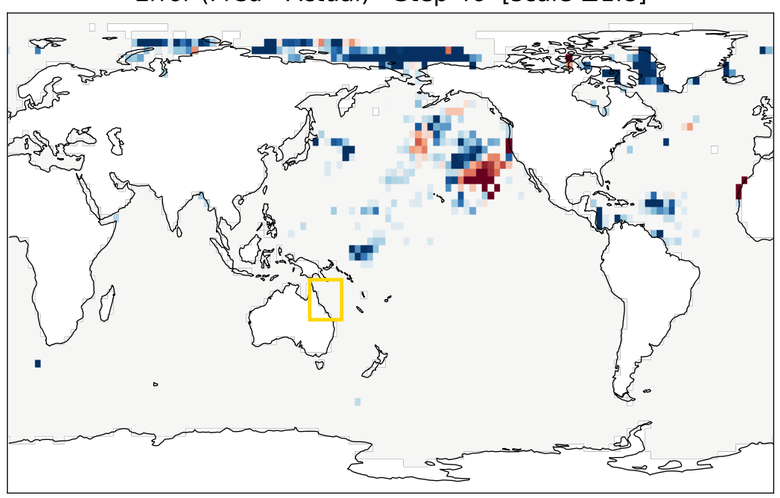
Actual DHW - Step 46



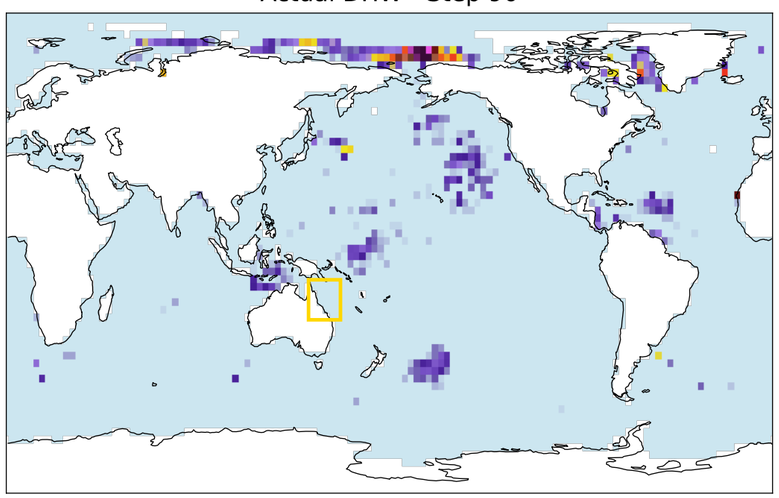
Predicted DHW - Step 46



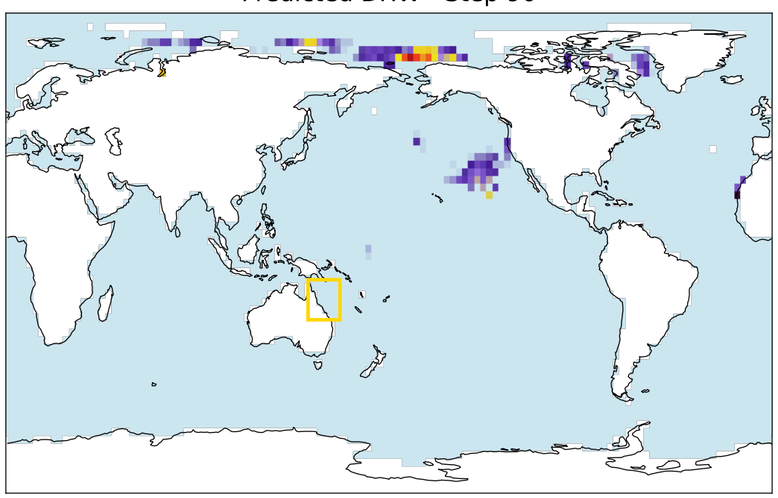
Error (Pred - Actual) - Step 46 [scale ±1.8]



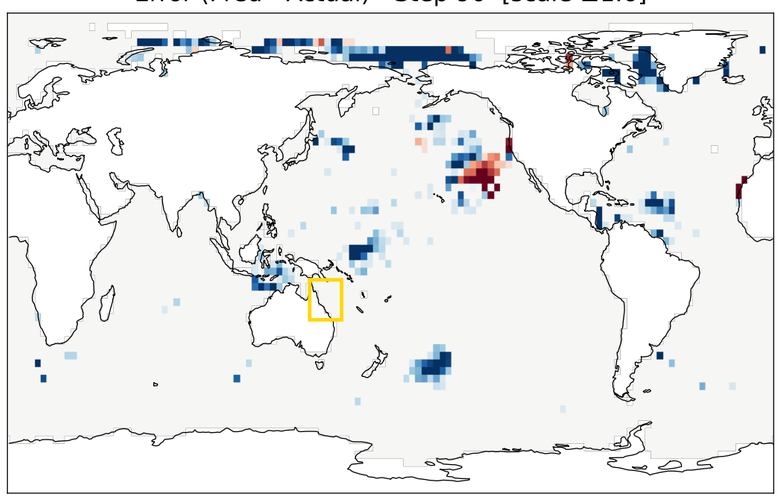
Actual DHW - Step 90

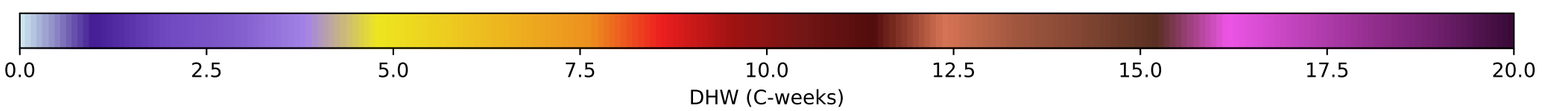


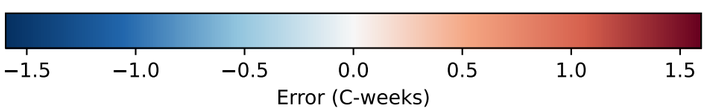
Predicted DHW - Step 90



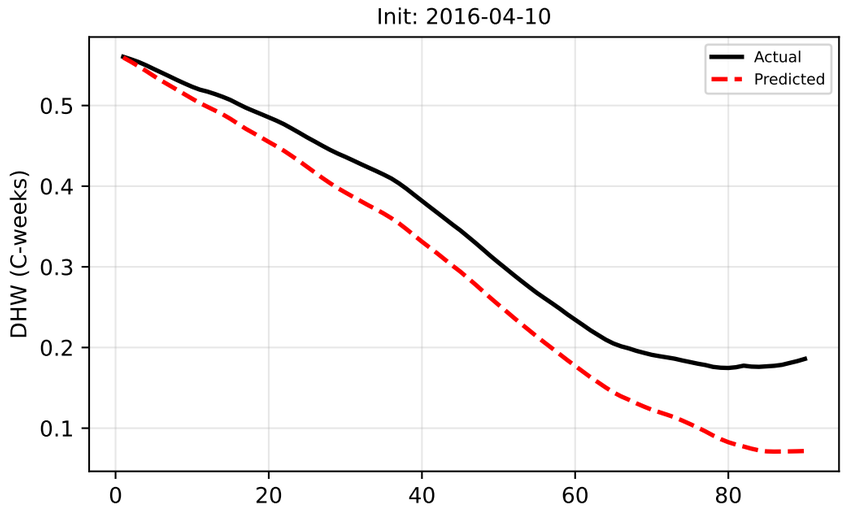
Error (Pred - Actual) - Step 90 [scale ±1.6]

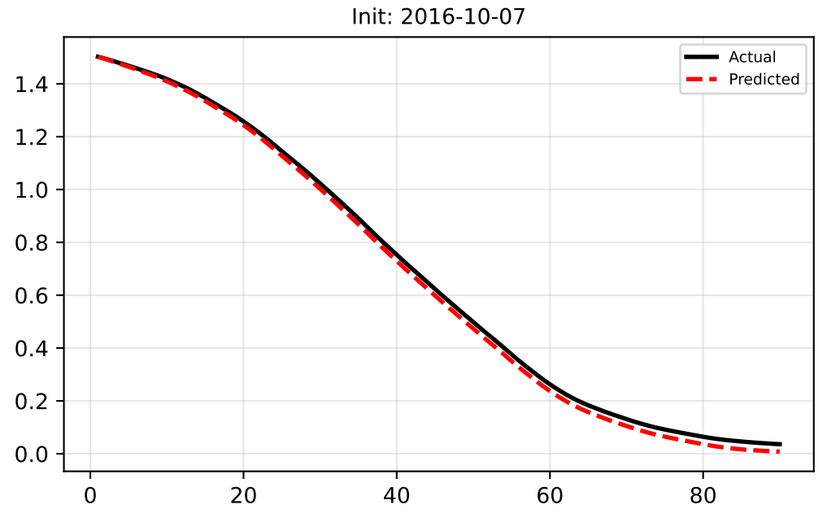


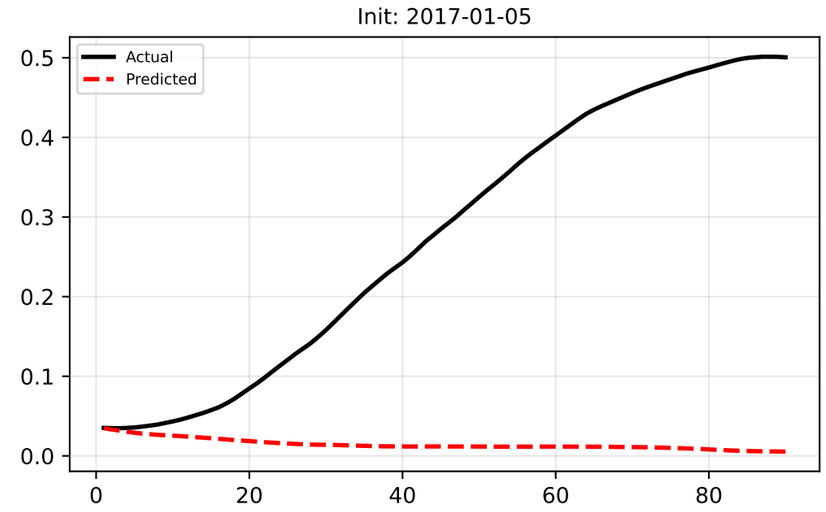


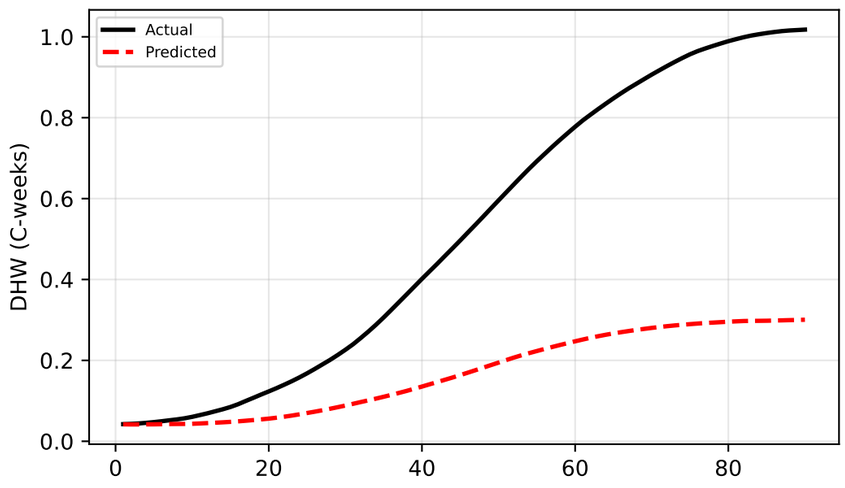


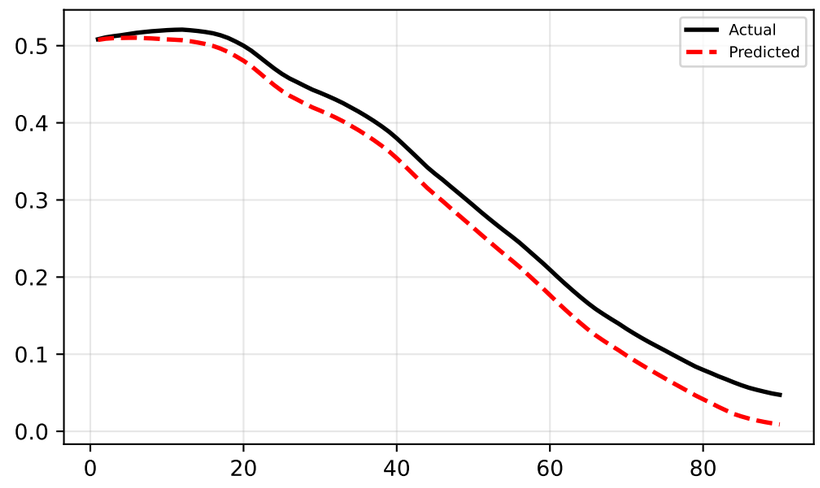
Global-Mean DHW - Val (C-weeks)

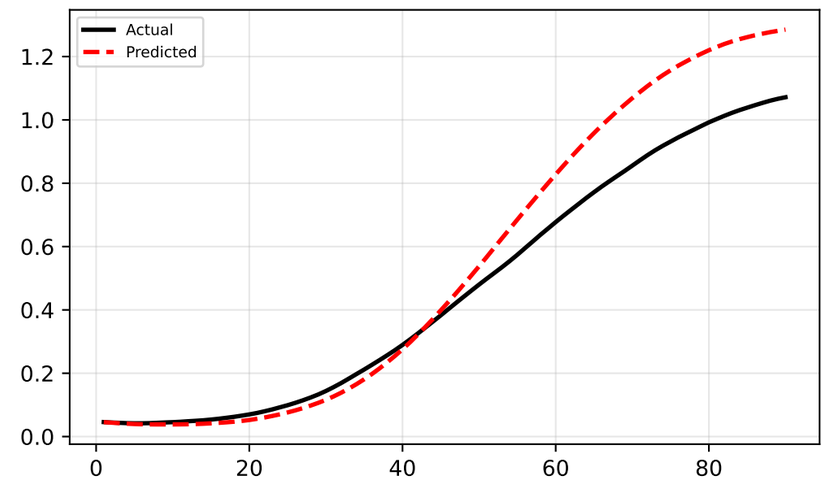


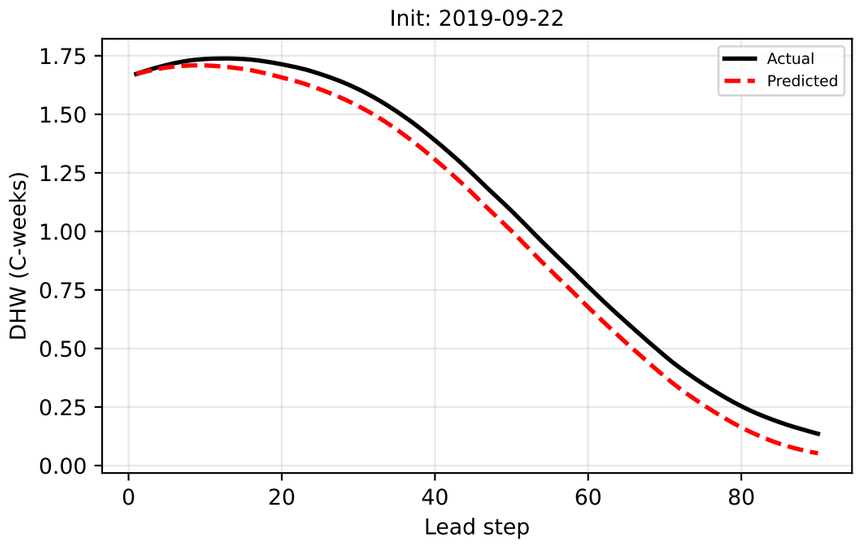




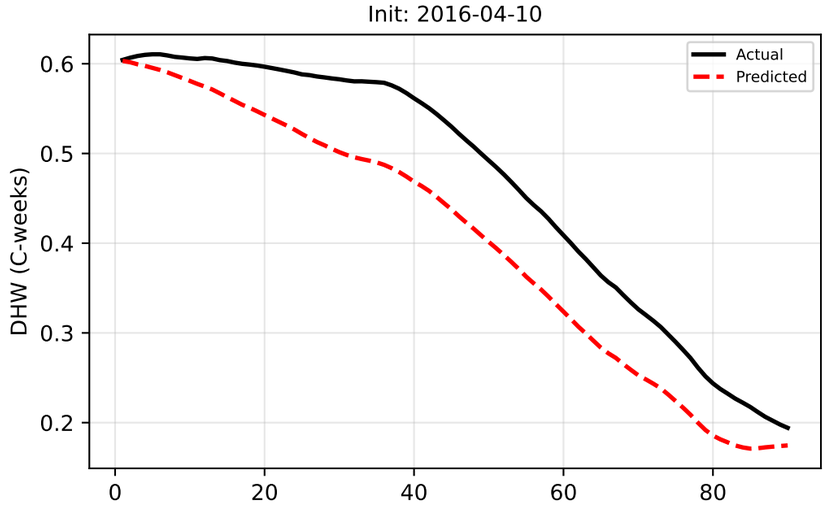


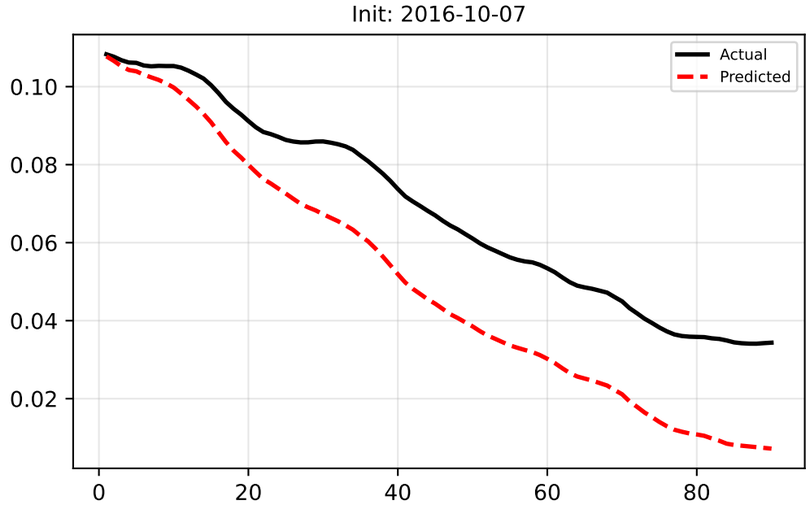


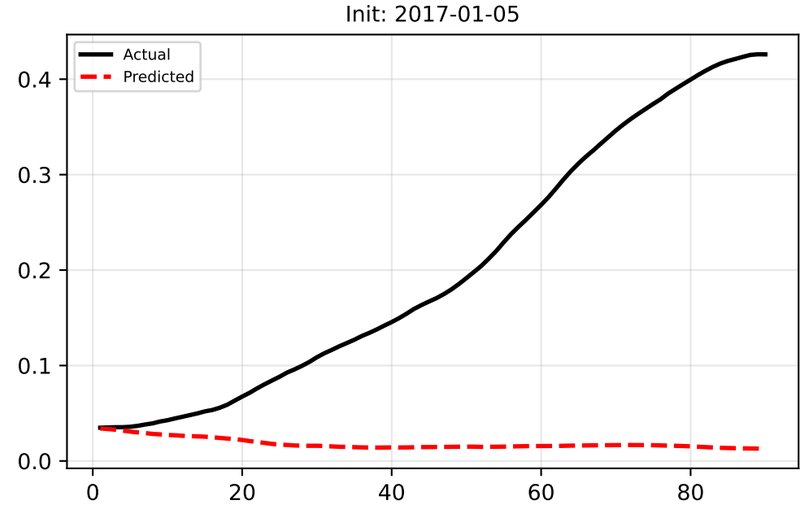


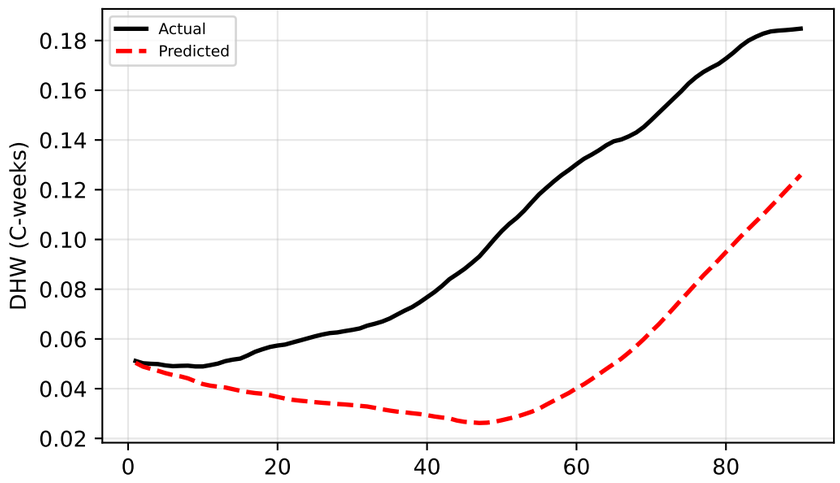


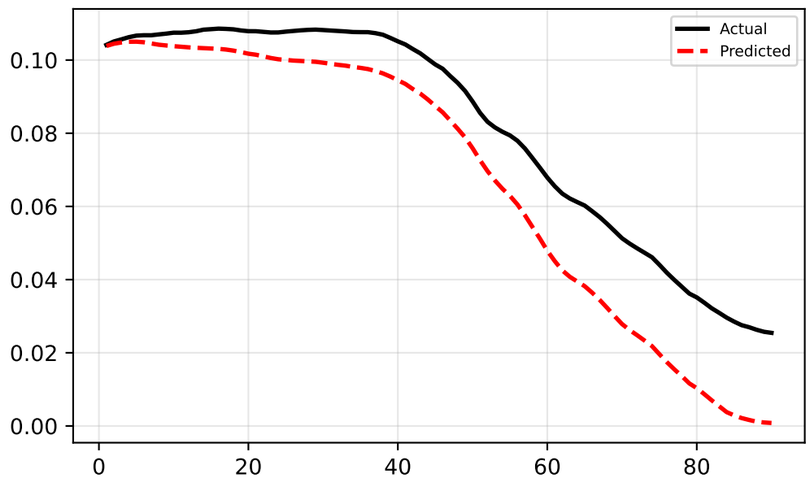
Tropical-Mean DHW (|lat|<=30) - Val (C-weeks)

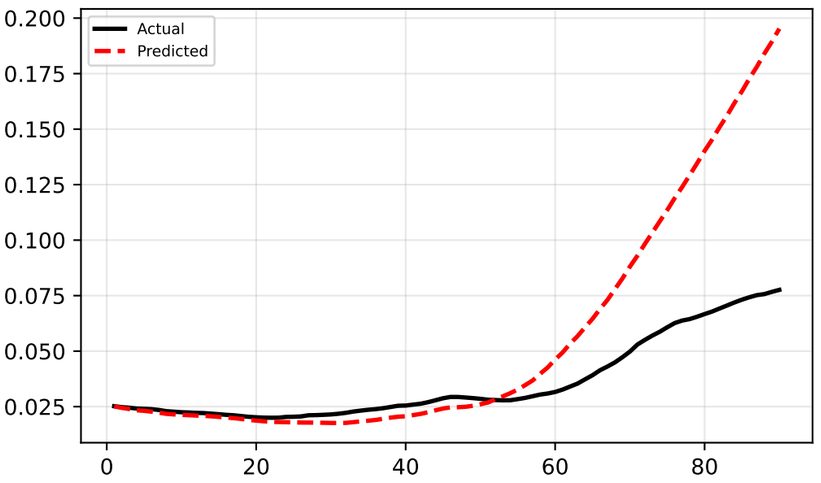


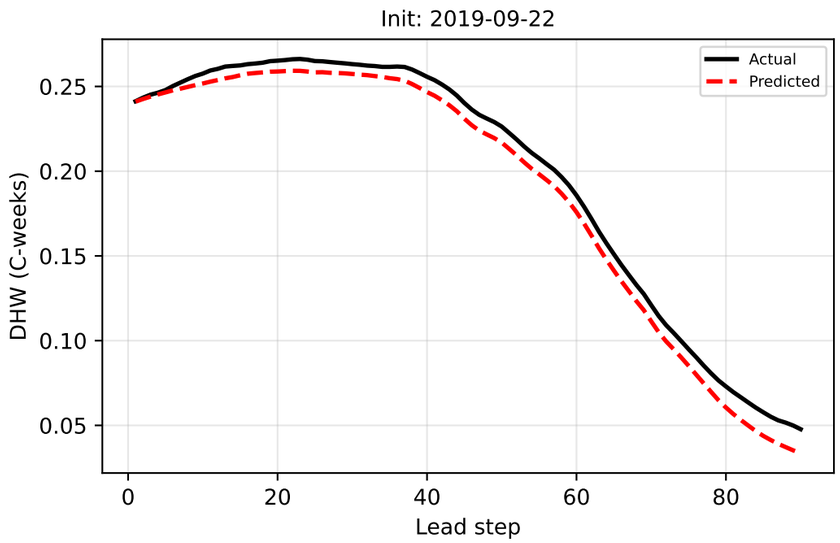




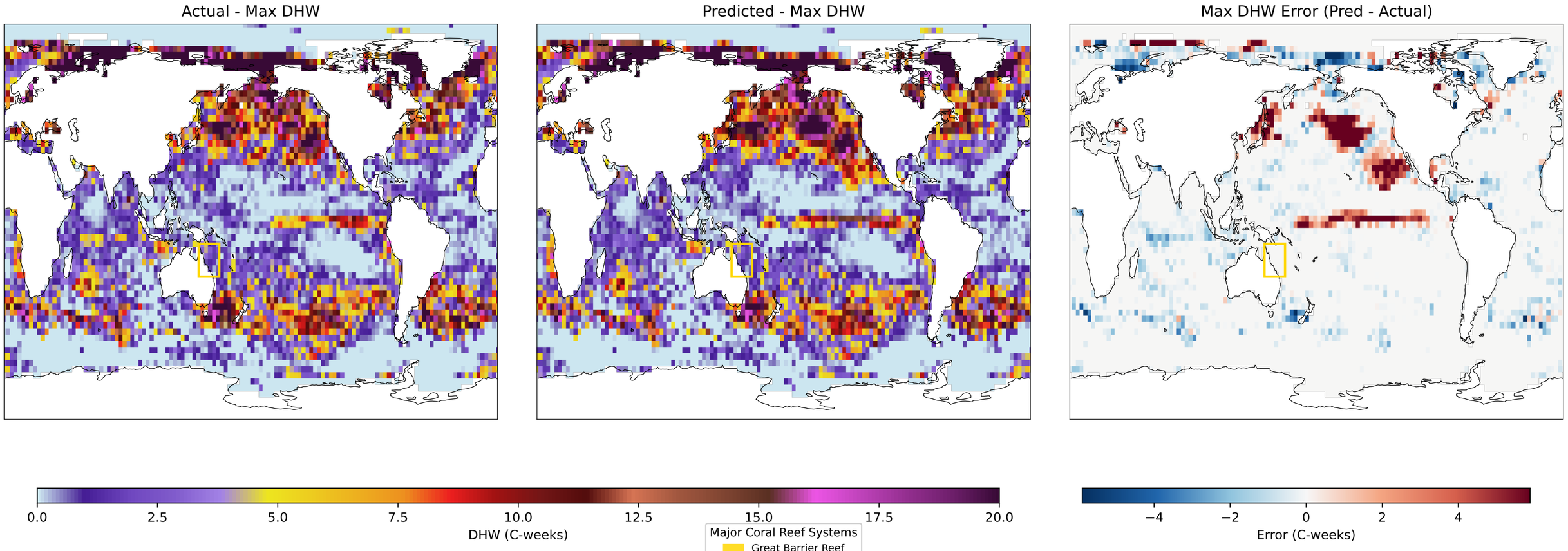


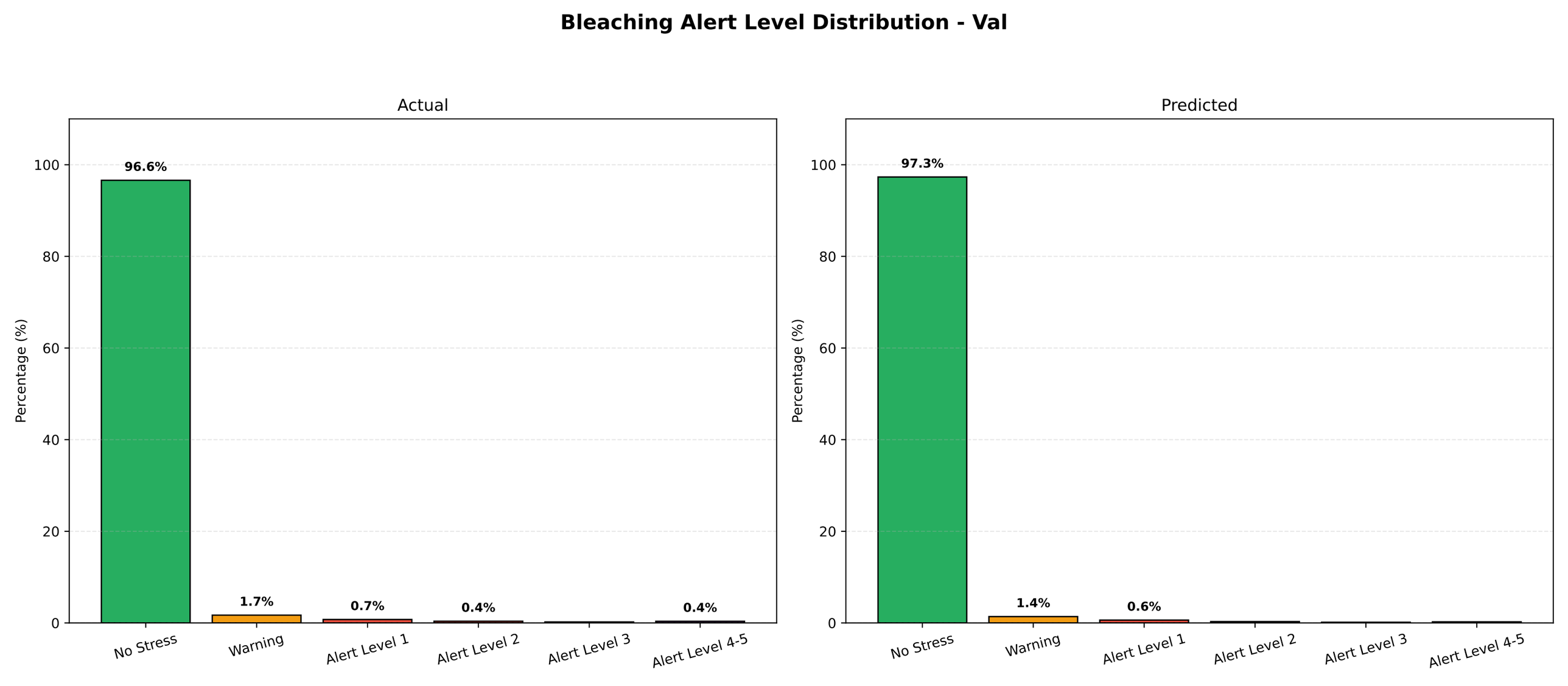






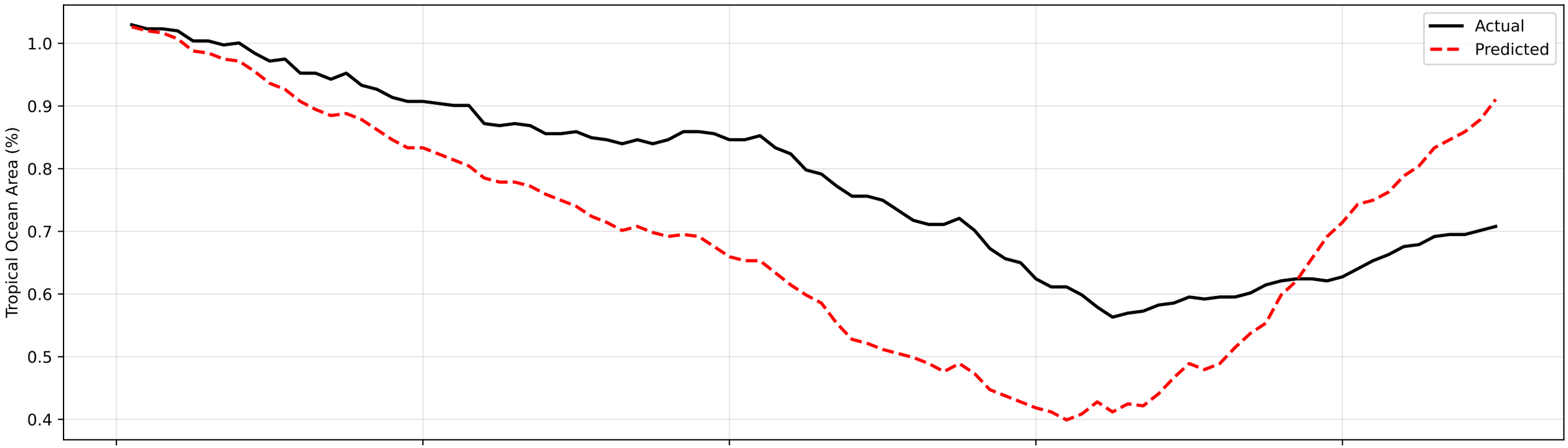
Peak DHW (max over all chains & leads) - Val (C-weeks)



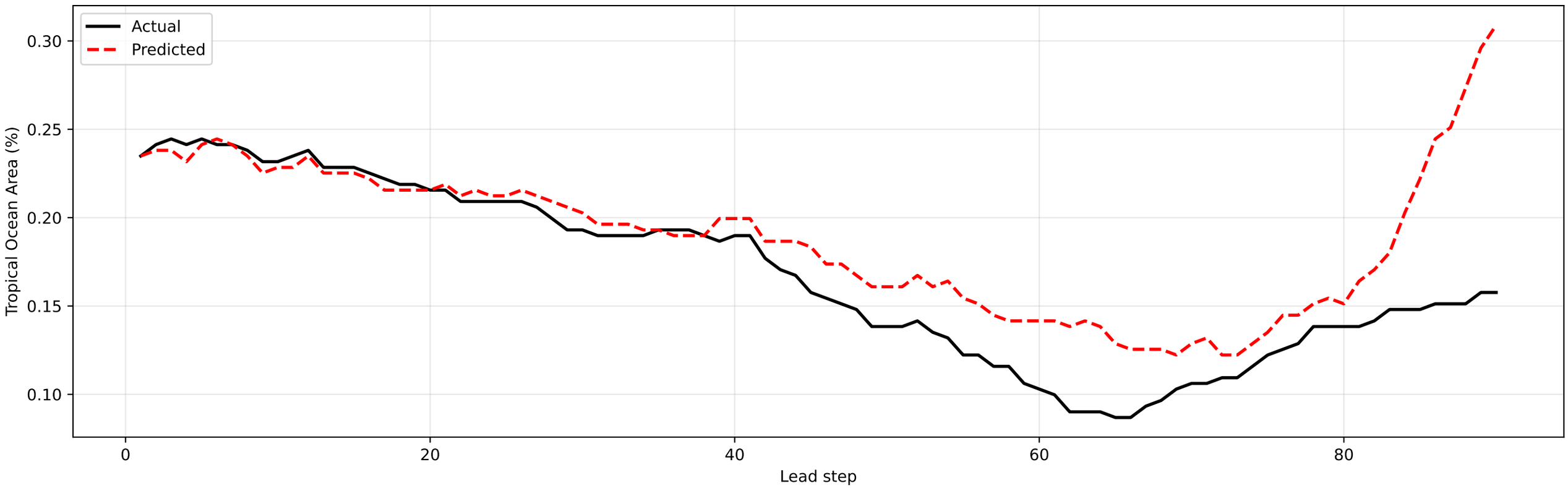

Bleaching Alert Level Distribution - Val
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.6%
1.7%
0.7%
0.4%
0.4%
97.3%
1.4%
0.6%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



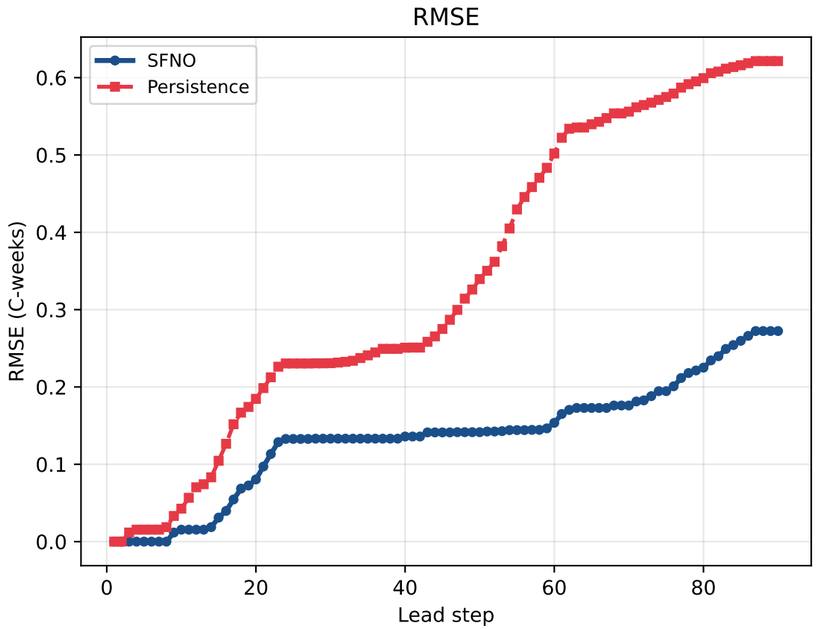
Great Barrier Reef Analysis - Val

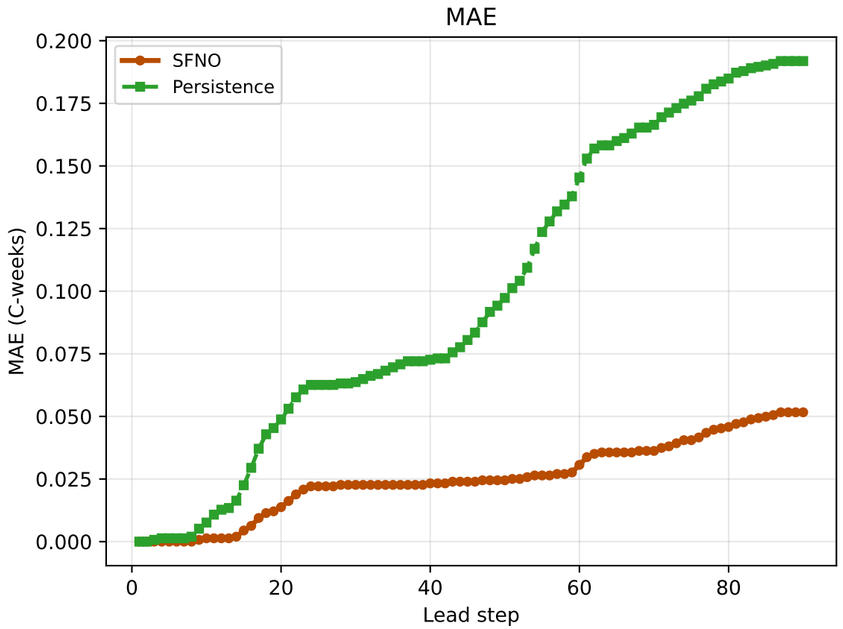
```
                    GREAT BARRIER REEF - VAL
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                    Grid: 5 lat x 5 lon points
```

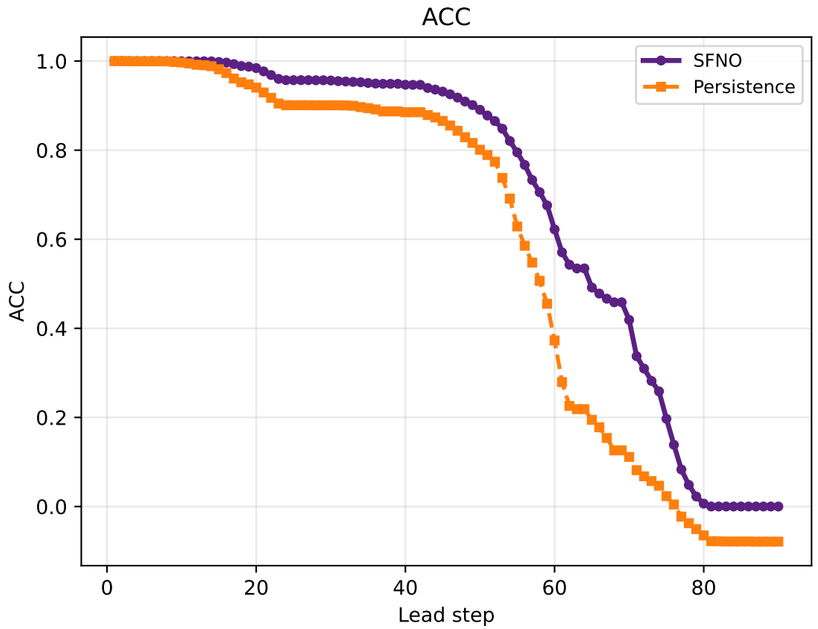
DHW Metrics by Lead Step - Val - GBR (15 chains scored at every lead)

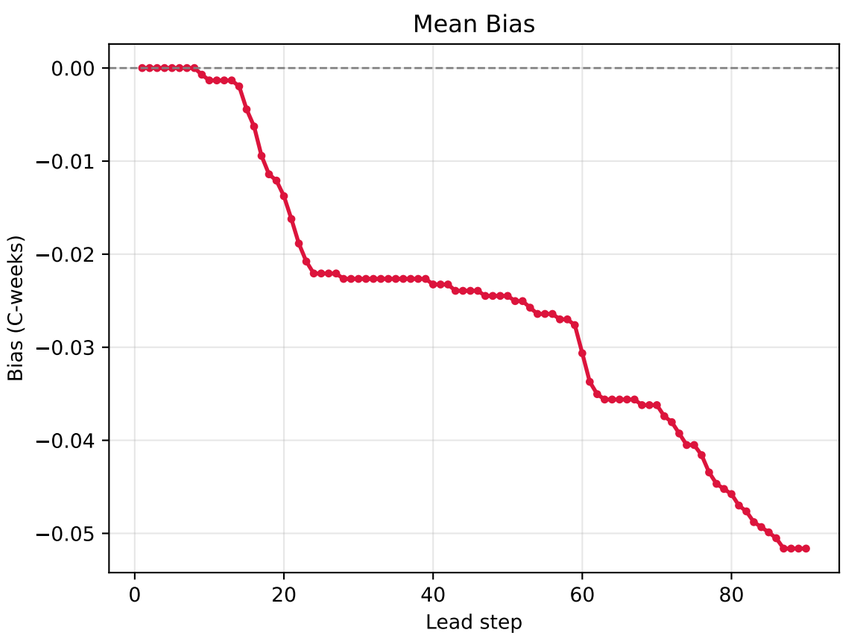
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0000	0.0000	n/a	0.0000	0.0000	0.0000	1.0000	1.0000	1.0000
2	0.0000	0.0000	n/a	0.0000	0.0000	0.0000	1.0000	1.0000	1.0000
3	0.0000	0.0117	100.0%	0.0000	0.0007	0.0000	1.0000	0.9998	1.0000
4	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
5	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
6	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
7	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
8	0.0000	0.0187	100.0%	0.0000	0.0020	0.0000	1.0000	0.9994	1.0000
9	0.0117	0.0332	64.6%	0.0007	0.0052	-0.0007	0.9997	0.9981	0.9995
10	0.0155	0.0426	63.7%	0.0013	0.0076	-0.0013	0.9996	0.9969	0.9991
11	0.0155	0.0567	72.7%	0.0013	0.0108	-0.0013	0.9995	0.9947	0.9991
12	0.0155	0.0703	78.0%	0.0013	0.0127	-0.0013	0.9995	0.9917	0.9991
13	0.0155	0.0742	79.2%	0.0013	0.0134	-0.0013	0.9995	0.9907	0.9991
14	0.0187	0.0832	77.5%	0.0020	0.0163	-0.0020	0.9993	0.9884	0.9987
15	0.0311	0.1046	70.3%	0.0044	0.0225	-0.0044	0.9980	0.9817	0.9962
16	0.0397	0.1265	68.6%	0.0063	0.0295	-0.0063	0.9966	0.9732	0.9935
17	0.0545	0.1518	64.1%	0.0094	0.0371	-0.0094	0.9932	0.9608	0.9871
18	0.0686	0.1668	58.9%	0.0114	0.0429	-0.0114	0.9889	0.9523	0.9789
19	0.0726	0.1742	58.3%	0.0121	0.0453	-0.0121	0.9874	0.9478	0.9761
20	0.0804	0.1846	56.4%	0.0138	0.0488	-0.0138	0.9843	0.9404	0.9703
21	0.0971	0.1986	51.1%	0.0162	0.0531	-0.0162	0.9770	0.9296	0.9566
22	0.1133	0.2125	46.7%	0.0189	0.0576	-0.0189	0.9688	0.9174	0.9416
23	0.1287	0.2262	43.1%	0.0208	0.0607	-0.0208	0.9600	0.9047	0.9259
24	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
25	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
26	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
27	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
28	0.1332	0.2306	42.3%	0.0227	0.0632	-0.0227	0.9570	0.9007	0.9203
29	0.1332	0.2306	42.3%	0.0227	0.0632	-0.0227	0.9570	0.9007	0.9203
30	0.1332	0.2308	42.3%	0.0227	0.0637	-0.0227	0.9564	0.9008	0.9192
31	0.1332	0.2316	42.5%	0.0227	0.0649	-0.0227	0.9550	0.9008	0.9167
32	0.1332	0.2325	42.7%	0.0227	0.0662	-0.0227	0.9542	0.9004	0.9152
33	0.1332	0.2339	43.1%	0.0227	0.0669	-0.0227	0.9533	0.8996	0.9136
34	0.1332	0.2374	43.9%	0.0227	0.0683	-0.0227	0.9525	0.8967	0.9121
35	0.1332	0.2409	44.7%	0.0227	0.0696	-0.0227	0.9509	0.8944	0.9093
36	0.1332	0.2447	45.6%	0.0227	0.0708	-0.0227	0.9496	0.8915	0.9068
37	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
38	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
39	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
40	0.1359	0.2508	45.8%	0.0233	0.0726	-0.0233	0.9469	0.8852	0.9019
41	0.1359	0.2509	45.8%	0.0233	0.0732	-0.0233	0.9466	0.8853	0.9015
42	0.1359	0.2509	45.8%	0.0233	0.0732	-0.0233	0.9466	0.8853	0.9015
43	0.1413	0.2584	45.3%	0.0239	0.0756	-0.0239	0.9396	0.8788	0.8887
44	0.1413	0.2654	46.8%	0.0239	0.0776	-0.0239	0.9361	0.8735	0.8822
45	0.1413	0.2750	48.6%	0.0239	0.0805	-0.0239	0.9315	0.8656	0.8740
46	0.1413	0.2869	50.8%	0.0239	0.0834	-0.0239	0.9255	0.8554	0.8631
47	0.1415	0.2997	52.8%	0.0245	0.0876	-0.0245	0.9180	0.8435	0.8499
48	0.1415	0.3142	55.0%	0.0245	0.0917	-0.0245	0.9092	0.8291	0.8344
49	0.1415	0.3260	56.6%	0.0245	0.0942	-0.0245	0.9013	0.8163	0.8207
50	0.1415	0.3395	58.3%	0.0245	0.0973	-0.0245	0.8907	0.8009	0.8025
51	0.1424	0.3503	59.3%	0.0250	0.1012	-0.0250	0.8778	0.7892	0.7809
52	0.1424	0.3620	60.7%	0.0250	0.1041	-0.0250	0.8653	0.7741	0.7605
53	0.1429	0.3821	62.6%	0.0257	0.1093	-0.0257	0.8482	0.7377	0.7322
54	0.1442	0.4049	64.4%	0.0264	0.1169	-0.0264	0.8206	0.6912	0.6890
55	0.1442	0.4296	66.4%	0.0264	0.1236	-0.0264	0.7950	0.6287	0.6501
56	0.1442	0.4457	67.6%	0.0264	0.1278	-0.0264	0.7668	0.5854	0.6088
57	0.1445	0.4585	68.5%	0.0270	0.1318	-0.0270	0.7330	0.5477	0.5614
58	0.1445	0.4705	69.3%	0.0270	0.1346	-0.0270	0.7057	0.5066	0.5248
59	0.1465	0.4834	69.7%	0.0276	0.1379	-0.0276	0.6760	0.4551	0.4861
60	0.1537	0.5019	69.4%	0.0306	0.1454	-0.0306	0.6222	0.3728	0.4186
61	0.1651	0.5223	68.4%	0.0337	0.1529	-0.0337	0.5707	0.2794	0.3594
62	0.1704	0.5339	68.1%	0.0350	0.1569	-0.0350	0.5428	0.2259	0.3295
63	0.1729	0.5356	67.7%	0.0356	0.1582	-0.0356	0.5345	0.2187	0.3212
64	0.1729	0.5356	67.7%	0.0356	0.1582	-0.0356	0.5345	0.2187	0.3212
65	0.1729	0.5397	68.0%	0.0356	0.1599	-0.0356	0.4915	0.1947	0.2746
66	0.1729	0.5429	68.2%	0.0356	0.1611	-0.0356	0.4781	0.1779	0.2597
67	0.1729	0.5476	68.4%	0.0356	0.1629	-0.0356	0.4666	0.1539	0.2469
68	0.1760	0.5538	68.2%	0.0362	0.1653	-0.0362	0.4585	0.1260	0.2369
69	0.1760	0.5538	68.2%	0.0362	0.1653	-0.0362	0.4585	0.1260	0.2369
70	0.1760	0.5561	68.3%	0.0362	0.1664	-0.0362	0.4189	0.1113	0.2001
71	0.1812	0.5617	67.7%	0.0374	0.1695	-0.0374	0.3375	0.0816	0.1329
72	0.1827	0.5646	67.6%	0.0380	0.1713	-0.0380	0.3096	0.0678	0.1116
73	0.1881	0.5677	66.9%	0.0393	0.1731	-0.0393	0.2819	0.0571	0.0915
74	0.1946	0.5713	65.9%	0.0405	0.1749	-0.0405	0.2585	0.0465	0.0756
75	0.1946	0.5750	66.2%	0.0405	0.1761	-0.0405	0.1969	0.0233	0.0384
76	0.2009	0.5794	65.3%	0.0416	0.1778	-0.0416	0.1385	0.0044	0.0092
77	0.2116	0.5871	64.0%	0.0434	0.1808	-0.0434	0.0831	-0.0226	-0.0142
78	0.2180	0.5916	63.1%	0.0447	0.1826	-0.0447	0.0485	-0.0374	-0.0277
79	0.2214	0.5952	62.8%	0.0452	0.1837	-0.0452	0.0225	-0.0507	-0.0367
80	0.2250	0.5994	62.5%	0.0458	0.1849	-0.0458	0.0064	-0.0649	-0.0418
81	0.2344	0.6058	61.3%	0.0470	0.1872	-0.0470	0.0000	-0.0779	-0.0419
82	0.2398	0.6079	60.6%	0.0476	0.1879	-0.0476	0.0000	-0.0782	-0.0411
83	0.2492	0.6117	59.3%	0.0488	0.1890	-0.0488	0.0000	-0.0785	-0.0399
84	0.2542	0.6137	58.6%	0.0493	0.1896	-0.0493	0.0000	-0.0786	-0.0391
85	0.2597	0.6160	57.8%	0.0499	0.1901	-0.0499	0.0000	-0.0786	-0.0383
86	0.2661	0.6187	57.0%	0.0505	0.1907	-0.0505	0.0000	-0.0785	-0.0374
87	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
88	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
89	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
90	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373

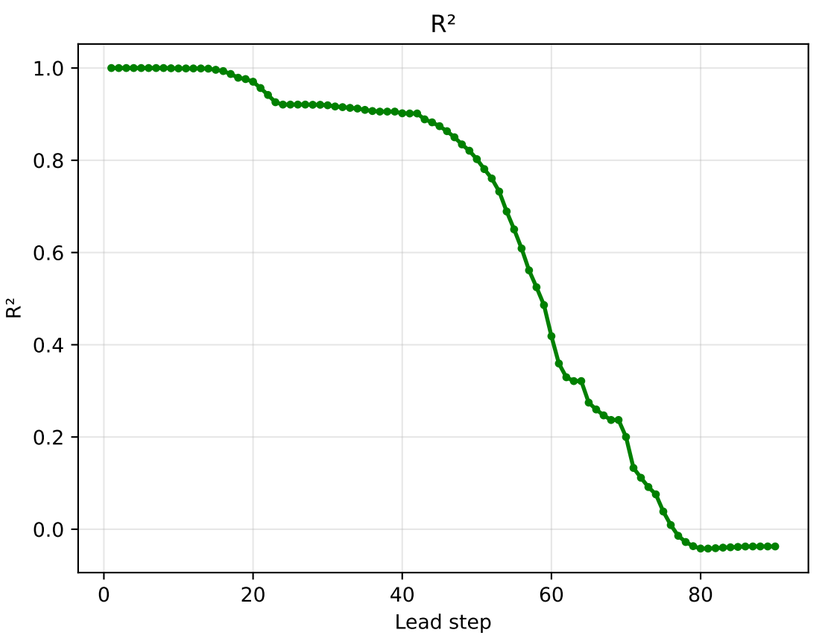
DHW Forecast Skill vs Lead Step - Val - GBR

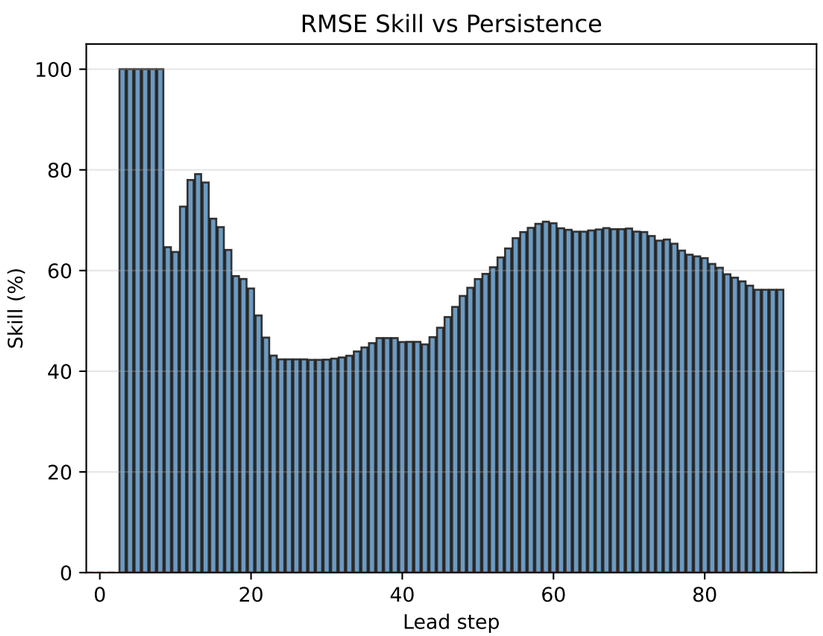


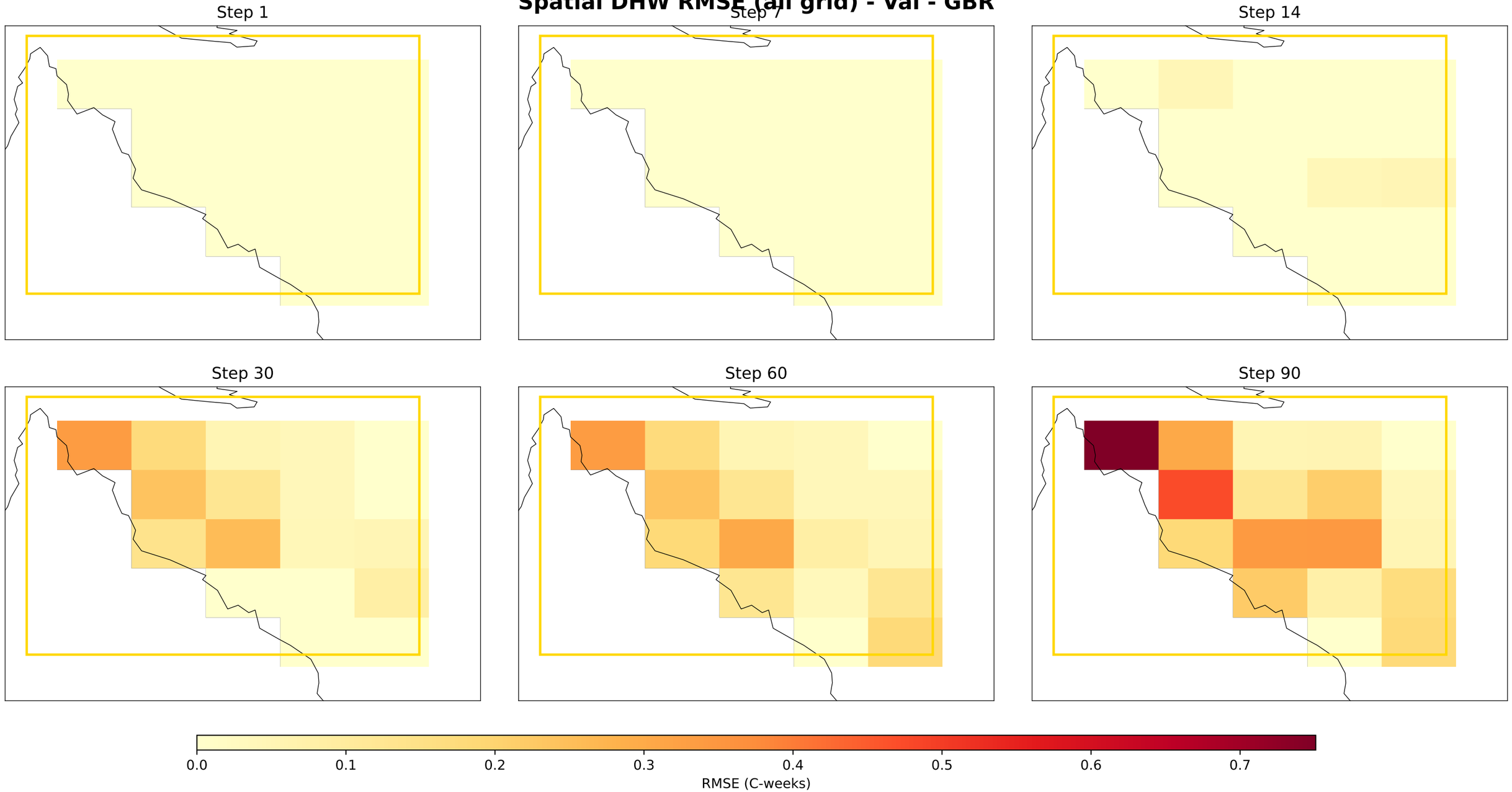


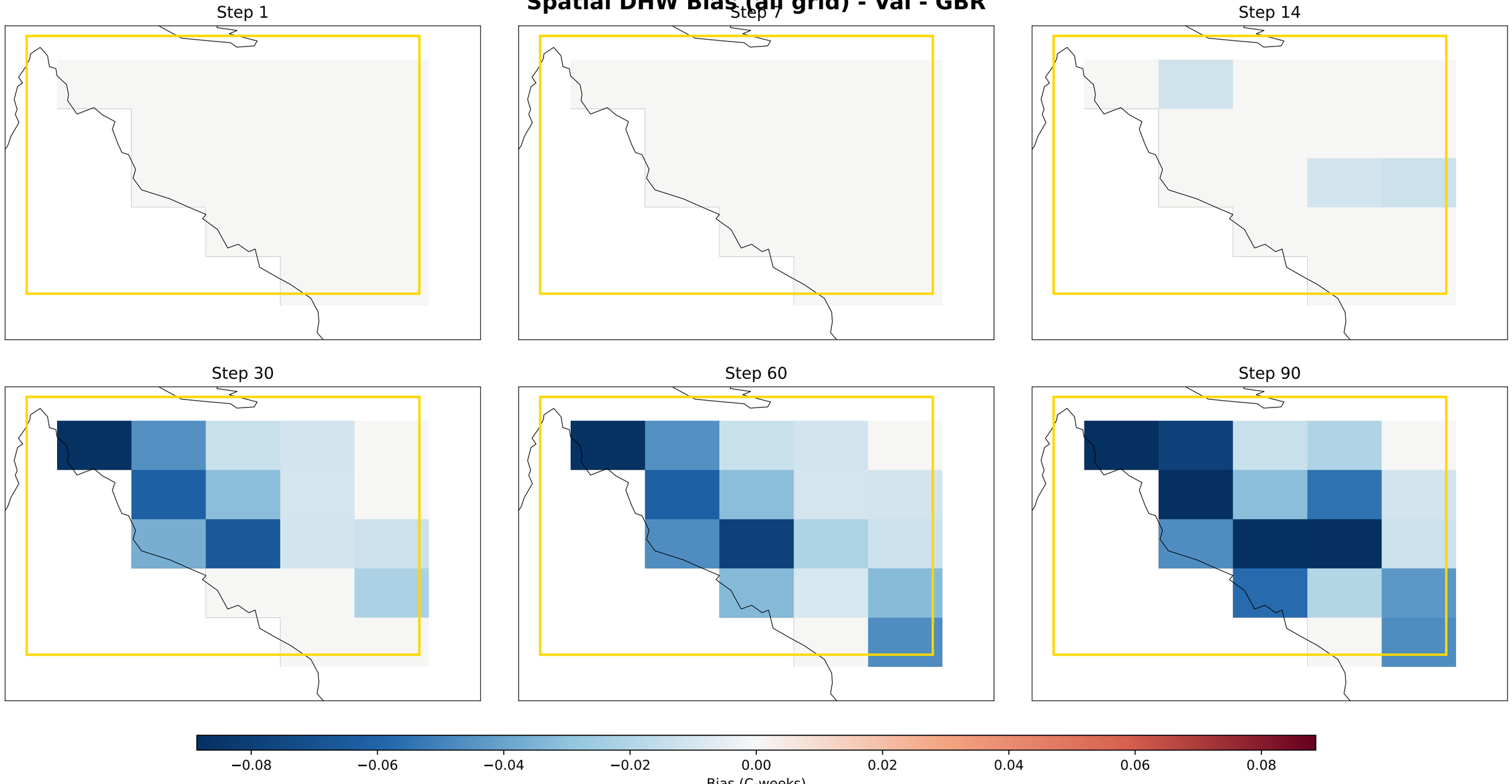


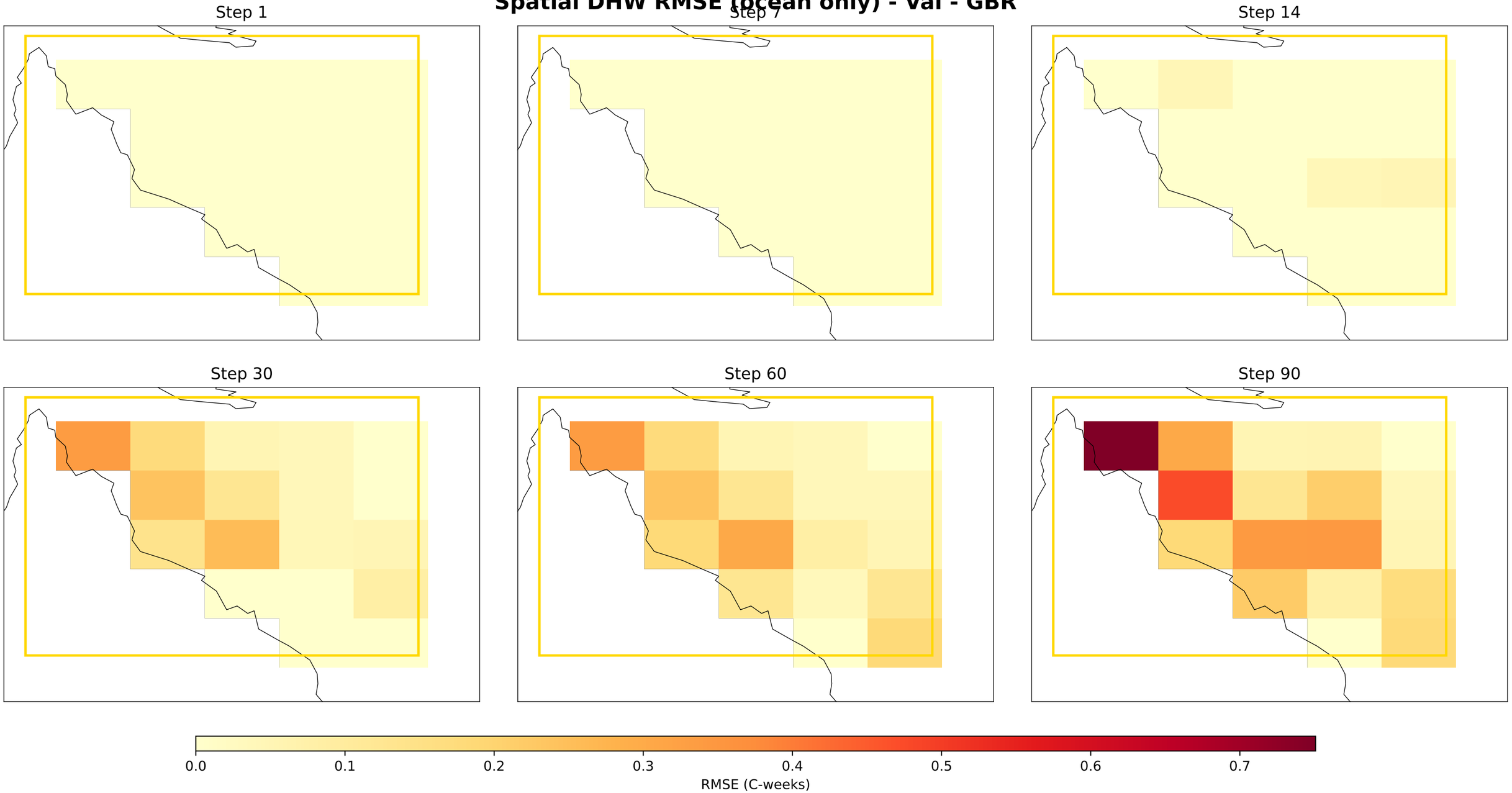


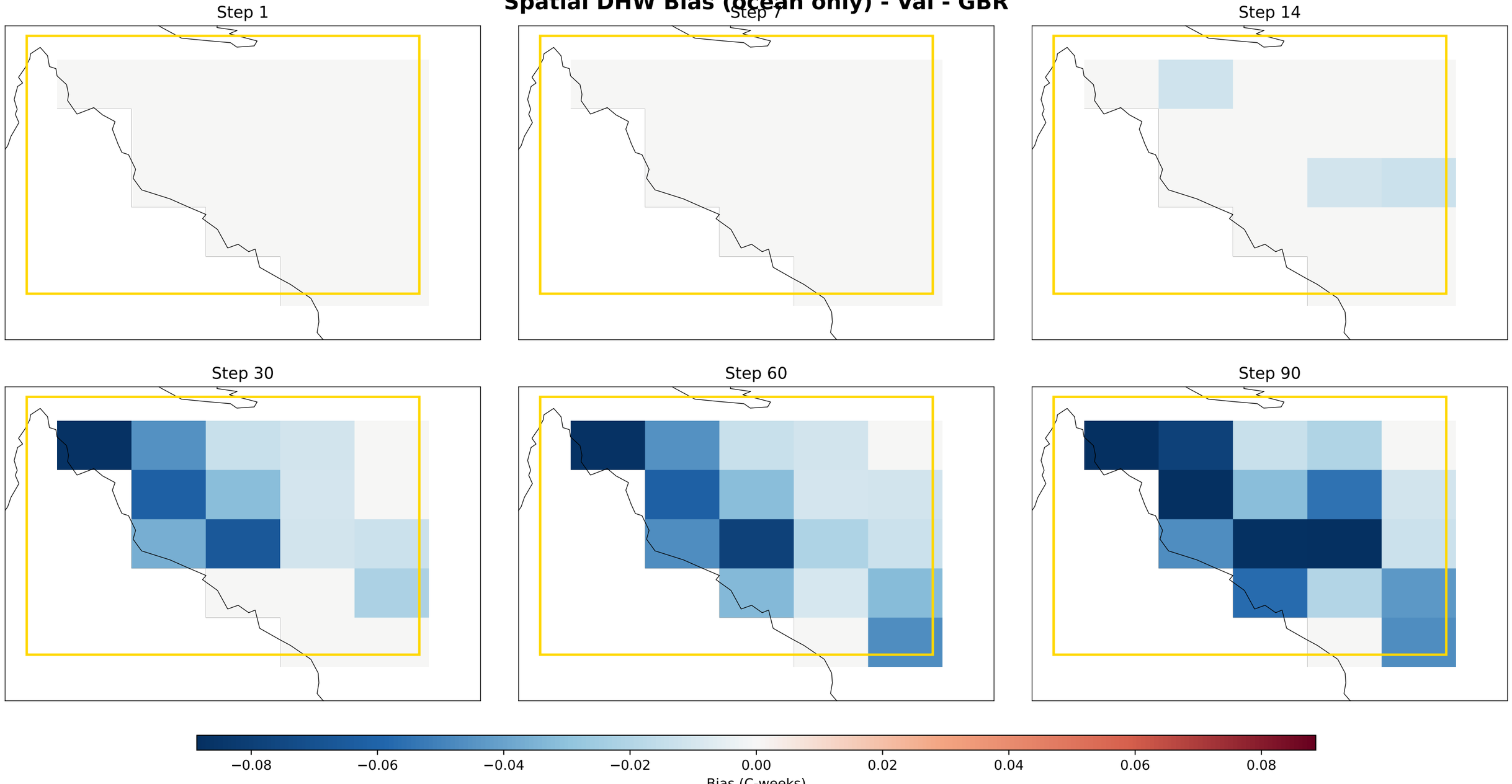





Spatial DHW RMSE (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
RMSE (C-weeks)

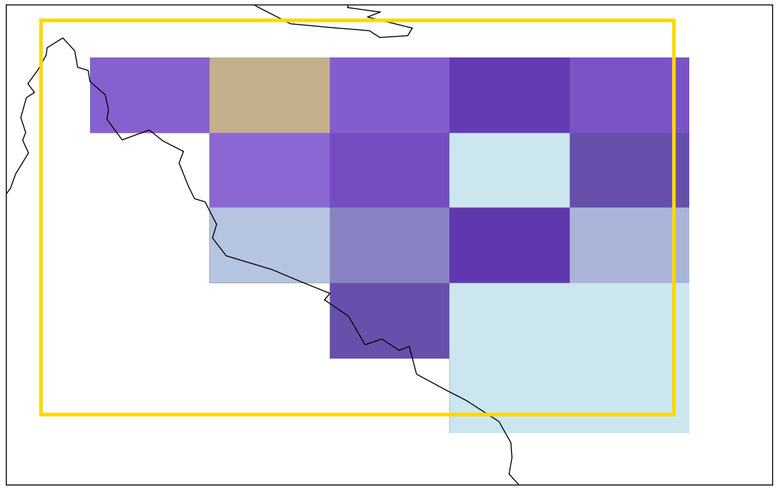

Spatial DHW Bias (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
RMSE (C-weeks)

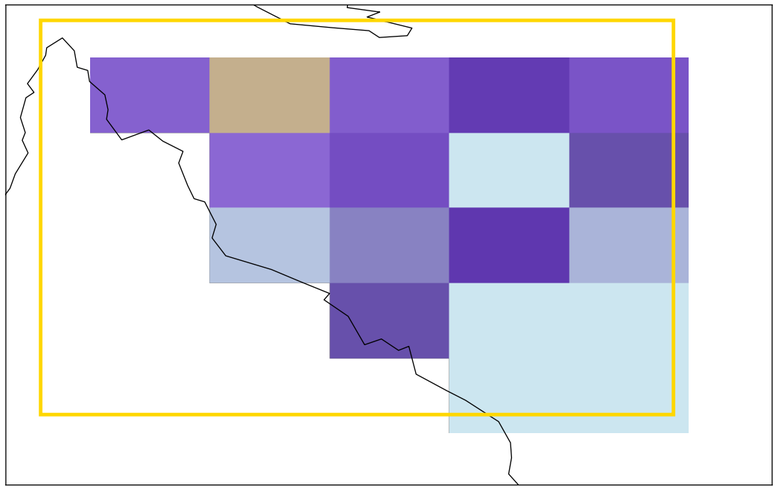

Spatial DHW Bias (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val - GBR (C-weeks)

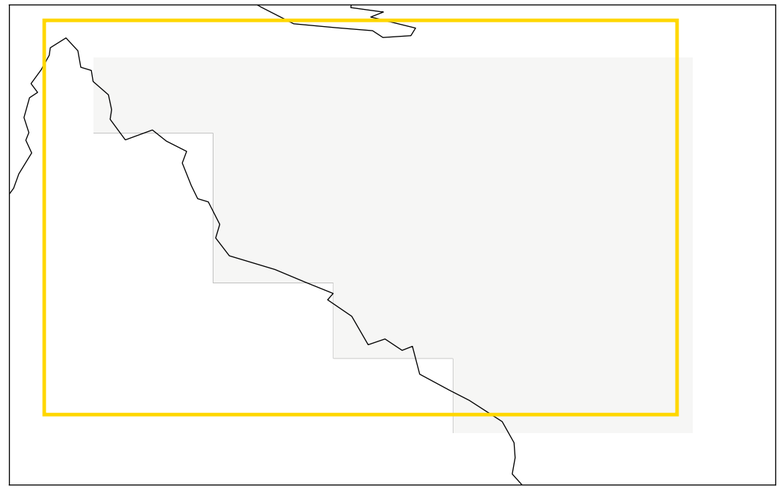
Actual DHW - Step 1



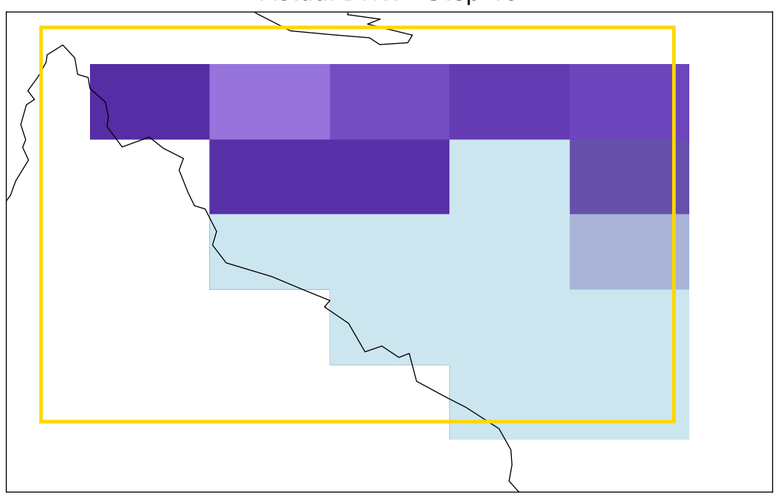
Predicted DHW - Step 1



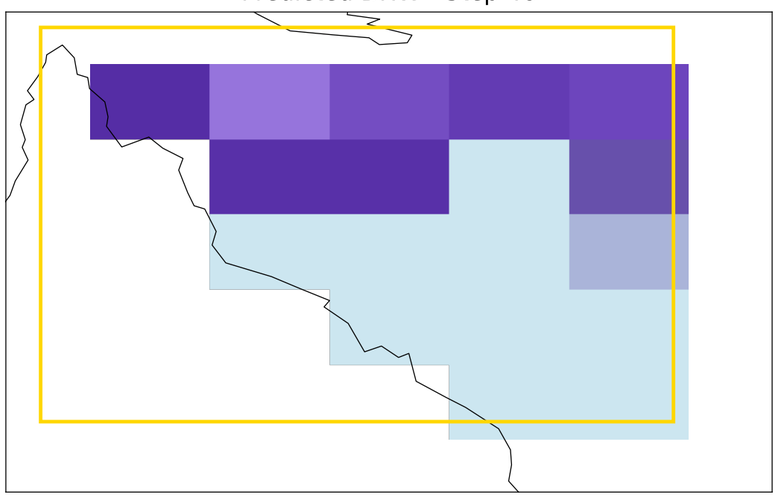
Error (Pred - Actual) - Step 1 [scale ±8]



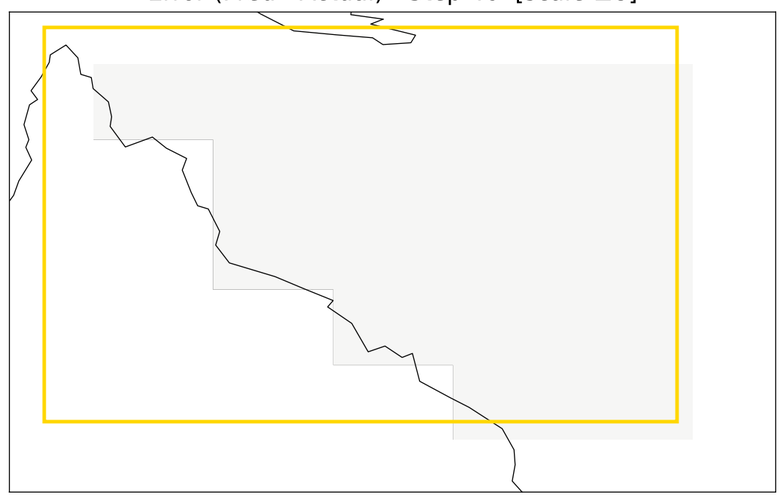
Actual DHW - Step 46



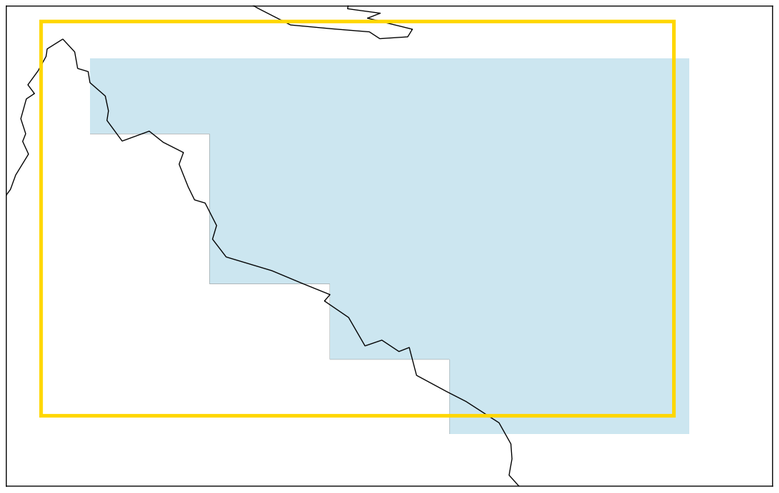
Predicted DHW - Step 46



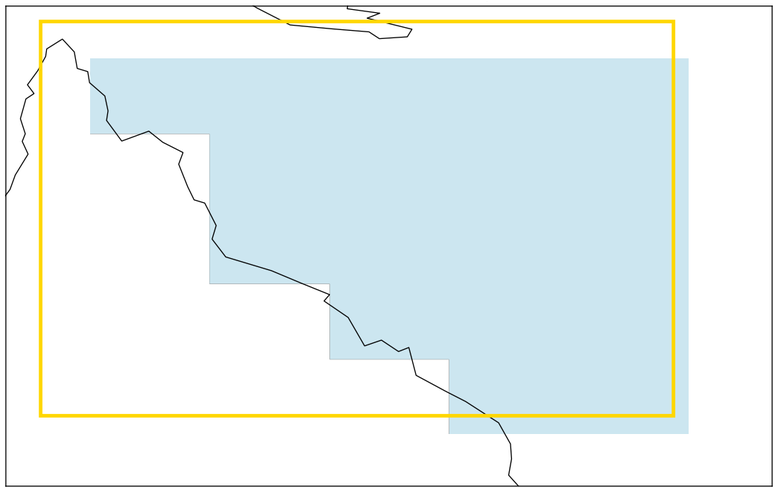
Error (Pred - Actual) - Step 46 [scale ±8]



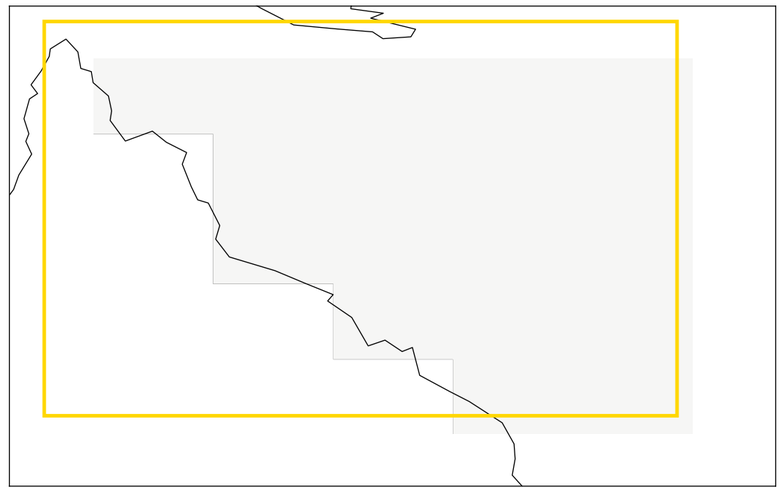
Actual DHW - Step 90

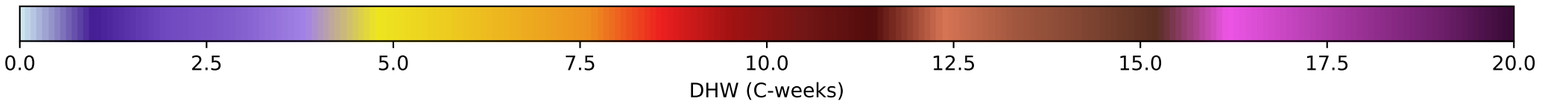


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

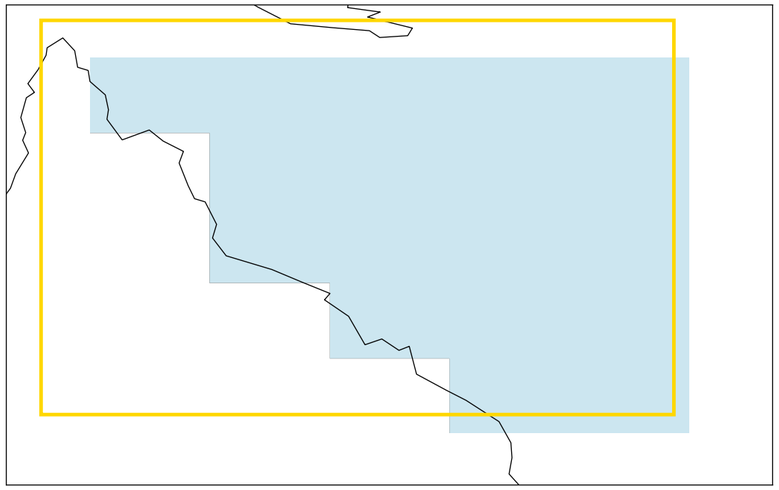




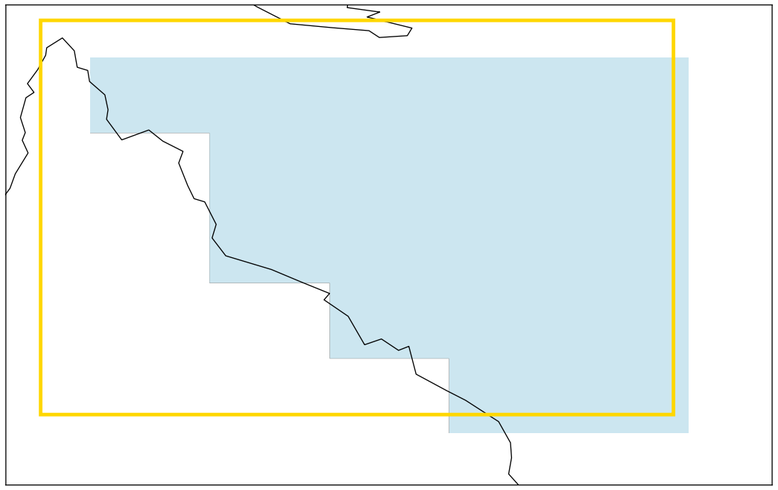


DHW Chain init: 2016-10-07 - Val - GBR (C-weeks)

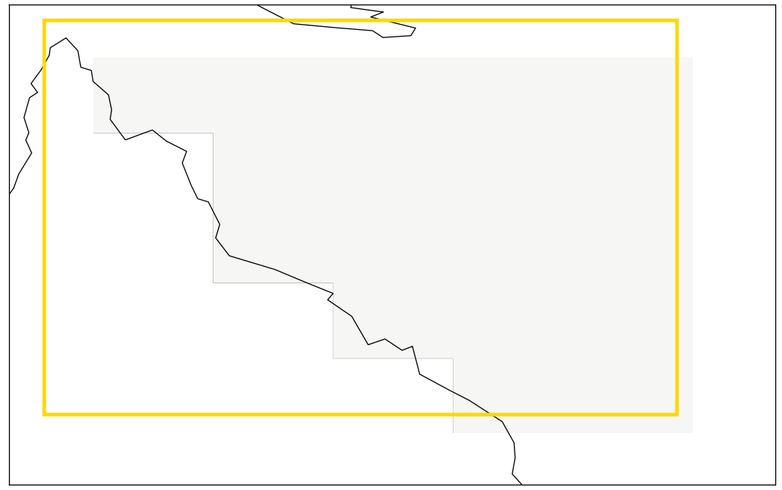
Actual DHW - Step 1



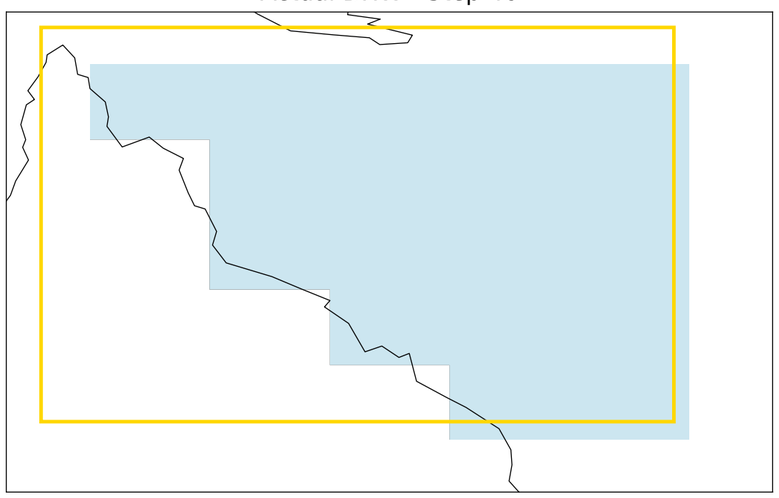
Predicted DHW - Step 1



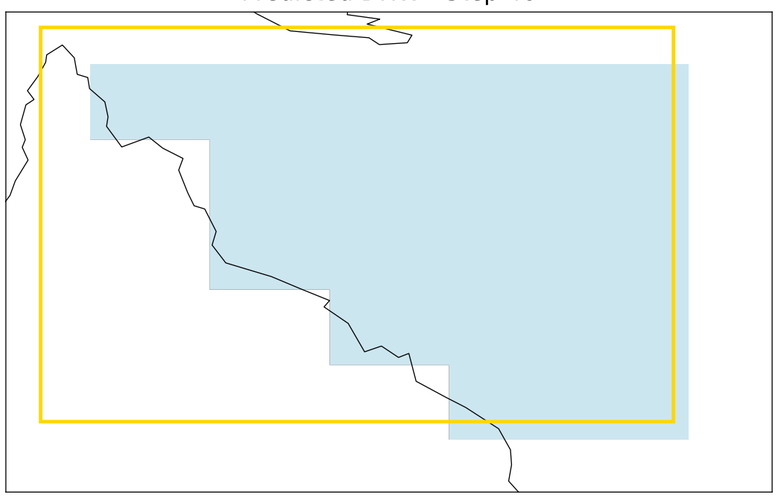
Error (Pred - Actual) - Step 1 [scale ±8]



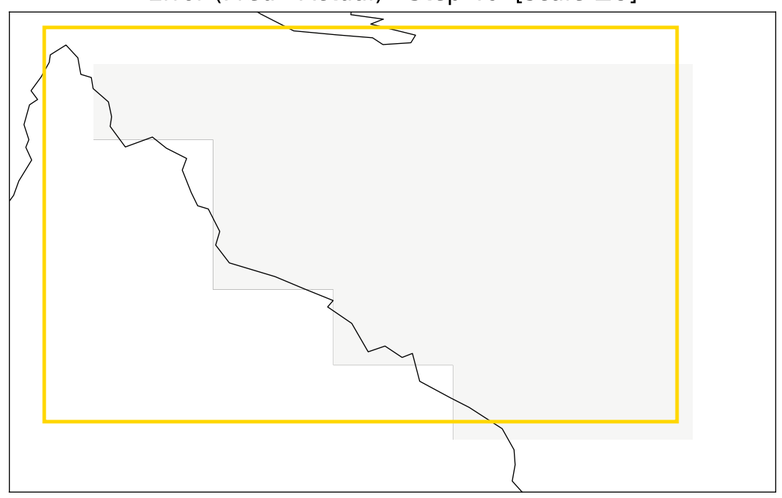
Actual DHW - Step 46



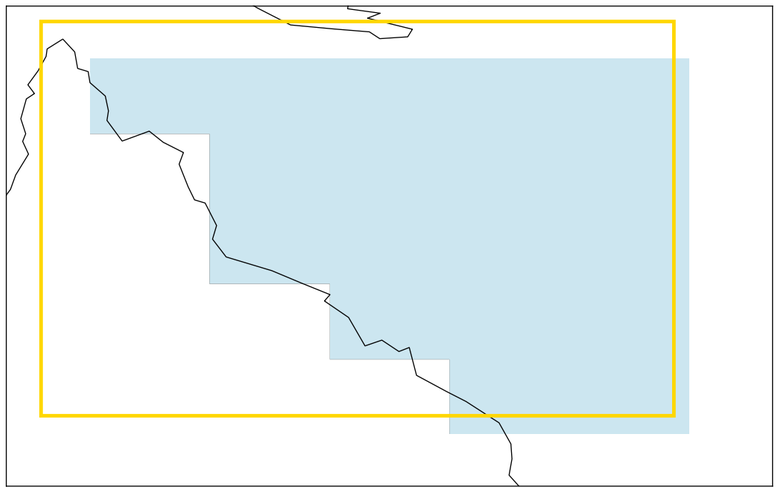
Predicted DHW - Step 46



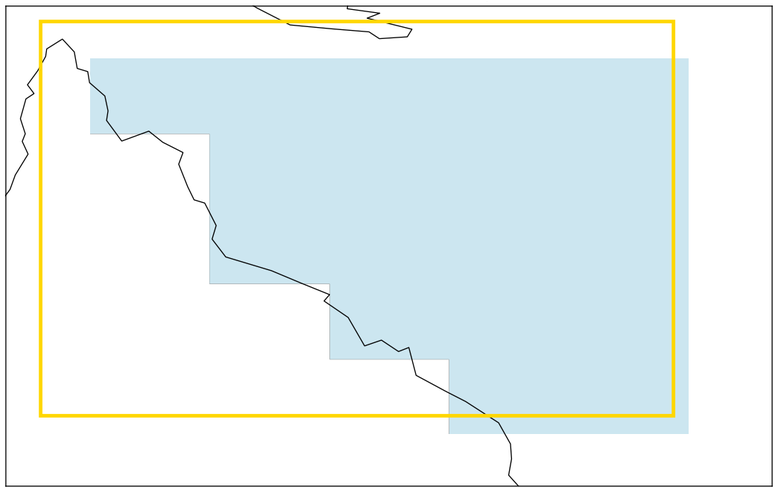
Error (Pred - Actual) - Step 46 [scale ±8]



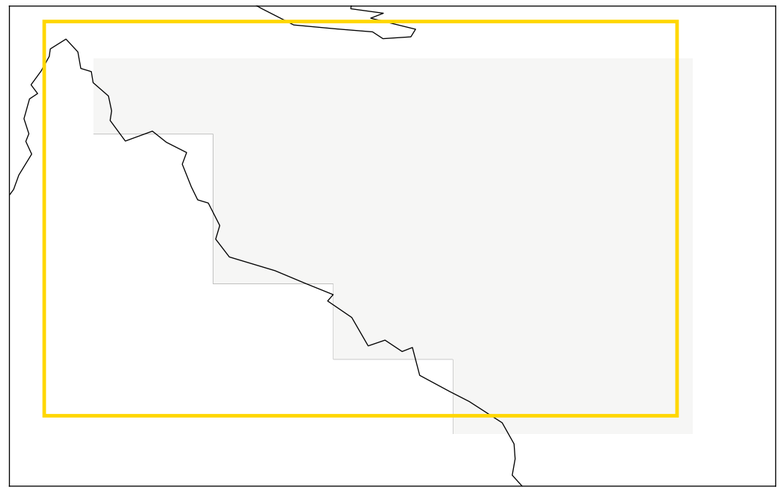
Actual DHW - Step 90

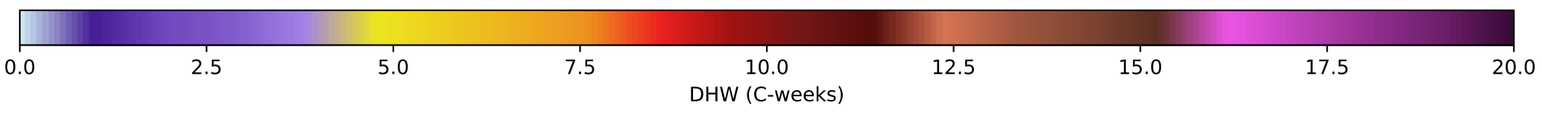


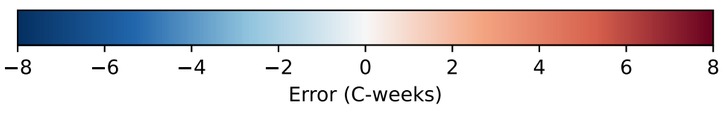
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

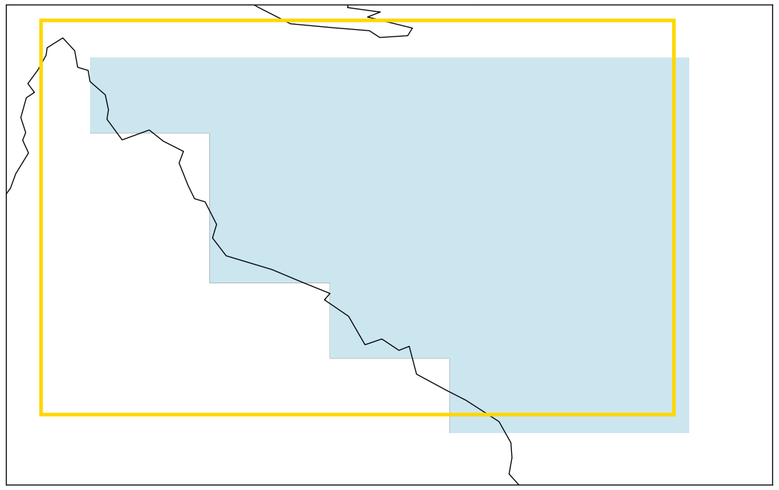




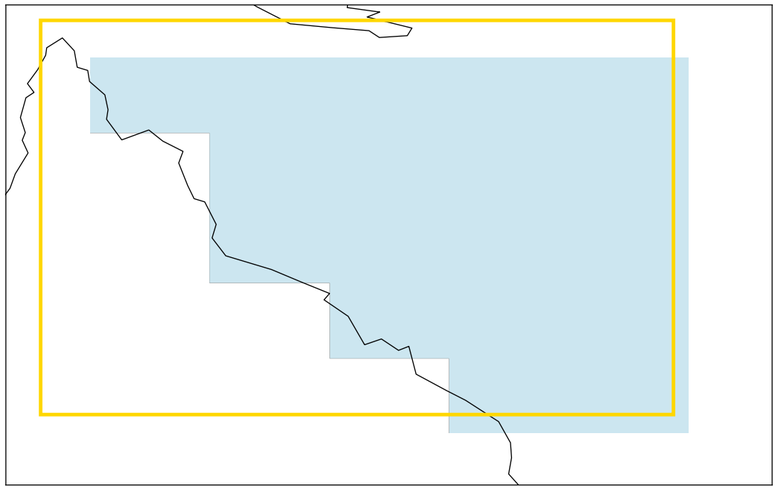


DHW Chain init: 2017-01-05 - Val - GBR (C-weeks)

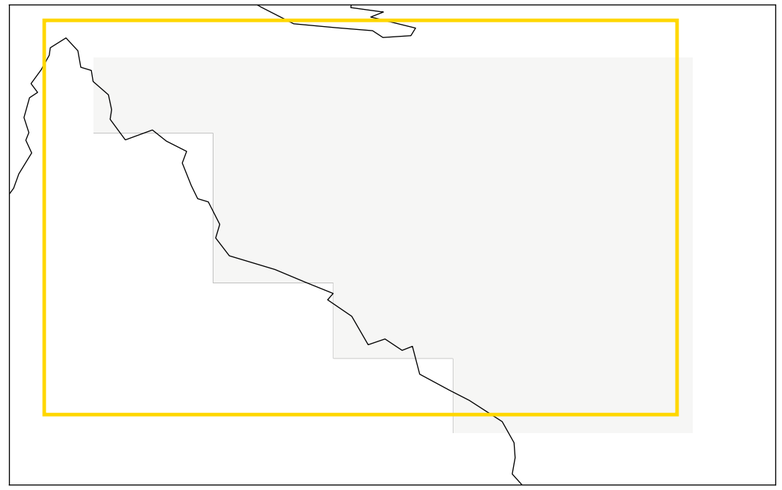
Actual DHW - Step 1



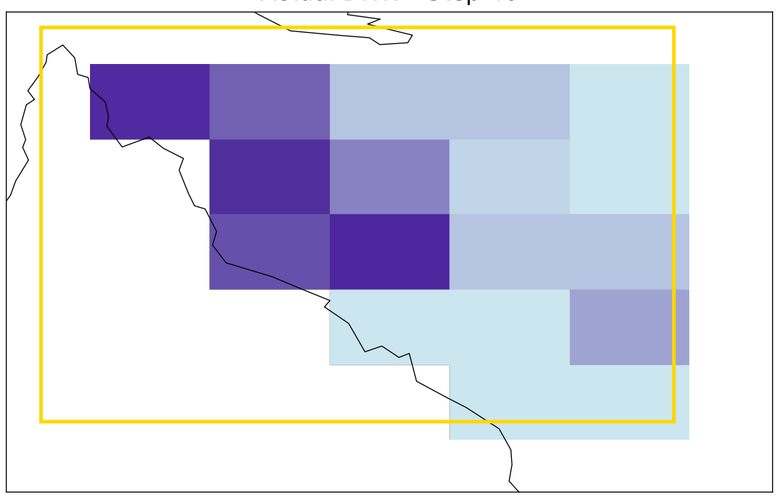
Predicted DHW - Step 1



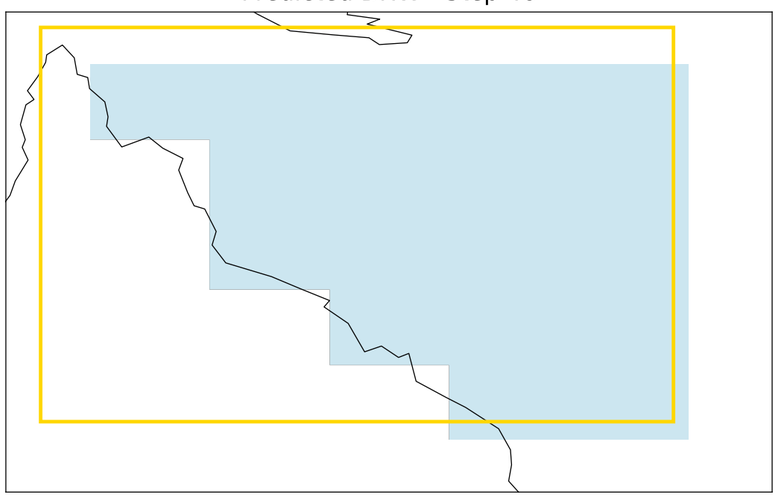
Error (Pred - Actual) - Step 1 [scale ±8]



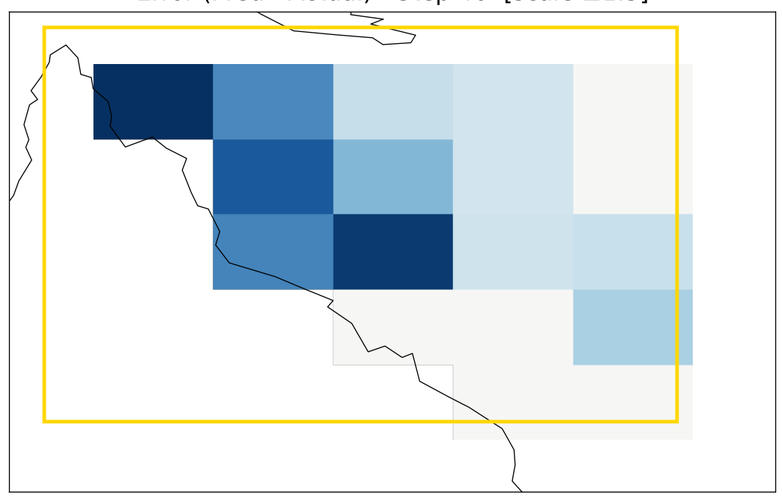
Actual DHW - Step 46



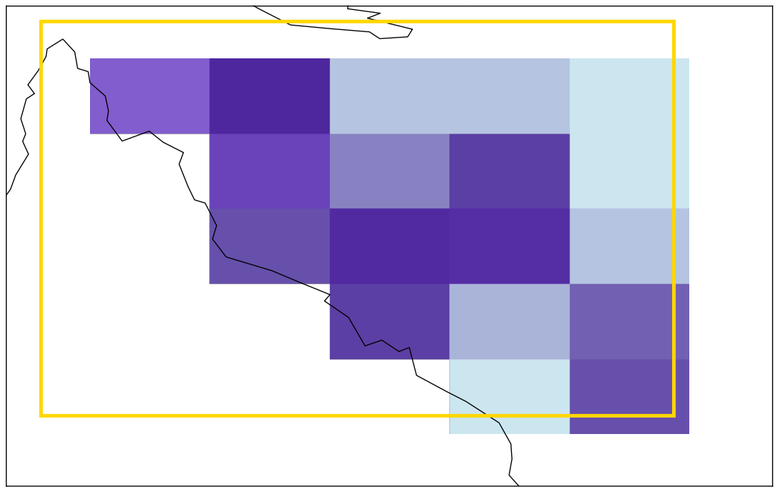
Predicted DHW - Step 46



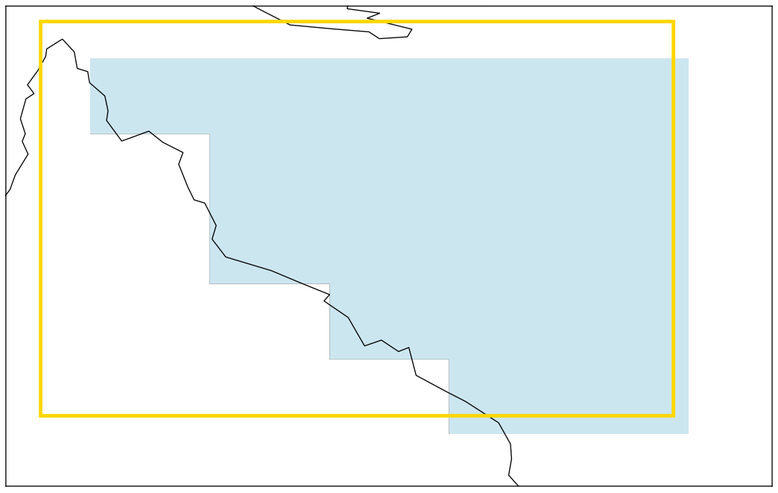
Error (Pred - Actual) - Step 46 [scale ±1.3]



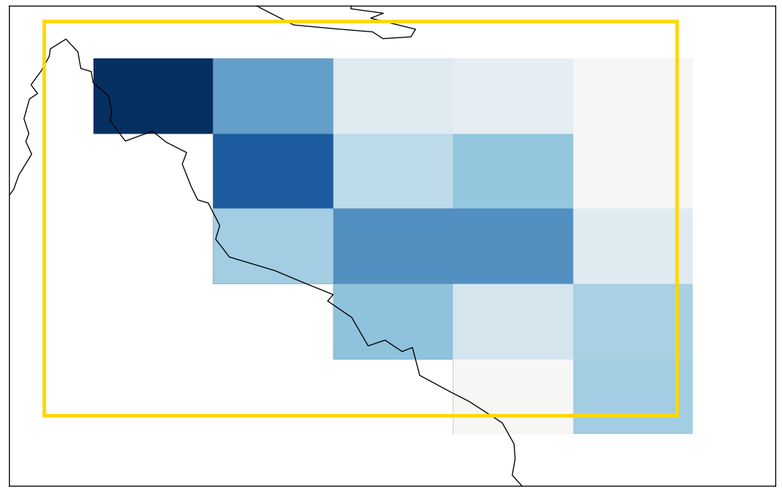
Actual DHW - Step 90

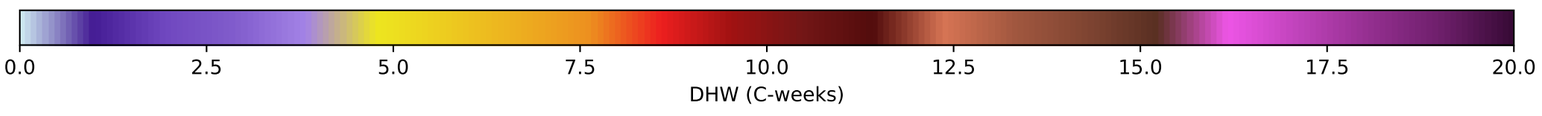


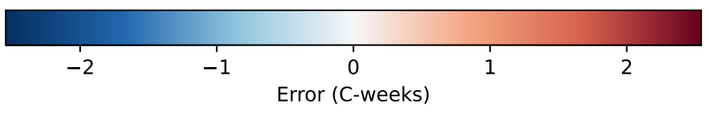
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.5]

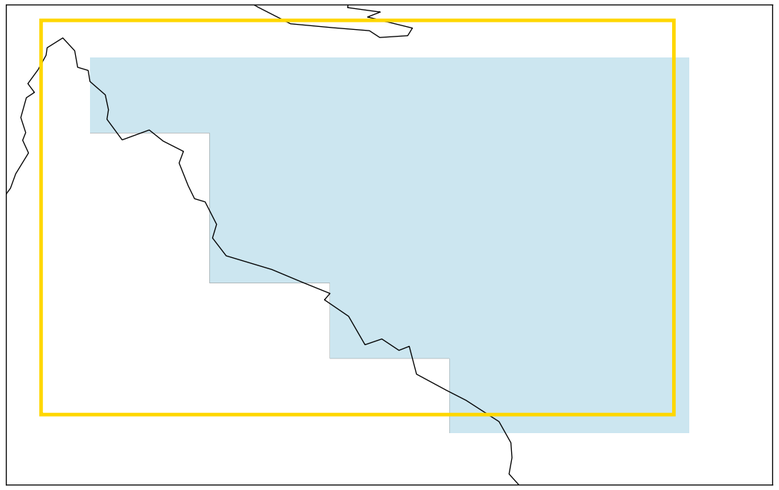




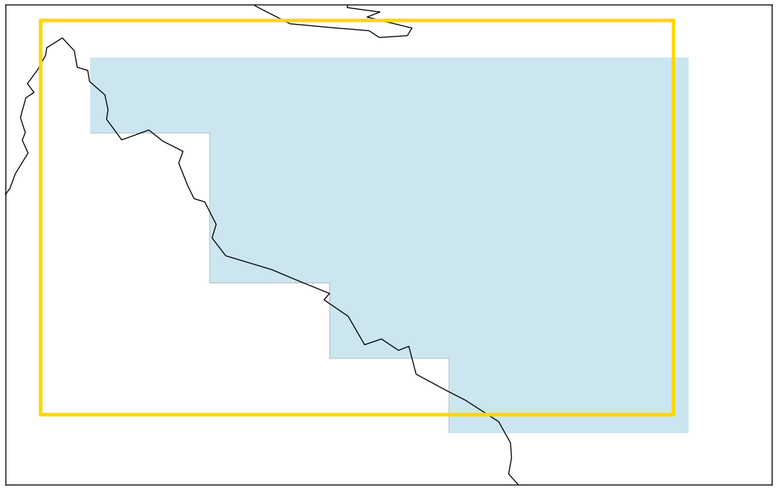


DHW Chain init: 2017-07-04 - Val - GBR (C-weeks)

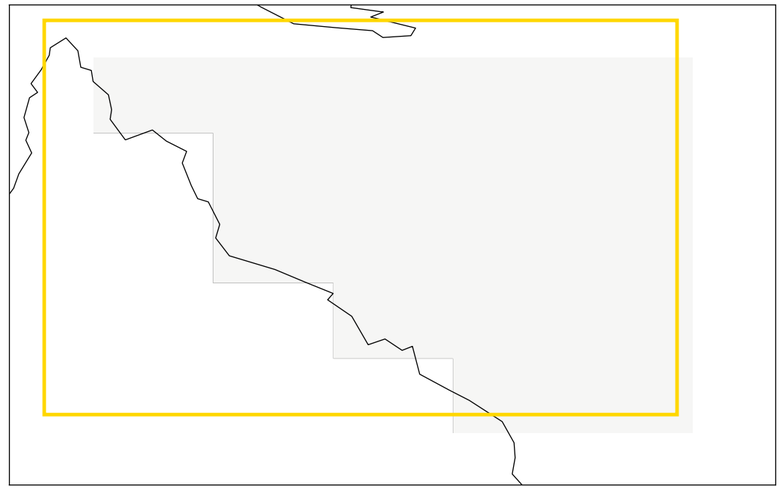
Actual DHW - Step 1



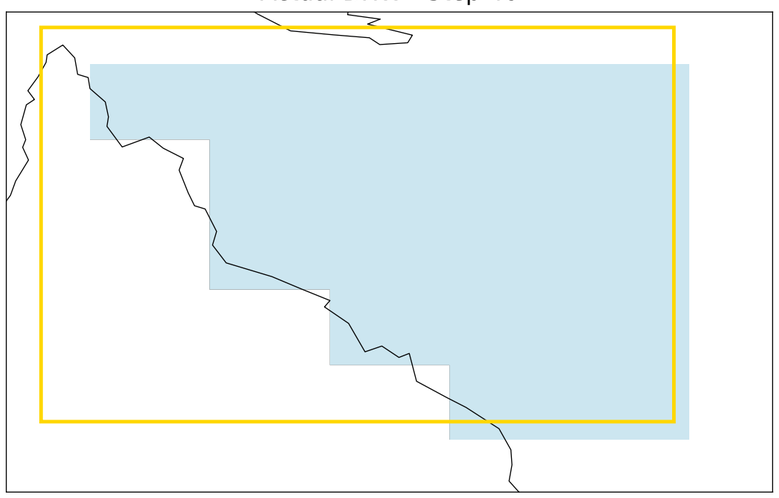
Predicted DHW - Step 1



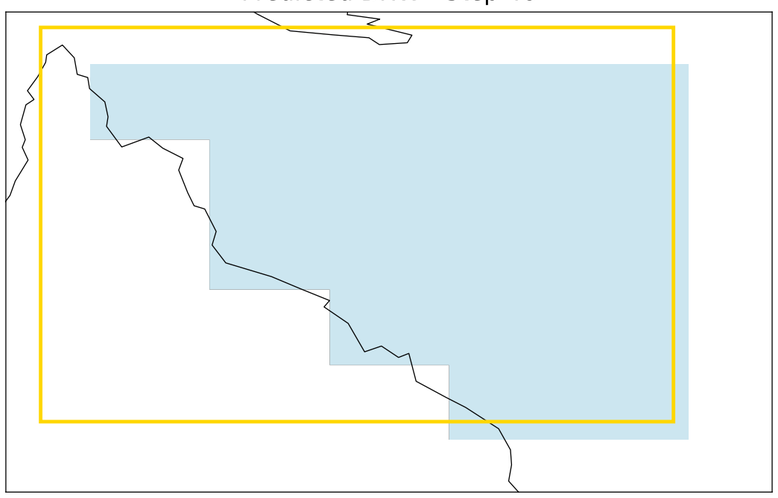
Error (Pred - Actual) - Step 1 [scale ±8]



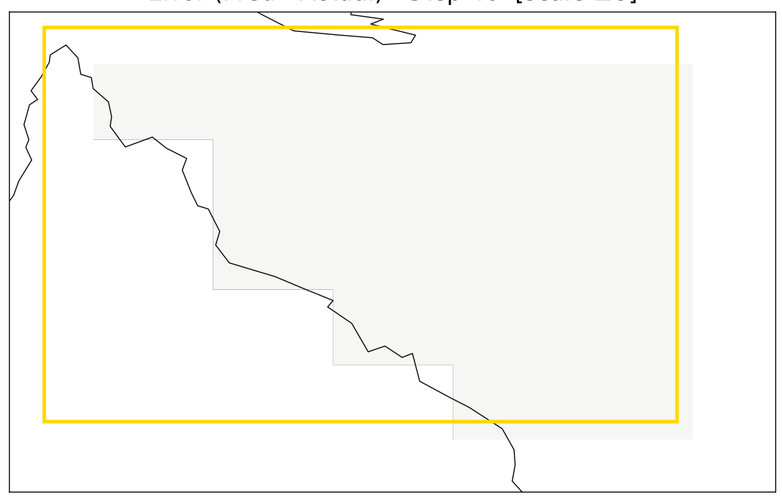
Actual DHW - Step 46



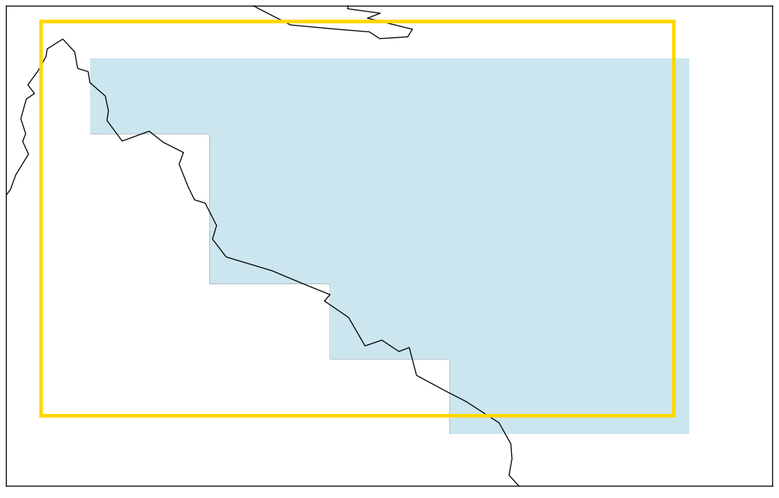
Predicted DHW - Step 46



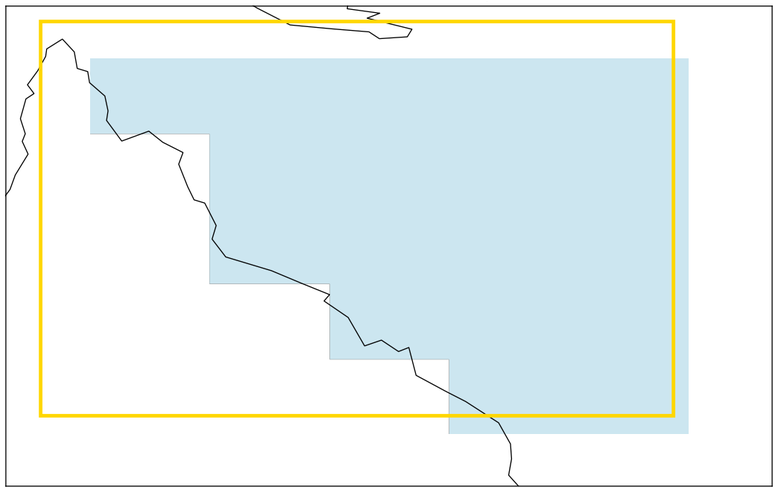
Error (Pred - Actual) - Step 46 [scale ±8]



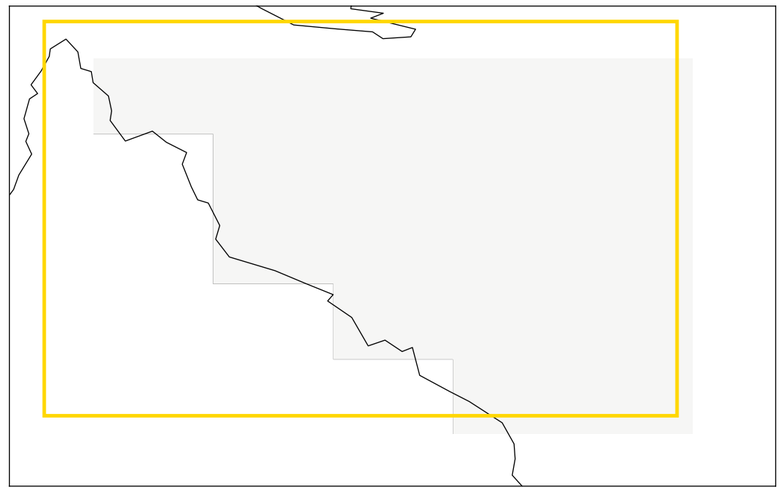
Actual DHW - Step 90

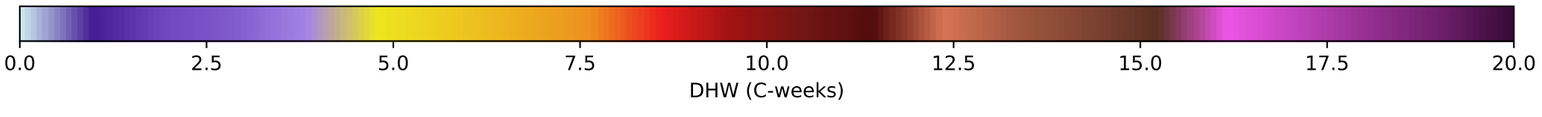


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

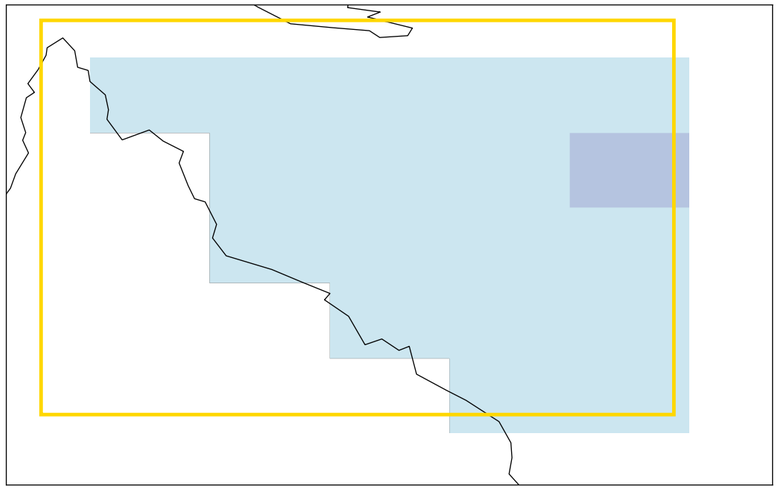




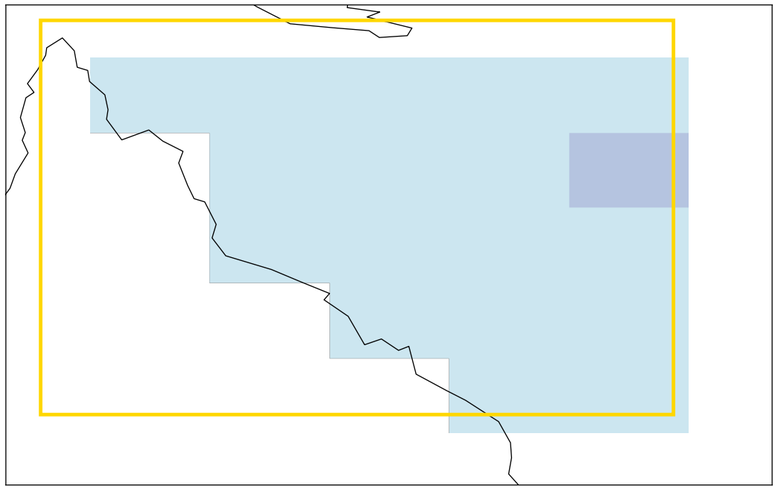


DHW Chain init: 2018-03-31 - Val - GBR (C-weeks)

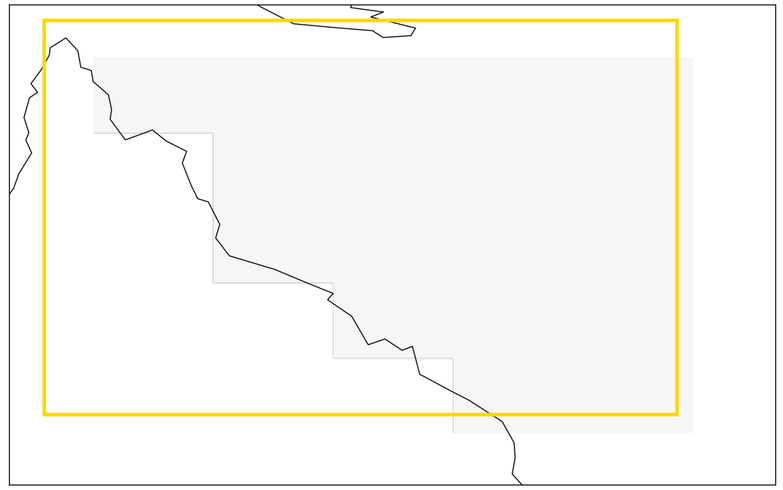
Actual DHW - Step 1



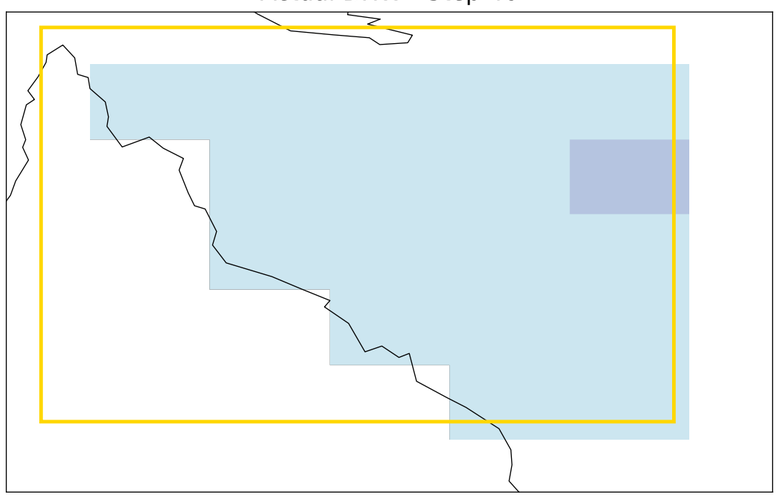
Predicted DHW - Step 1



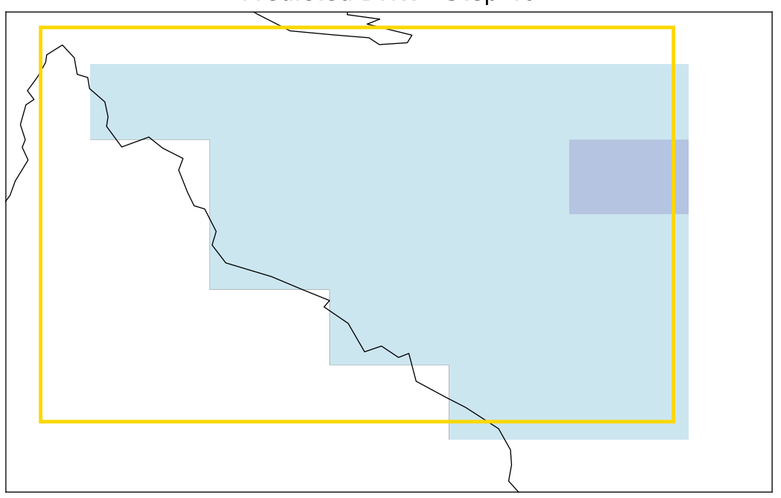
Error (Pred - Actual) - Step 1 [scale ±8]



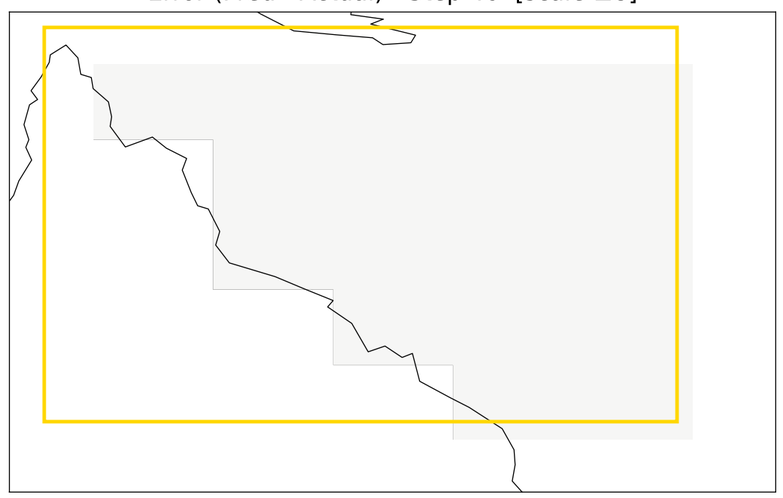
Actual DHW - Step 46



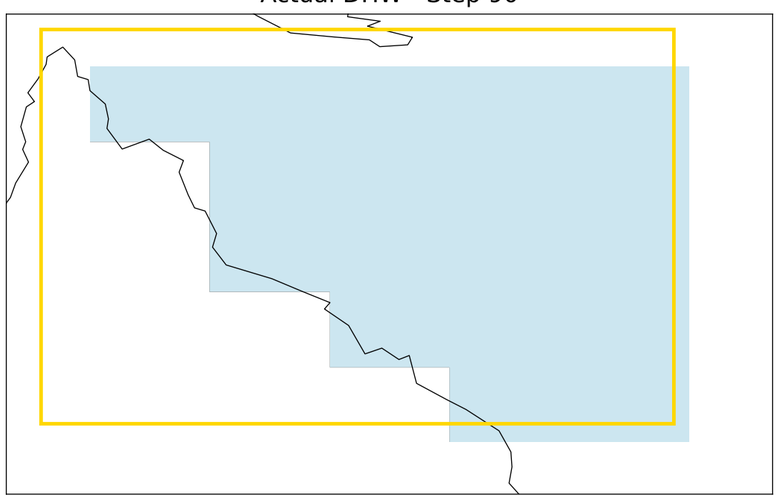
Predicted DHW - Step 46



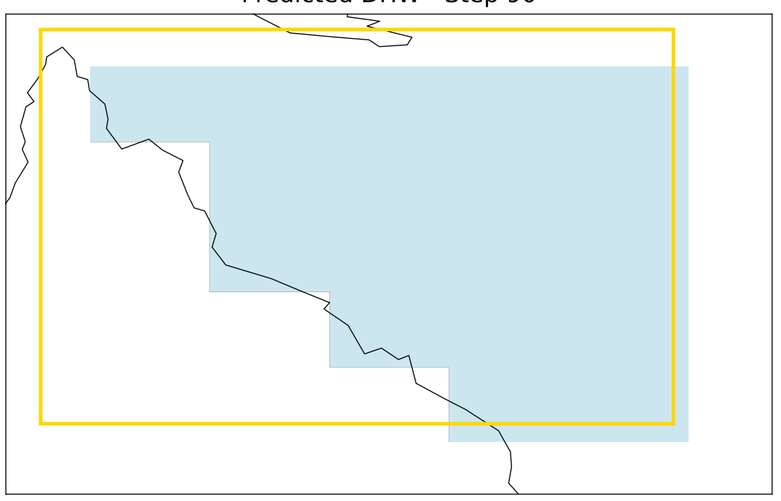
Error (Pred - Actual) - Step 46 [scale ±8]



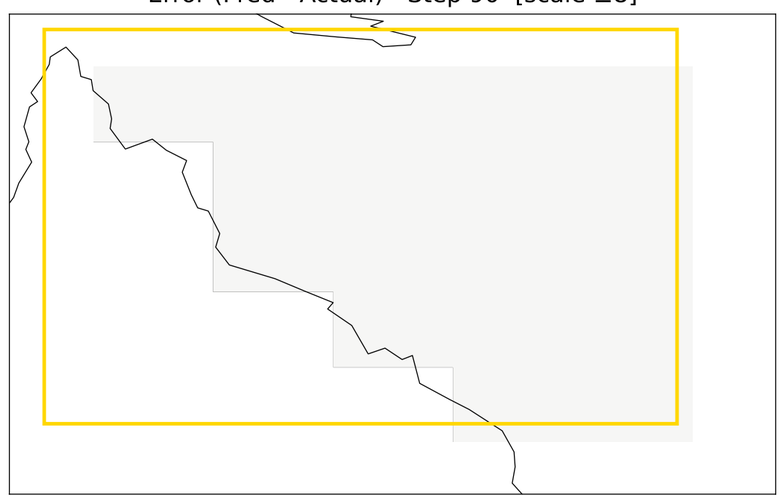
Actual DHW - Step 90

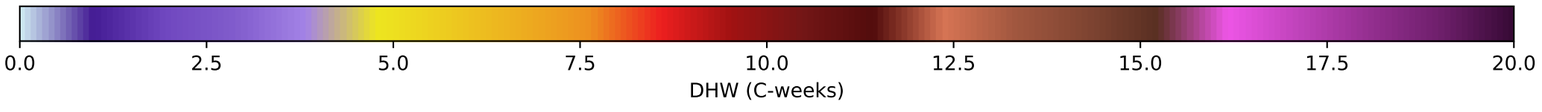


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

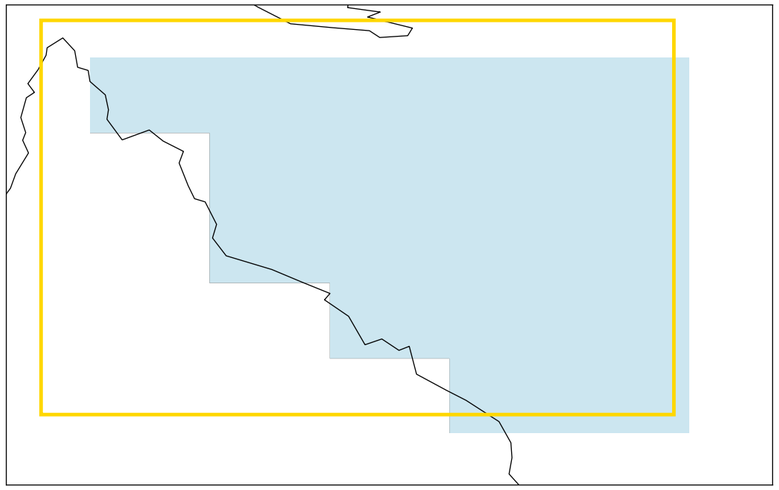




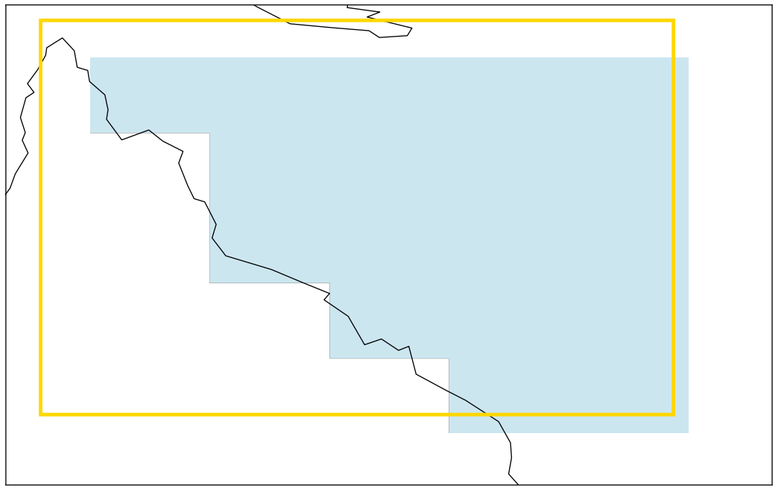


DHW Chain init: 2018-06-29 - Val - GBR (C-weeks)

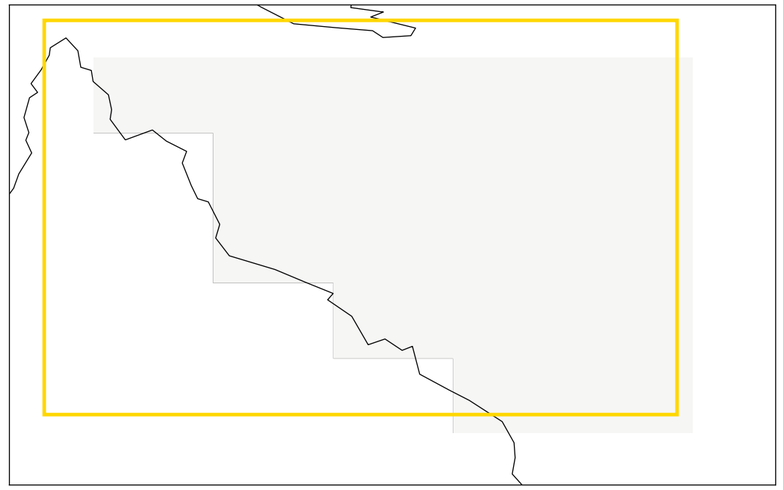
Actual DHW - Step 1



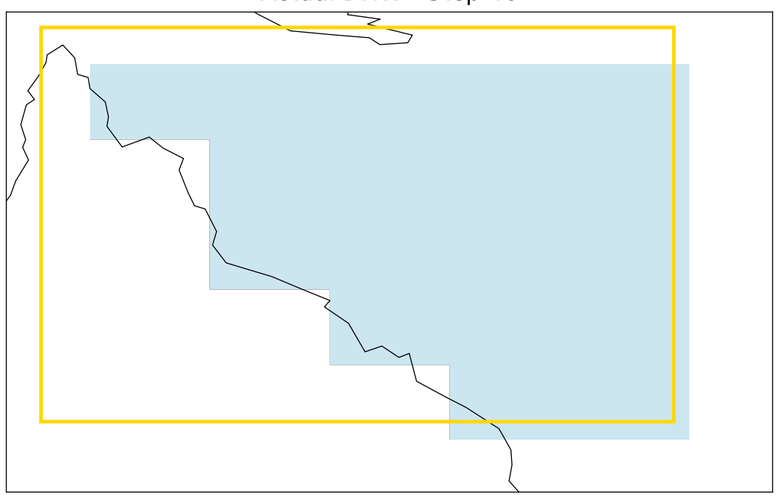
Predicted DHW - Step 1



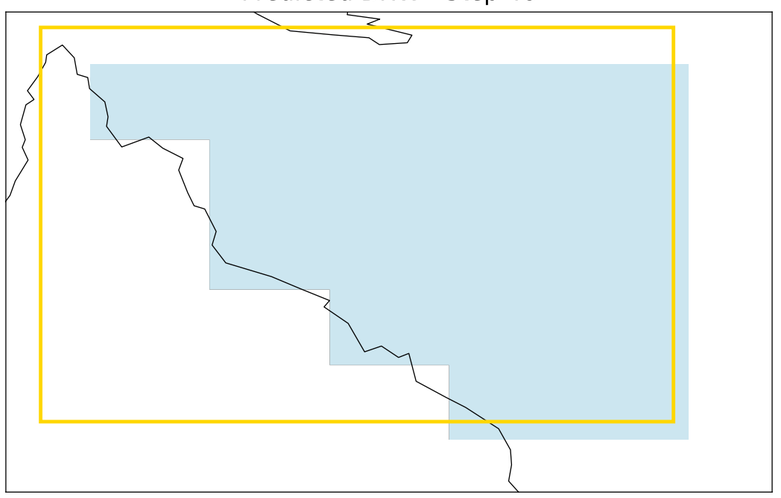
Error (Pred - Actual) - Step 1 [scale ±8]



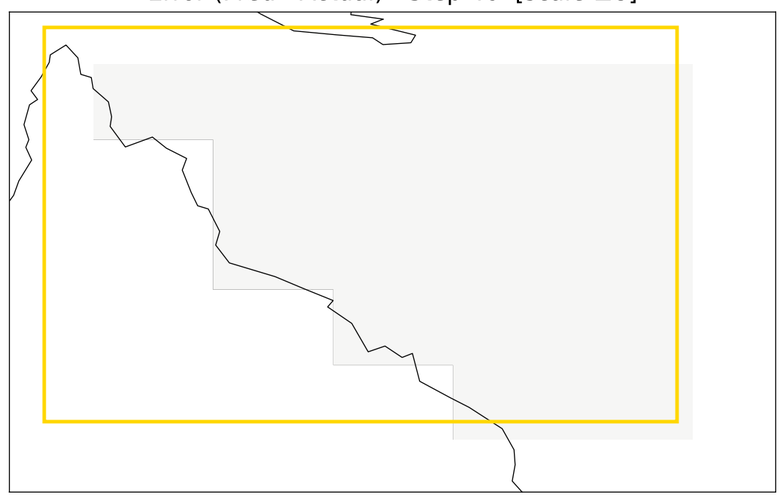
Actual DHW - Step 46



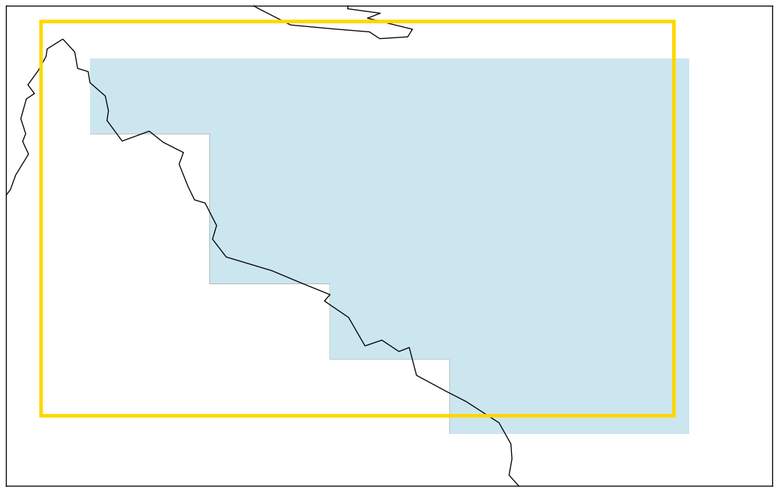
Predicted DHW - Step 46



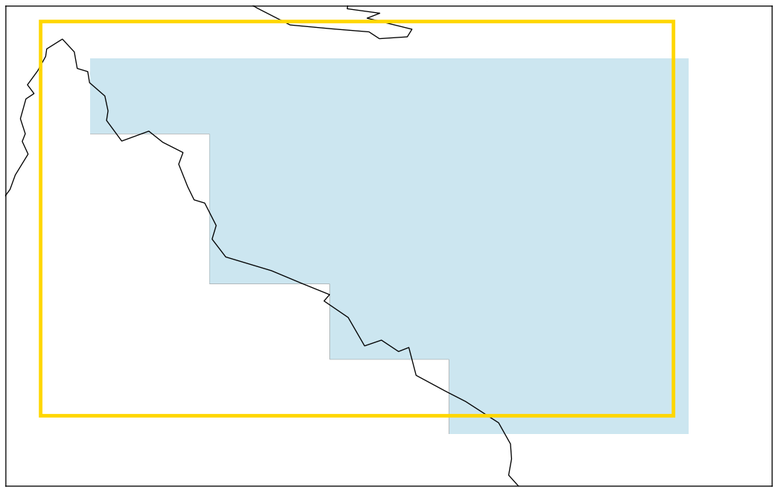
Error (Pred - Actual) - Step 46 [scale ±8]



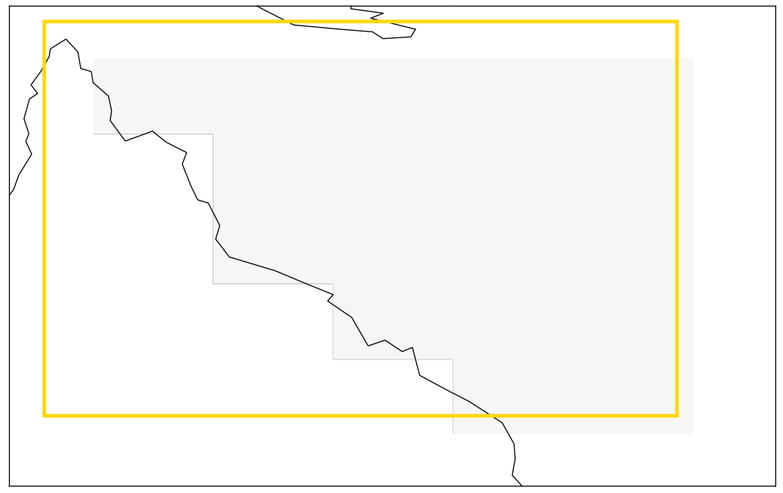
Actual DHW - Step 90

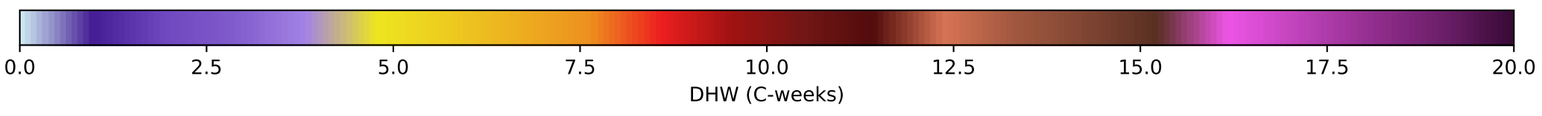


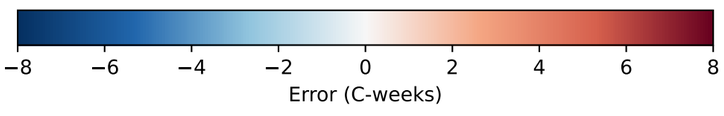
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

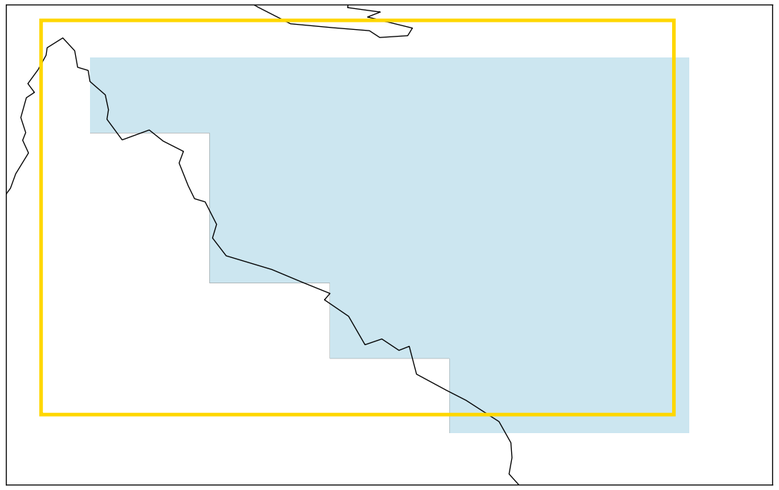




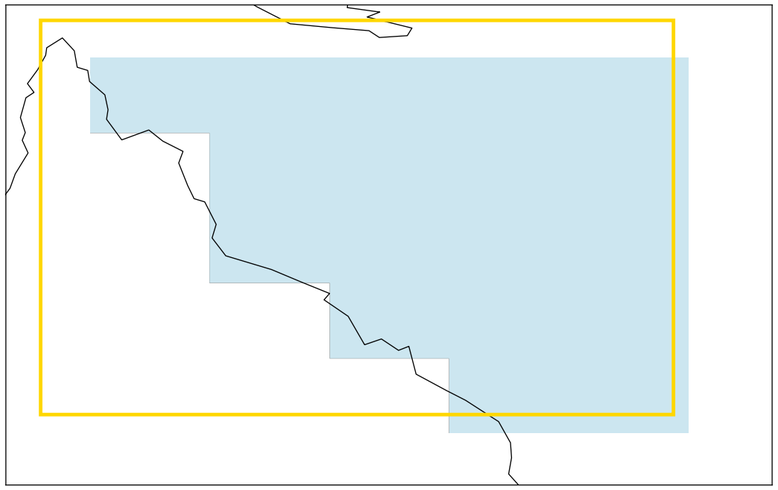


DHW Chain init: 2019-09-22 - Val - GBR (C-weeks)

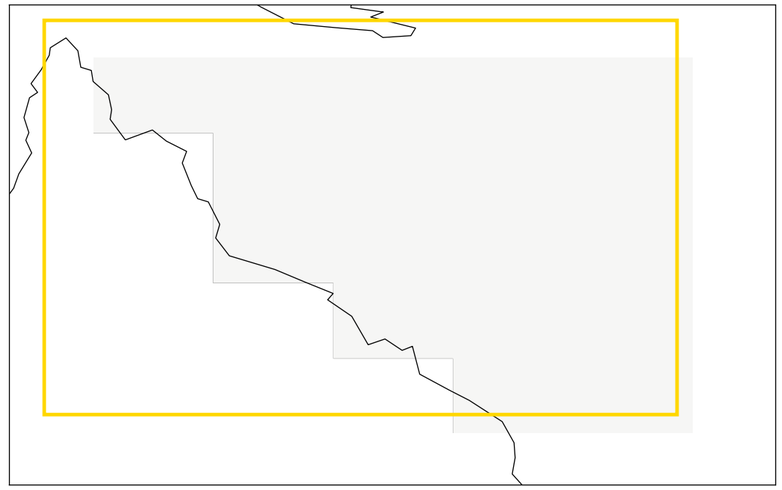
Actual DHW - Step 1



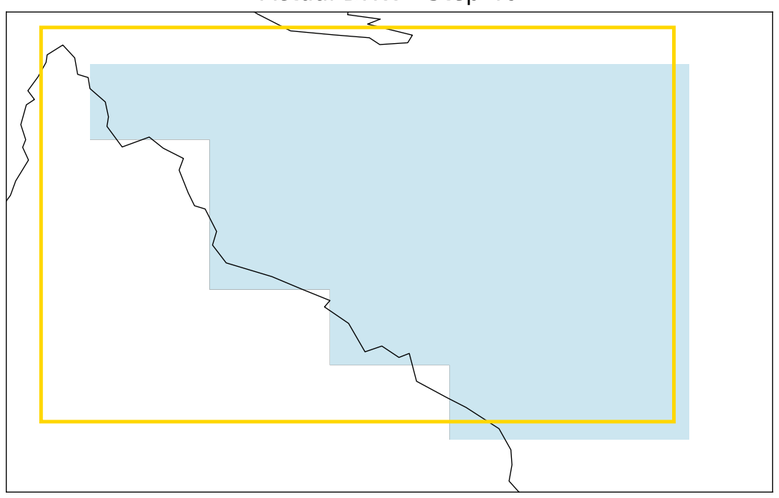
Predicted DHW - Step 1



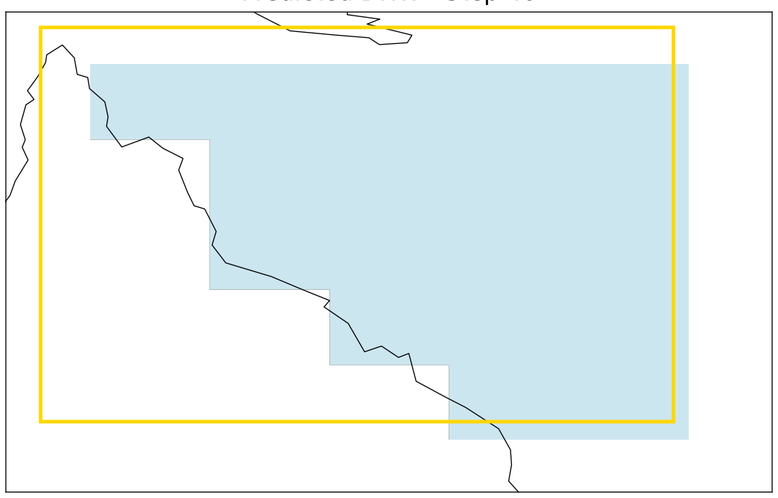
Error (Pred - Actual) - Step 1 [scale ±8]



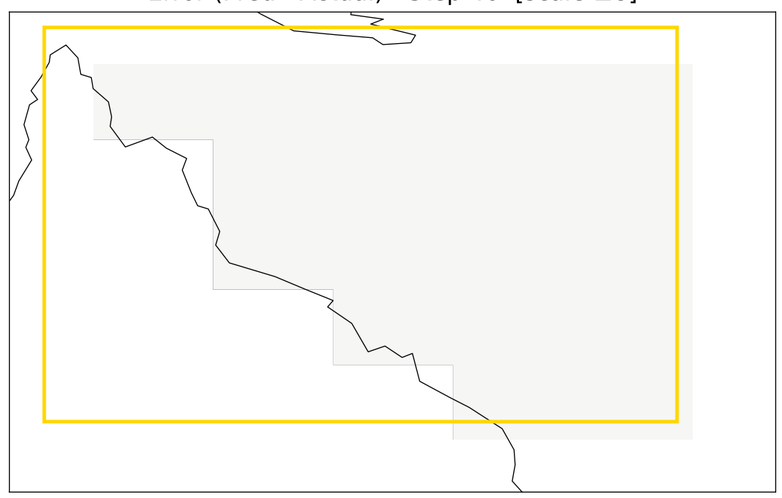
Actual DHW - Step 46



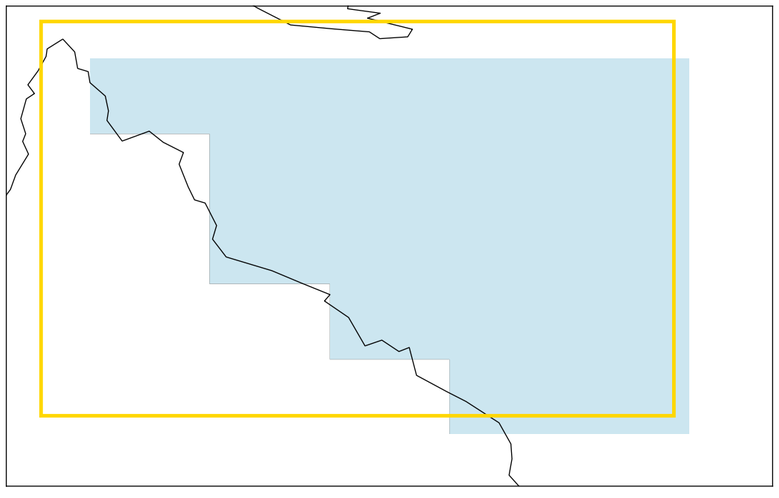
Predicted DHW - Step 46



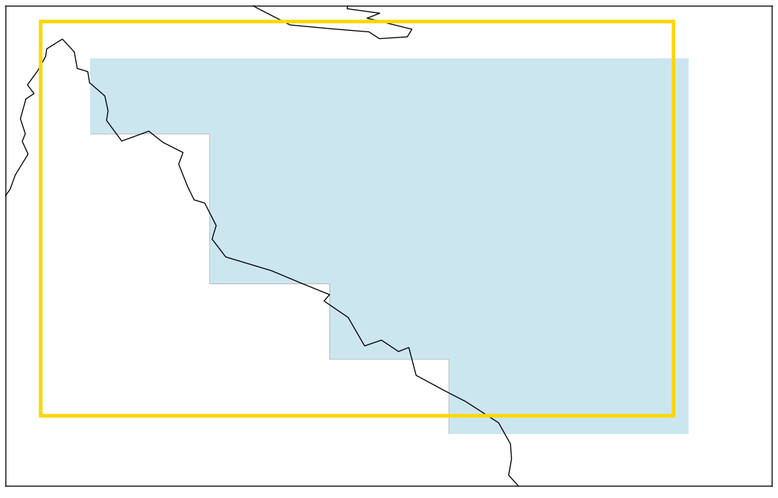
Error (Pred - Actual) - Step 46 [scale ±8]



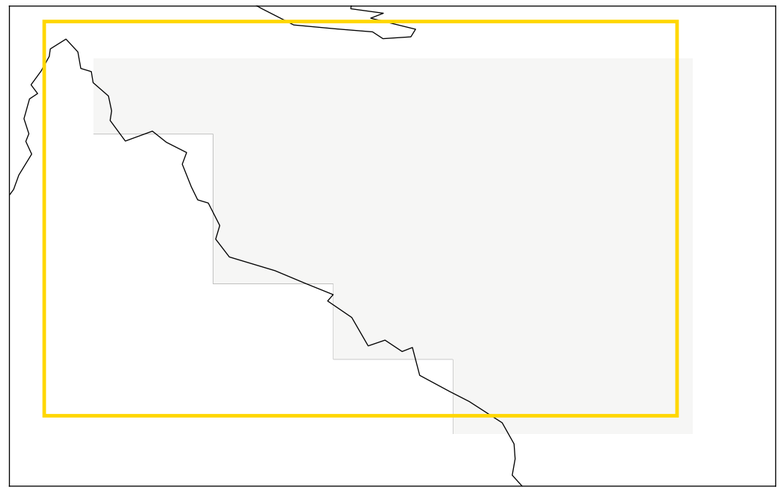
Actual DHW - Step 90

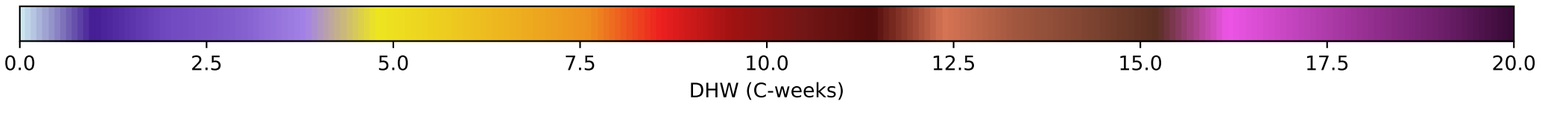


Predicted DHW - Step 90



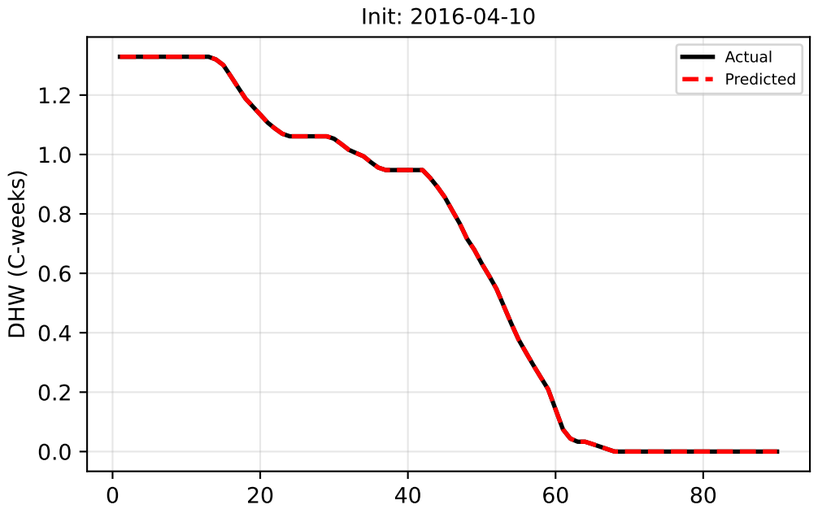
Error (Pred - Actual) - Step 90 [scale ±8]

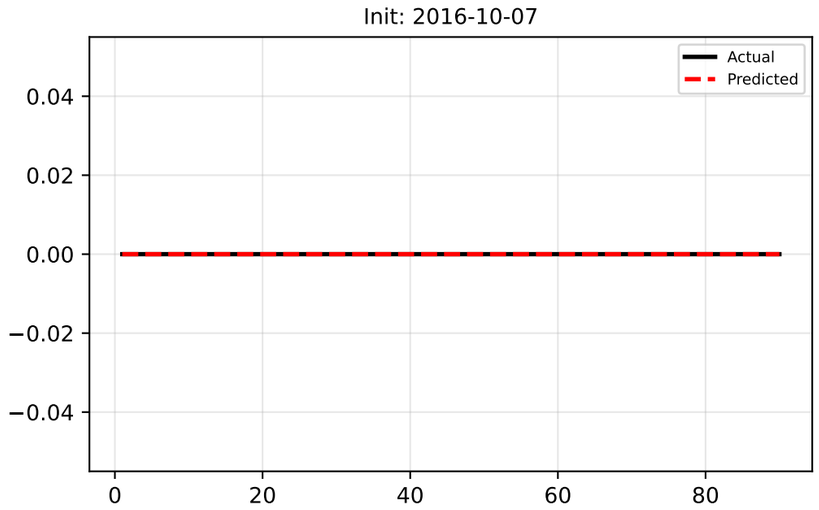


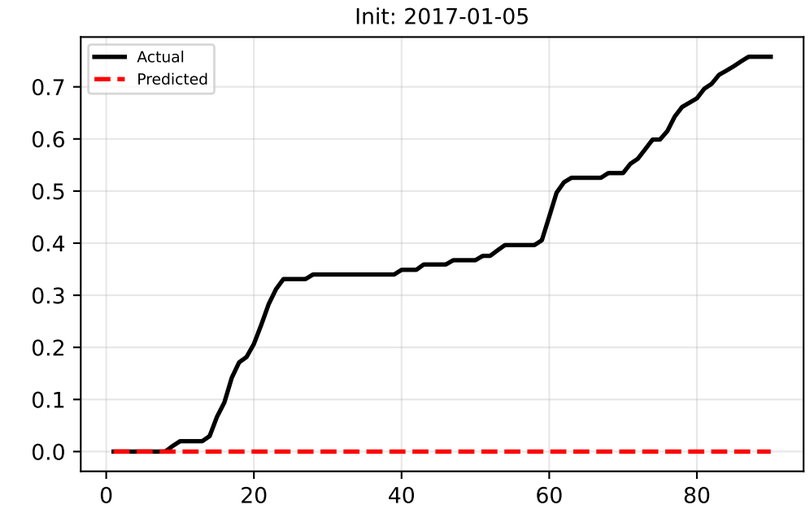


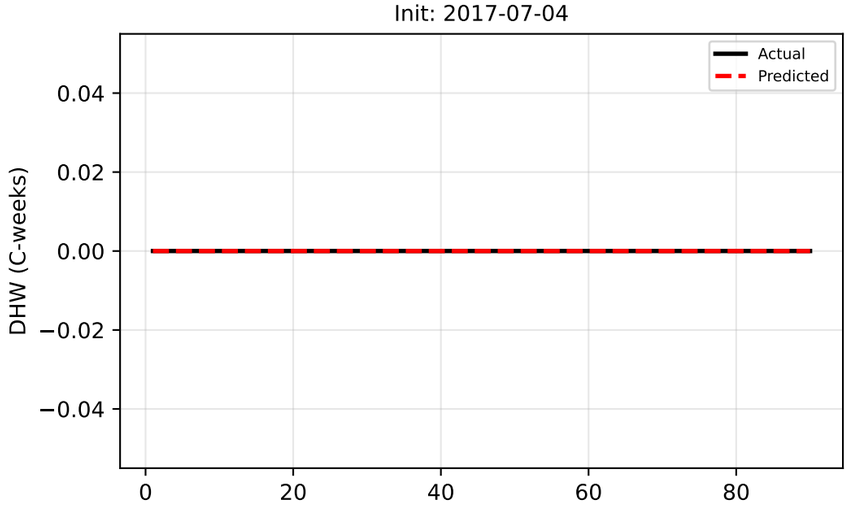


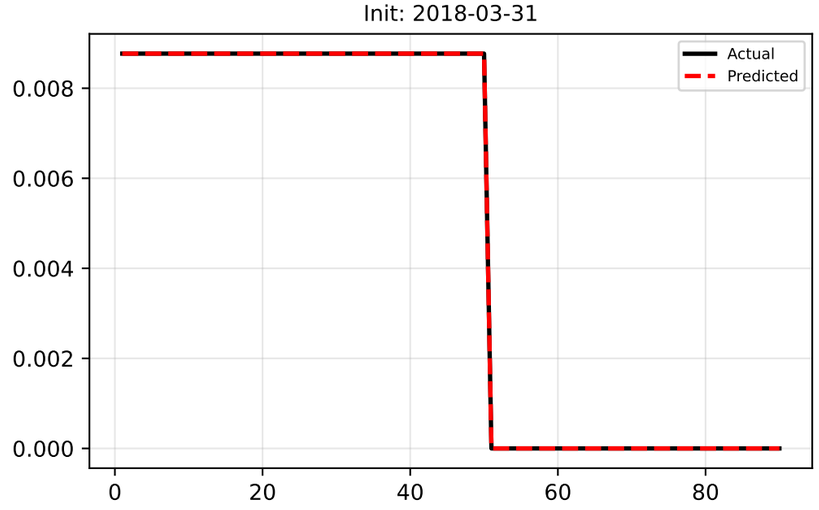
Global-Mean DHW - Val - GBR (C-weeks)

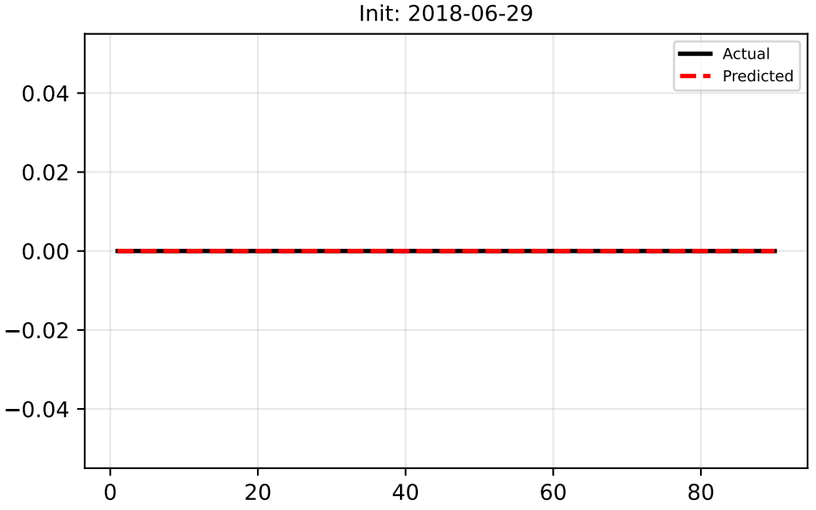


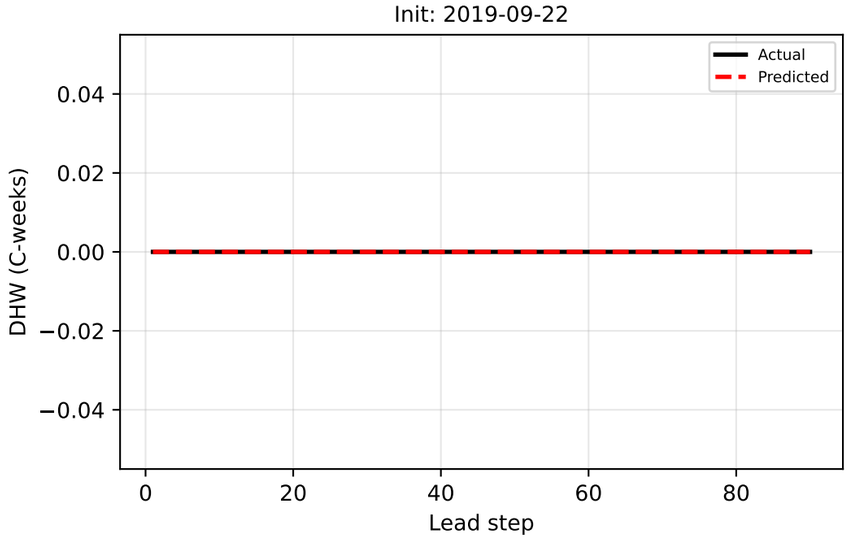


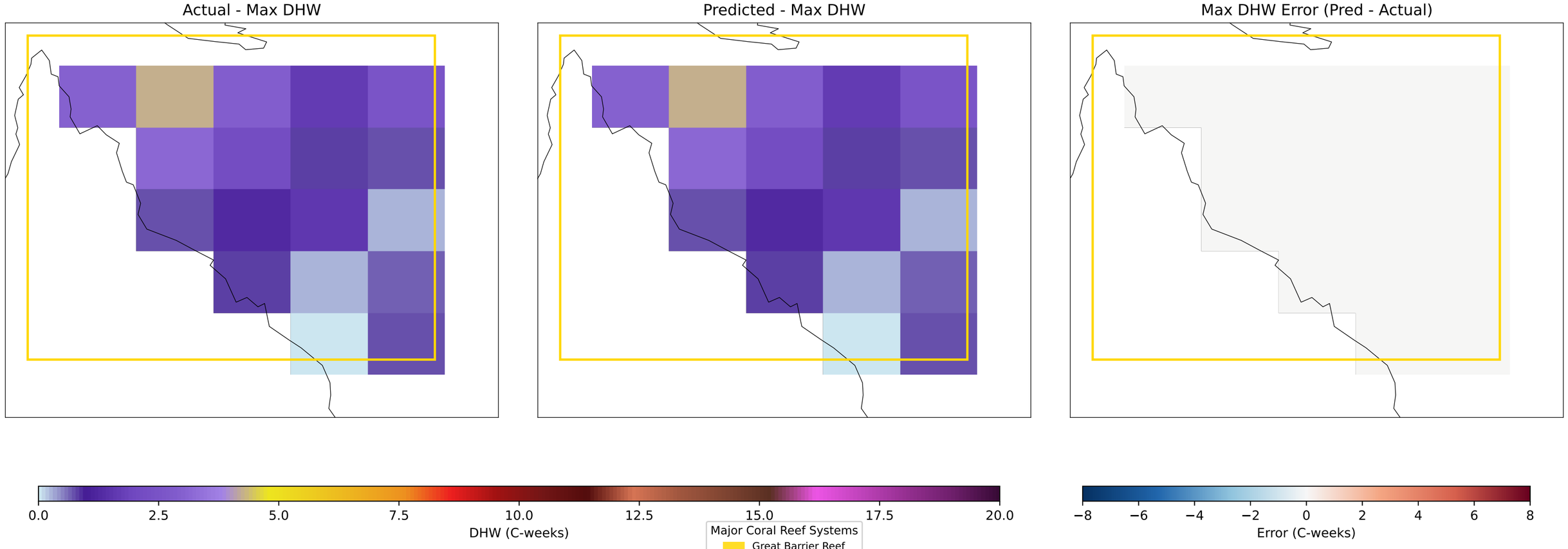


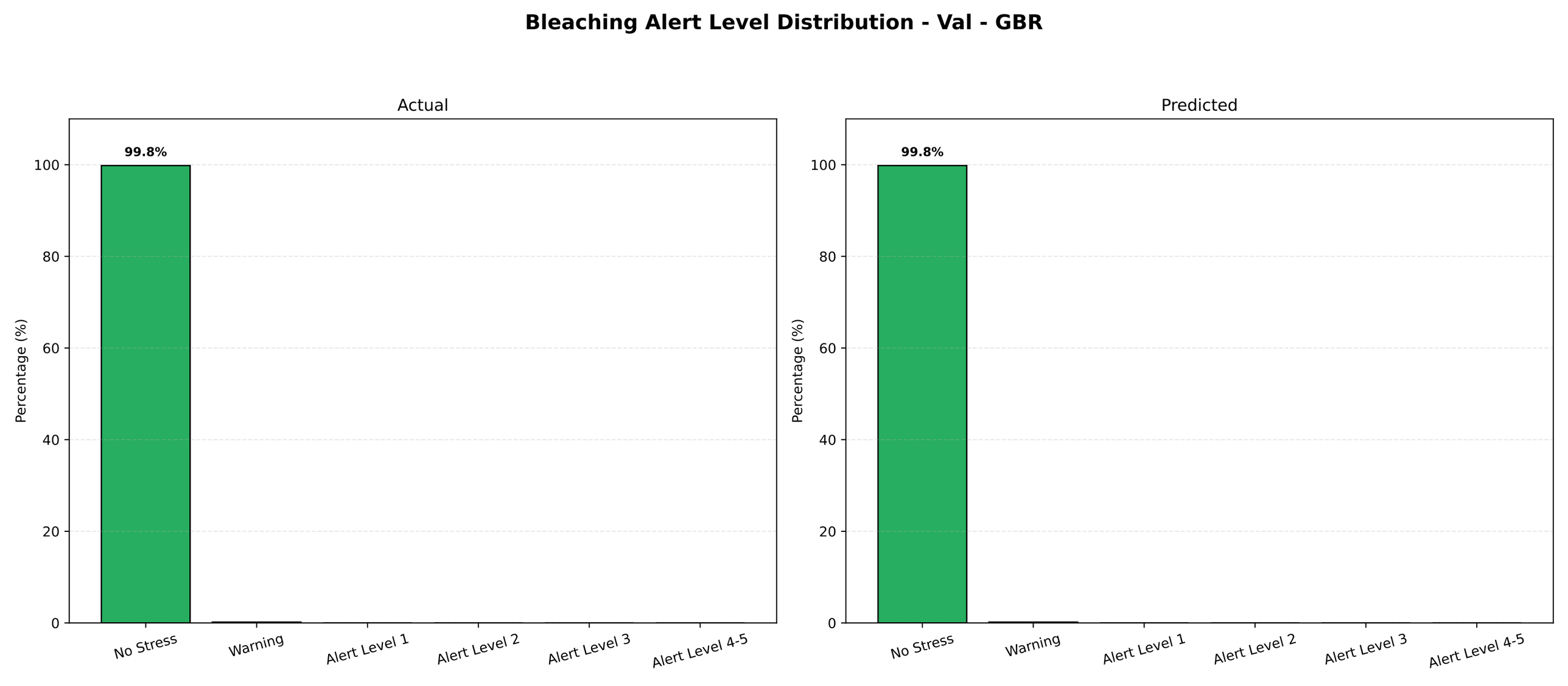






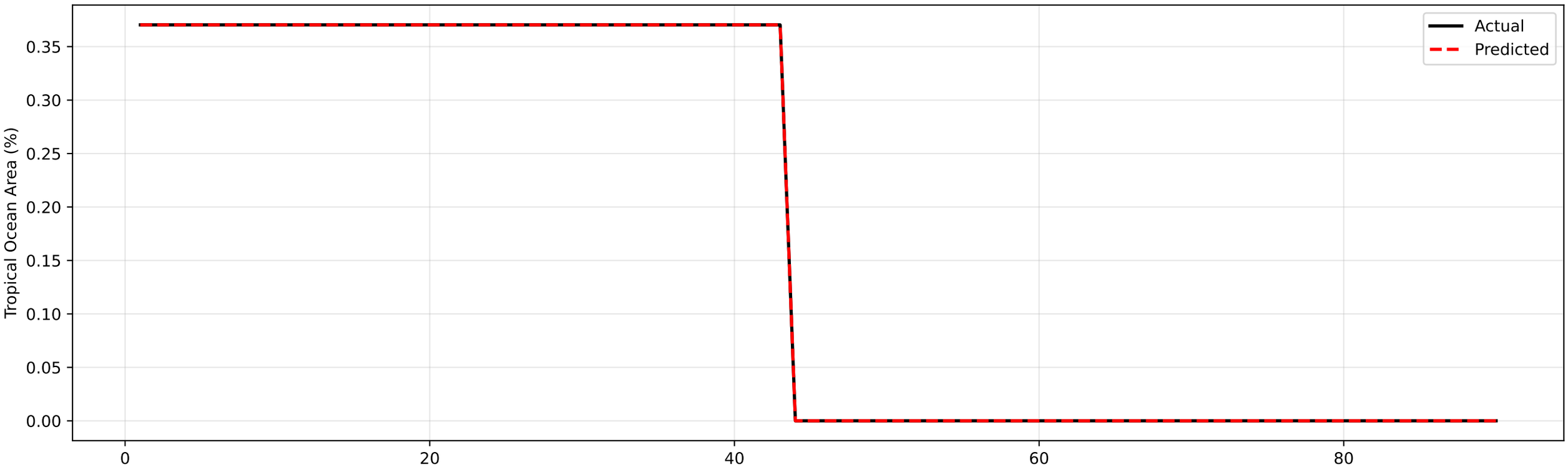



Peak DHW (max over all chains & leads) - Val - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−8
−6
−4
−2
0
2
4
6
8
Error (C-weeks)


Bleaching Alert Level Distribution - Val - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
99.8%
99.8%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)

